



**GREEN
CLIMATE
FUND**

Meeting of the Board
30 June – 3 July 2025
Port Moresby, Papua New Guinea
Provisional agenda item 11

GCF/B.42/02/Add.11

9 June 2025

Consideration of funding proposals – Addendum XI

Funding proposal package for FP269

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "Dairy Interventions for Mitigation and Adaptation (DaIMA)";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

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Disclaimer:

The designations and the presentation of the materials used in this document, including their respective citations, maps and references, have been included by the relevant Accredited Entity and do not imply the expression of any opinion whatsoever on the part of the Green Climate Fund concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Also, the boundaries and names shown, and the designations used in this document have been included by the relevant Accredited Entity and do not imply official endorsement or acceptance by the Green Climate Fund.

Funding Proposal

Project/Programme title:	Dairy Interventions for Mitigation and Adaptation (DaIMA)
Country(ies):	Kenya, Rwanda, United Federation of Tanzania and Uganda
Accredited Entity:	International Fund for Agricultural Development (IFAD)
Date of first submission:	[2024/11/28]
Date of current submission	[2025/04/18]
Version number	[V.5]



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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]”

ECT/PROGRAMME SUMMARY																																				
A.1. Project or programme	Programme	A.2. Public or private sector	Public and private																																	
A.3. Request for Proposals (RFP)	Not applicable																																			
A.4. Result area(s)	Check the applicable GCF result area(s) that the <i>overall</i> proposed project/programme targets below. For each checked result area(s), indicate the estimated percentage of GCF and Co-financers' contribution devoted to it. The total of the percentages when summed should be 100% for GCF and Co-financers' contribution respectively. <table border="1"> <thead> <tr> <th></th> <th>GCF contribution</th> <th>Co-financers' contribution</th> </tr> </thead> <tbody> <tr> <td>Mitigation total</td> <td>50 %</td> <td>40 %</td> </tr> <tr> <td>☒ Energy generation and access</td> <td>10 %</td> <td>10 %</td> </tr> <tr> <td>☐ Low-emission transport</td> <td><u>Enter number</u> %</td> <td><u>Enter number</u> %</td> </tr> <tr> <td>☐ Buildings, cities, industries and appliances</td> <td><u>Enter number</u> %</td> <td><u>Enter number</u> %</td> </tr> <tr> <td>☒ Forestry and land use</td> <td>40 %</td> <td>30 %</td> </tr> <tr> <td>Adaptation total</td> <td>50 %</td> <td>60 %</td> </tr> <tr> <td>☒ Most vulnerable people and communities</td> <td>50 %</td> <td>50 %</td> </tr> <tr> <td>Health and well-being, and food and water security</td> <td><u>Enter number</u> %</td> <td><u>Enter number</u> %</td> </tr> <tr> <td>Infrastructure and built environment</td> <td><u>Enter number</u> %</td> <td><u>Enter number</u> %</td> </tr> <tr> <td>☒ Ecosystems and ecosystem services</td> <td>%</td> <td>10 %</td> </tr> </tbody> </table>				GCF contribution	Co-financers' contribution	Mitigation total	50 %	40 %	☒ Energy generation and access	10 %	10 %	☐ Low-emission transport	<u>Enter number</u> %	<u>Enter number</u> %	☐ Buildings, cities, industries and appliances	<u>Enter number</u> %	<u>Enter number</u> %	☒ Forestry and land use	40 %	30 %	Adaptation total	50 %	60 %	☒ Most vulnerable people and communities	50 %	50 %	Health and well-being, and food and water security	<u>Enter number</u> %	<u>Enter number</u> %	Infrastructure and built environment	<u>Enter number</u> %	<u>Enter number</u> %	☒ Ecosystems and ecosystem services	%	10 %
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A.5. Expected mitigation outcome (Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)	2,169,485 tCO2 eq. (108,474 tCO2 eq./year)	A.6. Expected adaptation outcome (Core indicator 2: direct and indirect beneficiaries reached)	17,9 million (direct + indirect) 2,5 million (direct) 15,4 million (indirect) 1.3% (direct beneficiaries vis-à-vis total population ¹) 8.1% (indirect beneficiaries vis-à-vis total population)																																	
A.7. Total financing (GCF + co-finance)	358.26 million USD	A.9. Project size	Large (Over USD 250 million)																																	
A.8. Total GCF funding requested	150.00 million USD For multi-country proposals, please fill out annex 17.																																			
A.10. Financial instrument(s) requested for the GCF funding	Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8. <table border="1"> <tbody> <tr> <td>☒ Grant</td> <td>132.50 million</td> <td>☐ Equity</td> <td><u>Enter number</u></td> </tr> <tr> <td>☒ Loan</td> <td>17.50 million</td> <td>☐ Results-based payment</td> <td></td> </tr> <tr> <td>☐ Guarantee</td> <td><u>Enter number</u></td> <td></td> <td></td> </tr> </tbody> </table>			☒ Grant	132.50 million	☐ Equity	<u>Enter number</u>	☒ Loan	17.50 million	☐ Results-based payment		☐ Guarantee	<u>Enter number</u>																							
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A.11. Implementation period	6 years	A.12. Total lifespan	20 years																																	
A.13. Expected date of AE internal approval		A.14. ESS category	B																																	
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☒ No ☐																																	

A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐
A.19. Complementarity and coherence	<i>Does the project/programme complement other climate finance funding (e.g. GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1.</i> Yes ☒ No ☐		
A.20. Executing Entity information	In the four countries, the Executing Entities for the implementation of Components 1 and 2 will be: in the Republic of Kenya, acting through its National Treasury (for the channelling of the GCF Proceeds) and the State Department for Livestock Development (“SDLD”) of the Ministry of Agriculture and Livestock Development (the “MoALD”) (for the implementation of the Programme’s Activities); in the United Republic of Tanzania, acting through its Ministry of Finance (for the channelling of the GCF Proceeds) and its Ministry of Livestock and Fisheries (the “MLF”) (for the implementation of the Programme’s Activities); in the Republic of Rwanda, acting through its Ministry of Finance and Economic Planning (for the channelling of the GCF Proceeds) and the Rwanda Agriculture Board (the “RAB”) within the Ministry of Agriculture and Animal Resources (“MINAGRI”) (for the implementation of the Programme’s Activities); and in the Republic of Uganda, acting through its Ministry of Finance, Planning and Economic Development (for the channelling of the GCF Proceeds) and its Ministry of Agriculture, Animal Industry and Fisheries (the “MAAIF”) (for the implementation of the Programme’s Activities), in each case, with respect to the Activities to be implemented in the Host Country of the relevant Executing Entity, together the “National Executing Entities” or “NEEs”. The Executing Entities for the implementation of Components 3 and Output 3.1 will be: in the Republic of Kenya, acting through its National Treasury (for the channelling of the GCF Proceeds) and the Agricultural Finance Cooperation (for the implementation of the Programme’s Activities); in the United Republic of Tanzania, acting through its Ministry of Finance (for the channelling of the GCF Proceeds) and the Tanzania Agricultural Development Bank (for the implementation of the Programme’s Activities); in the Republic of Rwanda, acting through its Ministry of Finance and Economic Planning (for the channelling of the GCF Proceeds) and the Development Bank of Rwanda (for the implementation of the Programme’s Activities); and in the Republic of Uganda, acting through its Ministry of Finance, Planning and Economic Development (for the channelling of the GCF Proceeds) and the Uganda Development Bank (for the implementation of the Programme’s Activities); and in the case of Output 3.2, SDLD, MLF, RAB and MAAIF, in each case, with respect to the Activities to be implemented in the Host Country of the relevant Executing Entity. The Agricultural Finance Corporation, the Tanzania Agricultural Development Bank, the Development Bank of Rwanda and the Uganda Development Bank are together, the “Wholesale Banks”, (together, the “Executing Entities”).		
A.21. Executive summary			
1. Climate context. Observed and projected climate change scenarios (Representative Concentration Pathways (RCPs) 2.6 and 8.5) in Kenya, Rwanda, Tanzania, and Uganda indicate significant changes in precipitation and temperature patterns, as well as heat stress and other extreme weather events that will negatively impact the dairy sector. The frequency and severity of droughts are projected to increase, resulting in substantial losses of dairy livestock, affected pastures and rangelands. This combination of factors has profound implications for the dairy industry and dairy farmers in East Africa, leading to lower milk yields and production, reduced cattle health, deterioration in pasture and feed quality, and increased risks in milk losses along the value chains. In addition to the sector's existing vulnerabilities, such as farmers' lack of access to knowledge and new skills, insufficient milk collection facilities, limited access to quality inputs, natural capital and finance, climate change-related factors faced by dairy farmers in East Africa make adaptation strategies essential to its sustainability and resilience. On the mitigation side, methane (CH₄) emissions from the dairy sector dominate the greenhouse gas (GHG) emissions inventories of Kenya, Rwanda,			

¹ According to the World Bank statistics (<https://datacatalog.worldbank.org/search/dataset/0037712>), there are 54.2 million people in Kenya, 65.5 million in Tanzania and 47.2 in Uganda and 13.8 in Rwanda.

Tanzania, and Uganda. The expected growth of the dairy sector means that absolute emissions and emission intensities will continue to rise if no action is taken. The urgency of reducing the sector's GHG emissions and increasing its resilience to climate change is highlighted in national policies and plans, more particularly in the countries' Nationally Determined Contributions (NDCs). The IPCC recognizes "efficient livestock systems" as a feasible adaptation option that has synergy with mitigation. This is shown in IPCC AR6 Synthesis Report, Figure SPM.7 (4th row).

2. **Programme objectives and intervention approach.** The proposed Programme, which will be implemented in Kenya, Rwanda, Tanzania, and Uganda, aims to reduce methane and other GHG emissions from the dairy sector while increasing the resilience of livestock-dependent communities. The Programme will support a shift from low production efficiency to improved, gender-sensitive, integrated and NDC-aligned practices that reduce emissions at farm and food system levels (including interventions that reduce milk losses and increase milk collection, decarbonize processing with renewable energies, formalize dairy value chains and strengthening partnerships with the private sector), as well as interventions that promote circularity, and increase rangelands ecosystems resilience and services (including carbon sequestration in grasslands). The paradigm shift of the Programme will be accompanied by measures to strengthen policies, regulations, and institutional capacities for a low-carbon climate-resilient dairy sector at national and regional levels. To achieve this, the Programme comprises three components, each uniquely designed to promote and encourage climate change adaptation and mitigation within the dairy sector.

3. **Component 1** of the intervention focuses on initiating a paradigm shift towards a more productive, climate-resilient, and low-emission dairy sector in East Africa. The Component's strategic approach is to review and update national and regional regulations, policies, standards, and investment frameworks that promote public and private investments aligned with the NDCs, ensuring these promote gender equality and social inclusion. It also includes launching nutrition awareness campaigns on milk consumption and food safety, strengthening public and private veterinary and extension services, improving national capacities for measurement, reporting and verification (MRV) of emissions and climate risk analysis. In addition, a carbon certification system will be developed to give the sector access to carbon finance. With a focus on institutional strengthening and knowledge dissemination at the national and regional levels, the Component ensures that the sector is well equipped to manage and adapt to the changing climate landscape.

4. **Component 2** targets smallholder dairy farmers and cooperatives, with a view to a low-emission and climate-resilient dairy sector. It will focus primarily on farm-level interventions, where 90 percent of sector's emissions occur, and aim to reduce total emissions and reduce emission intensities in all dairy production systems, moving beyond the business-as-usual scenario. On these farms, it will improve the efficiency in milk production (through animal health, breeding, fodder, and feeding interventions), improve nutrient circularity and recycling through manure management (including biogas production). It will invest in pasture and grassland management to increase ecosystem resilience and carbon sequestration. Interventions also include capacity building for smallholder farmers, comprising of women, men and youth, and cooperatives, including access to markets, the use of emission reduction technologies (renewable energies) and milk aggregation to reduce losses. Primarily benefiting smallholder dairy farmers, Component 2 will encourage the adoption of productivity-enhancing and emission-reducing technologies, paving the way for these farmers to access finance through Component 3 for further growth and sustainability.

5. **Component 3** is key to driving the transformation of the East Africa's dairy sector towards sustainability and climate resilience. It encompasses two strategies: (i) launching the Green Dairy Financing Facility (GDFF) to provide green loans to climate-resilient dairy businesses, and (ii) provide high quality, well-targeted technical assistance to participating banks and the borrowing dairy-sector businesses to ensure the achievement of DaIMA's CC adaptation and mitigation targets and its sectoral transformation objectives. By addressing financial barriers, including those for women and the youth, and encouraging investment in sustainable practices, Component 3 aims to catalyze a significant shift in productivity, resilience, and reduced emissions in the East African dairy sector.

6. **Implementation arrangements.** As Accredited Entity (AE) for the Programme, IFAD will be responsible for overall management, which includes Programme appraisal, fiduciary and technical oversight, fund management, quality monitoring, and reporting to the GCF, resulting in the Programme completion and final evaluation. This role extends to IFAD's various offices, including its headquarters in Rome and regional office in Nairobi, as well as its country offices in Kenya, Rwanda, Tanzania, and Uganda. A Regional Steering Committee will ensure regional alignment and added value of the Programme. National Programme Steering Committees (PSCs) and Programme Coordination Units (PCUs) will be established in each country to oversee (for the former) and coordinate (for the latter) the implementation of the Programme. International and national entities, including ministries of agriculture and specialized entities (e.g. the International Livestock Research Institute (ILRI)) will provide targeted services to the national implementation teams.

7. **Programme costs and financing.** The total cost of the Programme is US\$ 358.26 million, of which: US\$ 58.20 million for Component 1, US\$ 234.98 million for Component 2, US\$ 45.0 million for Component 3 (Green Dairy Financing Facility), US\$ 13.36 million for Programme management, and US\$ 6.72 for monitoring and evaluation.

8. GCF will provide US\$132.50 million grants to finance Components 1, 2 and 3, as well as programme management and monitoring and evaluation (M&E) to the 4 countries, IFAD will contribute US\$193.28 million co-financing in the four countries through its national projects, while the Governments will contribute US\$14.98 million. Finally, for the Green Dairy Financing Facility under Component 3, a total of US\$45 million is allocated. Of this amount, US\$35 million will be for on-lending loan capital (GCF loan of US\$17.5 million, IFAD loan of US\$ 17.5 million).

9. **Beneficiaries and impacts.** The Programme aims to directly benefit 2.5 million people. The Programme aims to reduce the milk emission intensities by 29 percent, of which by 28 percent in Kenya, 50 percent in Rwanda, 29 percent in Tanzania and 33 percent in Uganda over a 20-year horizon. Over the same period, the Programme is expected to reduce total GHG emissions on the targeted dairy farms, while significantly increasing milk production by 34 percent, of which by 31 percent in Kenya, 73 percent in Rwanda, 36 percent in Tanzania, and 30 percent in Uganda. This increase in milk production will contribute to agricultural GDP growth in these four countries, marking a substantial impact on the economy and climate resilience of the region. Total GHG emissions from milk production are expected to decrease by 4.01 percent at the level of the Programme, or (i) by 2.02 percent (-61,967 tCO₂-eq) in Kenya, (ii) by 9.56 percent (-56,507 tCO₂-eq) in Rwanda, (iii) by 4.69 percent (-65,718 tCO₂-eq) in Tanzania and, (iv) by 5.22 percent (-82,372 tCO₂-eq) in Uganda. Methane from enteric fermentation is expected to decrease by 4.61 percent, while methane from manure management would decrease by 5.93 percent. The Programme is also expected to generate 54,435 tCO₂eq/yr savings from energy generation and access.

10. **Economic and financial analysis.** The financial analysis shows that the scenario's proposed for different types of dairy farms are financially viable, while increasing milk yields and reducing emission intensities per litre of milk. At regional level, the aggregated Economic Net Present Value (ENPV) of the countries' incremental net benefit streams, discounted at 13.25 percent (social discount rate), is US\$ 181.20 million, with an Economic Rate of Return (ERR) of 25.4 per cent for the baseline scenario. At a 6 per cent social discount rate, the ENPV is US\$ 518.0 million. Two additional economic scenarios were tested by integrating the economic value of the Programme's GHG emissions balance with a low carbon market price (LCP) and with a high carbon market price (HCP). Under the LCP scenario, the ERR is 27.90 percent and the ENPV is US\$ 217.87 million. Under the HCP scenario, the ERR is 30.30 percent and the ENPV is approximately US\$ 254.64 million. The sensitivity analysis confirms that DaIMA represents an economically worthwhile investment.

11. Programme will directly contribute to the following targeted results in alignment with the GCF's Updated Strategic Plan (USP-2); T1 (by supporting the Country's adaptation and mitigation goals), T4 Food (2,5 million beneficiaries adopting low-emission climate-resilient dairy production practices) and T5 Ecosystems (178,362 hectares of rangelands/pastures to be conserved, restored and sustainably managed) through T9 locally-led adaptation approach. The Programme will also support dissemination of climate information and MRV system, directly contributing to T3 -Climate Informational and Early Warning Systems (CIEWS). The Programme also aligns with T10 (by supporting resilient market chains) and T11 (by developing a Green Dairy Financing Facility).

B. PROGRAMME INFORMATION

B.1. Climate context

For more details, see Annex 2 (Feasibility Study)

12. Smallholder farmers dominate agriculture in East Africa. They rely heavily on livestock for their livelihoods², more particularly dairy, which are found in various production systems and diverse agroecologies, all driven by climate. The region is dominated by a tropical savannah climate, separated by a corridor of arid steppe and desert climates parallel to the Indian Ocean coastline and northwest of Kenya, as well as in the central parts of Tanzania. Tropical rainforest conditions are found in the Lake Victoria and temperate climates are present along the highest altitude areas of the Great Rift Valley bordering Kenya and Tanzania. Kenya is the driest among the four targeted countries, with a total annual rainfall ranging between 400 to 600mm in the northwest of Lake Turkana and in areas bordering Somalia. Conversely, Uganda and Rwanda have the highest total annual precipitation, ranging between 1000 and 1600mm. These conditions lead to a mosaic of climates and, consequently, to a wide range of agroecological zones.

13. The dairy sector in Kenya, Rwanda, Tanzania, and Uganda is highly vulnerable to climate change (CC) and is characterized by high GHG emissions, especially methane³. The four countries face significant CC challenges, including extreme weather events and changing rainfall patterns⁴. In the dairy sector, the impacts of CC include heat stress on dairy animals, prolonged and more recurrent droughts, intensifying heavy rainfall, flooding, and other extreme weather events that reduce dairy cow productivity. Indirect effects include changes in disease patterns and occurrence, decreased feed and water availability, and decreased pasture quality⁵. As a result, the Programme is cross-cutting and addresses both mitigation and adaptation. Section B.1 describes the adaptation needs following IPCC's AR5 Risk Conceptualization (compounded effect of observed and projected climate hazards, exposure⁶, vulnerability⁷, modulated by the existing adaptive capacity⁸), as well as the mitigation needs based on the GHG emission profile of the dairy sector.⁹

CLIMATE HAZARDS¹⁰

14. East Africa's dairy systems and their spatial distribution are diverse, exposing dairy farmers' livelihoods to a wide range of hazards. Climate change poses risks both because of the potential impacts of weather hazards and

² Shikuku, K. M., Winowiecki, L., Twyman, J., Eitzinger, A., Perez, J. G., Mwongera, C., & Läderach, P. (2017). Smallholder farmers' attitudes and determinants of adaptation to climate risks in East Africa. *Climate Risk Management*, 16, 234–245.

³ Due to its structure, methane traps more heat in the atmosphere per molecule than carbon dioxide (CO₂), making it 80 times more harmful than CO₂ for 20 years after it is released. Source: UNEP, 2022. Available online [here](#).

⁴ Ayugi, B. et al. (2022). Projected changes in meteorological drought over East Africa inferred from bias-adjusted CMIP6 models. *Natural Hazards*, 113(2), 1151–1176.

⁵ Godde, C. et al. (2021). Impacts of climate change on the livestock food supply chain; a review of the evidence. In *Global Food Security* (Vol. 28).

⁶ Exposure: presence of livelihoods, ecosystems, and assets.

⁷ Vulnerability: degree to which a system, livelihood might be negatively affected.

⁸ Adaptive capacity: capacity to respond.

⁹ A Climate Impact Potential Assessment was carried out during project preparation and more details can be found in the Feasibility Study.

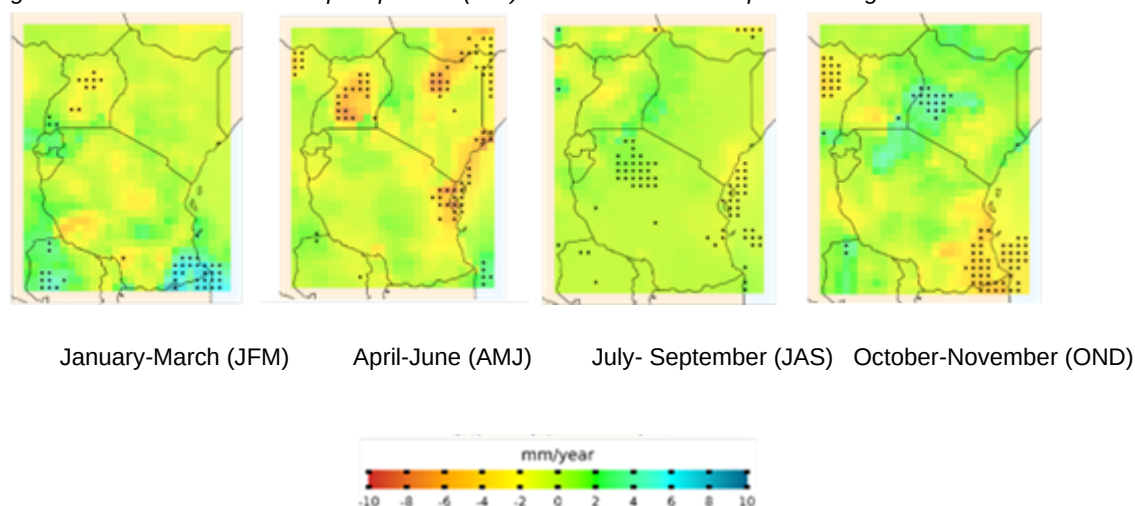
¹⁰ Cucchi, M. et al. (2020). WFDE5: bias-adjusted ERA5 reanalysis data for impact studies. *Earth System Science Data*, 12(3), 2097–2120. The following information is drawn from the W5E5 dataset for 1981–2016 (36-year dataset of daily precipitation, daily maximum and minimum temperatures, and relative humidity with a resolution of around 50 km

human responses to modulate their effects. The analysis of livestock and dairy exposure to observed and projected weather hazards is supported by geospatial datasets on flood and drought prone areas.

Observed climate change

15. **Precipitation** declined over most of the four countries during the 1950s (-6.7mm/yr.) and the 1960s (-21.8mm/yr.) but increased during the 1970s (+16.0mm/yr.)¹¹. There are also statistically significant differences on seasonal precipitation trends (p-value 0.05) across central Uganda between January and March, extending to the north of Kenya and northern Tanzania between April and June where seasonal precipitation rates have declined at a rate of -4 to -8 mm/yr. over 1981-2016. On the other hand, a precipitation increase has also been observed in the southeast of Tanzania during the austral summer (January to March) over 1981-2016 (Figure 1).

Figure 1. Changes in accumulated seasonal precipitation (mm) over the 1981- 2016 period using the W5E5 dataset¹²



16. **Temperature.** The annual average maximum temperature (mean Tmax) has increased significantly (p-value 0.05) across Kenya, Uganda, and Tanzania. In Uganda (Eastern and Northern region) and Kenya (north of Lake Turkana County), for example, the annual mean Tmax has increased by +0.030 to +0.035°C/yr., equivalent to +1.1 to +1.3°C during 1981-2016. In addition to having the highest mean Tmax increase, the Kenyan north Rift Valley has the warmest mean Tmax year-round among the four targeted countries. There is also a significant change in average daily minimum temperature (mean Tmin) in Kenya and Tanzania, but not in Rwanda or the west of Uganda. For mean Tmin, the increase is observed in the east of Uganda and the northwest of Kenya where temperatures have increased by +0.035°C/yr., equivalent to +1.3°C during 1981-2016.

17. **Mean relative humidity.** The Rift Valley in Kenya has the lowest average annual relative humidity (mean RH <50 percent), while central and southern Uganda, especially the areas bordering Lake Victoria, have the highest mean RH (75-85 percent). The Kenyan and Tanzanian coastlines are characterized by high mean RH (70 to 80 percent) all year round due to the incoming moisture from the Indian Ocean. A statistically significant decrease (p-value 0.05) in mean RH between -0.10 and -0.25 percent/year was observed in most areas of Uganda and Rwanda between 1981 and 2016. In addition, eastern and northeastern Kenya have experienced a statistically significant decrease (p-value 0.05), while central and southern Tanzania have experienced the largest decline (-0.10 to -0.15 percent/year).

18. **Rainfall variability.** The region's climatic seasons are influenced by a coupled transboundary ocean-atmosphere phenomenon called the Inter Tropical Convergence Zone (ITCZ). A long rainy season occurs from March to May (MAM), and a short rainy season from October to December (OND) when the ITCZ migrates southwards. There is high rainfall variability in OND due to large-scale oceanic-atmospheric patterns such as the El Niño Southern Oscillation (ENSO) and Indian Ocean Dipole (IOD), which frequently coincides with ENSO events. The relationship

¹¹ Ongoma, V., & Chen, H. (2017). Temporal and spatial variability of temperature and precipitation over East Africa from 1951 to 2010. *Meteorology and Atmospheric Physics*, 129, 131-144.

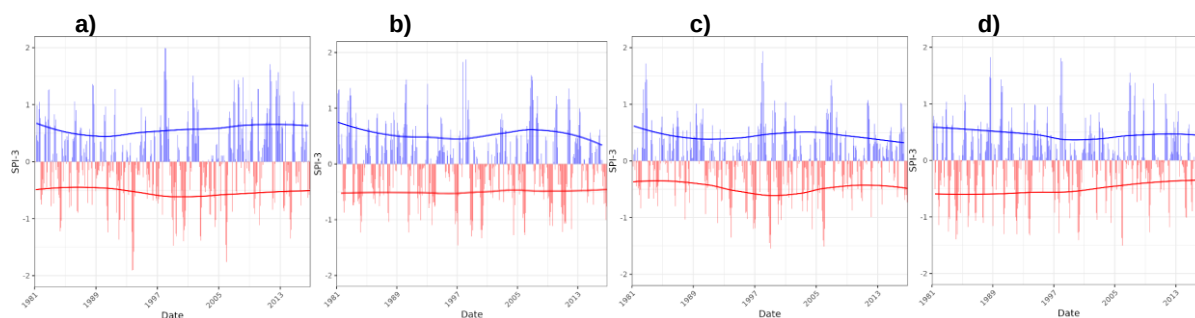
¹² Source: Annex 2: Feasibility Study

between the 3-month Standardized Precipitation Index (SPI-3) and ENSO is significant for understanding precipitation variability in East Africa (Figure 2). SPI-3 categorizes precipitation conditions, indicating whether they are normal (-0.99 and +0.99), moderately dry (-1.49 to -1.0) and moderately humid (+1.0 to +1.49), severely dry (-1.99 to -1.5) and severely humid (+1.5 to +1.99), and extremely dry (≤ -2.0) and extremely humid ($\geq +2.0$). During the strong El Niño of 1997-1998, SPI-3 values indicated very humid conditions, while the subsequent strong La Niña of 1999-2000 resulted in severe dry conditions over East Africa. The impacts of ENSO are apparent, since peak ENSO events occur between October and March, coinciding with the short (October to November) and long (March to May) rainy seasons in East Africa, further amplifying precipitation anomalies.

Extreme weather events

19. **Dry-spells and very wet days.** The average annual number of dry days ($R \leq 1\text{mm/day}$) in East Africa exceeds 300 days in the northern Rift Valley and northeastern Kenya. Conversely, the annual number of dry days decreases toward Lake Victoria (including Mount Elgon in Uganda and Kenya) and the tropical rainforest near the Democratic Republic of Congo (DRC). On average, Tanzania records 180 dry days/yr. along its border with Uganda and Rwanda and over 280 days in its interior. The Indian Ocean coastline has seen an increase in dry days by +1.0 days/yr., equivalent to +36 dry days per year since 1981. In eastern Uganda, the number of dry days by -0.50 days/yr., equivalent to +18 wet days on average each year since 1981. Trends on very wet days ($R \geq 20\text{mm/day}$) show statistically significant differences ($p\text{-value} < 0.05$) along the Great Rift Valley in Kenya and Tanzania, with an average annual increase of +0.15 days/yr. over the period 1981-2016, equivalent to an additional +5 very wet days on annual average since 1981 (Figure 2). In contrast, neither the interior of Kenya, Tanzania nor Uganda have seen an increase or decrease in the number of very wet days since 1981.

Figure 2. Spatiotemporal analysis of SPI-3-month over the 1981-2016 period in (a) Kenya, (b) Rwanda, (c) Uganda, and (d) Tanzania¹³



20. **Extreme temperatures.** Most of the change in the number of very warm days ($T_{\text{max}} \geq 35^\circ\text{C}$) on annual average is reported in the northern Rift Valley, where the number of very warm days has increased by +1.2 to +2.8 days/yr., which is equivalent to an additional 40 to 100 very warm days as an annual average since 1981. In addition, northern Uganda experienced an increase in very warm days at a rate of +0.8 days/yr. although this increase is not statistically significant.

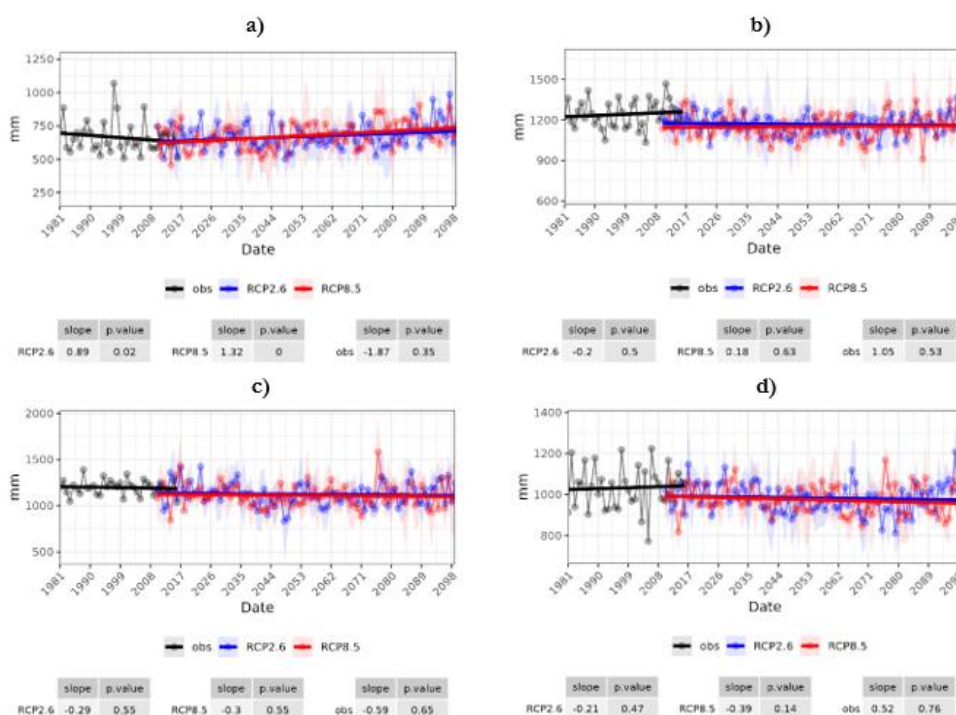
Projected climate change¹⁴

¹³ Source: Annex 2: Feasibility Study

¹⁴ Future climate projections were based on two representative concentration pathways (RCPs) namely RCP 2.6 (low emission scenario) and RCP 8.5 (high emission scenario). Climate and Agriculture Risk Visualization and Assessment (CAVA) is an approach to climate information with a focus on agriculture. Its goal is to provide long-term, climate services by democratising access and simplifying the use of past and future climate and impact data. CAVA analytics uses the W5E5 merged dataset over the 1976-2005 period to bias-correct the outputs of atmospheric models. More information is available here: <https://fao-cava.predictia.es/>

21. **Annual precipitation.** Some parts of East Africa may experience an increase in rainfall in the distant future (Figure 3) under various Representative Concentration Pathways (RCPs)¹⁵. Arid and semi-arid regions are expected to receive less precipitation, while the highlands and lake regions may receive more rainfall. Dry seasons in Kenya, Tanzania, and Uganda could see an increase in precipitation. Future projections indicate an eastward increase in Kenya under RCP 2.6 (+100mm), accentuated under RCP 8.5 (+100 to +200mm) in the distant future (2070-2099). In Rwanda, a decrease in rainfall is expected under RCP 2.6 (-0.20mm/yr.) and an increase under RCP 8.5 (+0.18mm/yr.). In Uganda, climate projections show a decrease in rainfall of -0.30mm/yr. under both RCPs (Figure 3).

Figure 3. Total annual rainfall (mm) projections under RCP 2.6 and 8.5 in (a) Kenya, (b) Rwanda, (c) Uganda, and (d) Tanzania¹⁶.



22. **Mean maximum temperatures.** Under RCP 2.6, the mean Tmax are expected to peak in the mid-term (2040-2069); whereas under RCP 8.5, the highest mean Tmax increase is expected by the end of the century (2070-2099). Under RCP 8.5, mean Tmax may increase by up to +4.5°C in central parts of East Africa by the end of the century (2070-2099). The rate of increase in both RCPs is slower (+1 to +2°C depending on the RCP) towards the ocean since water can hold more heat than air, thereby taking longer to warm up than continental parts. Accordingly, the coastal areas of Kenya and Tanzania may experience a mean Tmax increase of +0.5 to +1.0°C (RCP 2.6) and +2.0 to +3.0°C (RCP 8.5), respectively, by the end-of-century (2070-2099) compared to the baseline period (1976-2005). Similarly, because of its large size, Lake Victoria may warm more slowly than its surroundings and, thus, have a lower rate of mean Tmax increase over time. Overall, most models agree on a positive climate change signal for mean Tmax (Figure 4).

Figure 4. (Left) Projected Changes in Maximum Temperature (annual). (Right) Projected change in Mean Annual Minimum Daily Temperatures (mean Tmin; °C) over the 21st century from historical period (1976-2005)¹⁷

¹⁵ Ayugi, B. et al. (2022). Projected changes in meteorological drought over East Africa inferred from bias-adjusted CMIP6 models. *Natural Hazards*, 113(2), 1151-1176.

¹⁶ Source: Annex 2: Feasibility Study

¹⁷ Source: Annex 2: Feasibility Study



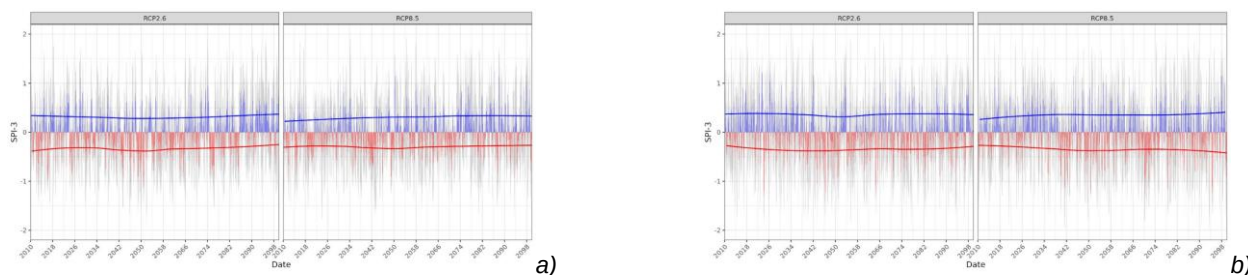
23. **Mean minimum temperatures.** The mean Tmin increase over East Africa is expected to follow the same spatiotemporal pattern as mean Tmax. However, the rate of change in mean Tmin is slightly higher to that of mean Tmax. Overall, mean Tmin could increase by +1.5 to +2.0°C (RCP 2.6) and by +4.0 to +5.0°C (RCP 8.5) inland and by less than +1.5°C (RCP 2.6) and +3.0°C (RCP 8.5) on the Tanzanian and Kenyan coast by 2070-2099 compared to 1976-2005.

24. The **Mean relative humidity** (mean RH) will change over the course of the century for RCPs 2.6 and 8.5. The annual mean RH (+2 %) is expected to increase along Kenya and Lake Victoria. Under RCP 8.5, however, the mean RH is expected to decrease on a yearly average (-2 to -8 %) in the central parts of Tanzania, with a high rate towards the south.

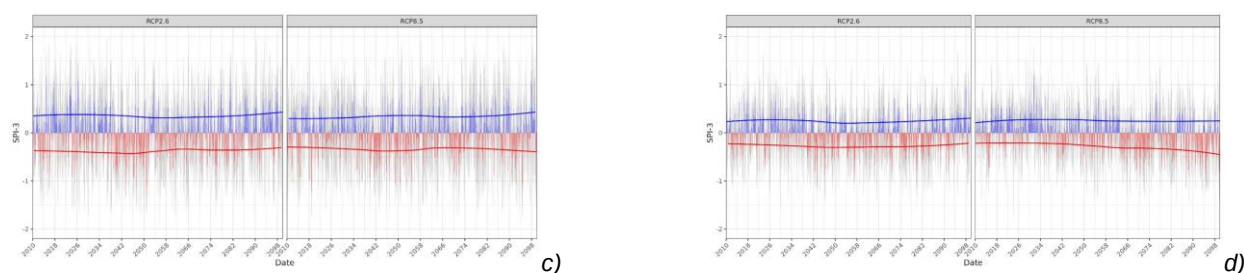
25. **Rainfall variability.** In all countries, positive SPI-3 values will remain lower under RCP 8.5 than RCP 2.6, implying fewer wet-years under RCP 8.5. For negative SPI-3 values (dry periods), negative trends are accentuated in Rwanda, Uganda, and Tanzania, implying that 3-month precipitation values are increasingly more dispersed from the mean.

26. **Dry spells.** The models show a positive agreement in the time of emergence on the number of dry days ($R \leq 1\text{mm/day}$), slightly earlier and widespread for RCP 8.5 (2030-2050) compared to RCP 2.6 (2040-2060). The positive change (increase in the number of dry days) is widespread throughout East Africa, particularly towards western Kenya, all of Uganda and Rwanda, and most parts of Tanzania. Conversely, the timing of the emergence of negative values (reduction in number of dry days) is projected from 2060 onwards for RCP 8.5 and a little earlier for RCP 2.6, mainly in eastern and southern Kenya. In addition, most of the increase in very wet days ($R \geq 20\text{mm/day}$) is projected in southwestern Kenya under RCP 2.6, where there could be an extra +1 to +2 very wet days on average per year by the end of the century (2070-2099). For RCP 8.5, the increase in very wet days is widespread over Uganda, Rwanda, Tanzania, and southwestern Kenya (Figure 5).

Figure 5. Analysis of SPI-3-month over the century under RCPs 2.6 and 8.5 in a) Kenya, b) Rwanda, c) Uganda, and d) Tanzania¹⁸

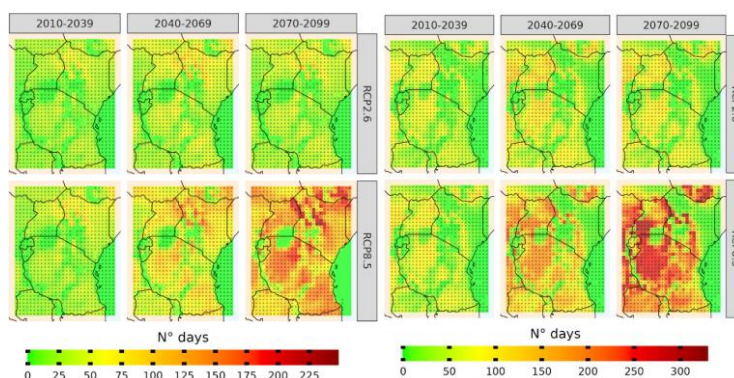


¹⁸ Source: Annex 2: Feasibility Study



27. **Extreme temperatures.** Very warm days ($T_{max} \geq 35^{\circ}\text{C}$) are expected to increase in frequency and intensity (Figure 6). In the low emission scenario (RCP 2.6), +25 to +75 additional very warm days on average per year are expected in Uganda, the Rift Valley of Kenya, and central Tanzania in the mid-term (2040-2069). Most of the increase in very warm days is projected by the end of the century (2070-2099) under RCP 8.5, showing an increase in the range of 100 to 200 additional very warm days on average per year compared to 1981-2010. On the contrary, the cooling effect of large water masses such as Lake Victoria and the Indian Ocean can significantly modulate the upward trend of very warm days over time.

Figure 6. Climate change signal in the number of (left) very warm days ($T_{max} \geq 35^{\circ}\text{C}$) on annual average and (right) days with tropical nighttime conditions ($T_{min} \geq 20^{\circ}\text{C}$) on annual average over the 21st century from historical period (1976-2005)¹⁹



EXPOSURE

28. Exposure is the presence of people, livelihoods, species, and ecosystems that could be affected by hazards, more particularly droughts and floods that cause disasters (Figure 7).

29. In East Africa, **droughts** tend to be more frequent, longer, and more severe in boreal spring and summer, as rainfall and water storage decrease overall. Table 1 shows how drought will change under different climate scenarios. Haile et al. (2020) show that the drought area in East Africa will increase, with extreme droughts increasing faster than severe and moderate droughts. For example, by the end of the century (2080s), severe droughts are expected to increase by 13.5 percent under RCP 8.5. In general, farmers can cope with mild droughts, but severe droughts lead to large-scale livestock losses, decreased pasture regeneration, and an increased prevalence of drought-related livestock diseases.²⁰ Overall, droughts affect the availability and quality of water for pasture regeneration, feed production, water and hygienic practices in the dairy production system.

¹⁹ Source: Annex 2: Feasibility Study

²⁰ Huho, J. M., & Mugalavai, E. M. (2010). The Effects of Droughts on Food Security in Kenya. The International Journal of Climate Change: Impacts and Responses, 2(2).

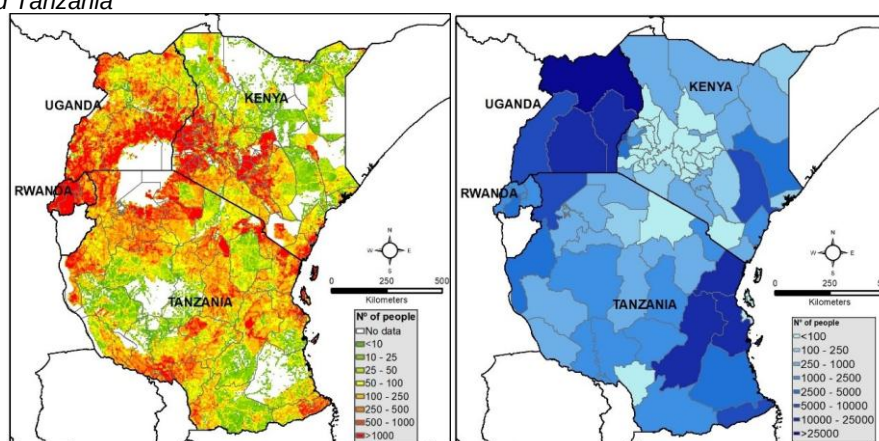
30. **Floods.** East Africa experiences frequent flooding due to climate variability and anthropogenic factors that exacerbate livestock exposure to flooding (Figure 7). Among its flood-prone areas, Uganda has the highest number of flood-related casualties, particularly in the Northern, Eastern, and Central regions along the White Nile River. Each year, floods affect between 10,000 and 25,000 people per administrative boundary in these regions of Uganda, with up to 36,000 people per administrative boundary in the Northern region. In Tanzania, floods affect 10,000 to 25,000 people per administrative boundary in Tanga, Morogoro, and Pwani. In contrast, in the central parts of Kenya, fewer than 100 people per administrative boundary were affected by floods on an annual basis. Due to the increase in population and increased rainfall in Lake Victoria under RCP 8.5, flooding along the White River Nile is likely to increase.

31. **Disasters.** In East Africa, between 1971 and 2023, more than 95 million people were affected by disasters caused by meteorological (storms), hydrological (floods, landslides) and climatological (droughts)²¹ events. Since 1971, 248 disasters have occurred, of which 78 disasters in Kenya, 70 in Tanzania, 60 in Uganda, and 40 in Rwanda. About 69 percent related to floods and 17 percent to drought. Over time, the number of disasters has increased exponentially, from 28 during 1971-1990 to 174 during 2001-2020. The most common disaster is flooding, followed by drought and landslides. Between 1900 and 2023, 176 floods were recorded in the targeted countries, most of them river floods caused by heavy rainfall. Evidence shows that women and men are affected by drought differently, and that gender inequalities diminish women's capacity to cope with these calamities.

Table 1. Projected changes (%) in drought over East Africa²²

Projected changes	RCP 2.6			RCP 8.5		
	2020s	2050s	2080s	2020s	2050s	2080s
Change in drought area (%)	9%	17%	16%	14.5%	41.5%	39%
Change in moderate drought (%)	7.5%	12%	12%	10%	17%	11.6%
Change in severe drought (%)	1.4%	3.8%	3.5%	3.4%	12.3%	13.5%
Number of drought events	3	4	4	3	5	3

Figure 7. (Left) Estimated number of people exposed to drought; and (Right) number of people affected by floods in Kenya, Uganda, Rwanda, and Tanzania



VULNERABILITY

32. CC also poses many challenges and threats to the East African dairy sector, which is subject to pre-existing physical and socio-economic vulnerabilities. By reducing these vulnerabilities, which will be exacerbated by climate change, the sector becomes more resilient. These vulnerabilities are presented below:

²¹ Centre for Research on the Epidemiology of Disasters (CRED) database (2023). 2022 Disasters in Numbers. Available at: [link](#)

²² Based on Haile et al. (2020)

- a) **Poverty and income inequality.** Kenyan, Tanzanian, Rwandan and Ugandan societies are marked by important poverty and income inequality. The Human Development index (HDI) 2022 ranking in Kenya (146th), Rwanda (159th), Uganda (159th) and Tanzania (160th) reflect low achievements in some of the key dimensions of human development, while high GINI indices (38.7 in Kenya, 43.7 in Rwanda, 40.5 in Tanzania and 42.7 in Uganda)²³ evidence strong income inequalities. The vulnerable – more particularly poor dairy farmers and women²⁴ - will be the most affected by climate change.
- b) **Degraded of rangelands and ecosystems.** The availability of feed depends on the quality of rangelands, which are degraded in the region. Figure 8 shows that degraded land accounts for 27 percent of total rangelands areas in Tanzania, 3.6 percent in Kenya, 2.1 percent in Uganda and 2.1 percent in Rwanda. The main factors leading to the degradation of rangelands and pasture are soil structure and erodibility, climate (rainfall intensity and temperature), reduction of vegetation cover and topography (influencing water runoff capacity). Degradation prevents rangelands from providing ecosystem services such as *provisioning* (production of above-ground biomass, such as grass, as well as water) and *regulating* (such as the capacity of rangelands to sequester carbon in the soils²⁵). It also has a negative impact on the forage balance, as well as on the livestock carrying capacity, which highlights the pressure that the land is under in terms of maximum number of animals that the ecosystem can sustainably support over time, without causing damages to the environment (*Feasibility Study, Annex 2*).
- c) **Limited access to quality inputs.** Feed that reduces enteric emissions, veterinary services and climate-resilient breeding stocks are difficult to access for dairy smallholder farmers. Limited veterinary services lead farmers to prefer local or crossbred animals that are not as high yielding, whereas exotic breeds are highly susceptible to local disease. In addition, limited access to improved breeds hinders genetic progress.
- d) **Limited water availability.** Climate change affects water resources with changes in precipitation patterns leading to water scarcity or excessive water supply in different regions. In food insecure areas, limited access to safe and reliable water resources, and limited water harvesting systems can hinder livestock and dairy production, contributing to food insecurity and increased vulnerability, especially for women who often provide labour for livestock.
- e) **Frequent disease outbreaks.** Foot-and-mouth disease, mastitis, and East Coast Fever (ECF) are common livestock diseases. Disease outbreaks result in a high mortality rate, reduced milk production, and increased veterinary costs. Poor disease control measures, inadequate quarantine facilities, and limited veterinary services contribute to the prevalence of diseases in the dairy farming sector.

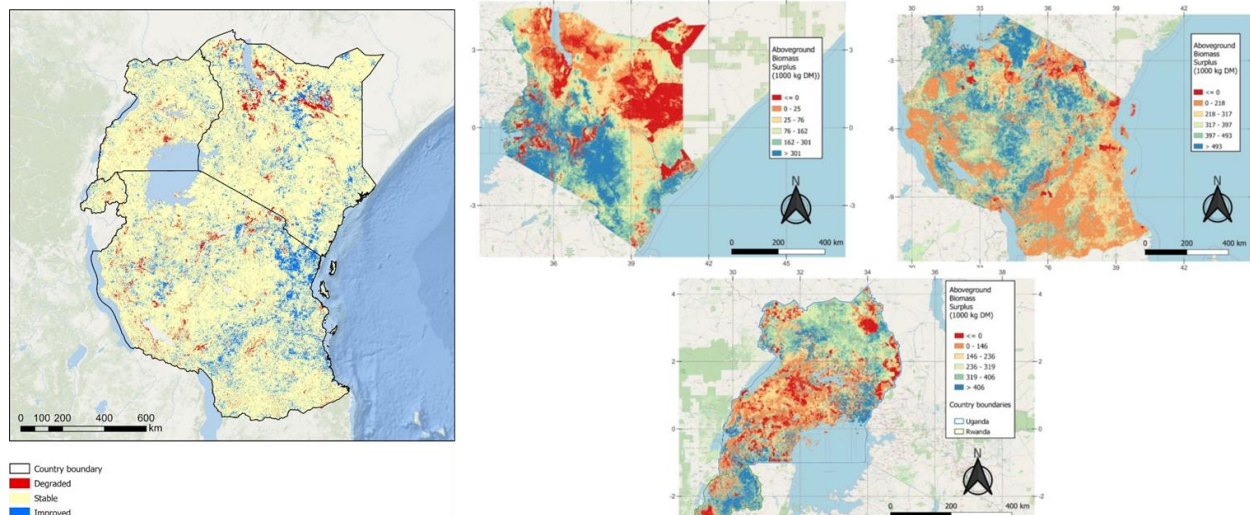
Figure 8. Level of degradation of rangelands and pastures in the four countries between 2012 and 2020 (left) and carrying capacity, expressed in above-ground biomass surplus (ton of dry matter) (right)²⁶

²³ World Bank, <https://data.worldbank.org/indicator/SI.POV.GINI>

²⁴ FAO. 2024. *The unjust climate – Measuring the impacts of climate change on rural poor, women and youth*. Rome.

²⁵ Sub-Saharan Africa has been estimated to have the highest potential for soil carbon storage (0.41 tC/ha/yr) compared to other regions of the world. The potential annual increase is estimated at 0.57, 0.53, 0.52, and 0.51tC/ha/yr, respectively, for Kenya, Tanzania, Rwanda, and Uganda

²⁶ As analyzed in Annex 2: Feasibility Study



- f) **Dairy processing facilities, energy supply and access to markets.** Dairy farmers in East Africa face inadequate infrastructure along the dairy value chain. The availability of reliable (and sustainable) energy supply is also irregular, which affects the ability to cool and process milk, and reduce losses. Additionally, price fluctuations and market volatility are common challenges. Most milk prices are determined by bulkers and processors. Access to stable higher-value-added markets is further hindered by limited outreach of milk collection centers (MCCs), dairy cooperatives and business alliances. More details are provided in the Market Assessment (Annex 2).
- g) **Limited access to credit and financial services.** Critical aspects of dairy farming, such as infrastructure, animal health, and feed, require access to credit and finance. Small-scale dairy farmers and pastoralists, especially women and young people, have difficulties accessing these services, due to lack of collateral, which limits investments in modern equipment, infrastructure, and improved livestock.
- h) **Land tenure.** In Kenya, Tanzania and Uganda, a main risk for intensive grazing is tenure insecurity, i.e. tenure rights or land ownership. Access to natural capital, particularly women's and youth ownership, management and control of land and land resources are major barriers to sustainability, and the adoption of climate-resilient and low-carbon practices (Annex 2).

CLIMATE ADAPTIVE CAPACITY

33. About 70-90 percent of East African agriculture is rain-fed, with small-scale farmers relying on seasonal rainfall for feed, fodder crops and pasture. As dairy systems are subject to significant seasonal, temporal, and spatial variations, dairy farmers face extreme difficulties in planning investments, which also become riskier under CC. The resilience of the dairy system depends on its ability to adapt to different climate conditions and to cope with the impacts of CC. Identifying adaptation needs facilitates prioritizing actions and allocating resources for addressing critical challenges.

34. Rising temperatures are disrupting agriculture, livestock and dairy, the economic backbone of East Africa. Water scarcity, pest outbreaks, and declining productivity threaten food and nutrition security. Droughts and deforestation exacerbate soil erosion. The intensification and frequency of droughts will hinder access to water under different climate scenarios. Areas affected by extreme drought will expand faster than those affected by severe and moderate droughts, especially under RCP 8.5. Storms and floods have become more frequent and intense, causing deaths, damage to infrastructure and loss of livelihoods. The dairy sector's adaptive capacity stems from the following:

- a) **Access to climate-smart²⁷ practices and technologies.** Improved dairy breeds, drought-tolerant forage varieties, and efficient milk processing equipment are some of the climate-smart technologies farmers need. These practices and technologies should be labor saving to reduce workloads for women.
- b) **Capacity building and knowledge transfer.** Farmers' technical knowledge and skills in dairy management can be enhanced by providing training programmes, Livestock Farmer Field Schools (L-FFS), and extension services on sustainable feeding practices, breeding strategies, disease control measures and water resource efficiency.
- c) **Financial and market support.** Affordable financing supports farmers' investments in productive assets, infrastructure, and value-addition. Dairy cooperatives and value chain linkages can improve farmers' incomes. Agribusiness can provide input and output markets that improve the productivity of smallholder dairy farmers²⁸.
- d) **Strengthening infrastructure.** Improving rural infrastructure, such as water harvesting systems, renewable energy, milk collection and processing facilities, will reduce transport costs and improve market access for dairy farmers, and reduce the drudgery associated with fetching water by women.
- e) **Climate information.** Informed decisions on feed management, breeding and disease control can be made through climate information, early warning systems, and advisory services. This will help dairy farmers anticipate and adapt to climate-related risks. These skills equip youth as agents and service providers in dissemination of new information.
- f) **Policy support and institutional strengthening.** To enable dairy farmers to thrive, it is imperative to develop policies, regulations, and institutions that are tailored to their needs, especially addressing barriers related to gender inequalities and youth, as well as strengthening research and extension actors. Key adaptation requirements require collaboration between governments, research institutes, dairy cooperatives, and financial institutions.

IMPACT OF CLIMATE CHANGE

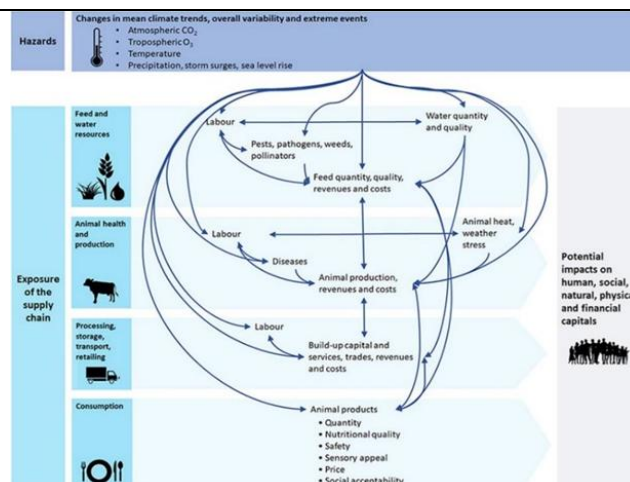
35. East Africa's dairy sector is one of the most vulnerable sectors to climate change, the impact of which are felt at all levels of the dairy value chain (Figure 9). The impacts at the dairy farm level are as follows: (i) a decline in milk yields, especially of high milk-yielding breeds; (ii) heat stress in dairy cattle; (iii) increased development of pathogens, diseases, parasite and vectors in dairy cattle; (iv) decreased feed production and productivity, due to reduced or excess soil moisture and fertility; (v) a decrease in the quantity and quality of pasture; (vi) deterioration of fodder in an unprotected warehouse. At the rangeland level, livestock owners face: (i) reduced access to rangelands resources; (ii) ecosystem degradation; (iii) a decrease in rangeland and pasture productivity and forage quality. At the dairy value chain level, high temperatures compromise the safety of milk and dairy products, higher risks of milk losses, and higher costs per unit of produce.

Figure 9. Potential impacts of climate hazards on the livestock land-based food supply chain²⁹

²⁷ According to FAO, climate-smart agriculture is "an approach to help the people who manage agricultural systems respond effectively to climate change" by pursuing "the triple objectives of sustainably increasing productivity and incomes, adapting to climate change, and reducing greenhouse gas emissions". For more details, please see <https://www.fao.org/climate-smart-agriculture/en/>

²⁸ Omondi, I. et al. (2017). Processor Linkages and Farm Household Productivity: Evidence from Dairy Hubs in East Africa. *Agribusiness*, 33(4)

²⁹ Godde, C. et al. (2021). Impacts of climate change on the livestock food supply chain; a review of the evidence. In *Global Food Security* (Vol. 28).



36. **Climate change and milk yields.** Temperature and humidity have a significant impact on milk production (*Annex 2, Feasibility Study, Chapter 2*). Significant milk production losses have been observed in Kenya, Uganda, and Tanzania's coastal regions between 1981 and 2016, from 4 to 8 kg/day for the Holstein-Friesian breed and between 0.5 and 1 kg/day for the Boran breed. Over the period 1981-2016, milk production declined less than 2 kg/day for the Holstein-Friesian breed in the high-altitude areas of Kenya, Tanzania, and Rwanda. Future milk production is expected to decrease up to 1 kg/day for Holstein-Friesian cows and up to 0.15 kg/day for Boran animals by the end of the century under RCP 2.6, and even up to 2.5 kg/day for Holstein Friesian cows and 0.5 kg/day for Boran cows under RCP 8.5³⁰. Heat stress also causes reproductive problems (missed mating or insemination).

37. **Ruminant feed and forage supplies are most affected by climate change due to changes in cereal, forage, and rangeland productivity.** Changes in temperature and CO₂ levels will affect the composition of pastures by altering the species competition dynamics due to changes in optimal growth rates³¹. In addition, availability of rainfall determines rainfed crop production. Feed and fodder are available in the rainy and early dry seasons and sparse in the dry season, leading to strong seasonal fluctuations in milk production and milk prices. Even during the rainy seasons, there is often not adequate forage (fodder and pasture) to be stored for utilization during the dry season³². Feed availability fluctuates throughout the year, challenging livestock production. It can also lead to conflicts between communities and competition with wildlife in pastoral areas. Pastoral communities are more vulnerable to feed shortages because of droughts, degradation of rangelands, and limited access to grazing lands. Both RCPs predict the greatest yield change in the north of Kenya and Uganda in the mid- and long-terms (2031-2060 and 2061-2090). Under RCP 2.6, maize yields, commonly used as feed, are likely to decline by 5 to 15 percent in Uganda and by 20 to 25 percent in Kenya under RCP 8.5. Napier grass yields are likely to decline mostly in Uganda (e.g., North region by more than 30 percent), the north of Kenya by 30 percent, and the southeast of Tanzania by 20 to 30 percent. Under RCP 8.5, Napier grass yield losses are up to 10 percent lower in the southwest of Tanzania both in the mid-term (2031-2060) and into the future (2061-2090). For maize, models show that Africa is likely to suffer up to 10 percent yield loss by 2055 due to CC.

38. **Climate change will affect pathogens, vectors, hosts, or reservoirs,** as well as human behaviour. Estimating climate change effects on livestock diseases is difficult due to the complex interactions between climatic and non-climatic factors. Temperature and humidity play a significant role in pathogen and parasite development. Wind patterns affect the movement of locusts and pathogens. Pathogen propagation may be limited by hot and dry conditions, changes in rainfall and temperature regimes which influence disease vector distribution, such as ticks, mosquitoes and flies. Vector dynamics are also influenced by soil and vegetation type, and topography. In most cases,

³⁰ For further details see section 7.2.2. of the Climate Impact Potential Assessment

³¹ Rojas-Downing, M.M., Nejadhashemi, A.P., Harrigan, T. and Woznicki, S.A., 2017. Climate change and livestock: Impacts, adaptation, and mitigation. *Climate risk management*, 16, pp.145-163. <https://www.sciencedirect.com/science/article/pii/S221209631730027X>

³² Mudavadi, O. et al. (2020). Effects of Season Variation on Water, Feed, Milk Yield and Reproductive Performance of Dairy Cows in Smallholder Farms in Eastern Africa. *Journal of Agriculture and Ecology Research International*.

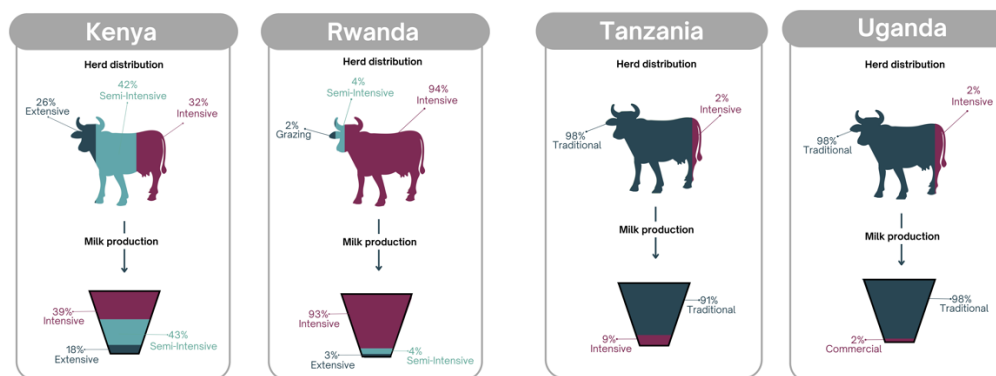
vectors and pathogens negatively affect livestock. Zoonotic diseases spread during droughts and feed shortages. Further details are available in Annex 2, Chapter 2.

39. **The above impacts of climate change on milk production at farm level will have a direct impact on national and household food and nutrition security.** At household level, less milk will be available for dairy consumption, with an immediate impact on nutrition and health of young children and women. It will also reduce revenues from sales of milk. In the four countries, small daily revenues from the sales of surplus milk, often 1 to 5 litres per household, are essential for rural poverty reduction. At national level, a reduction of domestic milk production will lead to higher imports from the international market and require monetary reserves. Also, a higher dependence on the international market to purchase feed, in particular maize and soybean, will impact the national balance of trade.

GREENHOUSE GAS EMISSION PROFILE OF THE DAIRY SECTOR

40. **Dairy production systems.** In Kenya, 39 percent of milk is produced by intensive systems or zero-grazing systems, 43 percent by semi-intensive systems, and 18 percent by extensive systems. In Rwanda, 93 percent of the milk is produced in intensive/zero grazing systems, 4 percent in semi-intensive and 3 percent in extensive. In Tanzania, dairy systems are traditional and intensive. The traditional system is characterized by meat-milk production and large herds of cattle grazing natural pastures, producing 91 percent of the country's milk. In Uganda, systems are also traditional and intensive (Figure 10). Traditional systems produce 98 percent of the country's milk. Further disaggregation of systems in Annex 2 (Chapter 2.2).

Figure 10. Livestock production systems in East Africa³³



41. **Absolute GHG emissions (baseline, present and future).** Methane (CH₄), nitrous oxide (N₂O) and carbon dioxide (CO₂) are released by ruminants (cattle, sheep, goats) during the enteric fermentation and through manure production.³⁴ East Africa represents about 5% of GHG emissions of the global cattle sector, but they are increasing. Enteric methane emissions from the dairy sector dominate the GHG emission inventories of Kenya, Rwanda, Tanzania and Uganda, followed by cattle manure. Enteric methane is not only a short-lived climate pollutant, but it represents a waste in the production process that doesn't result in milk or meat. In addition, cattle manure releases N₂O and CH₄. Over 90 percent of dairy sector emissions occur on farm³⁵. Land-use changes and degradation further contribute to GHG emissions due to the disruption in carbon stocks.

³³ Prepared under Feasibility study 1.

³⁴ The production of CH₄, a potent GHG with a 27 times higher global warming potential than carbon dioxide over the long-term - 100-year time period (IPCC, 2021), significantly contributes to the region's carbon footprint.

³⁵ See studies produced by Bain and Company (citing FAO), in the context of the project formulation studies.

42. In Kenya, GHG emissions have increased from 57 MtCO₂eq in 1955 to 94 MtCO₂eq in 2015³⁶. Agriculture produces half of all emissions. The GLEAM³⁷ tool of the Food and Agriculture Organisation of the United Nations (FAO), estimated that dairy cattle (4.5 million animals) emit 6.4 MtCO₂ eq (based on 2019 data). The dairy sector emits the most CH₄ and N₂O at 93 percent and 6.6 percent respectively. About 82 percent of CH₄ emissions are caused by enteric methane, while 11 percent are caused by manure management. While business-as-usual (BAU) emissions from agriculture are projected to reach 142 MtCO₂eq by 2030, FAO's GLEAM tool estimates that total milk production and enteric CH₄ emissions will respectively increase by 43 and 36 percent by 2050 under current practices, in a baseline scenario carried out at national level.

43. In Rwanda, total GHG emissions were estimated to be 6.7 MtCO₂eq in 2018, of which 3.3 MtCO₂eq, or 49 percent from agriculture. During the last decade, agriculture, forestry, and other land use (AFOLU) emissions increased by 34 percent. Methane emissions from the livestock sector represent about 46 percent of total GHG emissions. Enteric fermentation accounts for 74 percent of emissions, followed by CH₄ and N₂O emissions from manure management (15 percent and 11 percent, respectively). Dairy systems in Rwanda emit about 3.6 MtCO₂eq in 2020³⁸. Crossbred cattle contribute 58 percent to methane emissions in zero-grazing systems, while exotic dairy cattle contribute only 0.2 percent. Zero grazing practices account for 93 percent of these emissions, semi-grazing systems 2 percent, and open grazing 5 percent. FAO's GLEAM model estimates that by 2050 milk production and enteric CH₄ emissions will increase by 64 percent and 59 percent respectively, under a baseline scenario.

44. In Tanzania, total GHG emissions are estimated to be 286.5 MtCO₂eq³⁹, to which agriculture contributed 49.7 MtCO₂eq or 17 percent. Methane emissions represent 94 percent of total GHG emissions from dairy cattle systems. Enteric fermentation counts for 92 percent of GHG emissions, and manure management by 2 percent. Emissions from the dairy sector are expected to increase by mid-century, in response to the projected dairy herd growth to satisfy the future demand for dairy products, which is expected to grow from 1.4 billion litres of raw milk in 2020, to 2.0 billion litres in 2030, and 8 billion litres in 2050. Projections based on milk demand under a BAU scenario (from 2020 data), will be associated with an increase of enteric methane emissions by 43 percent, escalating from 19 to 27 Mt of CO₂eq between 2020 and 2030. FAO's GLEAM model estimates that enteric CH₄ emissions will increase by 117 percent by 2050, under a baseline scenario.

45. GHG emissions in Uganda totalled 95 MtCO₂eq in 2017, of which the AFOLU sector accounted for 79 MtCO₂eq⁴⁰ or 84 percent. The GLEAM tool estimated that the dairy sector in 2019 24.4 MtCO₂eq or 30 percent of the AFOLU sector's emissions. CH₄ emissions from enteric fermentation (78 percent, 19 MtCO₂eq) and CH₄ and N₂O emissions from manure management (4 percent or 2 MtCO₂eq) dominate GHG emissions from dairy cattle sector and mostly concentrated in the Eastern, Central, Southern and Northeast (Karamoja) regions. FAO's GLEAM model estimates that milk production and enteric CH₄ emissions will increase by 80 percent by 2050, under a baseline scenario.

46. Total GHG emissions from dairy cattle in Kenya, Rwanda, Tanzania and Uganda are presented in Figure 11. The pattern follows the distribution of the dairy herd in the country.

³⁶ Nationally Determined Contribution, communication to UNFCCC, December 2020. [unfccc.int/sites/default/files/NDC/2022-06/Kenya%27s First NDC %28updated version%29.pdf](https://unfccc.int/sites/default/files/NDC/2022-06/Kenya%27s%20First%20NDC%20updated%20version%29.pdf)

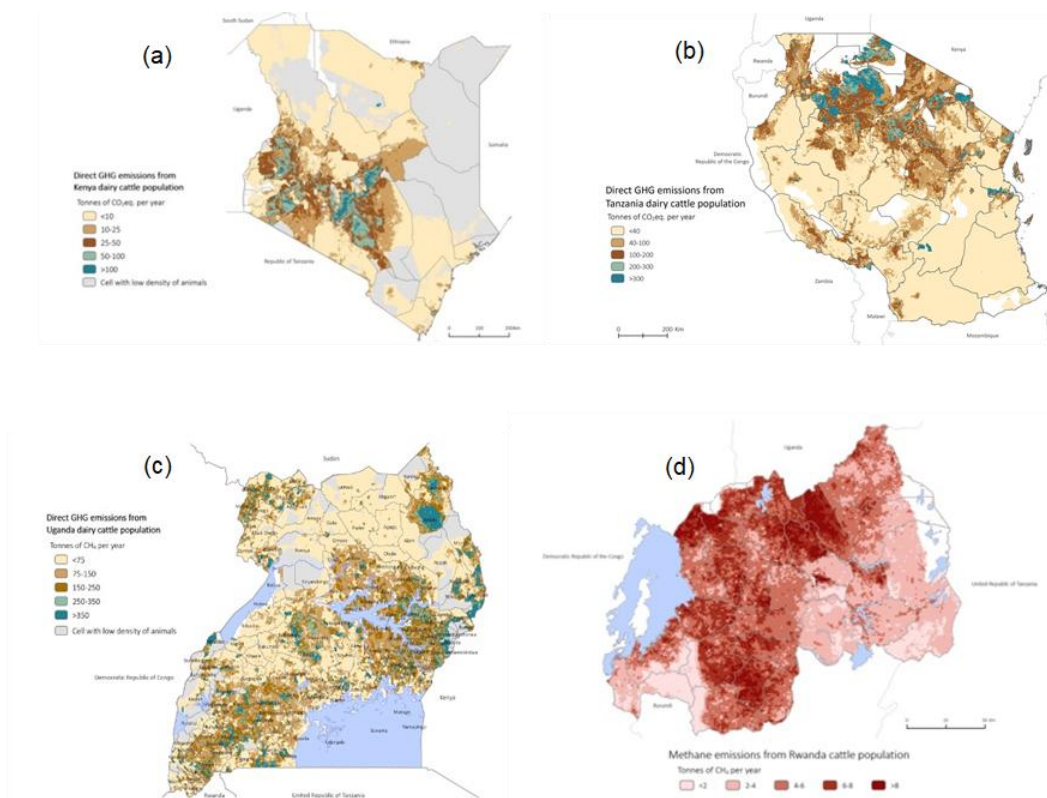
³⁷ The Global Livestock Assessment Model (GLEAM) is a modelling framework that simulates the interaction of activities and processes involved in livestock production and the environment. It is designed to analyze multiple environmental dimensions, such as feed use, greenhouse gas emissions, land use and land degradation, nutrient and water use and interaction with biodiversity. More information on GLEAM can be found here <https://www.fao.org/gleam/en/>

³⁸ The analysis of the direct GHG emissions was conducted using the Global Livestock Environmental Assessment Model (GLEAM), a geo-referenced model framework that simulates the bio-physical processes and activities along livestock supply chains based on the life cycle assessment and based on the IPCC Tier 2 methodology. More information on GLEAM can be found here <https://www.fao.org/gleam/en/>

³⁹ CAIT. (2022). Climate Watch data: Climate Watch. 2022. Greenhouse Gas Emissions. World Resources Institute. <https://www.climatewatchdata.org/ghg-emissions>

⁴⁰ MWE, Uganda National Communication 3, 2022. See <https://unfccc.int/sites/default/files/resource/Final%20TNC%20Uganda.pdf>

Figure 11: Direct GHG emissions from Dairy Cattle population (a) Kenya; (b) Tanzania; (c) Uganda; (d) Rwanda⁴¹



47. **Emission Intensities (EI).** The methane emission intensity, defined as the volume of methane emissions generated per kg of milk, varies widely across intensive, semi-intensive and extensive dairy production systems, with more productive systems having much lower EIs per kg of milk. In East Africa, EIs vary from 1.5 kg CO₂eq./kg milk for exotic dairy breeds in Kenya to 24.2 kg CO₂eq./kg milk for traditional systems in Uganda. This shows the range of performances between (but also within) systems and the potential for improvements, including by importance of increasing the productivity in dairy production and promoting intensive and semi-intensive practices from dairy systems with a low emission intensity per kg of milk. Details per country and per dairy production system are presented in Figure 12.

48. In Kenya, the enteric CH₄ EI is on average 2 kg CO₂eq./kg of milk produced, ranging from 1.4 kg CO₂eq./kg milk in intensive exotic systems to 5 kg CO₂eq./kg milk in extensive indigenous systems. Emissions intensities were on average higher for the indigenous systems (3.5 kg CO₂eq./kg milk) than for exotic systems (1.5 kg CO₂eq./kg milk).

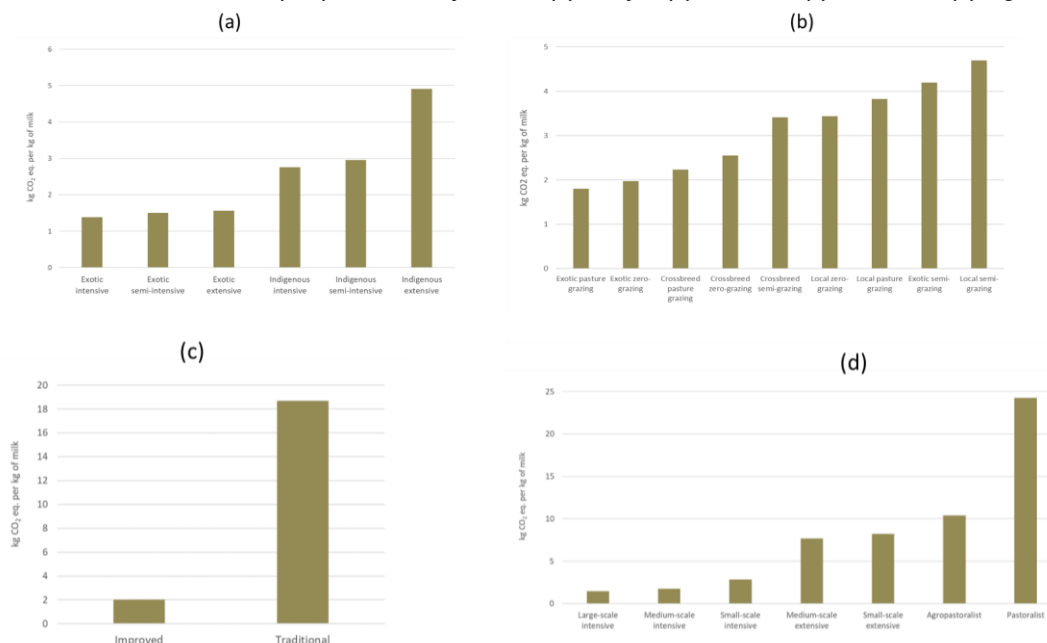
49. In Rwanda's dairy herd, crossbreeds account for 63 percent of enteric methane emissions, local breeds for 29 percent, and exotic breeds for 8 percent. The intensity of enteric CH₄ emissions lays between 1.8 (exotic/ crossbreed pasture grazing) and 4.6 kg (local semi-grazing) of CO₂eq. per kg of milk. This highlights the importance of mitigation strategies targeting the crossbreeds. Approximately 93 percent of the dairy sector emissions are attributed to zero grazing practices.

50. In Tanzania, enteric methane represents about 92 percent of total GHG emissions. About 97 percent of enteric CH₄ emissions are associated with traditional dairy cattle systems, whereas improved dairy systems contribute 3 percent. The intensity of enteric CH₄ emissions varies between 1.9 and 19 kg of CO₂eq. per kg of milk. Unlike CH₄ production, CH₄ intensity is notably elevated in the traditional system when compared to the improved system.

⁴¹ Source: FAO GLEAM, pre-feasibility study on GHG emissions baseline, scenarios and pathways analysis.

51. In Uganda, the enteric EI is on average 8.9 kg CO₂eq./kg milk produced. Emission intensities range from 7.7 to 24.2 kg CO₂eq./kg milk in traditional systems, compared to the range of 1.4 to 2.9 kg CO₂eq./kg milk for commercial systems. The highest emission intensity is observed for pastoral systems at 24.2 kg CO₂eq./kg milk.

Figure 12. Enteric emission intensities per production system in (a) Kenya; (b) Rwanda; (c) Tanzania; (d) Uganda⁴²



PROPOSED CLIMATE SOLUTIONS

52. The Programme's interventions will address the climate change adaptation and mitigation issues across the various dairy production systems and along the dairy value-chain, while addressing the barriers identified in the theory of change (ToC, see B.2). It will scale-up: (i) *farm and herd level practices* to increase productivity, reduce emission intensity, and decrease absolute emissions compared to a BAU scenario; (ii) *farm level* solutions for manure management and nutrient recycling, to improve circularity, resilience and reduce emissions; (iii) *landscape-level* interventions to improve the resilience and services of rangelands ecosystems, particularly soil carbon sequestration, and water and biomass supply for animal feed; and (iv) *value-chain level* interventions, to reduce milk losses (and corresponding GHG emissions) and increase the use of renewable energy. This holistic and sustainable approach, that will also address enablers such as access to markets, finance, knowledge and services, will lead to lower emissions from the dairy sector when compared to the business-as-usual scenario. A detailed table, linking the climate drivers with interventions and their benefits is available in Annex 2, Chapter 6.

53. **Interventions at farm and herd levels.** Interventions will focus on the various climate risks and impacts, namely the decrease in milk yields, heat stress of animals, the increase in cattle mortality, the change in dairy cattle diseases, the decrease in feed productivity, the deterioration of fodder in unprotected warehouses due to excess humidity. For all dairy systems, the Programme will encourage the combined adoption of climate adaptation and mitigation practices, including the enhancement of animal genetic resources (artificial insemination with locally-adapted improved and indigenous livestock breeds, sexed semen), improved herd health (e.g. vaccination against East Coast Fever), animal welfare, water access, recycling, harvesting and reuse for cattle watering. In intensive and semi-intensive systems, in addition to the above, interventions will improve the availability of forage grasses and legumes, feeding practices, and use of conserved silage and complementary feed. In intensive systems, interventions will include the establishment of forage grasses and legumes (*Calliandra*, *Leucaena*, etc.), the use of conserved silage

⁴² Ibid.

and urea-molasses multi-nutrient blocks (UMMB), the use of methane inhibitors (e.g. tannins rich Calliandra, lipids), supplementation (use of non-conventional feed resources, e.g. biochar or sweet potato vines), precision feeding and improving animal welfare and housing. *In extensive systems*, activities will include improving pasture management (e.g. rotational grazing), the use improved pasture species and management practices to improve forage quality and quantity, and use of multipurpose leguminous trees.

54. **On-farm solutions for manure management and nutrient recycling.** These interventions will address the CH₄ and N₂O emissions on dairy farms and limits related to the cost and availability of mineral fertilizers. Interventions include promoting simple circular manure management practices to reduce nutrient losses and GHG emissions (covering, compaction, composting, liquid-solid separation, production and addition of biochar). The Programme will also invest in biogas solutions to produce renewable energy for cooking and electricity, as well as nutrient-rich biofertilizer.

55. **Interventions at the landscape level** will address the negative impact of CC on ecosystem degradation (rangeland and pastures) that result in reduced soil carbon sequestration, decreased quantity and quality of pasture, reduced feed and water availability and livestock carrying capacity, all leading to lower milk yields. The interventions will focus on improved rangeland management, through participatory learning and promoting sustainable grazing practices. Land management activities will rehabilitate degraded rangelands to increase soil carbon sequestration, promote access to fodder and address the natural barrier to rangeland degradation. In the four countries, soil carbon impacts were identified as a strategy for carbon sequestration activities such as grazing intensity management, mowing frequency, organic inputs and fertilizers used in fodder production, and improved agroforestry systems associated with mixed crop-livestock production systems.

56. **Interventions at the dairy value chain level** will include the upgrading or equipping MCCs, milk collection points (MCPs) and processing centres. This will increase the supply of milk through formal aggregation and processing channels and create a stable market outlet for dairy farmers. Interventions will also aim to reduce milk losses and its carbon footprint, and reduce GHG emissions through the use of photovoltaic (PV) systems as a renewable energy (RE) source for milk collection, cooling and processing.

57. **Interventions to improve the enabling environment.** The focus will be on (i) a climate-sensitive policy and regulatory environment (institutional framework, dairy value chain governance, farmer awareness campaigns) for the dairy sector; (ii) strengthening proximity public and private services to the dairy sector, in particular expanding access to veterinary services and innovative gender-inclusive advisory services combined with input supply services; (iii) strengthening dairy cooperatives, MCCs and MCPs to encourage farmers to adopt the above- mentioned pathways. Improved MRV capabilities will provide more precise Tier2-emissions data and improve the quality of national inventories, as well as policies, planning, and investment decisions. Through regional cooperation, knowledge sharing and joint learning, coordinated strategies will be adopted and efforts of the individual countries will be amplified.

TARGET AREAS

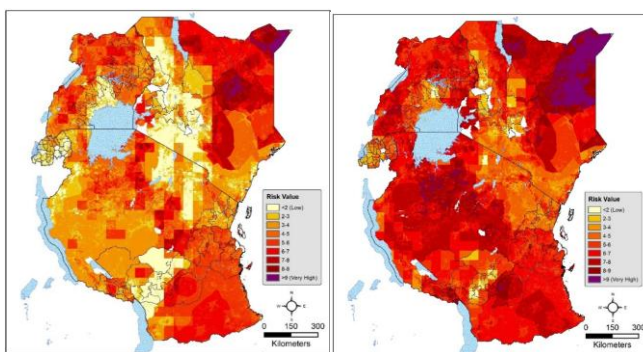
58. **Climate risk.** A climate risk mapping (Figure 13) was conducted by overlaying the sum of climatic hazards, exposure, vulnerability and subtracting the adaptive capacity. The maps show that most of the region has a high risk to climate hazards in the mid-term (2040-2069) and far future (2070-2099). More details, including the selected geospatial layers under each risk category, can be found in Annex 2, Chapter 4.

Figure 13. Spatial distribution of climate risks for RCPs 8.5 in (a) the mid-term (2040-2069) and (b) far-future (2070-2099)⁴³

(a)

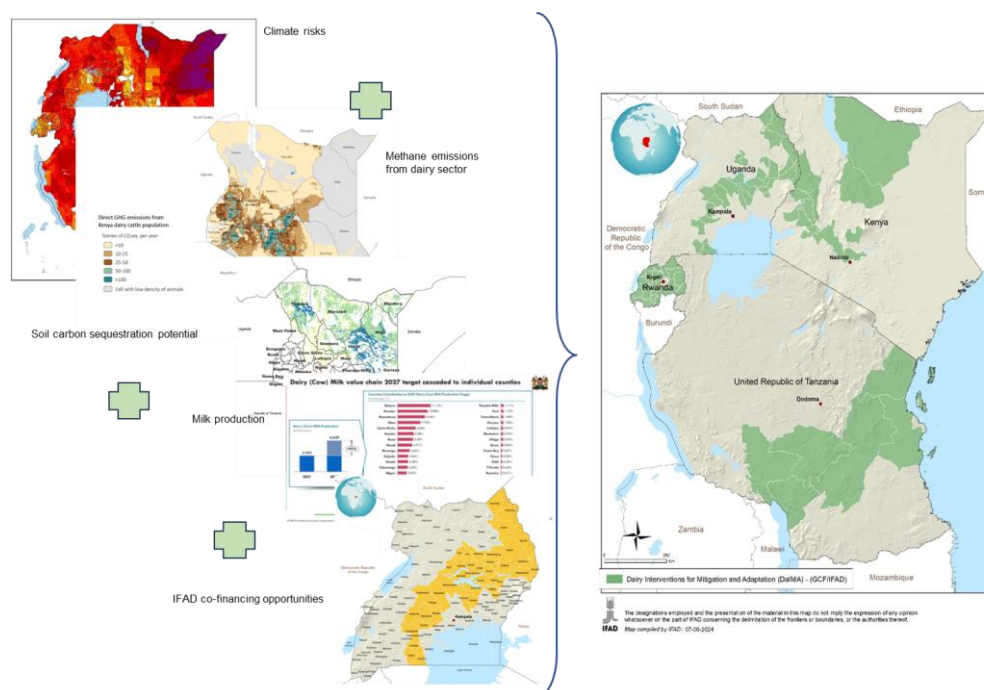
(b)

⁴³ Source: pre-feasibility study. Climate Impact Potential Assessment.



59. **Programme intervention areas are identified based on the integration of:** (i) the climate risk hotspots; (ii) absolute methane emissions from the dairy sector; (iii) soil carbon sequestration potential in pastures and rangelands; (iv) milk production by district/county; and (v) complementarity with other international financing institutions (IFI)-funded interventions, more particularly from IFAD (Figure 14).

Figure 14. Illustration of the methodology to select intervention areas (Districts/Counties)



60. The Programme will intervene in 94 counties/districts in the four countries. In **Kenya**, 12 counties⁴⁴ were selected and co-financing from the IFAD-funded Integrated Natural Resource and Environmental Management (INReMP) Project will be leveraged. **Rwanda** will implement the Programme in 27 rural districts⁴⁵, fully aligned with the IFAD-funded Rwanda Dairy Development Project, phase 2 (RDDP-2). In **Uganda**, the Programme will invest in

⁴⁴ Elgeyo Marakwet, Kakamega, Kericho, Kiambu, Nakuru, Nandi, Nyeri, Trans Nzoia, Uasin Gishu, West Pokot, Marsabit and Samburu. See Feasibility Study for more details.

⁴⁵ In the Eastern Province, Nyagatare, Gatsibo, Kayonza, Ngoma, Kirehe, Rwamagana, Bugesera; In the Northern Province, Gicumbi, Rulindo, Gakenke, Burera, Musanze; In the Western Province, Nyabihu, Rubavu, Ngororero, Rutsiro, Karongi, Nyamasheke, Rusizi; In the Southern Province, Kamonyi, Muhanga, Ruhango, Nyanza, Huye, Gisagara, Nyamagabe and Nyaruguru. See Feasibility Study for more details.

30 districts⁴⁶ along the cattle corridor, aligned with the ReLIV. In **Tanzania**, the Programme will be implemented in 9 regions (25 districts)⁴⁷, aligned with the Climate Smart Smallholder Dairy Transformation Project (C-SDTP).

COUNTRY OWNERSHIP

61. The NDCs, National Adaptation Plan (NAP) in the case of Kenya, and other national CC strategies outline the countries approaches to reduce GHG emissions and adapt to climate change (Table 2a). The proposed Programme is fully aligned with these national strategies, which propose measures such as: climate-resilient crops and livestock production, CC resilient animal breeds, improved husbandry, and windrow composting to reduce enteric fermentation emissions from livestock, and more efficient manure management systems. In addition, the Programme will support the Ministries responsible for the dairy sector to review and formulate national policies, strategies, and regulations that are essential for a sustainable transformation of the dairy sector aligned with the NDCs.

Table 2a. Alignment with key national climate priorities and livestock MRV status⁴⁸

Kenya	Rwanda	Uganda	Tanzania
National Climate Change Action Plan (NCCAP) (2018): Incorporates ways to reduce GHG emissions, enhance carbon sinks, and strengthen climate resilience into the National Climate Change Response Strategy (NCCRS).	National Climate Change and Environment Policy (2019) Mainstreams climate change concerns across all sectors and outlines the country's priorities for adaptation and mitigation measures.	National Adaptation Plan (NAP) for the agriculture sector (2016): A framework for adapting Uganda's livestock sector to climate change, including enhancing veterinary services and improving water management.	National Climate Change Strategy (2012): Provides a strategy for tackling climate change across all sectors, including climate change adaptation measures and measures for sustainable livestock management.
Kenya Climate Smart Agriculture Implementation Framework (2018–2027): Provides a roadmap for promoting climate-resilient agriculture, including improved animal feeding and watering, pasture management, and manure management.	National Adaptation Plan for the Agricultural Sector (2017): Provides a framework for adapting the livestock sector to climate change impacts, through drought-tolerant livestock breeds, improved water management, and veterinary services.	Climate Change Policy 2015): Encourages Uganda's livestock sector to incorporate climate change considerations. The policy includes measures to increase climate change resilience, reduce GHG emissions, and promote sustainable development.	Tanzania Climate Smart Agriculture Program (TCSAP) (2017): A framework for promoting climate-smart agriculture in Tanzania. Measures include increasing animal feeding and watering, managing pastures, and managing manure.
Kenya Livestock Master Plan (KLMP) (2013): Roadmap for the livestock sector to mainstream climate-resilient livestock management practices and livestock productivity.	Rwanda Livestock Master Plan (2017): Climate-resilient livestock management practices promoted along with strategies for improving livestock productivity.	Livestock Development Strategy and Investment Plan (LDSIP) (2015) Livestock sector roadmap, based on climate-resilient livestock management, enhanced livestock productivity and value chains.	Livestock Development Master Plan (LDMP) (2013): Livestock sector roadmap, based on improved livestock productivity, enhanced value chains, and adaptation to climate change.

⁴⁶ In the Central region: Kasaanda, Kayunga, Kiboga, Kyankwanzi, Kyotera, Nakasongola, Nakaseke, Gomba, Rakai, Sembabule; In the Eastern Region: Buyende, Kamuli, Kapelebyong, Katakwi, Kibuku, Kumi, Ngara, Pallisa, Serere and Soroti; In the Northern Region (this region includes what is sometimes referred to as Karamoja, with a distinct pastoralist livelihoods system): Amolotar, Amudat, Kaabong, Karenga, Kole, Kotido, Lira, Moroto, Nabitaluk and Nakapiripirit. See Feasibility Study for more details.

⁴⁷ Southern Highland: Mbeya (Rungwe, Mbeya, Chunya and Mbarali districts), Iringa (Mufundi, Kilolo, Iringa districts) and Njombe (Njombe, Makete and Wanging'ombe districts); Eastern Zone: Tanga (Lushoto, Muheza, Korogwe, Pangani, Tanga districts), Morogoro (Mvomero, Kilosa and Morogoro districts) and Pwani (Kibaha and Bagamoyo). Zanzibar: Unguja (West, Central and South districts) and, Pemba (South and North districts). See Feasibility Study for more details.

⁴⁸ Taken from Annex 2: Feasibility Study

Kenya Vision 2030: Emphasizes sustainable rangeland management to mitigate climate change impacts in pastoral areas, efficient resource use, value addition, research and innovation, and market access for livestock products.	Rwanda Green Fund (FONERWA) (2012): Finance for climate change mitigation and adaptation projects in Rwanda. Funds support renewable energy, sustainable agriculture, and ecosystem management projects.	National Delivery of Veterinary Services Policy (2003): Prioritizes a sustainable increase in the production of live animals and livestock products, such as milk, meat, and other by-products.	Tanzania Livestock Modernization Initiative (TLMI) (2020): Aimed at transforming the livestock sector through adoption of new technologies, including climate-smart feeds, heat-tolerant breeds, and improving disease management.
National Adaptation Plan (2016): Strengthen the adaptive capacity and resilience of livestock production systems, through drought-tolerant livestock breeds, improving water management, enhancing veterinary services.	National Dairy Strategy and Strategic Plan (2013): Guides dairy sector development, through policy support, research, extension services, infrastructure development, and capacity building.	National Organic Agriculture Policy (2019): Guidelines for organic livestock, poultry, fisheries, and related subsectors like feed, fodder, veterinary services, especially given the significant number of domestic livestock.	National Adaptation Plan (NAP) (2015): Framework for adapting livestock to climate change, including measures for enhanced livestock production resilience such as drought-tolerant breeds, veterinary services, and water management.
Livestock MRV status: currently using Tier 2 methodology for the dairy sector, and Tier 1 methodology with default emission factors for other livestock	Livestock MRV status: currently using Tier 1 methodology with default regional emission factors.	Livestock MRV status: currently using Tier 1 methodology with default regional emission factors.	Livestock MRV status: currently using Tier 1 methodology with default regional emission factors.

ADDITIONALITY OF THE PROGRAMME

62. **The Programme is a unique comprehensive framework to address concomitantly climate change impacts in, and GHG emissions from, the dairy sector.** As mentioned above, it will contribute to the NDCs, NAP, Nationally Appropriate Mitigation Action (NAMA), the Integrated Agriculture in National Adaptation Plans (NAP-Ag), the National Agriculture Policy, and contributes to National Development Plans on climate-smart dairy value chains. It is also fully in line with IFAD's climate adaptation and mitigation priorities outlined in the four (4) Country Strategic Opportunity Programmes (COSOPs). Through the introduction of climate-smart agriculture (CSA) practices and technologies, the Programme aims to trigger resilient low-carbon development of the dairy sector, with a focus on methane mitigation.

63. **Sources of methane from the dairy sector are among the main categories of anthropogenic GHG emissions from Agriculture, Forestry and Land Use (AFOLU) sector identified in national inventories (e.g., National Communication (NCs), Biennial Update Reports (BURs)) in East Africa.** Long-term low-emission development strategies in East Africa must address the main source of methane emissions, namely the dairy sector. In addition, the dairy sector is currently affected by several CC impacts, which requires more investment in the adaptation and resilience of the dairy herd. Rapidly reducing methane emissions from livestock will help make near-term gains in global efforts this decade. This strategy is regarded as the most effective in achieving the goal of limiting warming to 1.5°C.

64. **In this regard, the Programme is a flagship and pioneering investment vehicle to support the Global Methane Pledge (GMP).** The GMP was launched at the Conference of Parties (COP) 26 in November 2021 in Glasgow. Pledge participants agree to take voluntary action to contribute to a collective effort to reduce global methane emissions by at least 30 percent from 2020 levels by 2030, which could eliminate warming of more than 0.2°C by 2050. Participants also commit to using the highest tier IPCC good practice inventory methodologies, as well as continuously improve the accuracy, transparency, consistency, comparability and completeness of national

GHG inventory reports under the United Nations Framework Convention on Climate Change (UNFCCC) and the Paris Agreement, and to ensure greater transparency in key sectors. The Pledge aims to catalyse global action and strengthen support for existing international methane emission reduction initiatives to advance technical and policy work that will serve to support national actions. The Pledge recognizes the essential role that the private sector, development banks, financial institutions and philanthropy play to supporting the implementation of the Pledge and welcomes their efforts and commitment.

65. **In line with the GMP, Programme additionality also includes building, through a large-scale investment programme, long-term institutional capacity to deliver effective services that improve efficiency, climate resilience, and reduce GHG emissions by improving the enabling environment.** Capacities will focus on preparing and implementing climate-smart policies at national and regional levels, promoting supportive trade policies across East Africa to unlock climate-smart solutions, GHG MRV, climate risk analysis, and ensuring that the dairy sector fully contributes to the participating countries' NDCs.

SYNERGY WITH OTHER PROJECTS

66. At regional level, the Programme will be complementary with the GCF-funded Africa Rural Climate Adaptation Finance Mechanism (ARCAFIM, FP220, IFAD being the AE) for East Africa region. At the national level, synergies will be developed with IFAD-funded initiatives, and key stakeholders will be involved in coordination and implementation. In **Kenya**, implementation will be supported by the Kenya Dairy Board (KDB), the Kenya Agricultural and Livestock Research Organization (KALRO), the Kenya Animal Genetic Resources Centre (KAGRC), amongst others. The Programme will be closely linked to the Integrated Natural Resource Management Programme (INRMeP) and the "Transforming Livelihoods through Climate Resilient, Low Carbon, Sustainable Agricultural Value Chains in the Lake Region Economic Bloc" proposed by FAO to the GCF⁴⁹. It will also build on the lessons learnt of the GCF-funded "Towards Ending Drought Emergencies: Ecosystem Based Adaptation in Kenya's Arid and Semi-Arid Rangelands" (FP113). In **Uganda**, implementation will be supported by the Dairy Development Authority (DDA), the National Animal Genetic Resources Centre and Data Bank (NAGRC & DB), National Research Organization (NARO), Ministry of Local Government (MoLG). Synergy will be with the IFAD-funded Resilient Livestock Value Chain Project (ReLIV). In **Rwanda**, implementation will build on the existing networks established through an ongoing project, implemented by RAB, namely RDDP-2, as well as the Rwanda Environment Management Authority (REMA). Knowledge from the Transforming Eastern Province through Adaptation (GCF FP167) will also be used for landscape restoration interventions. In **Tanzania**, the Programme is an integral part of the C-SDTP. Implementation will be supported by the Tanzania Dairy Board (TDB), the Tanzania Livestock Research Institute (TALIRI). It will also leverage lessons learnt from the "Building climate resilience in the landscapes of Kigoma region" (GCF FP218). As DaIMA will be aligned with RDDP-2, C-SDTP, ReLIV and INReMP, IFAD's strategic objectives, more particularly its pro-poor, youth and gender-sensitive approach and investment strategy will be fully mainstreamed into DaIMA.

ALIGNMENT WITH GCF SECTORAL GUIDES

67. The Programme will invest in climate-resilient livestock systems, one of the two paradigm-shift pathways defined by the GCF Sectoral Guides on Agriculture and Food Security. The Programme improves and facilitates climate change adaptation. At the national and regional levels, the appropriate composition and functioning of management committees and knowledge platforms will ensure the meaningful participation of vulnerable groups, such as women, youth, and the rural poor. Due to the vulnerability of climate-sensitive factors affecting the dairy sector, the Programme supports integration of climate risks into business models involving the private sector in the dairy value chain. As a result, the Programme aims to implement a transformational approach that avoids locking dairy farmers into a high-carbon growth path.

B.2 (a). Theory of change narrative and diagram

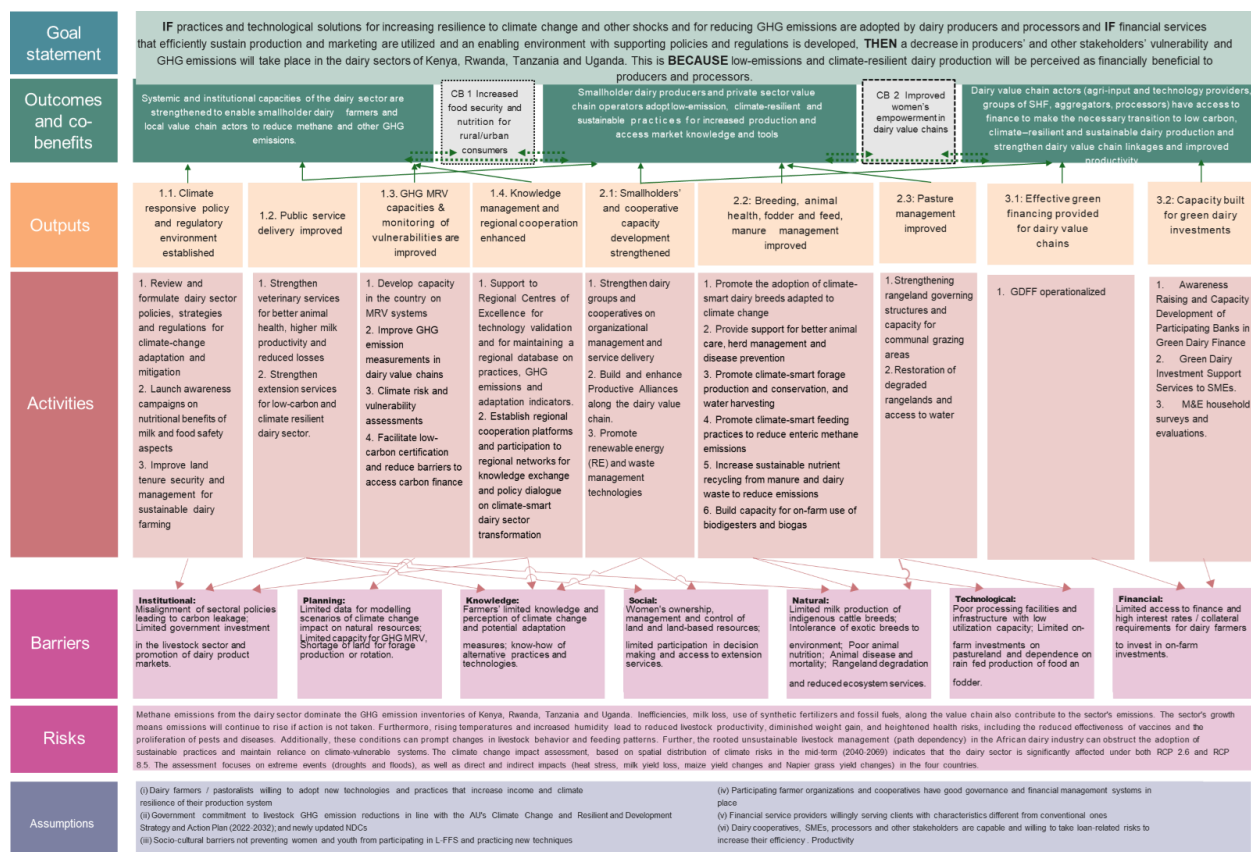
68. The aim of the proposed Programme, which will be implemented in Kenya, Rwanda, Tanzania, and Uganda, is to reduce methane and other GHG emissions from the dairy sector while increasing the resilience of livestock-dependent communities. The Programme will support a shift from low production efficiency to improved, gender-sensitive, integrated and NDC-aligned practices that reduce emissions at farm and food system levels (including

⁴⁹ Project to be tentatively approved by GCF Board in 2025.

interventions that reduce milk losses and increase milk collection, decarbonize processing with renewable energies, formalize dairy value chains and strengthening partnerships with the private sector), as well as interventions that promote circularity, and increase rangelands ecosystems resilience and services (including carbon sequestration in grasslands).

69. The ToC presented below outlines the intervention logic of the Programme (Figure 15a). This section describes the expected shift in the development trajectory as a result of the Programme's intervention. The Programme's Goal Statement is as follows: *"IF practices and technological solutions for reducing methane and other GHG emissions and increasing climate resilience are adopted by dairy farmers and processors, and IF financial services that efficiently sustain production and marketing are utilized, THEN a decrease in producers' and other stakeholders' vulnerability and GHG emissions will be achieved in the dairy sectors of Kenya, Rwanda, Tanzania and Uganda. This is BECAUSE the pathways to low-emissions and climate-resilient dairy production will be perceived as financially beneficial to farmers, pastoralists, and processors."*

Figure 15a. Overall Theory of Change (ToC) diagram



70. As highlighted in the ToC, the multifaceted barriers (institutional, planning, knowledge, social, natural, technological and financial) that hinder the achievement of the Programme goal are: (i) misalignment of multiple sector policies and limited capacity for GHG MRV and for inventories at Tier 2 level, and for climate risk assessments in the dairy sector, (ii) limited technical capacity to adapt to the changing climate in dairy farmers and extensions workers, (iii) insufficiently improved breeds and their adaptability to climate change, combined with sensitivity to diseases; (iv) dependence on rainfall for animal fodder and feed production, as well as rangeland degradation; (v) limited milk bulking, processing, transport and distribution; (vi) low access to finance; (vii) limited women participation in decision making, access to extension and control of resources. Adaptation and mitigation needs include: access to climate-smart technologies; capacity building and knowledge transfer; financial and market support; strengthening infrastructure and facilities; climate information systems; and policy support and institutional strengthening. More details on the barriers are found in the Feasibility Study.

Pathways to Low-Emission and Resilient Dairy Farming in East Africa

71. The dairy sector faces significant challenges related to climate change in East Africa. Dairy farming contributes substantially to GHG emissions, with CC threatening its sustainability. Inefficiencies at the dairy farm level result in excessive GHG emission intensities per kg of milk. In addition, many dairy farmers face barriers to accessing markets, procuring local feed and fodder, obtaining extension and financial services, and managing risks. Low genetic potential, herd management challenges, animal diseases, inadequate extension services, limited knowledge of climate-resilient and low carbon practices and technologies, rangeland degradation and gender inequalities further harm the dairy sector in these countries.

72. The Programme aims to address the following challenges: (i) strengthening climate resilience, (ii) reducing emissions across the dairy value chain; (iii) the reduction of GHG emissions per kg of milk; and (iv) improving the livelihoods of dairy farmers in East Africa. This requires a holistic approach to improving key aspects of the dairy sector and addressing barriers, from improving policy, institutional and service delivery environments to building the capacity of smallholder farmers, including women, youth and the marginalized. It also includes measures to facilitate access to finance and innovative technologies with mitigation and adaptation benefits. As a result, dairy farmers and processors will be motivated to adopt more efficient, climate-resilient practices that will increase incomes, reduce production costs, and help combat climate change. Central to this transformative approach is the symbiosis of private sector support and financial access within the dairy sector. This integration aims to foster market growth while promoting climate resilience and reducing GHG emissions.

73. The paradigm shift envisioned by the Programme is a more efficient, climate-resilient dairy industry that contributes to national and regional economic growth, improved smallholder farmers' livelihoods, and reduced GHG emissions. To achieve the above vision, the Programme aims to achieve the following three Outcomes:

Outcome 1: Systemic and institutional capacities in the dairy sector are strengthened to enable smallholder dairy farmers and local value chain actors to reduce methane and other GHG emissions.

74. Outcome 1 aims to foster an enabling environment for the transformation of the East Africa's dairy sector towards increased productivity, climate resilience and lower emission intensity. To achieve this, several outputs must be achieved.

75. The Programme will establish a climate-responsive policy and regulatory environment for the dairy sector (Output 1.1) by reviewing regulatory frameworks, supporting value chain governance initiatives, and work on food safety and dairy consumption. Through improved public services (Output 1.2.), namely the expansion of proximity veterinary services and innovative extension services, both male and female farmers will be supported to address animal health, feed quality, manure, and herd management issues. This will help scaling up climate-smart practices and addressing the lack of knowledge and perception regarding adaptation. Enhanced measurement, reporting and verification (MRV) capacities (Output 1.3.) will generate detailed Tier 2 emissions factors to improve national inventories, and inform evidence-based policies, planning and investment. Robust MRV and modelling will address data gaps and strengthen climate scenarios and assessments. Certification for low-carbon production and facilitating access to the carbon markets will create financial incentives for mitigation efforts. Regional cooperation and knowledge sharing (Output 1.4.) will ensure coordinated strategies, amplify country's efforts, and promote joint learning. This collaboration will be key to overcome institutional and knowledge barriers at national level.

Outcome 2: Smallholder dairy farmers and private sector value chain operators adopt low-emission, climate-resilient, and sustainable practices for increased production and access market knowledge and tools.

76. Outcome 2 aims to improve incomes and productivity of smallholder dairy farmers, dairy cooperatives and SMES by promoting sustainable and climate-resilient low-emission growth. The three Outputs leading to this Outcome will also have a positive effect on gender equality, increased food security and nutrition for rural/urban consumers (CB1) and women empowerment and inclusion of marginalized groups (CB2). More details on co-benefits are provided in the Feasibility Study.

77. Output 2.1 supports dairy cooperatives to ensure that smallholder dairy farmers have a stable market for their produce, reduce losses, and are interested in adopting the pathway innovations. Renewable energy solutions at cooperative level will also be supported. The promotion of integrated pathway-specific solutions (Output 2.2.) will include the introduction of better dairy cattle breeds, improved herd composition and animal health, increased availability of high-quality feed and fodder, better manure management for biogas production and nutrient recycling. This will increase productivity, improve circularity and reduce GHG emission intensity at animal, herd and farm levels.

This Output helps to remove technological and natural barriers by improving on-farm infrastructure and overcoming limitations related to animal nutrition, disease, mortality, and genetic potential. Improved pasture management (Output 2.3), through participatory learning and the promotion of sustainable grazing practices and land management practices will rehabilitate degraded rangelands, increase access to fodder and address rangeland degradation. Together, the three Outputs contribute to broader replication through increased productivity and income for smallholder dairy farmers, increased carbon sequestration in soils, resulting in reduced greenhouse gas emissions. This aligns with the desired paradigm shift towards a more productive, climate-resilient, and low-emission dairy sector, supported by sustainable practices and gender-transformative growth. In particular, the project will create equal opportunities for women and men and aim to: (i) Increase women economic empowerment through access to finance and skills; (ii) Increase women participation, voice and decision making in farmer organization and cooperatives (iii) Reduce and balance workloads with increased resilient production and greater efficiency of dairy production and post-harvest technologies; (iv) Influence positive gender norms and to prevent Gender based violence through the GALS methodology (v) Policy engagement to address gender barriers related in the sector. The result is a transformative change in the East Africa's dairy sector, which benefits not only dairy farmers, but also the wider community and environment.

Outcome 3: Financial institutions make available the appropriate investment resources to make the necessary transition to low-carbon, climate-resilient, and sustainable dairy production and strengthen value chain linkages and improved productivity.

78. By improving access to finance for climate-smart technologies and practices, Component 3 (Figure 15b) aims to facilitate dairy cooperatives and farmer-allied enterprises that are too small to qualify for GDFF investment, grant financing will be provided under Component 2. This dual approach ensures that all segments of the dairy value chain receive appropriate financial support, tailored to their size and capacity.

Figure 15b. GDFF Theory of Change (ToC) diagram



79. By improving access to finance for climate-smart technologies and practices, Component 3 aims to facilitate the larger-scale transition to sustainable, low-carbon, climate-resilient dairy value chains. To achieve this outcome,

dairy value chain actors (including agro-input and technology providers, well-organised dairy cooperatives of SHF, aggregators and processors) will be supported in overcoming technological and financial barriers to increase their productivity and green their operations.

80. To contribute to the envisaged paradigm shift, two Outputs will be delivered (Figure 15b). Under Output 3.1, a GDFF investment window will invest in climate-smart practices and technologies. The Investment Window will have a dual focus of (a) establishing long-term Green Dairy Partnerships with targeted medium-size companies to provide patient loans for dairy actors to invest in multiple productivity and climate-smart technologies and practices and (b) providing shorter-term Green Dairy Loan Products that meet specific financing needs defined by the smaller dairy sector businesses. Under Output 3.2, a GDFF Technical Assistance (TA) Facility will be established to maximize both impact and returns by enhancing the capacity of (a) the participating banks of GSFF in green lending to develop their green dairy sector portfolios and (b) borrowing dairy companies to implement effective green strategies, improve operational efficiencies and achieve growth. In doing so, TA will address non-financial barriers currently preventing greater climate-smart financing in the GDFF target regions. Together, these outputs contribute to the overall outcome of improved access to finance and the adoption of green technologies and practices. The result is expected to make a transformative change in East Africa's dairy sector, leading to increased productivity, climate resilience and reduced emissions.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Outcome number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Outcome 1	☐	☐	☐	☒	☒	☐	☐	☒
Outcome 2	☒	☐	☐	☒	☒	☐	☐	☒
Outcome 3	☒	☐	☐	☒	☒	☐	☐	☐
Co-benefit number	Co-benefit							
	Environmental	Social	Economic	Gender	Adaptation	Mitigation		
Co-benefit 1 Increased food security and nutrition for rural and urban consumers	☐	☒	☒	☐	☐	☐	☐	
Co-benefit 2 Improved women's empowerment in dairy value chains	☐	☒	☒	☐	☒	☐	☐	

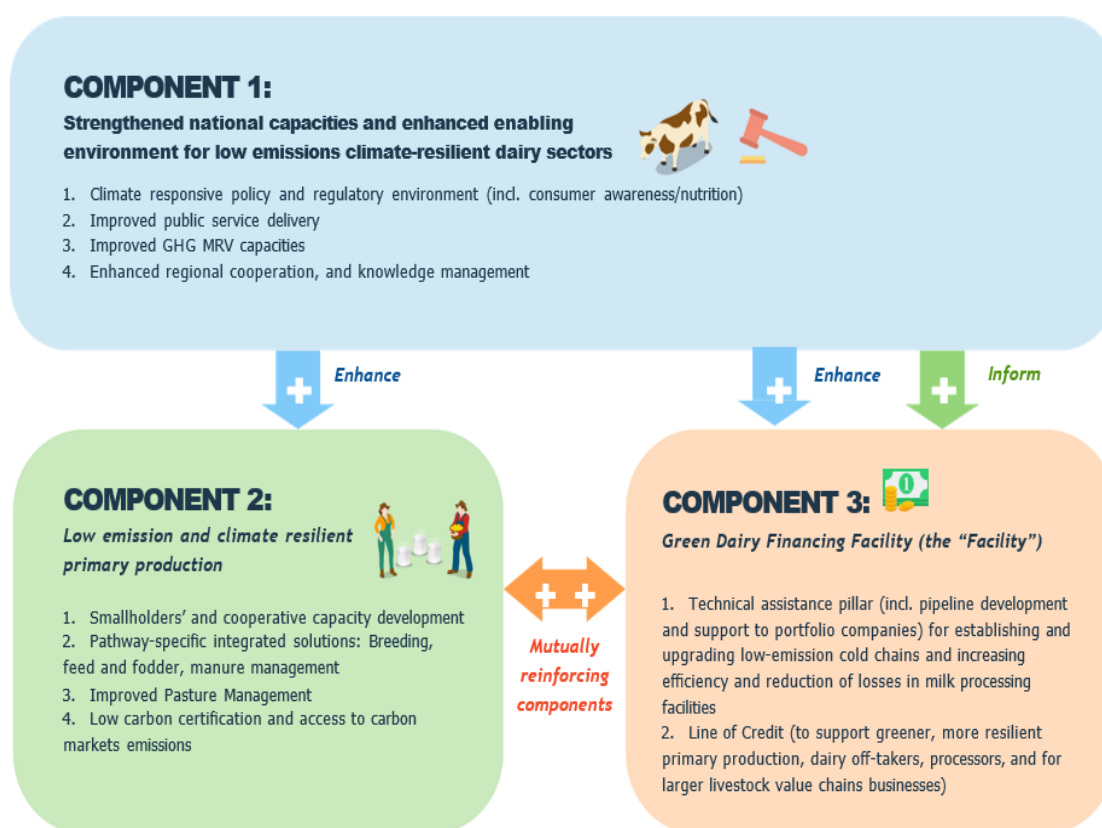
B.3. Programme description

DESCRIPTION OF COMPONENTS

81. The three Outcomes described in the ToC are delivered through 3 dedicated and interconnected components, namely: Component 1: Strengthened national capacities and enhanced enabling environment for low emissions climate resilient dairy sectors; Component 2: Enabling low-emission and climate-resilient dairy value chain development; and Component 3: Green Dairy Financing Facility (GDFF). Each Component consists of a combination of Outputs under full control of the Programme. They aim to result in specific outcomes, whether in behavioural or systems change. These changes/outcomes will take place between the completion of the program outputs and the achievement of impact.

82. All three components are interlinked. Component 1 creates an enabling national and regional policy and regulatory environment and governance of the dairy value chain, improving public service delivery, amplifying GHG MRV capacities, and ensuring regional knowledge management and exchange. This provides a solid foundation for the adoption of sustainable practices that improve emission reductions, climate resilience, and increased productivity. At the same time, Component 1 informs Component 3 by improving visibility and strengthening countries' GHG measurement capability, a critical element for accessing to more public and private climate finance. It also supports the creation and consolidation of support industries that can provide reliable and technically sound investment options by helping to regularise and standardise these industries. Similarly, Component 2 enhances Component 3 by strengthening the management capacity of smallholder farmers, improving their creditworthiness, and fuelling their demand for alternative technologies and loans to acquire them. Finally, Component 3 strengthens Component 2 by expanding smallholders access to finance through targeted financing and low-cost loans, providing carbon market revenue streams, and providing technical assistance to reduce emissions and increase efficiency. The synergies between the three components provide a holistic approach, addressing the environmental and economic needs of the targeted beneficiaries, enabling and incentivising them to implement the desired change towards a sustainable and climate-resilient landscape in the four target countries (Figure 16).

Figure 16. Complementarities and synergies between components



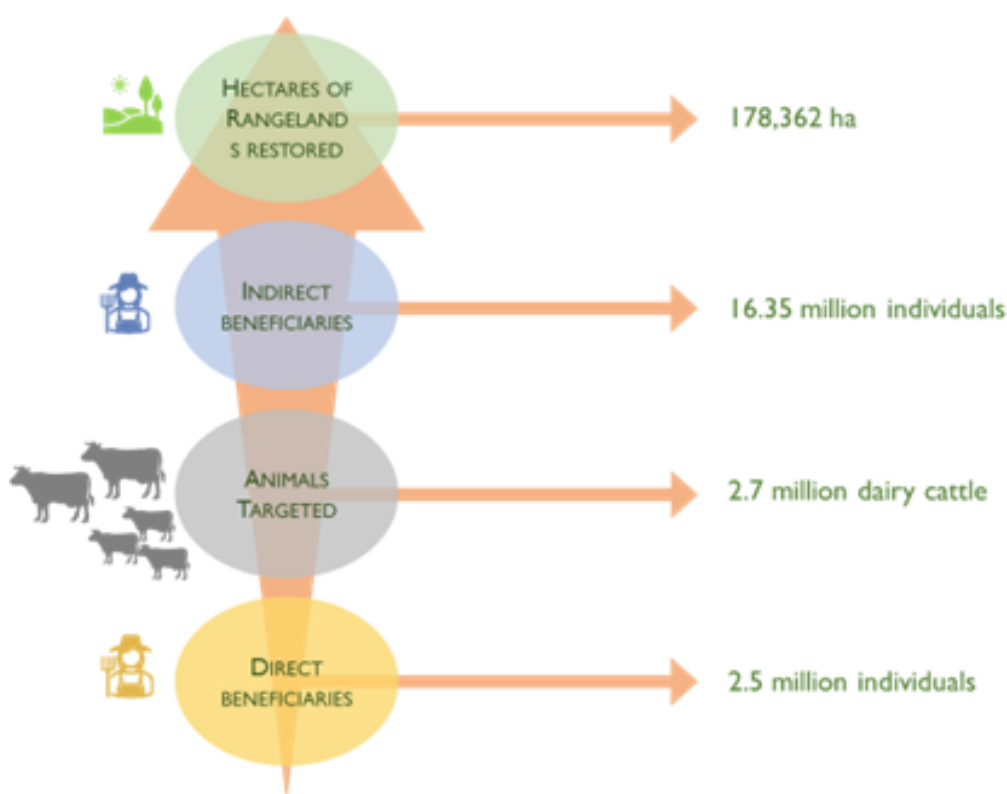
PROGRAMME BENEFICIARIES

83. The Programme will benefit about 2.5 million individual dairy farmers and dairy value chain actors, at least 40 percent of whom are women, 30 percent of whom are youth and 5 percent are marginalized groups (including people with disabilities). As the geographic and social targeting of DaIMA is fully aligned with the IFAD projects in the four countries (RDDP-2, ReLIV, INReMP, and C-SDTP), the pro-poor targeting approach, which will also include women and youth, is fully mainstreamed into DaIMA. These beneficiaries will mainly be reached through Component 2, but also through Component 3. The majority are dairy farmers in intensive and semi-intensive systems who currently sell or aim to sell milk in formal and informal markets. These smallholder dairy farmers have generally between 1 to 5 dairy cows (6 to 10 kg of milk/cow/day) in zero-grazing/intensive or semi-zero grazing/semi-intensive systems. They are the main target group and can be considered as rural poor. They are the essential segment of producers who need to be scaled to support the formalization of the dairy sector and foster sustainable intensification based on sound

investments and viable business development (see *Feasibility Study and Market Studies*). Dairy value chain support will also benefit medium and large commercial farmers (6 or more dairy cows, 5-9 kg of milk/cow/day in semi-intensive systems, 12 or more kg/cow/day in intensive systems). Under Output 2.3, the Programme will also benefit (i) pastoralists and agro-pastoralists, and (ii) medium- and large-scale extensive farmers through the rehabilitation/improved management of 178,362 hectares of rangelands. Entry points to reach dairy farmers will also benefit from the Programme's support, including dairy cooperatives and MCCs, L-FFS and Pastoralist field schools (PFS), farmers and community-based organisations.

84. Beneficiaries of Component 3 (Figure 17a) will be medium-sized farmer-allied dairy value chain actors, including aggregators and processors, agro-input and technology providers, and mature dairy farmers' cooperatives, selected through targeting and eligibility criteria as per the participating implementation partners, and based on a through screening process. These formal stakeholders will also be involved in the business linkages through Component 2. Component 1 supports public sector institutions with a mandate to regulate the dairy sector, as well as national dairy boards that participate in the governance of the dairy value chain. Businesses identified by Component 2 as suitable for GDFF financing will be proactively encouraged to apply for support under Component 3.

Figure 17a. Programme beneficiaries and outreach



COMPONENT 1: STRENGTHENED NATIONAL CAPACITIES AND ENHANCED ENABLING ENVIRONMENT FOR LOW EMISSIONS CLIMATE RESILIENT DAIRY SECTORS

85. This Component 1 will support the paradigm shift towards a more productive, climate-resilient, and low-emission dairy sector, supported by sustainable practices and gender-inclusive growth. It will create an enabling institutional environment for smallholder dairy farmers and value chain actors to reduce methane (CH₄) and other GHG emissions, while increasing resilience to CC. Therefore, it will also address the institutional, planning and knowledge barriers identified in the ToC, and foster an enabling environment for policy, institutional, and service delivery. The Component will contribute to the following GCF Results Areas: MRA4 "Forestry and land use", ARA1 "Most vulnerable people and communities", and ARA4 "Ecosystems and ecosystem services".

86. The Component 1 will have four Outputs, namely (i) an enabling policy and regulatory environment, including improved consumer awareness and nutritional balance; (ii) improved public services delivery to the dairy sector, in particular specialized livestock extension services and veterinary services; (iii) improved capacity to monitor GHG emissions and vulnerabilities; and (iv) strengthening regional cooperation and knowledge management. Component 1 activities will provide public service goods and will therefore be funded by grants.

87. **The activities carried out under the Component are sustainable.** The Programme will carry out a series of activities aimed at improving the enabling environment (policy, legal and regulatory) and long-term institutional capacity of key stakeholders of the dairy sector, to provide effective services for greater efficiency, climate resilience and GHG emission reductions. These capacities will revolve around the critical public responsibilities of preparing and implementing climate-smart policies at national and regional levels, enabling trade policies across East Africa to unlock affordable climate-smart solutions across the Region, GHG MRV, climate risk analysis, sector coordination with climate policies and to ensure that the dairy sector fully contributes to the NDC of the participating countries.

88. **The activities carried out under the Component have the potential for scaling-up and replication.** The Programme will provide innovative support to the countries in MRV of GHG in the livestock sector. GHG MRVs for dairy production in East Africa have been piloted by the FAO and ILRI, particularly in Kenya with a Gold Standard methodology⁵⁰. The MRV will build on this pilot methodology to achieve scale and replication in the Region. But further work is needed to obtain data on GHG emissions from East African livestock systems which remain scarce, as most estimates are based on proxies and methodologies that were developed for temperate zones in high-income countries and do not represent East African livestock systems well⁵¹. In addition, in the absence of specific data on the regional dairy typology, several inventories use broad African averages by default⁵². This creates a notable bias in national GHG inventories⁵³ and prevents African countries from tracking progress on mitigation in their livestock sectors. In addition, to ensure long-term feasibility and success of any CSA practice implemented at scale, East African countries need the data and capacity to set up and maintain MRV systems. In addition, innovative climate risk analysis support to extension services will strengthen the large-scale dissemination of climate information services to last-mile users including women, young dairy farmers through cooperatives and farmers' organizations. Moreover, regional platforms and South-South and Triangular Cooperation (SSTC) will help to export the concepts, successes and lessons learnt from the Programme to other sectors, regions, and countries.

Output 1.1: Climate-responsive policy and regulatory environment established.

89. **Activity 1.1.1: Review and formulate Dairy Sector policies, strategies and regulations for climate-change adaptation and mitigation.** In the four countries, the Programme will support the Ministries, Departments and aligned institutions in charge of the dairy sector to review and formulate national policies, strategies, laws and regulations that are essential for a sustainable transformation of the dairy sector, in line with the NDCs and NAPs. This includes aspects related to mainstreaming CC adaptation and mitigation, as well as adopting export and import regulations that favour sustainable development and private sector engagement and investment in the dairy value chain (taxation of imports of dairy produce, specialized equipment, inputs, such as germplasm, fodder seeds, feed additives, solar cooling, biodigesters, etc.). Resources will be provided to (i) mobilize national and international expertise for policy review, formulation and dialogue; (ii) facilitate stakeholder consultations to ensure that the policy review and dialogues are inclusive; (iii) validation processes through round tables, and (iv) the preparation of policy briefs and communication materials for popularization. This activity will be implemented by the relevant Executing Entities in each country (see sections B.3 and A.20), through service providers, including (i) in Kenya, the State Department of Livestock Development (SDLD), KDB; (ii) in Rwanda, RAB and MINAGRI; (iii) in Tanzania, MLF, TDB; (iv) in Uganda, DDA, and the National Animal Genetic Resources Centre and Data Bank (NAGRC & DB), in Uganda. The regional activities will benefit from the participation of East Africa Commission (EAC) and the African Union Inter-African Bureau for Animal Resources (AU-IBAR).

⁵⁰ FAO and ILRI, (2016) Smallholder Dairy Methodology: Methodology for GHG Emission Reductions from Smallholder Dairy Production Systems. Published by FAO, ILRI, KENYA State Department of Livestock and the Gold Standard. https://globalgoals.goldstandard.org/standards/405_V1.0_LUF_AGR_Agriculture-Smallholder-Dairy-Methodology.pdf

⁵¹ Graham MW, Butterbach-Bahl K, du Toit CJL, et al (2022) Research Progress on Greenhouse Gas Emissions from Livestock in Sub-Saharan Africa Falls Short of National Inventory Ambitions. *Frontiers in Soil Science* 2: <https://doi.org/10.3389/fsoil.2022.927452>

⁵² Intergovernmental Panel on Climate Change (IPCC), 2019

⁵³ Korir D, Ndung'u PW, Onyango AA, et al (2023) IPCC Tier 1 methodology overestimates the carbon footprint of smallholder cattle production systems in Kenya. 10 July 2023, PREPRINT (Version 1) available at Research Square. <https://doi.org/10.21203/rs.3.rs-3148140/v1>

90. **Sub-activity 1.1.1.1 Support the review and preparation of national dairy policies, strategies, and regulations to mainstream climate-change adaptation and mitigation.** Priorities in the four countries will be the following. In **Kenya**, (i) the Dairy Industry Strategy and Investment Plan to guide the sustainable transition of the dairy sector; (ii) the draft National Strategy and legal framework for Animal Feed and Fodder; (iii) a National Strategy for Animal Breeding and breeding regulations (iv) the draft Dairy Industry Act (1958); (v) the Kenya Agricultural Sector Extension Policy (KASEP), (vi) the development of compost and bio-slurry standards; and (vii) milk and milk product standards on food safety and quality. In **Rwanda**, the Strategic Plan for Agriculture Transformation 5 (PSTA5) provides a framework for a low-carbon and climate-resilient agriculture sector. Within this framework, the Programme will support the review of: (i) national climate and sectorial policies that are regulating the dairy sector; a new agriculture and livestock policy for Rwanda is currently under development which includes climate resilient livestock production; (ii) the regulatory framework and standard operating procedures (SOP's) for dairy value chain operators, the use of methanogenesis inhibitors as feed additives, in direct collaboration with Rwanda Food and Drugs Authority (FDA) and Rwanda Standards Board (RSB), for milk quality, standards and safety, feed formulation, energy efficiency, and CC; (iii) the 2016 Ministerial Order to address new dairy value chain realities, including the MCP-model and the need to regulate the raw milk market. Support will be given to the formulation of a national regulatory framework for the private veterinary practice including for Community Animal Health Workers (CAHWs) and L-FFS facilitators, with considerations of increasing women and youth participation; (iv) support will also be given to the Rwanda National Dairy Platform (RNDP) to participate in the formulation of the dairy value chain (DVC) regulatory law empowering the platform to conduct overall DVC regulation as in EAC member countries. In **Tanzania**, the Programme will support finalizing the draft national Livestock Policy, as well as (i) review of the Veterinary Act (2003) to update aspects related to private practitioners; (ii) review the Dairy Act to update regulations on raw milk trade; (iii) the formulation of disease contingency and control plans; (iv) prepare the Tanzania Dairy Development Strategy and roadmap, (v) finalization of the Animal Breeding Act, (vi) review of the Animal Diseases Act (2003), (vii) review of the Animal Welfare Act (2008), (viii) review of the Grazing Land and Animal Feed Resources Act (2010), and (ix) finalization of a Strategy for Formalization of Dairy Industry. In **Uganda**, the Programme will support: (i) finalizing the National Livestock Development Policy, integrating CC priorities (breeding, feeding, water), enabling policies for private sector investments, research strategies on climate-resilient feed germplasm to offset predicted yield declines, as well as rangeland ecology; (ii) the review of the Animal Disease and Health Act, and the Animal Breeding Act; (iii) support drafting of regulations for the Food and Agriculture Regulatory Authority, livestock identification and traceability, as well as dairy cattle breeding and livestock insurance to integrate CC mitigation; (iv) review land policies and regulation to mitigate land tenure insecurity for dairy farmers.
91. **Sub-activity 1.1.1.2: Support the regional policy dialogue to mainstream climate-change adaptation and mitigation in regional dairy policies, strategies, and legislations.** In coordination with regional and national dairy platforms, Ministries and stakeholders, the Programme will support a region-level assessment and proposals for further regional alignment on dairy sector regulations, including export and import regulations (common external tariffs), to accelerate regional convergence in dairy standards, regional trade and market integration, and to enhance technology transfer within and from outside the region for solar cooling and renewable energy and resource-efficient technologies. For this to happen, recurrent regional policy dialogue events will be organised bi-annually, on a rotational basis, by (and in) each of the 4 participating countries. Activities will be coordinated by regional coordination and policy dialogue experts recruited in each country to jointly support the implementation of regional activities. National stakeholders, including women and young people, will be supported to participate.
92. **Sub-activity 1.1.1.3 National dairy value chain governance and awareness creation on climate-change.** The aim of this activity is to enhance dairy value chain governance and to mainstream the importance of climate-change in this process. At national level, the national dairy boards (KDB, DDA, TDB, RNDP) and other relevant professional bodies of private stakeholders, will be supported to participate in the dairy policy dialogue, to create awareness and education/capacity building on climate, feed and dairy policies and regulations.
93. **Sub-activity 1.1.1.4 District/regional/county level dairy value chain governance and planning.** Local-level multi-stakeholder platforms involving delegates from dairy value chain stakeholders will participate in the annual planning and review of Programme activities. Participants will include dairy cooperative members, milk collectors, milk processors, service providers of artificial insemination (AI), veterinary services, extension staff and local government, ensuring participation quotas for women and youth.

94. **Activity 1.1.2 Launch awareness campaigns on nutritional benefits of milk and food safety aspects.** Support to milk consumption and nutrition awareness, accompanied by nutrition education and social behaviour change, targeting smallholder producer and consumer households, informal dairy value chain actors and milk handlers in the school milk programs aims at improving nutritionally vulnerable group's consumption of milk, therefore enhancing their individual's health and climate resilience. The Programme will support milk consumption and nutrition awareness through increased availability, accessibility, affordability, and consumption of high-quality and safe dairy products, accompanied by nutrition education and social behaviour change and communication (SBCC) to reach out to nutritionally vulnerable groups. This activity will be implemented by the relevant Executing Entities in each country (see B.3 and A.20), through service providers, i.e. (i) SDLD, KDB in Kenya; (ii) RNDP and the Single Programme Implementation Unit (SPIU) in Rwanda; (iii) TDB and the Tanzania Food and Nutrition Centre in Tanzania; (iv) DDA, together with the Ministry of Health (MoH), Ministry of Education and Sports, SNV and UNDP, in Uganda.

95. **Sub-activity 1.1.2.1 Public awareness on the contribution of milk to nutrition and resilience, with focus on climate vulnerable population.** The awareness raising will target women and men and use various community platforms such as extension, L-FFS, MCC/cooperatives. Topics will include improved milk treatment, food safety and nutritional benefits of dairy produce, pasteurized and processed milk. In Rwanda, it will include participation in dairy fairs/expos to showcase climate-friendly and energy efficient local processing and consumption of dairy products. In Tanzania and Uganda, the Programme will scale up school milk programs for improved nutrition of vulnerable children.

96. **Sub-activity 1.1.2.2 Build capacity of dairy inspection to ensure milk quality and safety and reduce waste.** Dairy inspection is essential for sector-level milk quality and safety, thus encouraging formal market growth and reduction of milk rejection and waste (less GHG emission intensities). The Programme will provide technical training and equipment to strengthen capacities of inspections to ensure development and enforcement of milk quality and safety standards. Women and young people who are often involved in the informal milk trade, will be included in the trainings.

97. **Activity 1.1.3 Improve land tenure security for sustainable dairy farming.** The Programme will adopt a fit-for-purpose approach to identify and mitigate tenure risks. Activities to mitigate this risk will vary depending on the type of land and type of governance. For pastoral and agropastoral systems, the focus will be on the identification and protection of rangelands and on supporting participatory planning and management. In Rwanda, all land is registered and individually owned, including women. The focus will be on optimizing land productivity through land consolidation and land use planning. Activities will include capacity building and support to inclusive and participatory planning processes to create local-level capacities. The activity will be implemented by the various EEs through service providers, i.e. (i) National Land Commission (NLC) and County Governments in Kenya; (ii) the districts, land registration centers, and specialized service providers in Rwanda; (iii) MLF, LGAs and national land use planning commission in Tanzania; and (iv) MAAIF and District Local Government (DLGs) in Uganda.

98. **Sub-activity 1.1.3.1: Mapping of local tenure risks and preparation of training materials.** The Programme will mobilize specialized technical assistance to: (a) implement a rapid tenure governance capacity assessment, including a tenure risk assessment for the Programme target areas; (b) develop training material to increase local authorities' capacity to secure inclusive tenure rights for dairy production and the public's capacity to identify and secure legitimate tenure rights. Based on the national tenure context, each dairy production system's tenure risks in terms of access, use and security of land will be identified and assessed, and appropriate mitigating strategies will be proposed.

99. **Sub-activity 1.1.3.2: Developing capacity on land tenure at decentralized level.** The focus will be on capacity building of local authorities and raising public awareness on options to incrementally increase tenure security, especially for women and youth. The Programme can support public awareness campaigns on tools to secure tenure rights.

Output 1.2 Improved service delivery to the dairy sector

100. **Activity 1.2.1 Strengthen veterinary services for better animal health, higher milk yields and reduced losses.** A key barrier to reduce GHG emissions and sustainably increase the productivity of dairy animals is animal health and welfare. Climate change exacerbates the emergence and re-emergence of diseases and infestation (East Coast Fever – ECF, helminths infestation, etc.). Healthier animals are more resilient to climate hazards and more productive, with less milk to be discarded, leading to a reduction of “unproductive” GHG emissions. The higher

revenues provide incentives for farmers to adopt animal health and welfare measures, especially if there is joint and equitable sharing of benefits from dairy at the household. To expand the outreach and capacity of veterinary services, the Programme will invest in proximity animal health support, as well as in national diagnostic capacity of veterinary laboratories. The activity will be implemented by EEs through service providers, i.e. (i) Kenya Veterinary Board and KALRO in Kenya; (ii) RAB, District veterinary extension services, and the Rwanda Council of Veterinary Doctors (RCVD) in Rwanda; (iii) Directorate of Veterinary Services (DVS) and Tanzania Veterinary Laboratory Agency (TVLA) in Tanzania; (iv) MAAIF (Directorate of Animal Resources, DAR), in partnership with service providers (e.g. VSFB), in Uganda.

101. **Sub-activity 1.2.1.1 Support the installation of both private and public veterinarians and veterinary paraprofessionals.** The Programme will implement the animal health support and disease control system, using the “One Health” approach. The approach will have three levels: (i) support private veterinarians and para-veterinarians, in particular young and female practitioners; (ii) support livestock extension workers at local level; (iii) provide additional training to lead farmers, including women, from the L-FFS for coaching their peers and assist private para-veterinarians and livestock extension workers. In **Kenya**, veterinary education, and training programs to both private and public veterinarians and veterinary paraprofessionals in the Programme area will be enhanced. In **Rwanda**, support will be provided for the installation of young veterinary doctors, veterinary technicians and Community Animal Health Workers (CAHWs) as private practitioners through the Veterinary Sanitary Mandate to enhance accessibility of services while creating jobs. In **Tanzania**, Livestock Extension Workers (LEWs) will be equipped and trained, and private male and female para-vets will be incentivised to ensure the last mile animal health and extension services. In **Uganda**, the Programme will build capacity of veterinarians, para-veterinarian technicians, and community-based animal health workers (CAHWs) on One Health principles, disease diagnosis, emerging and re-emerging diseases because of CC.

102. **Sub-activity 1.2.1.2. Strengthening diagnostic capacity of veterinary laboratories and capacity to test quality of animal feed.** In **Kenya**, the Programme will improve disease control (surveillance and monitoring, contingency planning, and treatment) of targeted diseases including modern diagnostic equipment in regional centres (located in Nairobi i.e. Kabete and Embakasi, Nakuru, Kericho, Eldoret, Karatina, Garissa and Mariakani), testing kits, telemedicine. In **Rwanda**, four (4) satellite laboratories will receive additional equipment for serology (ELISA) and molecular diagnosis (PCR) to enable rapid detection of infectious diseases, and treatment, as well as equipment for handling biological and hazardous waste and the establishment of environmentally friendly practices. The Programme will also strengthen the national veterinary laboratory (NVL) to handle advanced diagnostic capacity for viral diseases affecting livestock. Accredited laboratory services such as KALRO, private laboratories and research institutions will be included to provide laboratory services on testing of feeds. In **Tanzania**, the Programme will: (a) strengthen the Tanzanian Veterinary Laboratories Agency (TVLA) and the Central Veterinary Investigation Laboratory in Zanzibar; (b) support the establishment of district mobile laboratories that will serve LEWs local (para)vets; (c) train laboratory technicians for each mobile laboratory. In **Uganda**, veterinary laboratories will be strengthened (equipment, reagents, consumables for sample collection and mobile laboratory facility) for detection of the diseases related to CC. It will also establish a national data management system on emerging and re-emerging diseases because of CC (TAD and zoonosis). In each country, the Programme will: (i) enhance the capacity to undertake feed testing in the national research institutions (KALRO in Kenya, RAB and the College of Agriculture, Animal Science and Veterinary Medicine (UR-CAVM) in Rwanda, NARO in Uganda, TALIRI in Tanzania); this includes equipment, training of researchers, including young females, and inputs for testing feed samples; (ii) undertake on-farm feed sampling and laboratory quality testing (e.g., dry matter, carbohydrates, protein, fat, minerals, vitamins).

103. **Activity 1.2.2 Strengthen extension services for low-carbon and climate resilient dairy sector.** To reach out to dairy farmers, raise awareness and build their capacity in the field of CC mitigation and adaptation, the Programme will adopt innovative extension models and tools, including digital, and enhance the capacity of extension services. This activity will involve improving delivery of public and private extension services at farm and cooperative/aggregation levels, by local (district/county) livestock officers who coordinate the community-based extension system. To provide proximity advisory services on climate-smart practices, the Programme will provide support through L-FFS in mixed crop-livestock systems and PFS in grazing and pastoralist systems. This activity will be implemented by EEs through service providers, i.e. (i) KALRO and SDLD in Kenya; (ii) RDDP-2, RAB and REMA in Rwanda; (iii) MLF and the LGAs in Tanzania; (iv) MAAIF (Directorate of Agriculture Extension Service) and MWE and the Uganda National Meteorological Agency (UNMA), in partnership with implementers (e.g. FAO and SNV) in Uganda. International technical assistance will be provided by ILRI and FAO to develop the tools and enhance national capacity.

104. **Sub-activity 1.2.2.1. Preparation of training modules for extension in climate mitigation and adaptation measures.** Specific topics that are part of the proposed pathways under Outputs 2.2 and 2.3 and related to CC will be added to the modules of the L-FFS (or other extension approaches). The topics in the training modules related to climate-smart milk production and handling could include: fodder production, harvesting and conservation; use of conserved silage and UMMB; supplementation (use of non-conventional feed resources e.g. sweet potato vines or tannin-rich plants reducing and inhibiting CH₄ production) reproductive management; health management including One-Health aspects and management of anti-microbials, biogas and manure management, use of drought-resistant fodder, application of crop residues, seed production (certification, traceability, quality control), improved diet formulation to reduce emission intensities (use of feed formulation software, such as the innovative SNV-developed Rumen8® tool), and animal welfare, including housing in zero-grazing practices. Also guidelines will be prepared on improved fodder cultivar and climate-smart technologies, innovative management practices for pasture and fodder production, and agroforestry practices (e.g. leguminous fodder trees as standalone or line trees). The L-FFS modules will be developed with the support of national or international technical assistance. Also gender inclusive training on the implementation of guidelines will be carried out.
105. **Sub-activity 1.2.2.2 Training of master trainers (ToTs) and facilitators of extension systems in climate-change related topics.** The Programme will support the strengthening of existing L-FFS and the establishment of new L-FFS (aiming at 40 percent women and 30 percent youth). To support the sustainability and scaling up of the L-FFS, existing L-FFS/ PFS master trainers and facilitators will be trained on the specific topics listed above.
106. **Sub-activity 1.2.2.3 Promote innovative extension tools.** These innovative extension tools could include: (i) participatory extension approaches, such as the Participatory Integrated Climate Services for Agriculture (PICSA), which was piloted in Kenya, Rwanda and Tanzania, (ii) e-extension tools by using bulk SMS and the internet to reach more farmers, provide highly captivating and interactive packages, partner with other stakeholders in development of content and capacity and to improve ability to access and share knowledge and skills on farming technologies, (iii) TV, radio and social media networks. In addition, other innovative extension tools could include: in Kenya, the use of the agro-weather platform namely Kenya Agricultural Observatory Platform (KAOP), a digital platform developed through a joint initiative between the KALRO and the KMD, designed to provide weather information to support decision-making in the agricultural sector in Kenya to develop capacity of extension officers and producers on climate change adaptability. In Tanzania, the iCow app and or other innovative extension tool will be developed and promoted.

Output 1.3: Improved GHG MRV capacities & monitoring of climate vulnerabilities

107. To reliably track progress on the ambitious climate goals of the national NDCs, state-of-the-art systems of MRV of GHG emissions are needed within the participating countries, along with national capacity to manage such systems, more particularly in the livestock sector. This Output will enable countries to report with accuracy on emissions in the dairy sub-sector in their national GHG inventories at Tier 2 level, according to the 2006 IPCC Guidelines, and address the institutional, planning and knowledge barriers identified in the ToC. It will also enable countries to capture the impact of investments in dairy on GHG emissions in their NDC and therefore contribute to their climate commitments. Improved MRV systems will ensure that participating countries can conduct their own GHG calculations with cutting-edge methods, develop their own mitigation pathways and policies, and attract more public and private climate finance, including through carbon credits. This Output will also address the issues related to the data for climate vulnerability assessment and national feed balances, all contributing to improved climate resilience. The national research centers and Government Institutions will receive implementation support from ILRI (Mazingira Centre and research stations) to execute Activities 1.3.1 and 1.3.2.

108. **Activity 1.3.1: Develop capacity in the country on MRV system.** The Programme will identify critical capacity needs at national and local levels to improve the operation of MRV systems. This will help developing consistent MRV systems for the dairy sub-sector that allows systematic updates, so that the impact of interventions of the Programme, and beyond, can be monitored over time, and progress can be assessed against an established baseline.

109. **Sub-activity 1.3.1.1: Capacity assessment and development in capturing, recording, tracking, and reporting data on dairy GHG emissions.** This sub-activity will carry out in depth capacity assessments of

existing MRV tools and frameworks in the four countries, and the place of dairy in them. Based on these assessments, capacity development will be provided to design activity data questionnaires for Tier 2 and/or Tier 3 GHG emissions in dairy (e.g. animal numbers, weights, productivity, feed rations, animal health interventions, manure management and grassland management) and develop a data management system to ensure reliable flow of information between the various national and local agencies involved in the MRV system. Countries will receive capacity development on data collection protocols adjusted to their needs to relevant stakeholders (technicians, university staff, staff of national research institutes and ministries, and other relevant actors). ILRI⁵⁴ will provide dedicated digital data collection tools (e.g. ODK), data quality control, and calculation of GHG emissions for inventory reporting, following established protocols.

110. **Sub-activity 1.3.1.2: Developing MRV systems and engage in policy dialogue to align with national MRV for the livestock sub-sector.** After assessment of national MRV systems, this intervention will include baseline, measurement of individual mitigation activities, impact assessment on absolute emissions and emission intensities and institutional arrangements. To ensure sustainability and impact at scale, the sub-activity will update of national livestock census questionnaires to include activity data requirements for GHG inventories outlined by IPCC on other cattle and livestock. This will allow countries to track livestock productivity and to report on enteric CH₄ and manure CH₄ and N₂O emissions, and soil carbon storage in grasslands/rangelands⁵⁵, following international inventory guidelines, using IPCC Tier 2 methodology.
111. **Activity 1.3.2: Improve GHG emission measurements in dairy value chains.** On-farm data like animal live weights and milk yields are necessary to calculate Tier2 GHG emissions. Emission factors also need to be improved to reflect local conditions (e.g., animal metabolism, feed composition). This activity will comprise the following sub-activities:
112. **Sub-activity 1.3.2.1: Collect activity data on farm to calculate Tier 2 dairy emissions under various herd, feed, animal health and manure management situations.** The Programme will finance the collection on-farm activity data for baseline and “with-project” situation, using digital tools (e.g. livestock Tier 2-specific ODK tool). This includes data on animal numbers, weights, productivity, feed rations and manure management system. This sub-activity will be performed annually with the support of ILRI, in direct partnership with national research institutions and academia of the four countries, to ensure sustainability and replication through the transfer of capacity (see output 1.4). This activity will inform the Programme’s M&E system and provide data for baseline and “with-project” situation.
113. **Sub-activity 1.3.2.2: Conduct experiments and run models to generate Tier 2 emission factors (EF) for baseline and improved dairy, under various herd, feed, animal health and manure management situations.** Evidence shows that local feeding conditions and indigenous livestock breeds result in different emission factors for enteric methane⁵⁶ and manure CH₄ and N₂O emissions⁵⁷ than assumed by the IPCC. In this context, this sub-activity will develop local context-specific emission factors (EF) and inventory equations through lab experiments. Together with 4 countries and research/academia, ILRI will quantify the GHG mitigation potential of selected feed interventions using state-of-the-art cattle respiration chambers⁵⁸ to derive methane yields for enteric CH₄, as well as manure incubation setups to derive CH₄ and N₂O emission factors for manure management.

⁵⁴ DaIMA, in partnership with ILRI, will support the member countries in monitoring existing SOC stocks via targeted soil sampling campaigns, comparing different grassland management practices and degradation status along pedoclimatic gradients across the four countries. The Programme will inform on impacts of grassland management on soil carbon storage and sequestration. The Programme will contribute to the development and validation of novel remote sensing approaches for rangeland monitoring via drones and satellites. This also includes application of modern techniques such as Eddy Covariance (EC) towers for landscape-scale monitoring of ecosystem carbon exchange to show if the ecosystem is a carbon sink or source, and how carbon uptake and release are affected by seasonality, climatic extremes, and management. ILRI has two EC towers set up in their Kapiti research station. See Feasibility Study for more details.

⁵⁵ Ibid 30.

⁵⁶ Goopy JP, Goopy JP, Korir D, et al (2020) Severe below-maintenance feed intake increases methane yield from enteric fermentation in cattle. *British Journal of Nutrition* 123:1239–1246. <https://doi.org/10.1017/S0007114519003350>; Korir D, Marquardt S, Eckard R, et al (2022b) Weight gain and enteric methane production of cattle fed on tropical grasses. *Anim Prod Sci*. <https://doi.org/10.1071/AN21327>

⁵⁷ Zhu Y, Merbold L, Pelster D, et al (2018) Effect of Dung Quantity and Quality on Greenhouse Gas Fluxes From Tropical Pastures in Kenya. *Global Biogeochem Cycles* 32:1589–1604. <https://doi.org/10.1029/2018GB005949>; Leitner S, Ring D, Wanyama GN, et al (2021) Effect of feeding practices and manure quality on CH₄ and N₂O emissions from uncovered cattle manure heaps in Kenya. *Waste Management* 126:209–220. <https://doi.org/10.1016/j.wasman.2021.03.014>

⁵⁸ For more information, see <https://mazingira.ilri.org/infrastructure/>

114. **Sub-activity 1.3.2.3: Training on the Global Livestock Environmental Assessment Model – interactive (GLEAM-i) and IPCC assessment tool to calculate GHG emissions biannually, and to inform the national MRV system and inventories.** By using collected activity data and Tier 2 EF generated, Programme results will inform the national MRV system, using tools like the GLEAM-i⁵⁹ developed by FAO to calculate emissions biannually. Throughout the whole process, ILRI will transfer capacity on Tier 2 EF, data collection and GHG emission calculation from the Programme.
115. **Activity 1.3.3: Climate risk and vulnerability assessments.** Together with improvements to extension services under output 1.2, the Programme will seek to improve the accuracy and use of information (climate, feed availability) to last mile-users in the dairy sector. This will include building the capacity of devolved CC units and decentralized meteorological offices to develop and disseminate climate information services, including (i) impacts of CC on the dairy sector; and (ii) information on availability of feed, through tools like national feed balance sheets. This activity will be implemented by the relevant Executing Entities in each country (see sections B.3 and A.20), through service providers, i.e. national meteorological departments. These include: (i) the Climate Change Department in the Ministry of Water and Environment (MWE-CCD) in Uganda, (ii) the SDLD in MoALD and Climate Change Department (CCD) in Kenya, (iii) the National Carbon Monitoring Centre (NCMC) in Tanzania; and (iv) Rwanda Meteorology Agency.
116. **Sub-activity 1.3.3.1: Climate risk and vulnerability assessments, including impacts on the dairy sector, at local level.** It will consist of carrying out dairy sector climate risk and vulnerability assessments at local level, based on historical and projected climate data supported by impact-based models, analytical frameworks and machine learning. This will involve (i) the development of agro-climatic zone maps and Dairy Sector suitability maps based on the current and future climate, (ii) assessing the direct impacts of climate change on specific cattle breeds using the tropical humidity index (THI) and milk decline (MDEC), (iii) improved forecasting/projections of agricultural and pastureland drought through e.g. machine learning and artificial intelligence.
117. **Sub-activity 1.3.3.2: Review and enhancement of national feed balance mechanisms, and support to early warning systems.** The national feed balance mechanism and results will be reviewed and improved using the FAO guidelines, to prepare a minimum of two balances per year, with dairy specific results. Improvements will focus on seasonality and quality of feed and fodder resources (metabolizable energy and crude protein content), in addition to quantity (in dry matter DM).
118. **Activity 1.3.4 Facilitate low-carbon certification and reduce barriers to access carbon finance.** This activity will leverage MRV systems developed under activities 1.3.1 and 1.3.2 to track GHG emission reduction from improved practices. The aim is to support selected dairy value chain stakeholders to receive premium prices for low-carbon and high-quality milk production and engage on the voluntary carbon market. The Programme will pilot this initiative and raise awareness of processors and consumers of the environmental performance of dairy products.
119. **Sub-activity 1.3.4.1 Develop a carbon certification system for the dairy sector.** In order to reduce barriers to carbon finance, the Programme will mobilize technical assistance to: (i) review the alignment of national regulations for registration of carbon projects in East Africa (which facilitate foreign investments); (ii) develop/align monitoring and verification standards suitable for each national context (e.g. VERRA, Gold Standard) and specifically dairy sector projects – in cooperation with the Africa Carbon Market Initiative (ACMI) and research institutions; (iii) support the establishment of local monitoring and verification companies – to reduce costs of certification/verification; (iv) undertake a policy dialogue in strengthening the position of e.g. the Kenya Carbon Exchange in regional cooperation.

Output 1.4 Knowledge management and regional cooperation enhanced.

120. The Programme will facilitate transfer of knowledge through regional and SSTC, with the objective to replicate the transformative practices for climate-resilient and low-carbon dairy adopted by the Programme. ILRI Mazingira Centre, as a State-of-the-Art environmental and regional research and education Centre, will also contribute under direct agreement with the EE in each country. This Output will address the institutional, planning and knowledge barriers identified in the ToC. Shared knowledge will enhance future low-carbon planning in the dairy sector, minimize

⁵⁹ See Feasibility Study for more details on the GLEAM-i tool.

climate risks and enable participating countries to take advantage of available climate smart opportunities within the region.

121. **Activity 1.4.1. Support to Regional Centres of Excellence for technology validation and for maintaining a regional database on practices, GHG emissions and adaptation indicators.** To ensure sustainability and transfer of capacity to national institutions, the Programme will provide scientific support combined with capacity development to “regional centres of excellence” to export knowledge regionally and globally on CC and GHG emissions for the dairy value chain. National Centres of Excellence (or “knowledge hubs”) will include national agricultural research institutions (e.g., KALRO in Kenya, NARO in Uganda, TALIRI in Tanzania, RAB in Rwanda), local universities (e.g., Makerere and Busitema in Uganda; University of Nairobi, Eldoret or Maseno; University of Dar es Salaam or Sokoine University of Agriculture (SUA) in Tanzania), targeting in particular young researchers. It will build on the legacy of the East Africa Agricultural Productivity Programme (EAAPP, World Bank-funded), which has built Regional Centre of Excellence (RCoE) in the dairy sector in Kenya.

122. **Sub-activity 1.4.1.1: Support to regional Centres of Excellence in technology validation and dissemination.** The Programme will assist the above-mentioned institutions with (i) technical trainings and protocols for GHG measurements (emission factors, soil carbon storage) and estimation methods to build their own capacity for research, MRV and technology validation linked to climate change mitigation, (ii) research capacity for conducting innovative research such as optimizing the biogas process for enhanced energy and nutrient recovery and building efficient biodigester technologies, (iii) vocational training and development of materials, standards, university-level educational curricula and guidelines, and (iv) knowledge, information and visibility products, related to the four focus areas feeding, animal health, manure management and biogas, and grassland management and soil carbon sequestration.

123. **Sub-activity 1.4.1.2: Develop and maintain a regional database/ repository on practices, GHG emissions per sources and systems, and climate change adaptation.** This sub-activity will consist in developing, maintaining, and promoting a regional database on best practices, and their impact on GHG emissions per source and livestock production system, and adaptation indicators. ILRI has a large repository of training materials on its open access CGSpace platform⁶⁰, which will be leveraged, and made available to all actors in the Programme.

124. **Activity 1.4.2: Establish regional cooperation platforms and participation to regional/ global networks for knowledge exchange on climate-smart dairy sector transformation.** The cooperation platforms will be organized by each country to include various public and private stakeholders active at the regional level and within each of the DaIMA countries, including the AU-IBAR, IFAD, FAO, ILRI, the East and Southern Africa Dairy Association (ESADA), local government partners, centres of excellence, international organisations and non-governmental organisations (NGOs), and involve private sector platforms and networks (e.g. One Health Research and Education Centre in Africa – OHRECA; Global Research Alliance Manure Management Network; Global Dairy Platform; Nutrient Circularity Network; National Biogas Stakeholders Network in Kenya). The Inter-Governmental Authority on Development (IGAD), through its Center for Pastoral Areas and Livestock Development, will be invited to share its experience in facilitating appropriate and applicable knowledge, research, and technology development, including their domestication, adoption, and transfer through regional platforms.

125. **Sub-activity 1.4.2.1: Regional platform for knowledge sharing to enhance low-emission, climate resilient dairy development.** A regional platform – involving the above-mentioned entities - will be organised on a rotational basis in each of the participating countries to share experiences and scientific innovations and best practices implemented on farm. The regional platform will function as the knowledge management mechanism for the Programme. It will capture and disseminate good practices and lessons learned from both public sector, private sector, networks and the research and academic community. This sub-activity will provide evidence – generated from the Programme M&E system - on the GHG efficiency gains (from breeding, artificial insemination/ herd management, feed, animal health), circularity (manure management, recycling) and carbon sequestration. Success stories for replication (e.g. Kenya for Tier 2 MRV), scientific innovations (animal nutrition / TIMPs, manure management for continuous improvement) will be documented. It will finance annual regional workshops, conferences, seminars, tours (exchange visits including women and youth), meetings and policy briefs.

⁶⁰ CGSpace is a repository of CGIAR and partner research outputs and knowledge products.

126. **Sub-activity 1.4.2.2: Participation in regional and global climate policy mechanisms.** In practice, the sub-activity will consist of a joint participation of the 4 countries in global and regional climate policy events (global science and policy mechanisms on dairy and climate change) such as the Greenhouse Gas and Animal Agriculture Conference 2025, the African Climate Summit, the African Dairy Conference, and the UNFCCC COP⁶¹). The objective will be to present the progress and results from the Programme for scaling-up and replication, increase its visibility, and engage with other countries in sharing experience and learning.

127. **Sub-activity 1.4.2.3: South-South and Triangular Cooperation (SSTC).** This sub-activity will support the emergence of a global network of experience and expertise on reducing GHG emissions, especially methane, in dairy production and value chains, as well as financing solutions to support these reductions. This sub-activity will finance the participation in knowledge-exchange programs, including through IFAD's "Learning Routes" with e.g. Argentina, Brazil (Embrapa), Costa Rica and Uruguay on Tier2 analysis and climate impact assessments in the dairy sector. Other initiatives for SSTC will be identified and assessed by the Regional Steering Committee. It will finance study tours and participation of country delegations to workshops to share innovations and scalable practices generated by the Programme.

COMPONENT 2: ENABLING LOW EMISSION AND CLIMATE RESILIENT DAIRY VALUE CHAIN DEVELOPMENT

128. In line with the Programme paradigm shift, Component 2 aims putting dairy smallholders and private sector value-chain operators on a transformative pathway towards a low-emission and climate-resilient dairy sector. Climate mitigation and adaptation is critical on-farm, where 90 percent of the sector emissions arise. This Component will contribute to GCF MRA1 "Energy Generation and Access", MRA4 "Forestry and Land Use", ARA1 "Most vulnerable people and communities", and ARA4 "Ecosystems and ecosystem services". Insufficient availability and adoption of climate-smart technologies and practices have been identified in section B.1 and the ToC as key barriers to low emissions and climate-resilient dairy value chains in East Africa. The selected technologies respond to the needs of smallholder dairy farmers as well as medium and large-scale commercial farmers, dairy cooperatives, MCCs, MCPs and other SMEs in the value chain.

129. This Component will contribute to the reduction of total GHG emissions and lower GHG emission intensity per kg of milk compared to a "business-as-usual" scenario, and increase climate resilience of dairy farmers through a three-pronged approach, i.e. (i) improvement in efficiencies in dairy production at farm and post-farm gate levels; (ii) enhanced nutrient circularity, more particularly through manure management and bioeconomy interventions, and (iii) carbon sequestration and removal through improved pastures and grassland management. This strategy will be implemented through three Outputs, namely: (a) smallholders' and cooperative capacity development; (b) the adoption of integrated solutions for the dairy smallholder farm, including: animal breeding and health, feed and fodder, manure management, and biogas use; and c) improved pasture management, including access to water to restore ecosystems' functions and increase soil carbon sequestration.

130. Beneficiaries of the component are smallholder dairy farmers, operating in zero-grazing/ intensive or semi-zero grazing/semi-intensive systems. According to the Market Study, these smallholder dairy farmers are the critical segment to scale, to grow the formal dairy sector and drive sustainable intensification while addressing climate change adaptation and mitigation. The support to the dairy value chain will also benefit medium and large-scale commercial farmers organised in dairy cooperatives and SMEs. Under Output 2.3, the Programme will also benefit pastoralists, agro-pastoralists, and medium and large-scale extensive farmers. Entry points to reach the dairy farmers include dairy cooperatives and MCCs, L-FFS/ PFS and community-based organisations.

131. Given the focus on climate-vulnerable smallholder dairy farmers, Component 2 will utilise (non-reimbursable) grants (mechanisms depending on national practices and IFAD-funded projects co-financing DaIMA) to initiate the graduation of small-scale operators and strengthen their capacities to adopt productivity enhancing and emission-reducing technologies. If appropriate, grants will also support dairy cooperatives and MCCs to finance solutions that drive increased awareness and understanding. Once this grant approach (under Outputs 2.1, 2.2 and 2.3) is successfully deployed to demonstrate the impact of smallholders' livelihoods, producers will be able to graduate and access finance through Component 3's instruments. Where private sector engagement is stronger, highly

⁶¹ Several participants from the Programme preparation process in 2023 were present at COP27 and will be attending or speaking at COP28.

concessional loans (depending on the activity and the economic capacity of the recipient farmer) under Component 3 will be used.

132. **Activities under this Component are innovative, with scale-up and replication potential.** Transforming the dairy sector requires the comprehensive adoption of technologies and practices that sustainably increase production, reduce absolute emissions, and lower emissions intensities. The proposed pathways, as identified in the Feasibility Study (see Chapter 6), includes a range of well-selected and proven technologies that have significant mitigation and adaptation impacts. These technologies will be promoted through the L-FFS, as all required modules will be updated with all aspects related to the pathways. In addition, digital apps will be promoted to improve herd management.

133. **Activities under the Component are sustainable.** The strengthening of dairy cooperatives will significantly improve their operational and financial sustainability. This will be achieved through a combination of supports, including enhanced management capacity, business development services, expanding/upgrading the network of MCCs, investments in renewable energy to reduce the milk losses and energy expenditures, and building linkages with upstream and downstream dairy value chain stakeholders, as well as access to markets.

Output 2.1: Smallholders' and cooperative capacity development strengthened.

134. The aim of Output 2.1 is to provide dairy farmers with consistent and stable access to market outlets for their milk with fair prices and on-time payment. This will be achieved through the strengthening of dairy cooperatives and their network of MCCs and milk collection points (MCPs). Dairy cooperatives are an essential entry point to unlock the climate-smart milk production potential and driving adoption by dairy farmers of transformative climate-resilient practices and technologies. Cooperatives not only provide a stable outlet for milk, but also (i) work to comply with standards for milk; (ii) transmit technical knowledge and information on climate-smart practices to their members; (iii) structure the dairy value chain and engage in upstream and downstream supply-chain agreements; and (iv) facilitate access to markets, inputs, services, and financial services.

135. **Activity 2.1.1 Strengthen dairy groups and cooperatives on organizational management and service delivery.** This activity aims to form and/or strengthen cooperative, organizational, and technical capacities of dairy groups and cooperatives, and raise their climate-change and awareness of gender transformative approaches and social inclusion considerations in their management teams and membership. This will aim to increase women and youth representation, voice and decision-making in cooperatives. Support will also be provided to establish or enhance their network of MCCs and MCPs, and their business linkages. Activities 2.1.1 and 2.1.2 will be implemented by the relevant Executing Entities in each country (see B.3 and A.20), through service providers, i.e. (i) SLDL and KDB in Kenya, (ii) the Rwanda National Dairy Platform ((RNDP)) in Rwanda; (iii) the Dairy Development Authority (DDA), in Uganda; (iv) the Tanzania Dairy Board (TDB) and the Tanzania Cooperative Development Commission (TCDC). Implementation will also rely on specialized service providers, such as farmers' organisations (e.g. National Alliance of Agricultural Co-operatives in Uganda, NAAC), NGOs (e.g. Heifer International, SNV) and agritechs for digitisation activities.

136. **Sub-activity 2.1.1.1 Strengthen the governance and management structures of dairy cooperatives.** This sub-activity will include: (i) training on cooperative governance, management practices and management controls for all management, supervisory and general committees of dairy cooperatives; introduce incentive to encourage ethical practices in dairy cooperatives; (ii) the business coaching of cooperatives by business development service providers, (iii) where desirable, assistance for the formation and registration as a formal dairy cooperative; and (iv) the participation in national and regional events that are essential for dairy value chain governance, promoting inclusive policy, and business dialogues.

137. **Sub-activity 2.1.1.2 Strengthen capacity to reduce milk losses and ensure milk quality and safety.** At the aggregation stage, up to approximately 8 percent of the milk is lost via spillage and spoilage. Aiming at reducing unproductive GHG emissions due to milk losses, this sub-activity is to assist dairy cooperatives, MCCs and MCPs in upgrading their technical capacity in milk quality control, covering raw milk testing at milk collection, hygiene, and initial processing, in line with national and regional food safety standards. The support will include: (i) basic equipment for milk collection (heavy-duty bicycles, motorcycles, milk cans and can holders, testing equipment) and storage (cooling tanks), (ii) insulated milk transport vehicles (demonstration purpose); and (iii) technical assistance and training on milk quality, food safety and hygiene for dairy stakeholders along the value chain, e.g. farmers, milk transporters, staff of cooperatives, MCCs and MCPs; (iv) introduction of milk quality management systems.

138. **Sub-activity 2.1.1.3 Expand the outreach and capacity of dairy cooperatives, MCCs and MCPs.**

The expansion of market-oriented dairy operations and higher productivity of dairy cows, which leads to lower emission intensity of milk produced and sold, requires the nearby availability of milk collection capacity. Therefore, the Programme will support investments in MCCs and MCPs, as well as small processing equipment for cooperatives that are not linked / out of reasonable reach from existing formal market processors and have sufficient scale to engage in processing. Milk collectors will be facilitated to collect milk (at farm gate) and deliver to cooperatives and MCCs, for example with heavy-duty bicycles initially owned by the rural organisations, to be transferred to the milk collectors' ownership with proven use and maintenance over a period of 2 years.

139. **Sub-activity 2.1.1.4 Create awareness on gender transformative approaches among households of dairy farmers.**

Gender awareness have been integrated into the Programme interventions. The Enhanced Gender Action Learning System (GALS+) methodology will be used to address power relations and inequalities especially at household level, thereby enhancing behavioural change towards positive gender and social norms, attitudes and practices in dairy management for purposes of increasing gender equality and women's empowerment, and climate resilience. As men and women have diverse ways of coping and adapting to the adverse impacts of CC, GALS+ includes (i) an integrated climate component to strengthen the mitigation measures and adaptive capacities of households and communities to CC; (ii) nutrition and food safety aspects. Support will be given to: (a) gender awareness and targeting workshops for staff of implementation units and contractors; (b) training of GALS lead farmers and cooperative management as ToT; (iii) rolling out of GALS in L-FFS households through outreach; and (iv) national gender sensitization and training workshops for the management of dairy cooperatives. The Programme will also organize trainings on Gender-based Violence (GBV) and Sexual Exploitation Abuse and Harassment (SEAH) prevention. DaIMA will also invest into the Free Prior and Informed Consent (FPIC) approach to ensure stakeholders' and communities' buy-in for the various activities.

140. **Activity 2.1.2 Build and enhance Productive Alliances along the dairy value chain.** The Programme will provide support to productive alliances between farmer cooperatives and enterprises with farmer-allied engagement models, and scaling-up of dairy clusters. This will enable farmers to access affordable, high-quality inputs and services.

141. **Sub-activity 2.1.2.1 Strengthen cooperatives' capacity in business and financial management.**

The aim is to ensure that dairy cooperatives and their MCCs are run as viable business enterprises. Therefore, the Programme will support cooperatives in building out business plans and partnerships for expanding and enhancing the services they offer member farmers (e.g., facilitated feed and AI service purchases checked-off against milk sales) – serving as “dairy hubs”. For implementation, providers of business development services (BDS) will be mobilized. In addition, the members of dairy cooperatives and MCCs will receive: (i) financial literacy training, financial management training and equipment (software and IT equipment); (ii) gender-sensitive and inclusive business management and business plan development training (using participatory tools such as RuralInvest, developed by FAO⁶²).

142. **Sub-activity 2.1.2.2. Build linkages along the dairy value chain.**

The BDS service providers engaged to strengthen cooperatives will further provide capacity building for negotiation of offtake agreements between cooperatives and private off-takers (Productive Alliance model), as well as public procurement (e.g. school or hospital procurement) for better milk prices guaranteed by contract for selected cooperatives with high capacity. The aim is to support establishing Productive Alliances with cooperatives and in designing, enhancing, and scaling key components of their farmer engagement models, such as quality-based or volume-based price incentive terms offered by processors, extension provision, and direct or facilitated provision of inputs and services. Through this support, a pre-investment pipeline will be developed, and related financing needs identified.

143. **Sub-activity 2.1.2.3. Promote digitalization and traceability of dairy operations.**

The Programme will promote digitalization and traceability to ensure quality and entry into formal supply arrangements that come with services, and improved accounting (money and stock) for access to credit. Use of transformative end-to-end (E2) digital solutions is currently limited. Digitalized systems for monitoring on-farm activities, milk production disaggregated by farm (gender of farmer), aggregation, transportation, and marketing will increase the efficiency and transparency in the dairy value chain and cold chain. It will support off-takers and cooperatives to improve timely payments to farmers, traceability of the quality of milk and linkage to advisory services where farmers

⁶² <https://www.fao.org/in-action/ruralinvest/about/en/>

might need advisory interventions and access to finance. In the four countries, the Programme will enhance the capacity of milk aggregators and cooperatives to improve their record-keeping and management through a digital cooperative and MCC management information system (MIS) that will: (a) record the production of farmers, (b) track milk flow/volumes and quality, (c) manage records of farmer payments allowing the development of digital profiles and credit scores. Digital solutions developed by local agritechs and agri-fintechs (e.g. Emata in Uganda) or FAO (Collect Mobile App) will be scaled-up to enable business operations management and credit-scoring for better access to finance. A pilot will start with a limited number of cooperatives in each country and gradually upgraded. The Programme will finance the required IT equipment and services.

144. **Activity 2.1.3 Promote renewable energy (RE) and waste management technologies.** The decarbonization of the dairy value chain will be promoted. The Programme will focus on the substitution of fossil fuels and firewood by green energy sources such as solar or less wood-intensive energy sources for milk cooling and processing contributing to reduced workloads for women. Insulated vehicles and solar energy supply are typically more suitable solutions for larger MCCs (>3m³ capacity). For processing, cold chain and quality preservation solutions (e.g., cooling tanks, insulated / refrigerated trucks) are fundamental to business operations and contribute to reduced milk losses (and less unproductive GHG emissions), yet not always used by small-scale processors. Solar is the key mitigation solution and most applicable to relatively larger processors in the short-term given high upfront costs. Also the management of waste and wastewater will be promoted. There is limited adoption of waste management solutions along the value chain today. Processors only re-use waste (e.g., by-product whey as feed) selectively and recycling is not a priority. Beneficiaries of the support would be MCCs and SMEs. Activity 2.1.3 will be implemented by the relevant Executing Entities in each country (see sections B.3 and A.20), through service providers, i.e. (i) the KDB in Kenya, (ii) the SPIU and Rwanda Energy Group (REG) in Rwanda; (iii) the DDA, Ministry of Energy and Mineral Development, in Uganda; (iv) the MLF and TDB in Tanzania. Specialized service providers will be mobilized where needed.

145. **Sub-activity 2.1.3.1 Assess the need and viability of renewable energy and waste management solutions.** The aim will be to assess the clean energy needs of dairy cooperatives and propose RE solutions (photovoltaic – PV, biogas systems, etc.), energy and labour-efficient equipment, and solid and water waste management solutions. The assessment will look at scale and viability of such investments not just needs and potential tech solutions.

146. **Sub-activity 2.1.3.2 Support the acquisition of renewable energy and waste management solutions.** This will include the upgrading or equipping of dairy cooperatives with suitable RE, PV-powered technology for milk coolers for milk aggregation, milk transfer tanks, solar water heaters, water recycling systems, heat exchangers, milk pasteurizers, butter churners and cheese makers. Solar energy will complement the existing electricity from national grid or generated by standby generators to cool down milk to 4°C.

147. **Sub-activity 2.1.3.3 Organize capacity building in energy efficiency and waste management.** This training will focus on the management of the RE solutions in dairy cooperatives, SMEs and MCCs. This will include strategies to avoid waste (including e-waste) through repair of the installed equipment and the establishment of an equipment maintenance account in a local financial institution by each supported rural organisation.

Output 2.2: Breeding, animal health, feed and fodder, manure management improved.

148. As detailed in section B.1 and Feasibility Study (Annex 2, Chapter 6), the pathways to reduce farm-level GHG emissions from dairy cattle are based on the following building blocks⁶³: (i) promotion of dairy breeds that are resistant to heat stress and better adapted to CC; (ii) improving animal welfare, animal health and disease prevention to increase productivity of dairy cattle and reduce mortality, and herd management to optimize the composition of the dairy herd; (iii) affordable quality forage and fodder, and access to water; (iv) climate-smart feeding practices to reduce enteric methane emissions; (v) manure management and use to recycle nutrients and reduce GHG emissions; and (vi) biogas digesters at farm level to treat manure and recover methane for household energy use while providing clean energy for cooking.

149. **Activity 2.2.1 Promote the adoption of climate-smart dairy breeds adapted to climate change.** Breeding and genetic improvement will aim at improving productivity and heat tolerance of dairy cattle while preserving resilience of livestock production systems by crossing indigenous and exotic breeds through AI. Exotic breeds, such as Holstein-

⁶³ FAO and GDP. 2018. Climate change and the global dairy cattle sector – The role of the dairy sector in a low-carbon future. Rome. 36 pp. Licence: CC BY-NC-SA- 3.0 IGO

Friesian or Jersey, are more vulnerable to high temperatures, and consequently to climate change, but present a higher productive potential. Indigenous breeds, despite their lower productive potential and small size, are a reservoir for resilience traits such as hardiness, mobility, rendering them more adaptable to extreme climatic conditions and to certain diseases. Different generations of crossbreeds (exotic x native breeds) would contribute to maintaining the resilience traits of indigenous breeds, while also adding productive traits to the gene pool, increasing the offspring's productivity in milk production, therefore contributing to reducing GHG emission intensity. Available studies shows that improved dairy animals have a significantly lower GHG emission intensity per kg of milk due to their higher production when compared to indigenous breeds^{64,65,66}. This activity will be implemented by the relevant Executing Entities in each country (see sections B.3 and A.20), through service providers, i.e. (i) KALRO and the Kenya Animal Genetic Resources Centre (KAGRC) in Kenya; (ii) RAB and private in Rwanda; (iii) MLF, TALIRI, National Artificial Insemination centre (NAIC), in Tanzania; and (iv) MAAIF with support from NARO, NAGRC, and DDA in Uganda. Support will be given to scale-up the national breeding strategies with specific focus on climate resilience of dairy cows.

150. **Sub-activity 2.2.1.1 Make good quality semen for climate-smart breeds available, adaptable to extreme climatic conditions and to diseases.** The Programme will enhance availability and accessibility of germplasm of exotic that will be crossed with local⁶⁷. This will be achieved through (i) the identification of potential climate-smart cattle breeds; (ii) the import of sexed semen⁶⁸ and/or embryos, if needed; (iii) the procurement of high genetic merit bulls to produce sexed semen or no-sexed semen for national and regional use; (iv) the production and conservation of climate-smart breeds' germplasm, using laboratory techniques like in-vitro fertilization, or simply artificial insemination (AI); and (v) capacity development to produce sexed semen to increase production of dairy heifers.

151. **Sub-activity 2.2.1.2 Invest in breeding and artificial insemination (AI) centres.** To solve bottlenecks in AI and embryo transfer, the Programme will support scaling up capacities for production and distribution of liquid nitrogen, semen, and embryos. This includes the availability of liquid nitrogen, storage and quality control equipment, dry ice pilots for conservation of semen, and other specialized equipment. In addition, the following investment will be supported. In **Kenya**, the Programme will involve the Kenya Livestock Breeders Association, which is specialised in extension and training, breed inspection and registration, performance recording and reporting and feedback. In **Rwanda**, support will be given for the establishment of semen and liquid nitrogen storage and distribution centers, as well as to the existing bull centre in Songa Station to acquire micro-manipulators to enhance semen production operations. Moreover, RAB will be strengthened in record keeping and continuous updates of the Studbook to identify resilient and high productive crossbreeds. In **Tanzania**, support would be given to establish an AI centre in Southern Highland. Also, the feasibility of an LN unit in Zanzibar will be analysed. In **Uganda**, this investment will strengthen the NAGRC and Data Bank to improve the production and access to high-quality semen in small and medium size farms.

152. **Sub-activity 2.2.1.3 Scale up the insemination of dairy cattle to make the herd more productive and climate resilient.** The Programme will (i) train AI technicians, who can be private inseminators, para-veterinarians, lead farmers of L-FFS and District livestock officers (at least 30 percent youth and 40 percent women); (ii) scale up the insemination of dairy cattle in the Programme intervention area; (iii) include aspects of the interaction between dairy cattle husbandry and climate change in the L-FFS curriculum. In **Kenya**, the Programme will support the National Breeding Strategy and Action Plan, in collaboration with the Kenya Livestock Breeders Association and the Kenya Animal Genetics Resources Centre (KAGRIC) and ADC. Public and private partnerships (PPP) will be promoted to have AI services accessible to the farmers. In **Rwanda**, the breeding

⁶⁴ FAO & New Zealand Agricultural Greenhouse Gas Research Centre. 2017. Supporting low emissions development in the Ethiopian dairy cattle sector – reducing enteric methane for food security and livelihoods. Rome. 34 pp.

⁶⁵ FAO & New Zealand Agricultural Greenhouse Gas Research Centre. 2019. Options for low emission development in the Uganda dairy sector - reducing enteric methane for food security and livelihoods. Rome. 39 pp. <https://www.fao.org/3/ca3375en/ca3375en.pdf>

⁶⁶ FAO & New Zealand Agricultural Greenhouse Gas Research Centre. 2019. Options for low emission development in the Tanzania dairy sector - reducing enteric methane for food security and livelihoods. Rome. 34 pp. <https://www.fao.org/3/ca3215en/CA3215EN.pdf>

⁶⁷ Sahiwal, an indigenous breed that is a relatively high producing breed and resilient and has been adopted in Kenian pastoral communities as a means to improve milk production.

⁶⁸ Sexed semen is processed bovine semen that increases the likelihood of producing female calves, which are more valuable in dairy farming for their milk production. This technology is superior to conventional semen as it enhances dairy productivity by expanding milk-producing herds, improving genetic quality, and offering climate resilience benefits through higher returns and better herd management.

programme of RAB will be scaled up including the training of more private AI technicians through Veterinary Sanitary Mandate. In **Tanzania**, additional technicians will be trained and equipped with start-up kits. In addition, bull centers will be established in regions where AI is not available. In **Uganda**, more AI technicians will be trained by the NAGRC and will be equipped with AI kits focusing on the regions where AI services are still poorly developed.

153. **Sub-activity 2.2.1.4 Organize awareness campaigns and support breeders' organizations.** The campaigns will target members of dairy cooperatives and L-FFS groups to create awareness on the use of sexed-semen, thus impacting the herd's sex ratio, and further reduce absolute GHG emissions and the emission intensity per kg of milk produced mainly by replacing many low-productive local cows with a few adapted and high performant crossbreeds, thus reducing the herd size. In the four countries, support will be given to the breeders' associations and societies for animal registration and maintenance of breeding standards.

154. **Activity 2.2.2 Provide support for better animal care, herd management and disease prevention.** In East Africa, infectious diseases and parasites rank among the primary causes of livestock productivity and production loss. Production losses through reduced performance (growth rates, milk yield) or animal loss through morbidity, mortality, and abortion results in cows with relatively high GHG emission intensities. These losses affect people's resilience to CC, livelihoods and nutrition security and can affect human health through zoonotic disease transmission. In addition, the growing importance of zero-grazing systems has implications for animal welfare, thus farmers will be encouraged to consider animal well-being aspects. This activity will be implemented by the relevant Executing Entities in each country (see sections B.3 and A.20), through veterinary services providers like: (i) KALRO and County Governments in Kenya; (ii) RAB, RCVD and the LGAs in Rwanda through Veterinary Sanitary Mandate; (iii) MLF and LGAs in Tanzania; and (iv) MAAIF (Directorate of Animal Resources –DAR, District Local Government (DLGs) and the private sector in Uganda.

155. **Sub-activity 2.2.2.1 Organise trainings on animal husbandry, with a specific focus on animal health, animal welfare, herd management and disease prevention.** Updated modules of the L-FFS (and other applicable modules) training on animal health and welfare, and herd and reproduction management will be implemented. Dairy farmers, including women and youth who often act managers and animal handlers, will receive training to better care for the animals and to adopt biosecurity measures to prevent common diseases and pests that impact the production or suitability of the milk for human consumption. They will also receive support to better manage their herds, reproduction calendar, and animal nutrition to reduce animal mortality, heat stress and increase productivity, thus further reducing absolute CH₄ emissions and the emission intensity per unit of milk produced.

156. **Sub-activity 2.2.2.2 Scale up the use of digital apps to monitor animal health and herd performance.** Digital record keeping on-farm (e.g., using iCow, DigiCow Africa, My Fugo, i-EDSR, AMIS) will be scaled up to monitor the productive and health performances of the herd (e.g., records of calf mortality and causes, number of lame animals), to plan artificial insemination and guide herd management. The app will be promoted through the L-FFS in interested groups. The support includes: (i) training of L-FFS lead farmers on the use of the apps; (ii) guiding L-FFS farmers in the use of the technology; (iii) equip the farmers in the pilot with digital tools, including youth. To implement the sub-activity, the Programme will work with agritech companies and start-up to develop digital solutions that will enhance transparency in farm data and operations, monitoring of CH₄ emissions reduction and traceability of milk products.

157. **Sub-activity 2.2.2.3 Scale-up vaccination campaigns to mitigate the impact of climate change on the spread of diseases.** The Programme will scale up and extend the vaccination campaigns to target in priority dairy farmers in the Programme area, who are members of various cooperatives and L-FFS. The vaccination campaign will focus on: (i) the spread of East Coast Fever (ECF), Contagious Bovine Pleuropneumonia (CBPP) and Heartwater disease to new production areas through an extended vaccination campaign, in **Kenya**; (ii) ECF, RVF and FMD amongst others in **Rwanda**; (iii) ECF, FMD, RVF and CBPP in **Tanzania**; (iv) transboundary animal diseases (TADs), e.g. ECF, FMD, CBPP, and LSD in **Uganda**.

158. **Activity 2.2.3 Promote climate-smart forage production and conservation, and water harvesting.** The adoption of climate-resilient forage varieties and conservation technologies is essential to reduce enteric methane

emissions intensities in the dairy value chain, while increasing milk yields of dairy cattle^{69,70,71}. The benefits are multiple and encompass CC adaptation through adoption of a resilient forage variety (e.g., *Lablab purpureus*, *Brachiaria sp.*, *Cenchrus purpureus* (Napier grass), *Leucaena*, *Calliandra*, *Panicum maximum* (Giant Panicum)), improvements in soil fertility and increased capacity of farmers to meet forage production needs during dry periods and to respond to climate-induced pests and diseases. This activity will be implemented by the relevant Executing Entities in each country, through service providers, i.e. (i) KALRO and private seed multipliers in Kenya; (ii) RAB and LGAs in Rwanda; (iii) MLF, Tanzania Official Seed Certification Institute (TOSCI), Vikuge pasture seed farm, and LGAs in Tanzania; and (iv) MAAIF / NARO and private sector e.g. SNV in Uganda.

159. **Sub-activity 2.2.3.1 Make certified forage seed available to private and community multipliers.**

This sub-activity will (i) support the multiplication of breeder and foundation seeds for climate-smart, emission-reducing fodder by the national mandated public services; (ii) facilitate a seed certification scheme and sensitize the private sector to produce certified forage and fodder seeds.

160. **Sub-activity 2.2.3.2 Support multiplication of climate-resilient seed for fodder crops and pastures.** This sub-activity will include contracting community and private seed multipliers (including the youth and women) for multiplication, conservation of forage and pasture seeds. In each country, standard approaches will be used to promote fodder and pasture seed agri-business through facilitating multiplication of seeds and making them available to L-FFS, PFS, cooperatives, and other groups.

161. **Sub-activity 2.2.3.3 Build capacity of dairy farmers on forage production and conservation on intensive and semi-intensive dairy farms.** The L-FFS will be the knowledge-sharing vehicle to promote improved forage production and conservation. The L-FFS will focus on (i) training on good forage management practices (e.g., grass and legume intercropping, agroforestry, weed management, use of biochar); (ii) training and demonstration on small-scale irrigation to produce forage crops; (iii) training and demonstrations of mechanization of forage production with fodder choppers/cutters and conservation equipment; (iv) training and demonstrations on forage conservation practices, to respond to seasonal feed availability and quality (e.g., protein content, digestibility) variations. Youth groups will be supported to venture into appropriate commercial forage production⁷², e.g. hydroponics.

162. **Sub-activity 2.2.3.4 Access to water for animals and forage production in intensive and semi-intensive dairy systems⁷³.** The Programme will provide small-scale equipment for rainwater harvesting and recycling (e.g. individual dam sheets, 3m³ rooftop water tanks and water distribution kits).

163. **Activity 2.2.4 Promote climate-smart feeding practices to reduce enteric methane emissions.** Appropriate diet formulation is key to improve dairy cattle productivity, improve animal welfare, while reducing methane (CH₄) emissions. Balanced diets (more energy-dense or more digestible) decrease retention time in the rumen and reduce ruminal fermentation and methane yield (kg CH₄/kg dry matter intake). Better nutrient utilization (e.g., avoiding deficiencies or surpluses) reduces the environmental impact of livestock. The Programme will support the development of feed packages and plans, build national capacity on feed formulation, and promote these technologies through L-FFS and others. Implementation will be done by relevant EEs through service providers i.e. (i) KALRO, SDLD in Kenya; (ii) RAB and the University of Rwanda, College of Agriculture, Animal Science and Veterinary Medicine (UR-CAVM) and LGAs in Rwanda; (iii) MLF, TALIRI and LGAs in Tanzania; (iv) the MAAIF (NARO), in collaboration with NGOs (e.g. SNV, Heifer International) and private service providers in Uganda.

164. **Sub-activity 2.2.4.1 Develop production system-specific supplementation plans and feed packages,** based on available feed resources (energy or protein depending on the limiting nutrient, e.g. Urea Molasses Mineral Blocks⁷⁴). In each country, the national research and academic institutions will develop feed

⁶⁹ FAO & New Zealand Agricultural Greenhouse Gas Research Centre. 2017. Supporting low emissions development in the Ethiopian dairy cattle sector – reducing enteric methane for food security and livelihoods. Rome. 34 pp.

⁷⁰ FAO & New Zealand Agricultural Greenhouse Gas Research Centre. 2019. Options for low emission development in the Uganda dairy sector - reducing enteric methane for food security and livelihoods. Rome. 39 pp. <https://www.fao.org/3/ca3375en/ca3375en.pdf>

⁷¹ FAO & New Zealand Agricultural Greenhouse Gas Research Centre. 2019. Options for low emission development in the Tanzania dairy sector - reducing enteric methane for food security and livelihoods. Rome. 34 pp. <https://www.fao.org/3/ca3215en/CA3215EN.pdf>

⁷² The Uruhimi Kageyo Cooperative is a successful example.

⁷³ For access to water for extensive/ pastoral and agro-pastoral systems, please look at output 2.3

⁷⁴ FAO. Feed Supplementation Blocks. FAO Animal Production and Health Division. Paper 164. 2007

plans and packages for the main national dairy production systems. Biochar as an edible feed supplement will be promoted. Feed formulation will be included in the curriculum of the L-FFS. This sub-activity will benefit from the Programme's partnership with ILRI, as well as its regional knowledge-sharing platform to make existed training materials available.

165. **Sub-activity 2.2.4.2 Build capacity on appropriate feed formulation.** In each country, the Programme will support the capacity development of technical advisors, cooperative management, small feed processing plants and L-FFS lead farmers on sustainable animal feed formulations such as total mixed rations (TMR) to enhance productive efficiency and to mitigate GHG emissions.

166. **Sub-activity 2.2.4.3 Promote appropriate diet formulation that reduces enteric methane emissions.** In the four countries, the L-FFS and the dairy cooperatives will be the vehicle to reach out to dairy farmers and promote appropriate feeding of dairy cattle. The Programme will: (i) provide start-up kits for L-FFS to monitor diet formulation to reduce methane yield (e.g., Rumen8 tool by SNV); (ii) run L-FFS of feed formulation, including on the use of agroforestry leguminous fodder trees with methane inhibiting properties such as *Calliandra* sp. for improved feed supplementation and carbon sequestration. The Programme will also pilot and leverage on the existing innovative enteric methane inhibitors used as mixtures in feed to be adopted in zero-grazing and commercial/intensive systems⁷⁵.

167. **Activity 2.2.5 Increase sustainable nutrient recycling of manure and dairy waste to reduce emissions.** Livestock manure is a source of methane (CH₄) and nitrous oxide (N₂O) emissions, as well as of pollutants such as ammonia (NH₃) and nitrate (NO₃). At the same time, manure contains essential nutrients and is often the primary source of crop fertilization on smallholder dairy farms, as mineral fertilizers are too expensive or unavailable.⁷⁶ In addition, its costs have further increased due to recent events (Ukraine, Palestine and Lebanon wars).⁷⁷ Many smallholder farms are limited in nutrients, which limits crop yields^{78,79}. Circular management of agricultural nutrients via livestock manure is the best scalable option to overcome these limitations, improve soil health, and improve on-farm productivity. This has a positive effect on livestock productivity, as better manure management also improves the nutrient availability for feed and forages, which increases their growth, biomass yield and nutritional quality and ultimately improves milk production. In addition, manure application reduces soil nutrient mining and soil degradation and prevents additional CO₂ emissions from loss of soil organic carbon. This activity will be implemented by EEs, through relevant service providers, i.e. (i) SLDL, KALRO, KBP in Kenya; (ii) RAB and LGAs in Rwanda; (iii) MLF, TDB, LGAs in Tanzania; and (iv) MAAIF (NARO), ZARDI, and implementing partners (e.g. SNV, Heifer) in Uganda. Training materials have been developed by ILRI and NARO (Uganda).

168. **Sub-activity 2.2.5.1 Manure management in intensive and semi-intensive dairy production systems.** The Programme will create awareness and build capacity on low-cost manure management for soil health and soil microbiome improvement. The sub-activity will consist in providing trainings and equipment through L-FFS and demonstration farms, using existing manuals developed by NARO in Uganda, ILRI, etc. on scalable and already tested manure management practices (manure separator, manure ponds, use of biochar, manure covering with banana leaves, compaction, etc.) and manure use applications offering livelihoods opportunities.

169. **Sub-activity 2.2.5.2 Promote adapted animal housing.** The Programme will promote adapted animal housing with the possibility of separating solid and liquid manure. Good example of animal housing promoted by Heifer International can be scaled up.

170. **Sub-activity 2.2.5.3 Manure management in extensive systems** (pastoral and agro-pastoral dairying regions). The Programme will: (i) run L-FFS/PFS for promoting manure collection from communal kraals

⁷⁵ FAO. 2023. Methane emissions in livestock and rice systems – Sources, quantification, mitigation and metrics. Rome. <https://doi.org/10.4060/cc7607en>

⁷⁶ Sheahan, M., & Barrett, C. B. 2017. Ten striking facts about agricultural input use in Sub-Saharan Africa. Food policy, 67, 12-25. <https://doi.org/10.1016/j.foodpol.2016.09.010>

⁷⁷ Alexander P, Arneth A, Henry R, et al (2022) High energy and fertilizer prices are more damaging than food export curtailment from Ukraine and Russia for food prices, health and the environment. Nature Food 2022 4:1 4:84–95. <https://doi.org/10.1038/s43016-022-00659-9>

⁷⁸ Leitner S, Pelster D. E., Werner C, et al. 2020. Closing maize yield gaps in sub-Saharan Africa will boost soil N₂O emissions. Curr Opin Environ Sustain 47: <https://doi.org/10.1016/j.cosust.2020.08.018>

⁷⁹ Mueller ND, Gerber JS, Johnston M, et al. 2012. Closing yield gaps through nutrient and water management. Nature 490:254–257. <https://doi.org/10.1038/nature11420>

for use, sale/ income generation at community level; and (ii) establish community manure sheds to promote manure collection, storage and covering for better manure management and reduce emissions.

171. **Activity 2.2.6 Build capacity for on-farm use of biodigesters and biogas.** Biogas digesters utilize microorganisms to break down organic matter such as livestock manure or other organic waste, during which biogas and a nutrient-rich by-product (called digestate or bio-slurry) are produced. This technology offers a clean source of energy for people who do not have access to clean cooking fuels⁸⁰. In addition, it creates a biofertilizer (for use as organic fertiliser on nearby fields), reduces methane emissions (up to 70-90 percent compared to liquid manure storage)⁸¹, and ammonia and nitrous oxide emissions⁸². Biodigesters can reduce deforestation and empower women by reducing the time they spent collecting firewood⁸³. As part of the low-carbon pathways, the Programme will encourage the uptake of on-farm biodigesters for intensive and semi-intensive dairy farms. The activity will be implemented by (i) SLDL, Ministry of Energy and Petroleum (State Department for Energy) in Kenya; (ii) RAB and REG in Rwanda; (iii) MLU, the Centre for Agricultural Mechanisation and Rural Technology (CAMARTEC), Ministry of Environment in Tanzania; (iv) the MAAIF (NARO), Ministry of Energy and Mineral Development, private sector service providers to support installation (Biogas National Alliance, Bio Solutions Uganda), and other specialized implementation partners (e.g. SNV).

172. **Sub-activity 2.2.6.1 Support capacity building in biodigester construction and management.** The Programme will support the training and accreditation of biodigester technicians, with a particular focus on women and youth, in the target districts. In addition to installation-related aspects of biogas valorisation and quality management, including safe working standards.

173. **Sub-activity 2.2.6.2 Support investments in biodigesters at the household level**⁸⁴. The Programme will support investments in biodigesters for dairy farms in its intervention area.

Output 2.3: Pasture and grasslands management improved

174. Many dairy producers rely on pastures and grasslands as main feed resource, which are constrained by limited water availability and follow seasonal rainfall patterns. Well-managed and restored pastures and rangelands increases climate resilience through healthier ecosystems, improved water access, biodiversity conservation, and mitigate environmental risks – such as land degradation. In addition, to reduce emissions from dairy production systems, the Programme considers soil carbon sequestration as an important pathway of GHG emission removals to offset livestock and manure emissions. Well-managed rangelands not only improve the carbon balance of the livestock production systems, but they are also more productive (increased biomass yield) and improve the animal diet and reduce enteric CH₄ emissions.

175. **Activity 2.3.1 Strengthen rangeland governance structures for communal grazing areas.** This activity will finance interventions to improve rangeland management (including access to water), mainly in **Kenya, Tanzania and Uganda**. Engagement and awareness raising among local stakeholders involved in landscape management is essential for locally led adaptation and people-centred planning. The activity will be implemented by relevant EEs, through: (i) the SLDL, KALRO, County Governments and the NLC in Kenya; (ii) MLF and LGAs in Tanzania; (iii) MAAIF (DDA, NARO) with DLGs in Uganda. In each country, facilitators specialised in participatory rangeland management will be mobilized.

176. **Sub-activity 2.3.1.1 Participatory rangeland management planning.** In selected districts/counties, the Programme will (i) map and conduct a capacity assessment of existing community governance structures (including gender aspects), formal (e.g. village land committees, environmental committees, water use committees, local government, district) and informal entities (grazing committees, traditional committees, clan

⁸⁰ International Energy Agency (IEA). 2020. The Energy Progress Report, 2020. Paris. <https://www.iea.org/reports/tracking-sdg7-the-energy-progress-report-2020>

⁸¹ Steinfeld H, Gerber P, Wassenaar T, et al. 2006. Livestock's Long Shadow: Environmental Issues and Options. FAO <ftp://ftp.fao.org/docrep/fao/010/A0701E/A0701E00.pdf> 1–377. <https://doi.org/10.1007/s10666-008-9149-3>

⁸² Mohankumar Sajeev EP, Winiwarter W, Amon B. 2018. Greenhouse Gas and Ammonia Emissions from Different Stages of Liquid Manure Management Chains: Abatement Options and Emission Interactions. J Environ Qual 47:30–41. <https://doi.org/10.2134/JEQ2017.05.0199>

⁸³ Clemens H, Bailis R, Nyambane A, Ndung'u V. 2018. Africa Biogas Partnership Program: A review of clean cooking implementation through market development in East Africa. Energy for Sustainable Development 46:23–31

⁸⁴ Scaling up the results achieved by the Tanzania Domestic Biogas Programme part of the overarching Africa Biogas Partnership Programme, implemented by DGIS, Hivos and SNV

leaders); (ii) support community governance structures to produce pasture and rangelands management plans, including land tenure to secure tenure rights to carry out activities on rangelands, water access, as well as possible conflict resolution over land and water use, and (iii) support the development, revitalization and strengthening of laws/ charters and by-laws governing the use of communal resources (including water). For this sub-activity, the Programme will leverage lessons learnt from and (where appropriate) build on the legacy of TWENDE in Kenya (GCF FP113) and Transforming Eastern Province through Adaptation in Rwanda (GCF FP167).

177. **Activity 2.3.2. Restoration of degraded rangelands, including access to water.** This activity will finance interventions to restore rangelands. The sub-activity will be implemented by (i) the SLDL, KALRO, County Governments and the NLC in Kenya; (ii) RAB and REMA in Rwanda (iii) MLF and LGAs in Tanzania; (iv) MAAIF (DDA, NARO) with DLGs in Uganda. In each country, service providers specialised in rangelands restoration will be mobilized.

178. **Sub-activity 2.3.2.1 Training on rangeland and pasture management.** The Programme will facilitate L-FFS/PFS aimed at guiding and engaging farmers towards improving the management of rangelands and pastures. L-FFS will focus on (i) rotational grazing, (ii) management of grazing intensity (destocking and restocking i.e. sale of animals during the dry season and restocking during seasons of high food availability to reduce pressure on pastures)⁸⁵, (iii) integration of silvo-pastoral practices (multipurpose leguminous trees, including those that are medicinal or for fruits) for better carbon sequestration and reduction of livestock exposure to heat stress conditions, by providing shade for animals (more particularly around water harvesting points), and (iv) biochar production, elaborated from invasive bush species.

179. **Sub-activity 2.3.2.2 Rangelands and pastures restoration.** This sub-activity, focusing on grazing areas, will focus on: (i) reseeded and integration of legumes with grasses in natural pastures, (ii) integration of multipurpose leguminous trees and shrubs (improved agro-forestry systems) and control of invasive tree species, and (iii) implementation of soil and water conservation practices to increase water availability and biomass productivity.

180. **Sub-activity 2.3.2.3 Water availability.** Depending on the country, to enhance access to water for dairy cattle in extensive systems, the Programme will invest in water harvesting systems such as micro-dams, small-scale valley dams, charco dams and rock-catchments schemes. Feasibility studies will be carried out before the start of the work to assess sustainability. The Programme will also enhance the capacity of the community-based water supply organizations (COWSOs) to better manage and operate all water harvesting and storage infrastructure. The design of water infrastructure will follow the guidelines of the local government authorities, making gender considerations such as the provision of animal watering and washing bays for women.

COMPONENT 3: GREEN DAIRY FINANCING FACILITY (GDFF)⁸⁶

181. To support the achievement of the programme objectives and impact, DaIMA will establish a Green Dairy Financing Facility (GDFF) that will operate in the four targeted countries. GDFF aims to enhance access to finance for the target group and support the implementation of climate-smart technologies and practices, thereby facilitating the transition to climate-resilient, sustainable, and low-carbon dairy value chains. This component will specifically address the financial and technological barriers identified in the ToC as key obstacles in greening the sector's operations. Under Component 3, medium-sized farmer-allied dairy value chain actors, including aggregator- and processor-SMEs, agro-input and technology providers, and farmers' cooperatives, will be eligible to access DaIMA-supported financing. Businesses identified by Component 2 as suitable for GDFF financing will be proactively encouraged to apply for support under Component 3.

182. **GDFF Financial Resources.** DaIMA will provide catalytic international financial resources of at least US\$ 35 million, including US\$ 17.5m from the GCF and minimum US\$ 17.5m from IFAD and other partners, to be shared between the selected four Wholesale Banks for on-lending through the Retail Banks to the programme target group. The target in later negotiations with the participating Retail Banks is to raise an additional investment capital injection of US\$ 8.75 million (25 percent of retail loans) from the Retail Banks' local private sector resources. In addition,

⁸⁵ Feasibility Study, chapter 6 for details on the various strategies to be promoted.

⁸⁶ Component 3 includes proposed activities that are subject to change or adjustment and estimates for loan volume, interest rate, and number of borrowers that are based on assumptions which are subject to change.

substantial grant financing is included under Component 3 to support the GDFF lending and dairy sector development operations through a US\$ 10m technical assistance facility.

183. The GDFF credit capital from DaIMA resources will be lent by IFAD to the four Governments in USD and by the governments, through the EEs, to the local implementing partners in local currencies, and all the lending operations to the end borrowers will be conducted in local currencies. The credit capital provided by the international financiers will revolve in the GDFF operation during the period of the DaIMA project loans and will be paid back by the executing partners to the Governments according to the repayment schedule of the IFAD DaIMA loans.

184. To achieve its development objectives, Component 3 activities are structured into two outputs: (i) Output 3.1: Effective Green Financing for Dairy Value Chains; and (ii) Output 3.2: Capacity Building for Green Dairy Investments.

Output 3.1: Effective Green Financing for Dairy Value Chains

185. **Activity 3.1.1: GDFF Operationalized.** In each of the four participating countries, a GDFF line of credit will be established to channel green funding to eligible dairy sector borrowers. This line of credit will be managed by a local Public Development Bank (PDB), which will act as a Wholesale Bank for the GDFF intervention. The Wholesale Banks will at national level manage the financial DaIMA services to the programme target group based on eligibility criteria and other terms and conditions, agreed upon in the programme design and included in the financing agreements between IFAD, the Governments, and the Wholesale Banks. Based on IFAD's rural finance experience in East-Africa and reviews of financial markets in the targeted countries as well as interviews with several candidate banks, the following PDBs were selected as candidates for the Wholesale Banks for GDFF: Agricultural Finance Corporation (AFC) for Kenya, Uganda Development Bank (UDB) for Uganda, Development Bank of Rwanda (DBR) for Rwanda, and Tanzania Agricultural Development Bank (TADB) for Tanzania. The criteria for Wholesale Bank identification included: (i) status of a leading PDB in the targeted countries, (ii) solid institutional and financial conditions, (iii) strong interest in agriculture and rural finance operations, (iv) corporate objective to develop and scale-up the bank's climate finance portfolio, (v) corporate mandate and operational capacity to provide wholesale financing products, and (vi) strong expressed interest to manage the DaIMA wholesale financing operations in the dairy sector.

186. The **Wholesale Banks** will provide bulk loans for on-lending to the eligible Retail Banks, which can be commercial banks or such microfinance banks and commercial development banks that are prudentially supervised. Financial resources provided as bulk loans by the Wholesale Banks to Retail Banks will be exclusively on-lent to eligible climate-smart investment projects of dairy sector SMEs. Operating with the targeting guidelines and other conditions of the wholesale loan agreements between the Wholesale Banks and Retail Banks, the actual lending methods used for retail loans under GDFF financing will follow each Retail Bank's operational structure and network, rural and agriculture finance strategy, and their digital tools development approach.

187. **Retail Banks** for each country will be selected after programme effectiveness as a joint activity between the Wholesale Banks and DaIMA/IFAD. Key criteria in the selection of Retail Banks include: (i) operate under prudential supervision with solid institutional and financial conditions as per central bank standards, and prudentially supervised, (ii) strong interest in financing operations of rural small and medium size businesses, (iii) corporate objective to develop and scale-up the bank's climate finance operations, (iv) an appropriate network to provide services in the DaIMA programme area in the target country, and (v) a strong expressed interest to partner with DaIMA in the planned dairy sector financing operation and co-finance 25 percent of each investment. During the period leading to DaIMA effectiveness, the DaIMA PCUs and the Wholesale Banks will closely co-operate to ensure a smooth take-off of the programme activities. To support and co-ordinate the start-up phase, experienced Climate Finance Specialists will be recruited and hosted by each Wholesale Bank (financed by the TA Sub-component 3.2). At the same time the DaIMA PCU and the Wholesale Banks would finalise the preparations for the selection and training of appropriate Retail Banks to take part in the GDFF lending and develop the framework and action plans for capacity building at both the Retail Bank and the borrowing SME level. Each Retail Bank selection will also go through the IFAD no-objection process.

188. **Targeted Borrowers and Key Lending Products.** As indicated, the GDFF lending operations under Component 3 target medium-sized farmer-allied dairy value chain actors, including aggregators and processors, agro-input and technology providers, and mature dairy farmers' cooperatives. In all four targeted countries, various projects supported by donors including IFAD have special lending windows for individual dairy farmers operating in these dairy value chains. Furthermore, the proposed large Africa Rural Climate Adaptation Finance Mechanism (ARCAFIM) covers the same countries as DaIMA and targets smallholder farmers with CC adaptation finance services including

the dairy sector. The GDFF financing support to dairy value chain companies will be planned and operated in close co-operation with these other green financing interventions in the four target countries.

189. Based on the market and demand surveys conducted during the DaIMA design (see attachments to this Funding Proposal), the following two types of lending approaches/products can be included in GDFF lending, depending on the scale and maturity of the dairy value chain operations of the borrower (more information on the type of client companies for each loan category in the Feasibility Study in Annex 2):

190. **Green Dairy Investment Partnership Loan Product.** These are longer-term partnerships between mature, well-organised dairy businesses and the GDFF Retail Banks. The GDFF will offer tailored lending packages to dairy companies at favourable interest rates, with substantial TA to develop the companies. The financed greening opportunities could include - but are not limited to - to proactive expansion and modernising of the smallholder outgrower networks of the companies, supporting smallholder farmers to access more productive cows or energy-efficient fodder, solarising cooling or processing equipment, and developing the milk transport networks. These investments would require long-term debt finance, supported by working capital injections. While the exact loan conditions will depend on the identified needs and objectives of the businesses, the GDFF loans are expected to range between US\$ 250,000 and US\$ 550,000 with investment loan tenors of between 5 and 9 years, with the average loan size at US\$ 400,000. Eligible businesses will need to exhibit clear ambitions to invest, develop or distribute climate-smart technologies and productivity-enhancing technologies, and have a track record and solid plans for intensifying their smallholder partnerships. Based on the market surveys, it is anticipated that businesses in this borrowing category will be with annual turnover above US\$ 800,000, and investment decisions will be based on a thorough credit assessment by the Retail Banks using lending criteria agreed for the GDFF operations.

191. **Green Dairy Loan Product.** These are shorter-term, smaller-sized working capital or capital expenditure loans that seek to meet a specific financing need of businesses of smaller SMEs and well-organised dairy cooperatives. The Retail Banks in GDFF will provide standardised Green Dairy Loan Products with the maximum loan size at US\$ 250,000, an estimated average ticket size of US\$ 50,000, and loan tenor of up to three years. Similarly to the Green Dairy Investment Partnerships loans, eligible businesses will need to exhibit clear ambitions to invest, develop or distribute climate-smart technologies, productivity-enhancing technologies and have a track record and clear plans of strong smallholder partnerships. However, recognising that the majority of the dairy sector SMEs will not be operating at a significant scale, these Dairy Loan Products will target smaller businesses, likely with an annual turnover of less than US\$ 800,000. Investees will be eligible to receive pre-investment TA to translate investment needs into investment requests, and they can be connected to other DaIMA components to access post-investment TA support as required.

192. **End-use of GDFF Finance and Green Dairy Taxonomy.** To achieve the development objectives and impact of DaIMA and its Component 3, it is critically important that there is a clear joint understanding and agreement on what types of investments can be considered for the end-use of green dairy finance. A significant amount of work has been carried out in recent years on rural green finance taxonomy in East Africa, particularly during the IFAD/GCF design process for ARCAFIM and during the start-up phase of the Green Financing Facility of IFAD's Rural Kenya Financing Facility project. These taxonomy development processes, including for dairy and livestock sectors, has progressed through CC risk assessment and adaptation and mitigation models mapping to the assessment of the bankability of the related investment options for the private sector financial institution implementation. Furthermore, there are various other green finance taxonomy policy processes ongoing in the targeted countries.

193. The **Taxonomy of investments** for the GDFF lending will build on the findings of the Market Study carried out as a part of the DaIMA design (full Study attached to the Funding Proposal), and will include both agri-input supply investments (e.g. cooling tank producers, AI service producers, dairy meal producers, PV/solar solution producers, agri-fintech providers, fodder producers, and biodigester providers) and investments in technologies that reduce GHG emissions and improve productivity (e.g. milking machines, solar milk coolers, bio-digesters, milk pasteurizers, UTH treatment equipment, milk transfer tanks, renewable energy solutions and water recycling technologies), as well as biochar production.

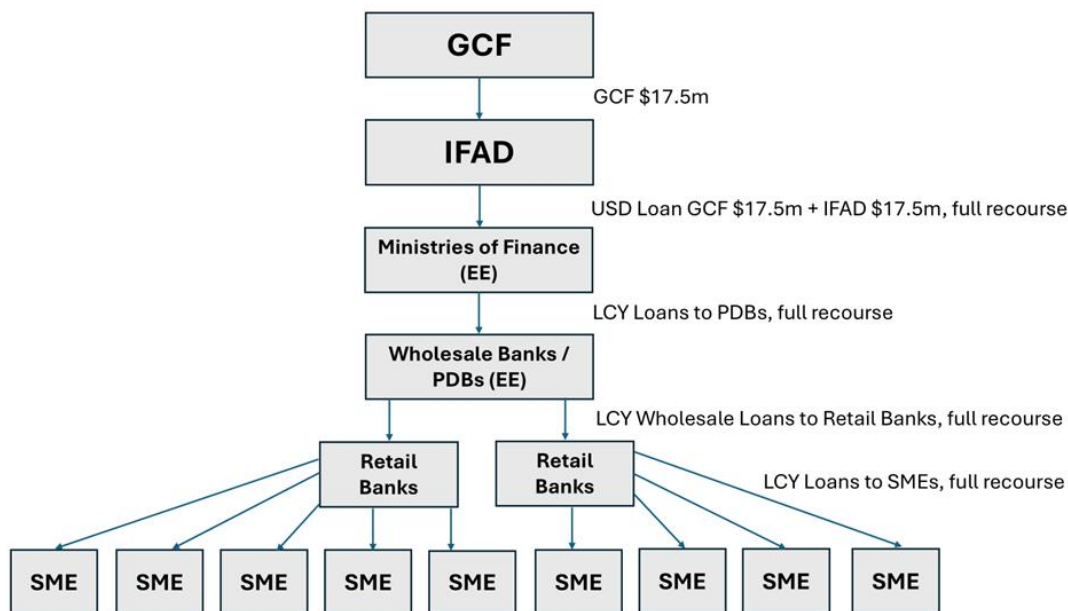
194. The guidelines on appropriate dairy sector greening options for private sector finance in the participating four countries will be included in the Loan Agreements and Implementation Guidelines of the GDFF credit lines. They will form a core part of the training and capacity building activities with the participating banks and the borrowing SMEs (see more below in Sub-Component 3.2). As a key development objective during the DaIMA implementation, the development of the dairy sector green financing taxonomy will continue, based on the documented experiences from

the GDFF credit line operations. The target will be to distribute the results for wider audience, to support the mainstreaming of green finance activities in private banking institutions in Africa. It should be noted that the parties are the PDBs as lenders and retail financial institutions as the recipients of the loans.

195. **Lending Terms to End-Users.** The DaIMA/GDFF financing package will proactively encourage the Retail Banks to expand their operations in green dairy financing with the programme target group using appropriate and favourable loan products. The long loan periods of the IFAD loans to the Governments and via the Governments to the GDFF implementing partners will make it possible for the Retail Banks to offer appropriate loan and grace periods to support the SME investments. This will be important particularly in innovative capital investments that improve the CC resilience and mitigation but require longer repayment periods to attract investors from the business community.

196. The **GDFF financing chain** is illustrated in the diagram (Figure 17b). Concerning the interest rates throughout this GDFF financing chain from GCF to end-borrower loans, (i) GCF's USD-lending highly concessional terms (0.25 percent annual interest rate including fee) apply for Rwanda, Tanzania and Uganda, and the less concessional (1.25 percent annual interest rate including fee) apply for Kenya; (ii) The estimated interest rate on DaIMA funding from the Governments to the GDFF Wholesale Banks, before final negotiations, is projected at 2-5 percent on local currency per annum fixed over the programme period; (iii) With this low price for on-lending capital, it is expected that the interest rates offered by the Wholesale Banks to the Retail Banks would be at the range of 5-8 percent on local currency per annum; (iv) This will enable the Retail Banks' pricing to end borrowers to be highly competitive, below the lowest market-based interest rates charged by commercial banks for this type of loans in the target countries.

Figure 17b. GDFF financial chain diagram



To define the appropriate market-based interest rate for GDFF lending at SME level, in collaboration with the GCF/IFAD ARCAFIM project, an interest rate survey will be carried out periodically during the programme implementation, IFAD annual DaIMA performance assessments will include a review of the materialised GDFF interest rate structure and provide recommendations for any required revision of the GDFF interest rates for the future operations. To support the start-up of GDFF, the existing ARCAFIM interest rate survey summary table (Table 2b) gives indication of the region's interest rate levels, and the full survey is available at the ARCAFIM Funding Proposal annexes. It should be noted, the lowest indicated interest rates are supported by blended finance instruments similar to what DaIMA GDFF proposes.

Table 2b. Interest rates

Country	Commercial banks	Microfinance institutions	International and national - project facilities	SACCOs and Fintech
Kenya	- Average 13% - annually - Maturity up to 5 years	- Range from 18-24% - annually - Maturity up to 48 months	- Up to 10% annually - Grace period 3 months - Up to 5 years	- On lending annual interest rate of 8-10% - Retail annual interest rate 10.5% to 13%
Uganda	- Range from 10% to 18% - annually - Maturity up to 15 years	- Range from 12-48% - annually - Maturity up to 3-36 months	- Up to 13% annually - Grace period 3 years - Maturity from 6 months to 3 years	- Interest rate of 30% - Short term - unspecified
Tanzania	- Range from 9% to 25% - annually - Maturity up to 15 years	- Range from 18-96% - annually - Short term - several months (seasonal)	No data/ not applicable	- Interest rate range from 24%-36% - Short term - unspecified
Rwanda	- Range from 15% to 18% - annually - Maturity up to 15 years	- Range from 18-25% - annually 6 months to 10 years	- Interest rate subsidy of 8% - Maturity up to 10 years	- Interest rate range from 22%-24% - Short term - unspecified

Output 3.2: Capacity Building for Green Dairy Investments

197. GDFF will proactively use the TA facility to (i) build the capacities of the Wholesale Banks and Retail Banks to understand and implement green dairy sector projects, (ii) maximise both the impact and returns of the financed investments by enhancing the capacity of borrowers to implement effective strategies, improve their operational efficiencies, and achieve sustainable growth; and (iii) support the establishment of an effective M&E system and impact assessment surveys, to enable reporting of the GDFF progress, results, lessons learnt and best practices. In doing so, the TA facility will address non-financial barriers currently preventing greater climate-smart financing in the GDFF target countries.

198. The technical support activities of Sub-Component 3.2 with both the participating banks and the SMEs will be managed and implemented by Technical Assistance Service Providers, to be identified and selected through regional competitive bidding by the EEs through the PCUs in each of the four DaIMA countries and supervised by IFAD through the standard no-objection process. The TA will be guided by a Green Dairy Finance Lending Toolkit.

199. Operating with a total TA budget of US\$ 10 million (financed by a GCF grant), output 3.2 includes three activities: (a) Awareness Raising and Capacity Development of Participating Banks in Green Dairy Finance, (b) Green Dairy Investment Support Services to SMEs, and (c) M&E system and household surveys and evaluations.

200. **Activity 3.2.1 Awareness Raising and Capacity Development of Participating Banks in Green Dairy Finance.** The activity will provide top end expertise to the national Wholesale Banks and their Retail Bank partners, to ensure that the credit provided with GSFF funding will deliver real, effective, and transformational adaptation and mitigation impact and is channelled to the intended purposes and beneficiaries. Thus, the criteria for the selection of the beneficiaries can be found in the feasibility study (Annex 2, section 5). The participating Wholesale and Retail Banks will be sensitized to the importance of adaptation and mitigation taxonomy and targets and receive a demand-driven package of technical support and innovation services, with the aim to develop and offer appropriate green dairy financial services, manage risks of this lending, and report on portfolio performance and impact. Initiation of the Technical Assistance to the Retail Banks is a condition to access to GDFF financing for on-lending.

201. Before the GDFF lending operations commence, the Wholesale Banks will receive international and local TA support to introduce green dairy wholesale financing products into their mainstream financing operations. The Wholesale Banks will also receive systematic support to establish effective GDFF management systems, to coordinate the programme implementation and to report on the performance. Through cost-shared continuous TA, in medium term, the aim is that DaIMA Wholesale Banks will strengthen their internal capacities to independently mobilize and

implement similar green finance projects to support green transformation in dairy and other sectors of the rural economy.

202. Participating Retail Banks involved in Gdff financing activities will receive technical support packages tailored for their specific needs under green dairy finance. A preliminary demand analysis for the services was conducted during the DalMA design and earlier, during the ARCAFIM design in the same countries, to enable TA action planning and detailed budgeting. The support will be launched against expressions of demand by Retail Banks after the DalMA start-up. Potential technical support areas will include: (i) development of green dairy finance strategy and costing of its implementation, (ii) development or strengthening of green dairy lending products targeting medium and small businesses, (iii) support to rural outreach and social inclusion strategies, including CC adaptation and mitigation finance advisory capacities, (iv) strengthening of Digital Finance Services by inclusion of functions for climate finance, and (v) support to improved Management Information Systems (MIS) to follow up and report progress achieved in Gdff lending. Furthermore, the PCU experts, in particular the Climate Finance experts, as well as the Wholesale Bank will work with the Retail Banks to develop technical assistance packages to be delivered to small and medium dairy businesses with Gdff grant financing.

203. **Activity 3.2.2. Green Dairy Investment Support Services to SMEs.** The aim of the Gdff capacity building services to potential and actual SME borrowers is to strengthen the knowledge of rural businesses of viable CC mitigation and adaptation options, benefits, and related financial services, and to develop their business planning, investment start-up, and operational and financial management capacity. These actions aim to increase the demand for green dairy finance and improve the investment success rate, at the same time lowering the investment risk for the Retail Banks. With this TA support, bankability assessments and the related loan applications can be completed for agribusinesses, helping to inform the Retail Banks of the real investment opportunities and risks. The capacity building packages for dairy sector SMEs will be demand-driven and tailor-made, with the aim to ensure that Gdff financing will target well planned, productive, and screened investments in the dairy sector, to ensure both the economic growth impact and the highest possible positive impact on climate change resilience and mitigation.

204. **Activity 3.2.3. M&E, household surveys and evaluations.** In order to assess the outputs, outcomes and impacts of the Line of Credit and the TA activities, a dedicated M&E system, incorporating inputs from all partners, will be established. In addition, one midline and one endline household survey will be conducted, along with collection of other relevant M&E data as part of in-depth evaluations at the mid-term point to allow for course corrections, as well as the completion point to assess effectiveness and capture learning.

B.4. Implementation arrangements

Accredited Entity

205. IFAD will be the Accredited Entity (AE) for the Programme. In this capacity, IFAD will be responsible for the overall management of the Programme, including: (i) all aspects of Programme appraisal; (ii) administrative, fiduciary and technical oversight and supervision throughout the implementation of the Programme towards achievement of the expected Outputs and Outcomes; and (iii) Programme closure and final evaluation. IFAD will carry out these responsibilities in accordance with the detailed provisions set out in the Accreditation Master Agreement (AMA) between IFAD and the GCF, from relevant offices and divisions in IFAD Headquarters located in Rome, Italy (HQ), the IFAD East and Southern Africa regional office located in Nairobi, Kenya and the IFAD Country Offices in Kenya, Rwanda, the United Republic of Tanzania, and Uganda.

206. To fulfil the AE functions, IFAD will set up a dedicated Programme Task Force (PTF) consisting of IFAD experts from within its HQ, regional and country offices.

Regional Programme Governance

207. The Regional Steering Committee (RSC) will be constituted by Permanent Secretaries from the Executing Entities. Each national project steering committee will appoint 6 members to the RSC, including 3-4 Government representatives and 1 or 2 representatives of beneficiaries and private sector or civil society. The RSC will meet annually, on rotational basis between countries. Chairmanship will be rotational for one year period, entailing preparing for the physical RSC meeting in terms of agenda, reports to be reviewed and other preparatory/secretarial functions as required. A knowledge management event for technical officers will be organized in conjunction with the meeting. More frequent virtual meetings may be held on a needs basis. The RSC will specifically oversee (i) the progress

realized of regional aspects of the Programme and that regional added value is achieved; (ii) timeliness and quality of reporting to GCF, both consolidated annual work plan and budgets (AWPBs), progress reports and M&E reports; (iii) collaboration and oversight with ILRI, who will be providing technical assistance to all four countries and support regional initiatives around MRV reporting, and (iv) regional M&E, knowledge management and reporting activities, to ensure that, for example, data collection is carried out in a streamlined manner and that coherent and high quality programme level reports are being produced for reporting to IFAD and GCF. The RSC will specifically monitor progress on Output 1.1 (regional policy harmonization and dairy market integration, harmonization of dairy market protection), Output 1.3 (MRV GHG capacity building) and Output 1.4 (regional exchange platform, regional knowledge management and exchange). Regarding Output 1.4, the RSC will oversee the development of learning materials and knowledge products and support the establishment of a knowledge platform specific to each country. This platform will provide opportunities for exchange visits (farmers, private sector, and ministries), development and sharing of studies, technical work, trainings, and financial initiatives between the four countries.

208. A senior regional coordinator and policy dialogue expert would be recruited in each country, with similar TORs and responsibilities to support the functions of the RSC and any regional activities. As such, the group of the four regional coordinators will act as a secretariate to the rotating RCS, with particular focus on activities with a regional focus and where regional harmonization is beneficial.

National Programme Steering Committee

209. In each country, a National Programme Steering Committee (PSC) will be established. The PSCs will: (i) provide guidance and direction to the Programme at national level, ensuring that it remains within specified constraints, and escalate issues to RSC as needed (in case a national PSC decision has implications for the strategy of other countries); (ii) address issues related to the Programme raised by the national Programme Coordinator; (iii) monitor Programme risks and the effectiveness of mitigation measures, and provide guidance on emerging risks of the Programme, and agree on possible countermeasures and management actions to address specific risks; (iv) review the progress and effectiveness of the Programme, and provide guidance and recommendations to ensure that the agreed deliverables are satisfactorily produced in accordance with plans; (v) review and approve the AWPB and provide strategic guidance for its implementation; (vi) appraise the annual report on the implementation of the Programme, including the quality assessment report; (vii) make recommendations for subsequent work plans to build on achievements and address any shortcomings; (viii) provide *ad hoc* guidance and advice in exceptional situations; (ix) ensure coordination/ synergies with other Programmes and Projects focusing on sustainable livestock in the same areas, and (x) identify lessons learned and good practices. In each country, the PSCs will have to meet formally at least once every 12 months. Regular meetings will be organized by the National Programme Coordinator; extraordinary meetings may be requested by any of its members. The composition of each PSC is presented below.

210. In **Kenya**, the same PSC as INReMP will be used. The Inter-Ministerial/Agency PSC under the leadership of the Permanent Secretary (PS), SDLD will provide overall policy guidance to implementation. The other members of the PSC will come from the National Treasury and Economic Planning; State Department for Cooperatives; State Department of Micro, Small and Medium Enterprises Development; State Department of Environment and Climate Change; the Agricultural Finance Cooperation, the National Environmental Management Authority as NDA, KDB, KALRO, and the Council of Governors.

211. In **Rwanda**, the same PSC as RDDP-2 will be used. The PSC will be chaired by the PS of MINAGRI, co-chaired by the Director General of RAB. The Programme will add additional members (e.g. Ministry of Finance and Economic planning, Development Bank of Rwanda, the Rwanda Environmental Management Authority as NDA, Meteorology services, Ministry of Environment, Ministry of Infrastructure, REMA, RNDP, private sector, civil society) to the RDDP-2 PSC.

212. In **Tanzania**, the same PSC as C-SDTP will be used. It will be chaired by Permanent Secretary of the MLF, and co-chaired by Principal Secretary, MAINL, Zanzibar. It will be composed of the Permanent Secretaries of the Ministries of Finance and Planning, the Vice President's Office (VPO) and the President Office-Regional Administration and Local Government (TAMISEMI) for Mainland; and the First Vice President Office in Zanzibar (VP1), Ministry of Finance and Planning President's Office for Zanzibar, the Tanzania Agricultural Development Bank, TDB, as well as representatives of private sector **Official Seed Certification Institute** and farmers' organizations. Additional members could be added (e.g. Ministry of Environment as NDA, Meteorological Authority, private sector, civil society).

213. In **Uganda**, the same PSC as ReLIV will be used. It will be co-chaired by the Permanent Secretary of the EE (MAAIF) and Ministry of Finance, Planning and Economic Development (MoFPED), the GCF National Designated Authority (NDA). The PSC will comprise of the Director of Animal Resources (DAR), Executive Directors of MAAIF

Agencies (DDA, the National Animal Genetic Resources Centre and Data Bank (NAGRC&DB) and NARO); Uganda Development Bank, Director responsible for climate change from the Ministry of Water and Environment as NDA, Ministry of Energy and Mineral Development, MoLG and UNMA.

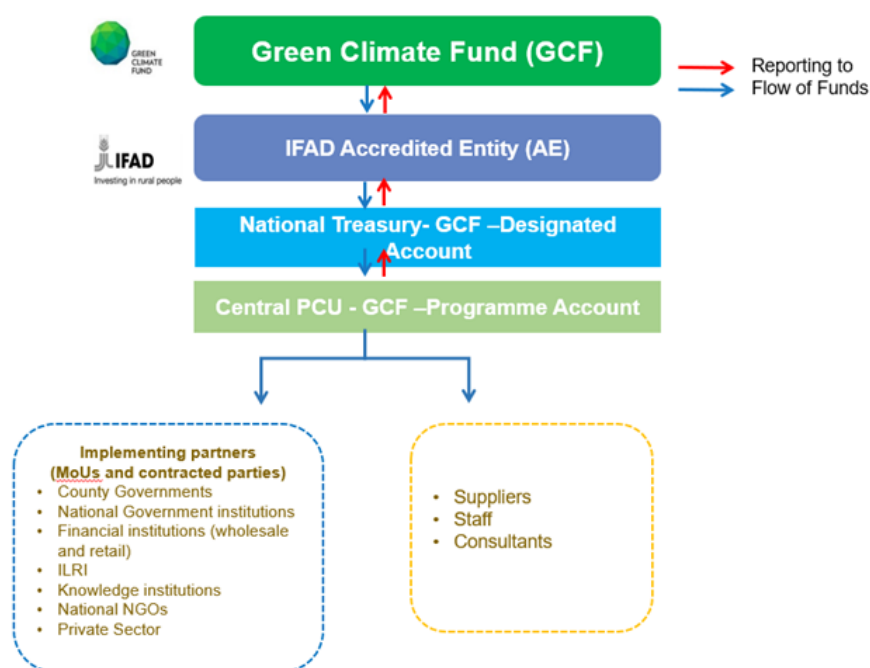
National Executing Entities (EEs)

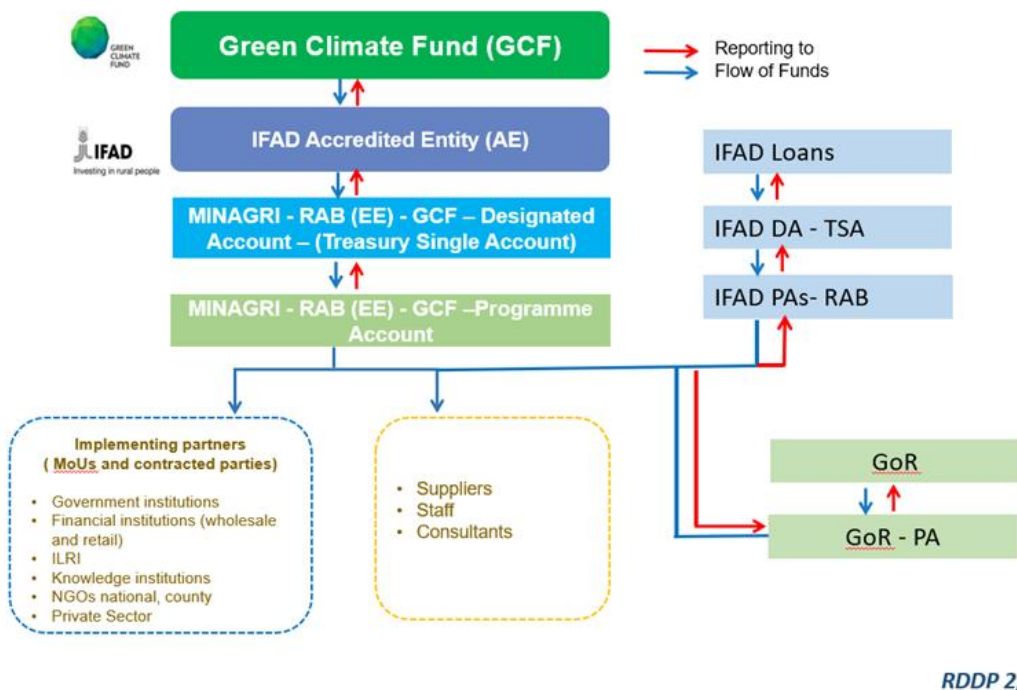
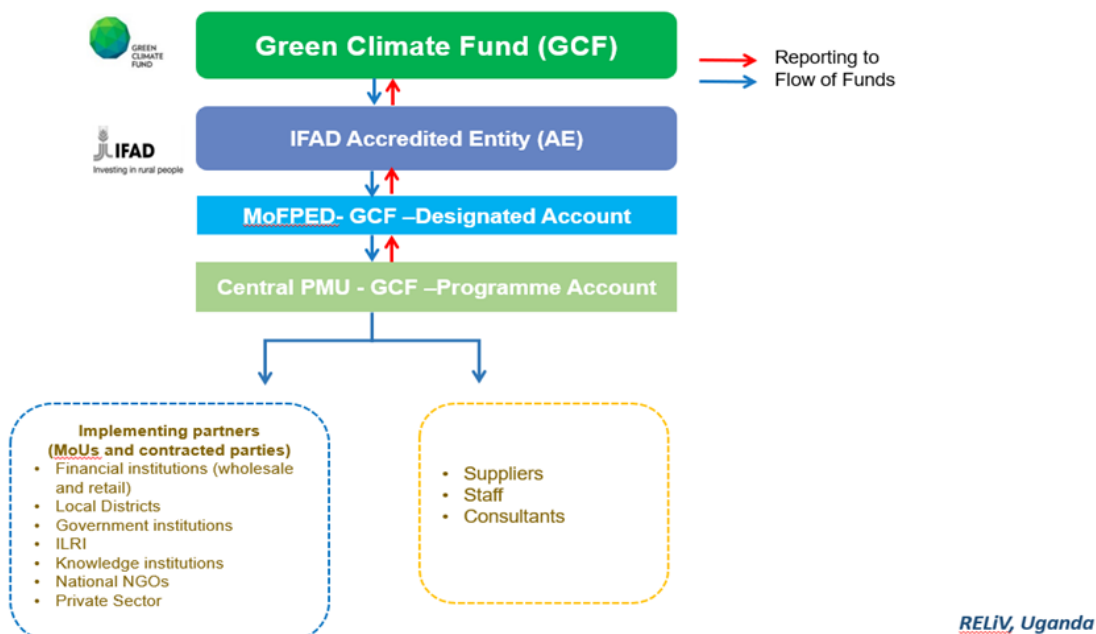
214. The EEs (or “lead agencies”): in **Kenya** National Treasury and Economic Planning, MoALD and the Agriculture Finance Cooperation; in **Tanzania**, the Ministry of Finance, MLF and the Tanzania Agricultural Development Bank; in **Rwanda**, the Ministry of Finance and Economic Planning, RAB under MINAGRI and Development Bank of Rwanda; and in **Uganda**, Ministry of Finance, Planning and Economic Development, MAAIF and Uganda Development Bank (for more information see section A.20)

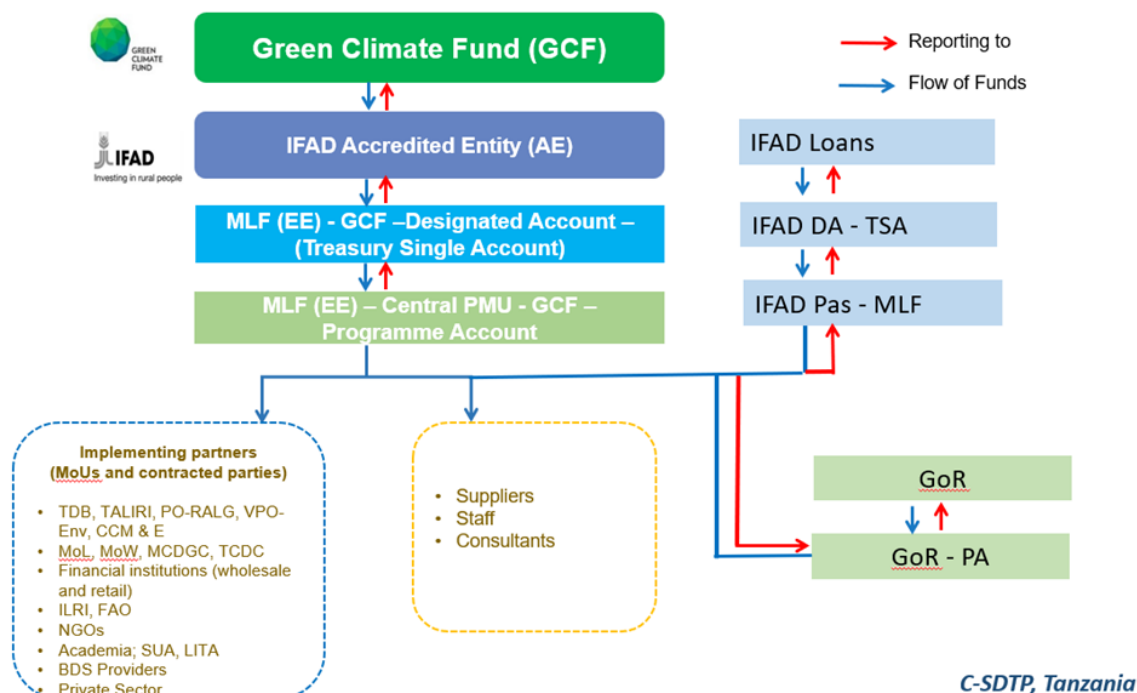
215. In areas where international technical expertise and regional experience is needed, one or more Centers of Excellence and International Organizations (such as ILRI) would be recruited as service providers by each EE in each country, with preference to those that have proven capacity and regional presence to ensure regionality of the interventions.

216. Flow of funds and legal arrangements are described in section G.3 and flow of funds for each country will be as follows (Figure 17c):

Figure 17c. Flow of funds in the project countries







National Programme Coordination and Management

217. In the four countries, the Programme will be managed by national Programme Coordination Units (PCUs), either stand-alone for the Programme or jointly for other projects or programmes, comprising competitively recruited Government staff with appropriate project management skills, subject to IFAD No Objection at the key stages of developing terms of reference, shortlisting, interviewing and contracting. These units will be responsible for overall coordination and day-to-day management of the Programme, as well as facilitating a conducive environment. Specifically, the units' responsibilities will include: (i) development of AWPBs; (ii) the coordination of Programme activities implementation among EEs and implementing agencies and partners; (iii) the development of operational strategies and establishment of effective tools (e.g. MIS systems) for Programme implementation, (iv) the fiduciary management of the Programme; (v) the mobilization, contracting and coordination of implementation partners (vi) the monitoring, evaluation, knowledge management and learning (M&E–ML) -including dissemination of lessons learned and good practices and providing guidance for effective Programme implementation, (vii) management of the environmental and social management framework (ESMF), (viii) the procurement of goods and services in accordance with financing agreements, (ix) communication and visibility of the Programme.

218. In **Kenya** (Figure 18), at national level, a PCU will be established within SDLD in the MoALD. To steer implementation at County level in accordance with the Government decentralization principles, a County Programme Coordination Committee (CPCC) and a County Programme Implementation Team (CPIT) will be established with the responsibility to steer and lead implementation within each County, in accordance with the MoUs established with the national PCU. The CPCC and CPIT will also ensure coordination with other Projects, for example synergies with the "Transforming Livelihoods through Climate Resilient, Low Carbon, Sustainable Agricultural Value Chains in the Lake Region Economic Bloc" Project proposed by FAO to the GCF. At the sub-county level, a Subcounty Programme Implementation Team (SCPIT) composed of officials from the sub-counties of the relevant departments, as outlined for the CPIT will be established to coordinate the implementation of activities.

219. In **Rwanda** (Figure 19), the Programme will be an integral part of the RDDP-2. It be implemented through the SPIU under the authority of RAB. The SPIU will build on its long-standing partnership with IFAD. At local level, the Programme will create partnerships with the Local Government Authorities in each targeted District, building on the existing networks established through the ongoing RDDP-2.

220. In **Tanzania** (Figure 20), the Programme is an integral part of the C-SDTP, which will establish a Programme Management Unit (PMU) within the MLF. At the District level, the Programme will adhere to the existing C-SDTP structures by working with the District Facilitation team. The activities will be carried out by implementing partners contracted by the PMU.

221. In **Uganda** (Figure 21), the Programme will be an integral part of the ReLIV project under MAAIF and DaIMA will be managed within the same Programme Management Unit (PMU) under MAAIF. At district level, a District Coordination and Implementation Unit will be formed and chaired by the Chief Administrative Officer. Members will include relevant Departments such as Production and Marketing, Veterinary, Environment, the Department responsible for gender, youth and social development, representatives of farmers' organization, the private sector, and NGOs.

222. The detailed responsibilities and mandates of the partners, as well as composition of the teams involved in the coordination and management of the Programme are presented in the Feasibility Study (Annex 2, Chapter 7).

Figure 18. Programme implementation and coordination arrangements in Kenya

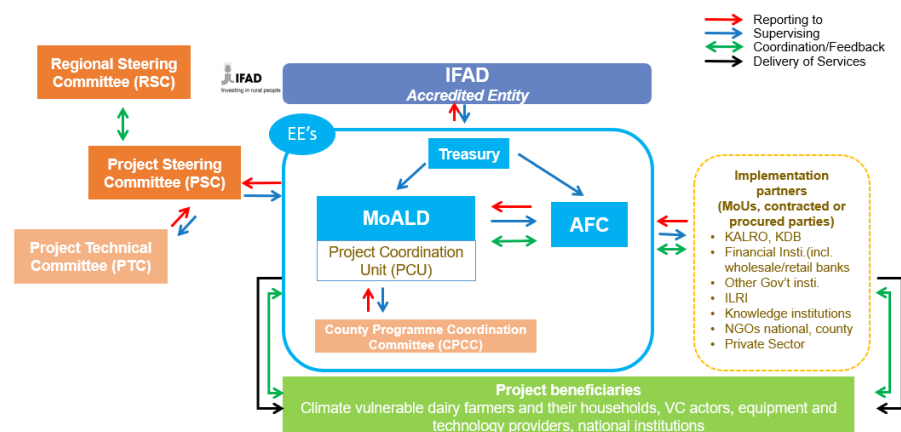


Figure 19. Programme implementation and coordination arrangements in Rwanda

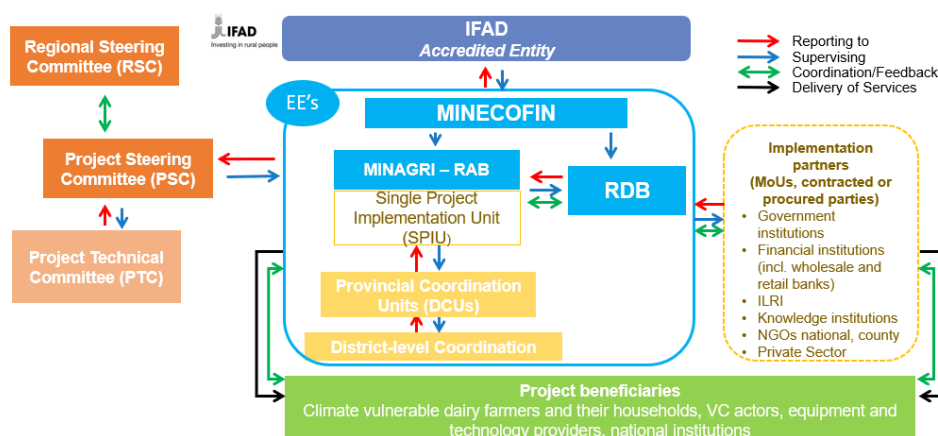


Figure 20. Programme implementation and coordination arrangements in Tanzania

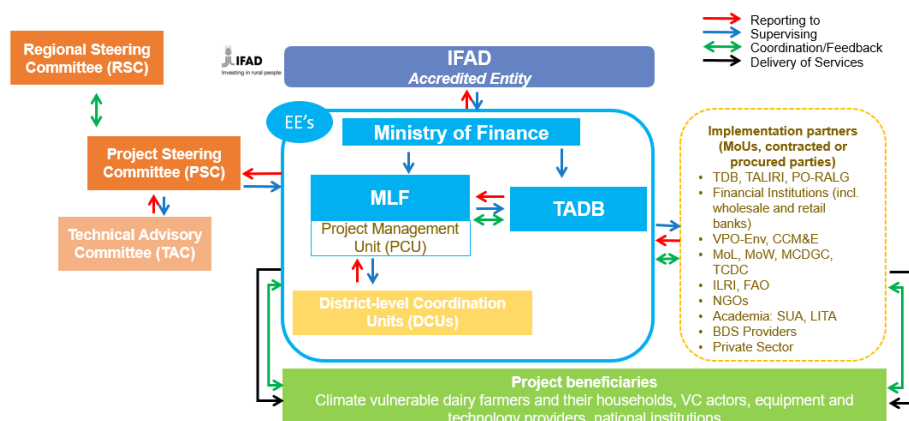
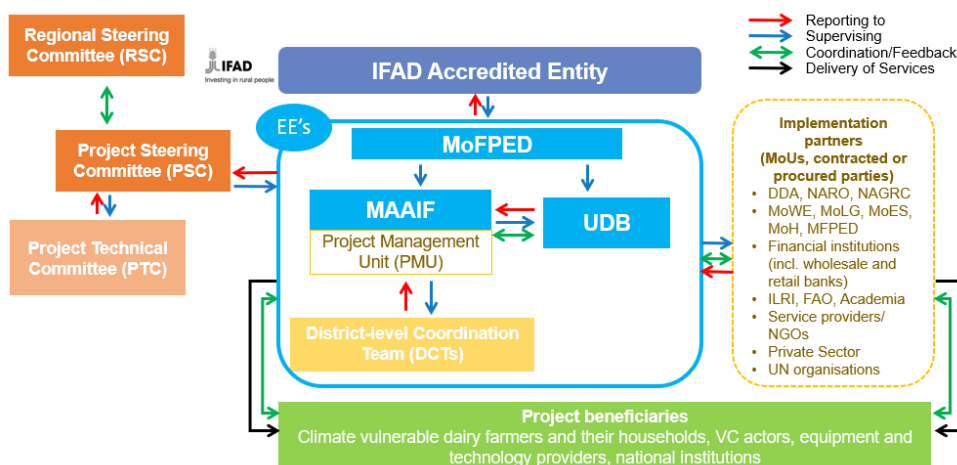


Figure 21. Programme implementation and coordination arrangements in Uganda



223. **Implementation of Component 3.** This component will be **implemented as four separate lines of credit in the target countries, under the coordination and supervision of DaIMA regional management and the National PCUs. The green dairy sector financing operations** will be managed by a local Public Development Bank (PDB), which will act as a Wholesale Bank for the GDFF intervention. The Wholesale Banks will at national level manage the financial DaIMA services to the programme target group based on eligibility criteria and other terms and conditions, agreed upon in the programme design and included in the financing agreements between IFAD, the Governments, and the Wholesale Banks. The Wholesale Banks will report quarterly, semi-annually, and annually on the financing progress of GDFF to the PCUs and IFAD in each country based on a pre-defined reporting schedule, indicators, and formats.

224. From the bulk loans provided by the Wholesale Banks, on-lending to the eligible borrowers will be conducted and managed by Retail Banks, which can be commercial banks or such microfinance banks and commercial development banks that are prudentially supervised. Operating with the targeting guidelines and other conditions of the wholesale loan agreements between the Wholesale Banks and Retail Banks, the actual lending methods used for retail loans under GDFF financing will follow each Retail Bank's operational structure and network, rural and agriculture finance strategy, and their digital tools development approach.

225. **The technical support activities of Sub-Component 3.2 with both the participating banks and the borrowing SMEs will be managed and implemented by experienced Technical Assistance Service Providers, to be identified and selected through regional competitive bidding organised by DaIMA PMU staff for each of the four DaIMA countries and supervised by IFAD through the standard no-objection process.** Practices used and experiences gained from using private TA service providers in the four target countries in IFAD-supported rural finance programmes will be fully utilised when organising the bidding processes. In the planned ARCAFIM operation,

mapping of service providers with direct experience in climate finance TA provision will provide guidance to this selections process.

Partnerships

226. During implementation the Programme will involve relevant and specialized government agencies, academia and research institutions. Several partners (NGOs/ non-profit organizations, service providers, private sector (including informal/formal milk collectors and processors), financial intermediaries, ILRI, FAO, and other international development organizations, etc.) will also participate in the Programme to ensure complementarity with their activities, provide technical assistance and/or the provision of goods, works and services. The PCUs, on behalf of the EEs, will enter into Memoranda of Understanding (MoUs) and/or subsidiary agreements with the implementing partners (also referred as “project parties”) for implementation of the respective activities. Pre-identified Government and private entities are provided below; others will be identified and/or procured on a competitive basis, based on performance, needs and comparative advantage. The list of implementing partners is described in the components, in the Feasibility Study.

227. In **Kenya**, implementation will be supported by KDB, the Kenya Agricultural and Livestock Research Organization (KALRO), the Kenya Animal Genetic Resources Centre (KAGRC), among others.

228. In **Uganda**, implementation will be supported by DDA, NAGRC&DB, NARO, the Ministry of Local Government (MoLG), universities, private sector and NGOs, amongst others.

229. In **Rwanda**, implementation will be supported by RAB, RNDP, REMA, RCVD, the Ministry of Environment, the University of Rwanda; Faculty of Veterinary Medicine and Animal Sciences (UR-CAVM), among others.

230. In **Tanzania**, implementation will be supported by TDB, TALIRI, the Vice President Office, Regional Administration and Local Government (PO-RALG), the Ministry of Lands, the National Land Use commission (MoL NLUC), the Ministry of Water (MoW), the Ministry of Community Development, Gender and Special Groups, NGOs, universities, the Livestock Training Agency.

Programme Co-financing

231. The co-financiers (Table 3a) are responsible for reporting of co-financing activities and the amount of their disbursement to the AE in accordance with the detailed provisions outlined in the GCF policies as well as in AMA, the Funded Activity Agreement (FAA) between IFAD and GCF and the co-financing agreement signed between the co-financier and IFAD in its capacity of AE. The amounts of co-financing are detailed in Section C.1.

232. In **Kenya**, the Programme will be co-financed by INReMP to the amount of US\$24.75 million from IFAD. IFAD funds will be used to cofinance Component 1 and 2, as well as Programme management, and M&E.

233. In **Rwanda**, the Programme will be co-financed by RDDP-2 for US\$ 26.61 million provided by IFAD. In addition, Government of Rwanda will provide US\$14.75 million cofinancing to Outputs 2.1 and 2.2. IFAD funds will be used to cofinance Component 1, 2 and 3, as well as for Programme management and M&E. This will include: under Component 1 the activities related to the development of veterinary and extension services, dairy consumption and food safety aspects, and knowledge management. Under Component 2, IFAD funds will be used to finance cooperative development and business linkages. Under Component 3, credit lines will be provided to finance Output 3.1.

234. In **Tanzania**, the Programme will be co-financed by the C-SDTP in an amount of US\$ 50.81 million provided by IFAD. These IFAD funds will co-finance Components 1 and 2, as well as for Programme management and M&E. This will include: under Output 1.1, the activities related to policy review, stakeholder participation, as well as dairy consumption and food safety aspects; under Output 1.2 the activities related to the development of veterinary and extension services (except for the development of specific training modules and training of facilitators); under Output 2.1, the development of dairy cooperatives (except for the activity related to renewable energy); under Output 2.2, the vaccination campaign, as well as the investments in improved cowsheds. Programme Management will be financed using IFAD and Government Funds.

235. In **Uganda**, the Programme will be co-financed by the IFAD-funded ReLIV project for a total amount of US\$ 94.23 million. The IFAD proceeds will be used to cofinance Component 1 and 2, as well as Programme management and M&E. In addition, IFAD will cofinance Output 3.1.

Table 3a. Programme activities and sub-activities by financier in Kenya, Rwanda, Tanzania and Uganda

Component	Output	Activity	Kenya	Rwanda	Tanzania	Uganda
C1: Strengthened national capacities and enhanced enabling environment for low emissions climate resilient dairy sectors	Output 1.1: Climate-responsive policy and regulatory environment established.	1.1.1	GCF+IFAD	GCF+IFAD	GCF+IFAD	GCF+IFAD
		1.1.2	GCF+IFAD	GCF+IFAD	GCF+IFAD	GCF+IFAD
		1.1.3	IFAD	GCF	GCF	GCF
	Output 1.2: Improved service delivery to the dairy sector	1.2.1	IFAD	GCF+IFAD	IFAD	IFAD
		1.2.2	IFAD	GCF+IFAD	IFAD	GCF+IFAD
	Output 1.3: Improved GHG MRV capacities & monitoring of climate vulnerabilities	1.3.1	GCF	GCF	GCF	GCF
		1.3.2	GCF	GCF	GCF	GCF
		1.3.3	GCF	GCF	GCF	GCF
		1.3.4	IFAD	GCF	GCF	GCF
	Output 1.4: Knowledge management and regional cooperation enhanced	1.4.1	GCF+IFAD	GCF	GCF	GCF+IFAD
		1.4.2	GCF+IFAD	GCF+IFAD	GCF+IFAD	GCF
C2: Enabling low emission and climate resilient dairy value-chain development	Output 2.1: Smallholders' and cooperative capacity development strengthened	2.1.1	IFAD	IFAD+Gov	IFAD	IFAD
		2.1.2	IFAD+GCF	IFAD	IFAD	IFAD
		2.1.3	GCF	GCF	GCF	GCF
	Output 2.2: Breeding, animal health, feed and fodder, manure management improved	2.2.1	IFAD	IFAD+Gov	GCF+IFAD	IFAD
		2.2.2	GCF	GCF+IFAD+Gov	GCF+IFAD	IFAD
		2.2.3	GCF	GCF+IFAD	GCF+IFAD	GCF+IFAD
		2.2.4	GCF+IFAD	GCF	GCF	IFAD
		2.2.5	GCF	GCF	GCF+IFAD	GCF+IFAD
		2.2.6	GCF	GCF	GCF	IFAD
	Output 2.3: Pasture and grasslands management improved	2.3.1	IFAD	GCF	GCF	IFAD
		2.3.2	GCF	GCF+Gov	GCF	GCF+IFAD
C3: Green Dairy Financing Facility (GDFF)	Output 3.1: Effective green financing for dairy value chains	3.1	GCF	GCF+IFAD	GCF	GCF+IFAD
	Output 3.2: Capacity building for green dairy investments	3.2	GCF	GCF	GCF	GCF

B.5. Justification for GCF funding request

236. The Programme will be implemented in four countries in East Africa, three of which are classified Least Developed Countries (LDCs). They have limited budgetary space to devote to the public service activities of a Programme of this nature. These LDCs are the most vulnerable to the adverse effects of climate change, as their economic development, growth and food security are highly dependent on climate-sensitive sectors, such as livestock and dairy. The wars in Ukraine, Lebanon, Palestine and Syria have deeply affected these countries, through contractions in supply and demand both in domestic and export markets. The public debt-to-GNI ratio has risen considerably in all four countries. Half of public debt is external. Between 2015 and 2021, external debt relative to gross national income GNI (%) increased from 31.3 to 37.9 percent in Kenya; from 30 to 48.3 percent in Uganda; from

41.2 to 82.9 percent in Rwanda; and from 39 to 42.2 percent in Tanzania⁸⁷. High debt limits their ability to finance the investments to meet climate change mitigation and adaptation needs in the dairy sector.

237. The implementing of NDCs, NAPs and NAMAs will remain slow without GCF support, due to few alternative options for climate finance. Low-income Countries received only 8 percent of all climate finance, compared to 70 percent invested in Middle Income Countries (UNEP 2022 Adaptation Gap Report). In addition, according to the Climate Policy Initiative, climate finance in Africa stands at US\$ 29.5 billion annually, well below the US\$ 277 billion needed each year for countries to meet their NDCs by 2030. In East Africa, annual flows amount to US\$ 10 billion against a demand of US\$ 82 billion. Climate finance on the continent is more balanced between adaptation and mitigation than in other regions: mitigation activities amounted for 49 percent (US\$ 14.6 billion) of climate finance flows, compared to 39 percent (US\$ 11.4 billion) for adaptation and 12 percent (USD 3.5 billion) for activities generating dual benefits. Adaptation finance would need to be increased six-fold to meet the estimated annual needs of US\$ 62 billion. Private sector investments, which are urgently needed to scale adaptation solutions, accounted for only 14 percent (US\$ 4.2 billion) of climate finance in Africa. This is lower than in other regions and was invested primarily in mitigation projects (81 per cent)⁸⁸.

238. The public service activities of the Programme – e.g., regulatory reform, improvements in veterinary and extension services, knowledge management, capacity development – will be financed by GCF grants. The proposed level of GCF grant support is highly justified due to the significant unmet demand for CCA&M financing among the target groups of the Programme, and the need to develop new technical approaches that can be replicated elsewhere to significantly increase private sector participation in the dairy sector.

239. Furthermore, as noted in the Market Study (Appendix to Feasibility Study), the adoption of low-carbon and climate-resilient solutions and practices by dairy SMEs is constrained by challenges in access to finance paired with technical barriers (Table 3b). Dairy value chain actors are typically cash constrained that makes them reliant on debt capital to invest in climate-smart technologies. The Market Study finds there is a lack of affordable debt within the market with high collateral needs that smaller businesses cannot fulfil. Alongside this, cash-constrained businesses are more likely to prioritise investments that improve productivity over climate mitigation or adaptation. The GDFF is designed to overcome these challenges through its affordable local currency loan funding, providing result-based incentives to businesses to invest in climate-smart technologies in the form of concessional loans.

Table 3b. Summary of barriers to accessing finance for climate-smart practices and technologies and GDFF's response

	Barrier	GDFF's solution
Financial barriers	Real and perceived high risks of financing investments into dairy value chain actors lead to high costs of borrowing passed onto the borrower through high interest rates and high collateral requirements	- Impact-focused investment loans with a mandate to invest in dairy value chain actors⁸⁹ - Competitive interest rates
	Financial products are standard and not tailored to the needs of the dairy value chain actors	GDFF Green dairy loans are developed in collaboration with businesses to ensure debt financing terms enable achievement of long-term greening ambitions, with adequate loan and grace periods
	Enterprises are operating at low margins with cashflow challenges that reduce project invertibility with low ability to repay	Longer tenor loans with repayment holidays to reduce immediate burden of repayment for businesses. Two kinds of products, smaller loans for shorter-term investment and longer-term holistic greening partnerships
	Internationally financed operators often do not offer a local currency debt option. Especially smaller entities cannot absorb the costs of FX volatility if they are selling in domestic markets.	Lending in local currency to all SME borrowers, with the GDFF absorbing the costs of FX hedging.

⁸⁷ See The World Bank, international debt statistics, <https://datatopics.worldbank.org/debt/ids/countryanalytical/tza/counterpartarea/wld>

⁸⁸ ARCAFIM Funding Proposal. [gcf-b37-02-add11-funding-proposal-package-fp220.pdf](https://www.greenclimate.fund/gcf-b37-02-add11-funding-proposal-package-fp220.pdf) (greenclimate.fund)

⁸⁹ There is currently no dairy-specific funding facility in the market focussing on green investments direct to businesses specifically. Recently Pearl Capital Partners have launched a fund lending to SACCOs and MFIs in dairy.

Technical barriers	Agri-businesses require high levels of engagement and support per investment which drives up the transaction costs of lending to them	- Local presence by local implementing partner financial institutions (see implementation) - Specialist technical assistance support with dairy-specific expertise
	For businesses prioritizing the use of debt capital, investments into climate-smart practices and technologies are often deprioritized compared with fast productivity-enhancing investments	- Provide effective pre-deal technical assistance to support businesses to identify investment opportunities and understand the productivity benefits of climate-smart investments - Better understanding in the participating banks and borrowing SMEs about CC options an incentive to invest in greening the dairy value chain by providing affordable finance in return for delivery of outcomes

240. A central offering that is critical to the delivery of the GDFF's impact is lending in local currency to dairy SMEs. This is a significant challenge faced by enterprises in the sector, and it poses risks to SME viability and sustainability⁹⁰. The capital protection within the GDFF's finance structure enables it to assume the currency risks on behalf of dairy SMEs, with a two-fold justification: (a) without FX hedges, it allows for lower interest rates for the all borrower SMEs, and (b) it helps crowd investors into serving smaller, domestically oriented SMEs that are unable to borrow in foreign currencies.

241. **Justification for technical assistance grant:** The proposal includes a US\$ 10 million technical assistance grant request that is critical for GDFF for supporting businesses to effectively adopt and sustain the adoption of climate smart technologies. This equates to 28 percent of the loan capital allocated to GDFF. With this funding, GDFF will proactively use the TA facility of Sub-Component 3.2 to (a) build the capacities of the Wholesale Banks and Retail Banks to understand and finance green dairy sector projects, and (b) maximise both the impact and returns of the financed investments by enhancing the capacity of borrowers to implement effective strategies, improve their operational efficiencies, and achieve sustainable growth. In doing so, the TA facility will address non-financial barriers currently preventing greater climate-smart financing in the GDFF target countries.

B.6. Exit strategy

242. The design of the Programme includes several elements aimed at improving the sustainability of its results and maintaining the paradigm shift after implementation. The exit strategy and sustainability are based on: (i) the full engagement and ownership of governments, regional institutions, national dairy boards, and other stakeholders (private sector, development partners); (ii) policy solutions to create an enabling environment for low-carbon, climate-resilient investments in the dairy sector, in line with NDCs and NAMAs; (iii) capacity development of key institutions, public services, dairy processors and dairy farmers on practices (animal health, feeding, breeding, manure management) that effectively reduce GHG emissions and build climate resilience, without compromising milk productivity; (iv) instruments to mobilise more public and private climate finance for the dairy sector, such as GHG MRV systems; (v) value chain strengthening to enhance service delivery, avoid losses and increase dairy consumption for better nutrition; and (vi) sustainable access to green finance through innovative blended-finance instruments and products under the GDFF for the benefit of the private sector. Thus, in relation to point (vi), it is anticipated that equipment purchased with GCF proceeds for groups of smallholder farmers (cooperatives), e.g., Milk Collection Centers and Milk Collection Points, will be sustained through improved access to finance, i.e., the GDFF financing.

243. In terms of engagement and ownership, the Programme was designed with the participation of a wide range of national and regional public institutions, private sector representatives and operators, civil society, and development partners. The validation of the GCF Project Preparation Facility (PPF) at the COP27 also raised awareness of the Programme globally and at high level, and engagement of parties beyond the participating countries. The Programme's Stakeholder Engagement Plan (Annex 7) provides detailed information on how these stakeholders have been and will continue to be involved during implementation. M&E and knowledge management and learning (KM&L) will ensure that lessons learned, documents prepared, and activities financed during the implementation of the Programme are archived (in the Programme monitoring information system), disseminated and replicated for the benefit of similar Programmes, including in Latin America and Asia.

⁹⁰ <https://acumen.org/blog/we-need-new-fx-solutions-in-africa-now/>

244. Regarding the enabling environment, by deeply integrating low carbon, climate-resilient dairy interventions into national and regional policies and regulations, the legacy of Component 1 will be an enhanced enabling environment and enable the replication and scaling of CCA&M solutions through other projects and programmes– for impact and sustainability at scale. In particular, the policy and regulatory review (Output 1.1) ensures an enabling framework for dairy value chain development and market integration at national and regional level. Improvements in veterinary and extension services (Output 1.2) strengthen the capacity of public services to continue to support similar extension interventions within and outside the targeted areas, beyond the duration of the Programme. In addition, the increase in MRV capacities (Output 1.3) will enable Governments and private operators to effectively monitor GHG emissions, to improve inventories and reporting to the UNFCCC, and to mobilise more public and private finance to address climate change.

245. The Programme will strengthen the capacity of dairy cooperatives to provide dairy farmers with consistent and stable access to outlets for their milk with fair prices and timely payment (Output 2.1). They will also receive support to formulate better business plans and financing applications for climate resilient and low-carbon investments. When dairy farmers have access to a stable market for their milk, they will be less risk-averse and more willing to adopt the technology packages, needed to make the dairy value chain climate-resilient and profitable. Output 2.2 will build on the well-established L-FFS approach to encourage farmers to adopt practices in the long term. L-FFS is one of FAO's flagship innovations, implemented sustainably in more than 90 countries. An estimated 20 million farmers worldwide have participated in FFS since their inception. The four participating countries are not an exception, and the extension methodology is being implemented in many projects sponsored by the IFIs and development partners such as FAO. The Programme will take advantage of the approach and combine it with digital and extension innovations, so that dairy farmers have direct and technical knowledge of the different low carbon practices being promoted.

246. The GDFF will mobilize long-term capital for additional climate financing together with related top end technical support to participating banks for climate-resilient investments in support of the dairy value chain over its 12-year life, supporting the adoption of climate-smart technologies and practices beyond the DaIMA programme period. By using GDFF funding, in the long term, the GDFF will reduce the perceived risk of the banks in investing in dairy SMEs and catalyze additional private capital into the sector. The GDFF will bridge the gap between demand and supply of finance for climate-resilient and low-carbon dairy and create a sustainable and scalable mechanism that can be replicated across other value chains and regions. The GDFF's embedded M&E function (see Annex 2, Chapter 10 for detail on GDFF's M&E) will ensure that learnings are captured throughout delivery, which will facilitate replication and scaling.

FINANCING INFORMATION						
C.1. Total financing						
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	150.00			million USD (\$)		
GCF financial instrument	Amount	Tenor	Grace period	Pricing		
(i) Senior loans	17.5	Enter years	Enter years	Enter %		
(ii) Subordinated loans	0					
(iii) Equity	0					
(iv) Guarantees	0					
(v) Reimbursable grants	0					
(vi) Grants	132.50					
(vii) Results-based payments	Enter amount					
(b) Co-financing information	Total amount			Currency		
	208.26			million USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
IFAD in Kenya	Senior Loans	24.75	million USD (\$)	Enter years Enter years	Enter%	Options
IFAD in Rwanda	Senior Loans	22.24	million USD (\$)	Enter years Enter years	Enter%	Options
IFAD in Tanzania	Senior Loans	50.81	million USD (\$)	Enter years Enter years	Enter%	Options
IFAD in Uganda	Senior Loans	77.98	million USD (\$)	Enter years Enter years	Enter%	Options
IFAD, Uganda, POLG	Loan	13.125	million USD (\$)	Enter years Enter years	Enter%	Options
IFAD, Rwanda, POLG	Loan	4.375	million USD (\$)	Enter years Enter years	Enter%	Options
Government Kenya	In kind	0.23	million USD (\$)	Enter years Enter years	Enter%	Options
Government Rwanda	In kind	14.75	million USD (\$)	Enter years Enter years	Enter%	Options
(a) Total financing (c) = (a)+(b)	Amount			Currency		
	358.26			million USD (\$)		
(d) Other financing arrangements and contributions	N/A Note: POLG is the Programme of Loans and Grants					
C.2. Financing by component ⁹¹						

⁹¹ Tentative budget subject to revisions

Component	Output	Indicative cost million USD (\$)	GCF financing		Co-financing		
			Amount million USD (\$)	Financial Instrument	Amount million USD (\$)	Financial Instrument	Name of Institutions
Component 1	Output 1.1	\$17.50	\$3.47	Grants	\$14.03	Loan	IFAD
	Output 1.2	\$26.32	\$3.75	Grants	\$22.57	Loan	IFAD
	Output 1.3	\$8.03	\$7.16	Grants	\$0.87	Loan	IFAD
	Output 1.4	\$6.35	\$2.90	Grants	\$3.45	Loan	IFAD
Component 2	Output 2.1	\$83.72	\$18.78	Grants	\$64.94	Loan /in kind	IFAD /GoR
	Output 2.2	\$98.79	\$40.63	Grants	\$58.15	Loan /in kind	IFAD /GoR
	Output 2.3	\$52.47	\$41.29	Grants	\$11.18	Loan /in kind	IFAD /GoR
Component 3	Output 3.1	\$35.00	\$17.50	Loan	\$17.50	Loan	IFAD
	Output 3.2	\$10.00	\$10.00	Grants	0	Loan	
Monitoring and Evaluation	M&E	\$6.72	\$1.46	Grants	\$5.26	Loan	IFAD
Programme Management Cost	PMC	\$13.36	\$3.05	Grants	\$10.31	Loan /in kind	IFAD /GoK
Indicative total cost (USD)		\$358.26	\$150.00		\$208.26		

C.3 Capacity building and technology development/transfer

C.3.1 Does GCF funding finance capacity building activities? Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer? Yes ☒ No ☐

247. Capacity building and dairy CC adaptation and mitigation technology transfer are integral parts of the Programme. Firstly, GCF funding is used to insert the CC aspect into the L-FFS training modules and to train trainers and lead farmers in these technologies, in addition to organizing large-scale technology transfer to smallholder dairy farmers. These technologies include all the ones that are needed for a sustainable pathway to increase productivity of dairy cows, while reducing emission intensity per kg of milk produced. Secondly, the management of dairy cooperatives will be trained in the use and management of renewable energy sources, in addition to waste management. Thirdly, stakeholders in the dairy value chain will receive TA support and capacity building to prepared business plans for green investments. Fourthly, substantial capacity building support are at the core for making the GDFF financial services to reach objectives and impact and will target both the participating banks and borrowing businesses.

EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

D.1. Impact potential

248. The Programme will guide smallholder dairy producers and private sector value-chain operators on a transformational path to a low-emission and climate-resilient dairy sector. As part of this transformation, dairy production and value chains will be streamlined for maximum efficiency, financial services will be made available, and the sector's institutional framework will be enhanced. This transformation will result in increased dairy production, dairy productivity, climate resilience of the sector, reduced emissions intensity of milk, and reduced total emissions.

249. To achieve mitigation targets, the Programme will provide technical assistance for the establishment and improvement of low-emission dairy processing, the optimization of efficiency and the reduction of losses in milk processing facilities, as well as the promotion of adaptive livestock breeding, integrated rangeland and pasture management, and animal health systems. Moreover, the Programme aims to facilitate the establishment of the GDFF, aiming to blend GCF proceeds with other investors, including IFAD's. In this way, businesses and dairy farmers receive financial incentives to reduce their carbon emissions, providing them with potential sources of income.

250. The Programme objective of promoting climate resilience and sustainable livestock production will be achieved by supporting local private sector actors, such as dairy cooperatives, aggregators and milk processors, to overcome technological and financial obstacles. By expanding rural employment opportunities and strengthening the resilience of local households to climate shocks, the Programme will help them increase their productivity and expand their businesses. By building the management capacity of smallholder farmers, improving their creditworthiness, fueling their demand for alternative technologies, and providing them with access to finance through the GDFF, smallholders will be able to significantly reduce their emissions and maximize their efficiency. These intertwined components contribute significantly to the creation of a climate-resilient and sustainable landscape in target countries. The Programme is expected to make a significant contribution to GCF MRA 4 "Forestry and Land Use," ARA 1 "Most vulnerable people and communities" and ARA 4 "Ecosystems and ecosystem services". It will directly contribute to results of the GCF's Updated Strategic Plan (USP-2), i.e. T1 (by supporting the Country's adaptation and mitigation goals), T4 Food and T5 Ecosystems, through T9 locally-led adaptation approach. The Programme will also support dissemination of climate information and MRV system, directly contributing to T3 CIEWS. The Programme also aligns with T10 (by supporting resilient market chains) and T11 (by developing GDFF).

251. The total number of direct beneficiaries will be about 2.5 million individuals and value chain actors vulnerable to climate change (467,105 in Kenya, 614,013 in Rwanda, 757,652 in Tanzania and 665,097 in Uganda). The Programme will invest in a total of 94 districts/counties (12 in Kenya, 27 in Rwanda, 25 in Tanzania, and 30 in Uganda). A total of about 2.7 million dairy cattle will be reached. Approximately 178,362 ha of rangelands and natural pastures will be brought under improved management. 15.4 million individuals (indirect beneficiaries) will be likely to receive an adaptation benefit due to strengthened veterinary services (for better resilience of dairy farmers in the context of diseases exacerbated by climate change), enhanced extension and breeding services, and improved access to climate information. See Feasibility Study (sections 5 and 6) for more details.

252. The efforts of the Programme to strengthen adaptation to climate change will result in a sustainable increase in milk production in East Africa. Within the same horizon of 20 years and relative to the baseline, the Programme is expected to significantly increase milk production of the targeted dairy farms 34 percent, of which 31 percent in Kenya, 73 percent in Rwanda, 36 percent in Tanzania, and 30 percent in Uganda (see Annex 22). This increase in milk production will contribute to the agricultural GDP in the four countries, marking a substantial impact on the economy and climate resilience of the region.

253. As a mitigation benefit, over a 20-year horizon and relative to the BAU, the Programme targets a decrease of milk emission intensities by 29 percent, of which 28 percent in Kenya, 50 percent in Rwanda, 29 percent in Tanzania and 33 percent in Uganda.

254. Over the same period, total GHG emissions on the targeted dairy farms are expected to decrease (relative to BAU) by about 4.01 percent at the level of the Programme, or (i) by 2.02 percent (-61,967 tCO₂-eq) in Kenya, (ii) by 9.56 percent (-56,507 tCO₂-eq) in Rwanda, (iii) by 4.69 percent (-65,718 tCO₂-eq) in Tanzania and, (iv) by 5.22 percent (-82,372 tCO₂-eq) in Uganda. Methane from enteric fermentation is expected to decrease by 4.61 percent overall, while methane from manure management would decrease by 5.93 percent. The Programme is also expected to generate 54,435 tCO₂eq/yr savings from energy generation and access. More details about the analysis, which used – among

others - the Global Livestock Environmental Assessment Model interactive (GLEAM-i), are provided in Annex 22. Impact metrics can be found in E.3 and Annex 11.

D.2. Paradigm shift potential

255. The Programme's comprehensive approach to the paradigm shift in the dairy sector will reduce GHG emissions while improving the livelihoods of dairy producers in East Africa. By improving policy, institutional, and service delivery environments, building climate adaptation capacity, and enabling access to finance and climate-smart solutions for smallholder dairy farmers, the Programme aims to transform all aspects of the dairy value chain (refer to section B.2.(a)). This holistic and transformative approach, providing support to the private sector and financial access to the dairy sector has the potential to foster sustainable, low-emission growth in the dairy market to create a lasting impact beyond the initial investment in the Programme. This is due to the scalability of its components, the generation and dissemination of knowledge, the creation of an enabling environment for the widespread adoption of climate-resilient solutions, and the improvement of national regulatory policies in Kenya, Uganda, Tanzania, and Rwanda.

256. Through the integration of two key strategies into its programming, the Programme has a strong potential for **scalability**. Firstly, it improves access to veterinary services and offers gender-inclusive advisory services, enabling male and female farmers to improve animal health, feed quality, and herd management, thereby increasing the adoption of climate-smart practices. Second, it improves access to finance and alternative technologies, enabling local private sector actors such as dairy cooperatives and milk processors to overcome technological and financial constraints (refer to section B.3. for more details). The dual approach not only improves productivity, but also facilitates the transition to sustainable, low-carbon livestock production, demonstrating both substantial and scalable impact.

257. The two main components of the Programme are expected to contribute significantly to the creation of **knowledge and lessons learned**. First, the Programme generates detailed data on TIER2 emissions by enhancing national MRV capacities, which are critical to improving national inventories and informing evidence-based policies, planning, and investments. Through the implementation of this robust MRV framework, data gaps will be addressed, and climate scenarios and assessments will be strengthened. In addition, the Programme will facilitate knowledge transfer through regional and SSTC to replicate transformative practices to build a climate-resilient and low-carbon dairy sector. A key component of this effort is supporting Regional Centres of Excellence for technology validation, as well as maintaining a regional database on practices, GHG emissions, and adaptation indicators. In addition, the creation of regional platforms for cooperation as well as participation in global networks facilitate the exchange of knowledge and the development of policy initiatives (refer to section B.3. for more details).

258. In addition, the Programme plays an important role in **creating a conducive environment for the adoption of climate-resilient solutions**, including through the strengthening of market structures. To this end, a comprehensive review of the dairy sector will be conducted in addition to conducting farmer awareness campaigns and improving supply chain traceability. Through these efforts, market formalization and development will be fostered. The provision of tailored financial products and technical assistance will facilitate this process and improve access to financing and affordable credit for on-farm investments. The Programme also facilitates access to the carbon market and supports the development of a certification program for low-carbon production. This not only creates financial incentives for mitigation efforts but also accelerates the transition to low-carbon practices by offering potential revenue streams for dairy farmers and businesses that achieve low-carbon certification (refer to section B.2(a)).

259. In addition, the Programme will **advance regional, national, and local legislative and regulatory frameworks** that encourage investment in low-emission or climate-resilient development (refer to section B.2(a)). In Rwanda, it will focus on the regulatory framework and standard operating procedures for dairy value chain operators, working directly with the Rwanda Food and Drugs Authority as well as the Rwanda Standards Board to address milk quality, standards, safety, and energy efficiency issues. As part of the Programme, the formation of a national regulatory framework for private veterinary practice and the development of a Dairy Value Chain (DVC) regulatory law will be supported. Tanzania and Uganda will benefit from the program by assisting in the review and approval of policies such as the Animal Health Act and the Dairy Act, which will focus on aspects such as raw milk trade regulations and disease control plans (refer to section B.3. for more details). This will allow for the finalization of policies aligned climate change priorities, which will enable private sector investment policies and research strategies (e.g. on the development of climate-resilient feed germplasm, the improved genetic resources of dairy breeds, etc.).

D.3. Sustainable development

260. The Programme will contribute significantly to social and health benefits in the targeted regions. It encompasses a comprehensive approach, such as market formalization, awareness campaigns and product traceability to improve nutrition and health awareness, with a focus on the consumption of safe and high-quality dairy products, thereby

improving the individual health of rural and urban consumers (sustainable development goal (SDG) 3: Good Health and Well-being). In addition, by strengthening veterinary services, the Programme aims to improve animal health, which will lead to higher milk productivity and reduced losses, which will decrease GHG emissions and increase farmers' incomes. This will incentivize farmers to adopt better measures on animal health and welfare (SDG 12: Responsible Consumption and Production). By improving nutrition and food safety through increased consumption of high-quality dairy products, the Programme contributes to ending hunger, achieving food security, and improving nutrition.

261. In addition, the Programme contributes to positive environmental impacts, in particular soil quality and conservation. A key element is the promotion of biodiversity conservation and healthier ecosystems, through better management and restoration of rangelands. This approach is essential to maintain biodiversity and combat climate change (SDG 15: Life on Land). The holistic strategy to build climate resilience and reduce emissions along the dairy value chain includes support for rangeland management and restoration. This restores carbon in soils and encourages farmers and processors to adopt more efficient and climate-resilient practices. The Programme also focuses on improved pasture management and circular economy principles, with a focus on pathway-specific integrated solutions such as improving manure management for biogas production and nutrient cycling, which improves soil health. In addition, the Programme advocates for the circular management of agricultural nutrients through livestock manure, improving soil health and on-farm productivity (SDG 12: Responsible Consumption and Production). These strategies collectively reinforce the Programme's commitment to environmental sustainability.

262. The Programme is expected to generate significant positive economic impacts, including job creation, poverty reduction, attracting private finance, increasing productivity, and strengthening income-generating capacities in the dairy sector. Smallholder dairy producers and local private sector value chain actors will have access to essential financial services, enabling them to transition to low-carbon, climate-resilient dairy production, and value chain development. This approach aims to increase dairy farmers' incomes, motivate them to adopt more efficient practices, while reducing production costs. In addition, the Programme will facilitate the adoption of climate-smart technologies, leading to increased productivity and incomes of smallholder farmers (SDG 9: Industry, Innovation, and Infrastructure). Capacity building and adoption of integrated solutions will improve sustainable and resilient production, thereby improving the incomes and productivity of smallholder dairy farmers (SDG 1: No Poverty).

263. The Programme reduces gender inequalities. By actively promoting women's participation in L-FFS and building gender awareness among dairy farming households through the GALS+, the Programme addresses knowledge, social, and gender barriers by equipping male and female farmers with the tools to understand the impact of climate change and adopt alternative technologies and practices. By focusing on gender-inclusive practices and promoting inclusive participatory planning for rangeland management, the Programme plays a crucial role in mitigating the disproportionate impacts of climate change on women, thereby contributing to gender equality in climate action (SDG 5: Gender Equality).

D.4. Needs of recipient

264. The Climate Impact Potential Assessment, conducted under the Programme indicates that East African countries such as Kenya, Rwanda, Tanzania, and Uganda face significant challenges related to the dairy value chain⁹². The impacts of climate change threaten the sustainability of dairy farming as it contributes significantly to GHG emissions. Smallholder farmers make up most dairy producers, yet they face limitations due to low productivity, inadequate training, limited supplies, and climate vulnerability.

265. Kenya is the driest among the four countries, with annual rainfall ranging from 400 to 600mm in some areas. It has the highest number of climate-related disasters, with 78 recorded since 1971. The central parts of Kenya have the lowest number of people affected by floods each year, with less than 100 people per administrative boundary affected. However, exposure to drought is more significant in the southwest, especially around cities like Kisumu, Nakuru, and Nairobi.

266. In Uganda, people are most affected by flooding in the northern, eastern, and central regions along the course of the White Nile. The annual average number of people affected varies from 10,000 to 25,000 per administrative boundary, rising to 36,000 in the Northern region. Uganda, along with Kenya and Tanzania, falls under the "alert THI class" for most areas, indicating increased exposure and vulnerability of animals to climate extremes.

267. Rwanda is one of the countries with the highest total annual rainfall (1000 to 1600mm). Due to the high population density, there are also many people exposed to drought, especially in areas with high population density.

⁹² Kirungu, C., Alvar-Beltrán, J., & Soldan, R. (2023). DaIMA (GCF-IFAD) - Study 3 - Climate Impact Potential Assessment. Office of Climate Change Biodiversity and Environment (OCB), Food and Agriculture Organization (FAO).

Despite this, strategies based on centuries of exposure to droughts and floods are becoming less effective due to difficulties in implementing coping strategies.

268. In Tanzania, the provinces of Tanga, Morogoro, and Pwani have the highest number of people affected by flooding, with 10,000 to 25,000 people affected per administrative boundary on average per year. Exposure to drought is comparatively lower in places such as the Kizigo Rungwa and Usungu Game Reserves, where population density is lower.

269. Each of these countries suffers from a significant development deficit, which negatively impacts their ability to steer the dairy sector on a path to low-emission and climate-resilient development. In all four countries, public fiscal space is limited (see section B.5), and governments rely heavily on official development assistance (ODA) and concessional financing to fund their dairy sector programs. As a result, vulnerable populations, such as smallholders, women, and youth in the dairy sector are highly exposed to climate risks due to technological, organizational, and policy constraints.

270. The lack of funding for climate-smart agriculture and climate-resilient practices is a significant barrier in the region. Small-scale dairy pastoralists struggle to access credit and financial services, which is crucial for investing in modern equipment, infrastructure, and better farming practices. This limitation exacerbates the vulnerability of the dairy sector to climate-induced changes like unpredictable weather patterns, prolonged droughts and increased exposure to disease, leading to reduced productivity and environmental degradation.

271. In addition, poor communication between climate information suppliers and agricultural extension services, limited institutional arrangements, insufficient resources to maintain observation networks, gaps in technical skills, and underdeveloped communication networks in East Africa hinder the effective dissemination of climate information and response to climate-related challenges in the region.

D.5. Country ownership

272. Country ownership of the Programme is ensured through alignment with national policies and ongoing GCF investments, as well as a participatory design process of the Programme.

273. Participating countries' NDCs focus on reducing GHG emissions and increasing resilience to CC. Support to the dairy sector can benefit a variety of factors, including climate and environment, nutrition, sustainable rural livelihoods, and social inclusion. Therefore, the Programme aligns with the national strategies of the participating countries by supporting and complementing existing plans, policies and strategies related to agriculture, livestock, and climate change. In Kenya, for example, the Programme aligns with the Agriculture Sector Development Strategy (2019-2029), National Climate Change Action Plan (NCCAP), Kenya Climate Smart Agriculture Implementation Framework (2018-2027), and Kenya National Strategy on Climate Change and Low Carbon Development. In Uganda, the Programme aligns with the Livestock Development Strategy and Investment Plan (LDSIP), NAP, and National Climate Change Strategy. In Rwanda, the Programme aligns with Rwanda's NDC, National Climate Change and Environment Policy, Rwanda Livestock Master Plan, National Adaptation Plan for the Agricultural Sector, and National Livestock Policy.

274. In addition, the Programme aims to integrate climate change adaptation and mitigation by reviewing and developing national dairy policies, such as the Dairy Industry Strategy and Investment Plan, and the Dairy Industry Act (1958) in Kenya. The Dairy Act to update regulations on raw milk trade; the formulation of disease contingency and control plans; the draft Tanzania Dairy Development Strategy in Tanzania. In Rwanda, the Programme will work closely with Rwanda Food and Drugs to review the regulatory framework and standard operating procedures (SOP's). In Uganda, the program will engage DDA to review the Dairy Industry Act and the Animal Disease and Health act and the Animal Breeding Act (refer to section B.3.).

275. The Programme complements several planned and ongoing GCF programs in East Africa. At the regional level, the objectives of the Programme complement the GCF regional program, Strengthening Climate Information Systems for Climate Change Adaptation in the Greater Horn of Africa (GCF Concept note), aimed at strengthening climate resilience in eight countries (IGAD members) through the establishment of early warning and climate dissemination information system. It is also aligned with the ARCAFIM project, which covers the same countries. At a country level, the Programme is aligned with several ongoing projects described in section B.1. (section Synergy with other projects).

276. The Programme was designed following an extensive consultation process with local stakeholders, ensuring that the needs of recipients are considered and ensuring full country ownership of the Programme. The NDAs in each

target countries were directly involved in all stages of Programme ideation, concept, preparation. In each country, National Programme Preparation Teams (comprising a wide range of stakeholder from ministries of livestock and dairy boards) were established in 2023, at the regional workshop in Kigali (September 2023). Through regional (Arusha, February 2023; Kigali, September 2023) and national workshops (July and September 2023), stakeholders engagement meetings were organised to ensure that the preparation of the Funding Proposal (CN process included) involved a wide array of national and regional public institutions, private sector representatives and operators, civil society, research institutions and development partners. The Stakeholder Engagement Plan (Annex 7) details the list of stakeholders and their involvement in the Programme design. The Programme benefited from high-level exposure and engagement with Parties beyond the target countries during the validation of the GCF Project Preparation Facility (PPF) at COP27.

D.6. Efficiency and effectiveness

277. The ex-ante financial and economic viability of the Programme is demonstrated by an economic and financial analysis (EFA). Economic analysis looks at the profitability of investments for the entire Programme from the perspective of society, while financial analysis considers the profitability of these investments from the perspective of Programme participants. The Livestock Sector Investment Policy Toolkit (LSIPT) was used to develop the dairy production models in all four countries. All the models estimate the “with” and “without” Programme situation over a 20-years period. For cattle models, the ECORUM module of the LSIPT for simulating bio-economic performance of herd of tropical domestic ruminants have been used. The parameters of the dairy herd models reflect the activities proposed by the Programme. Five dairy herd models were developed for each of the countries. The national parameters and ponderation between models are derived from the characteristics and national structure of the dairy herd in the Programme intervention area, as presented in the pre-feasibility studies. The dairy herd models were: (i) small scale intensive system/ zero grazing with cross breed and exotic animals; (ii) small scale intensive system/ zero grazing with exotic animals; (iii) semi-intensive with cross breeds; (iv) semi-intensive with local breeds; and (v) extensive systems with local breed. In addition, financial models were developed for MCPs; MCCs; dairy processing units, all powered by solar panels, as well as service providers such as para-veterinary services. The Programme GHG mitigation impact was evaluated using the Global Livestock Environmental Assessment Model, Interactive – GLEAM-i (Annex 22).

278. The Programme net cash flows are based on the incremental approach, which results from comparing the With Programme Situation and Without Programme situation⁹³. The financial models have been calculated at 17.5 percent discount rate⁹⁴, the average lending rate in Tanzania, Uganda, Rwanda, and Kenya. The selection criterion for the IRR is to accept all Programmes for which the IRR is above the opportunity cost of capital. The financial analysis indicates that the investments proposed by the Programme are profitable for the dairy farmers and cooperatives.

279. At the regional level, the analysis shows that DaIMA represents an economically viable investment for the society. Economic performance indicators for Tanzania, Uganda, Rwanda, and Kenya, individually confirm economic viability. At regional level, the aggregated ENPV of the countries’ incremental net benefit streams, discounted at 13.25 percent (social discount rate), is US\$ 181.20 million, with an ERR of 25.0 percent for the baseline scenario. Additionally, two additional economic performance scenarios are included, by integrating the economic value of the Programme’s GHG emissions balance with a low carbon market price (LCP) assumption and a high carbon market price (HCP) assumption. Under the LCP scenario, the ERR is 27.9 percent and the ENPV is US\$ 217.87 million. Under the HCP scenario, the ERR is 30.3 percent and the ENPV is US\$ 254.64 million. The sensitivity analysis confirms that DAIMA represents a robust economically worthwhile investment. For more information see Annex 3.

⁹³ IFAD, 2015, Economic and Financial analysis of rural investment projects, basic concepts, and rationale.

⁹⁴ World Bank data -average 2017 to 2020.

LOGICAL FRAMEWORK

E.1. Programme Focus

- ☒ Reduced emissions (mitigation)
- ☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Livestock, especially cattle, are the main source of methane emissions in the East Africa region, representing about 70 percent of total emissions from livestock and the second highest source, representing about 14 percent from agriculture, forestry and other land uses. Without a major shift to low-carbon practices and mitigation pathways across all dairy production systems, trends in absolute and emission intensities will continue to rise by 2050, in the context of a growing demand and growth of the dairy cattle population in the region. Exacerbated climate hazards are reported to cause serious impacts on and risks to the dairy sector with direct and indirect effects being heat stresses, droughts, flooding and other extreme weather among others affecting productivity, changes in disease patterns, availability and quality of animal feeds, and ecosystem services (degradation of pastures for grazing, reduced soil carbon sequestration).	Medium	Paradigm shift will involve a move away from the current state of growing absolute emissions, low production efficiencies, informal and climate-vulnerable dairy sector, with limited access to services, including extension, finance and milk markets. Shift from low production efficiencies to improved, gender-sensitive, integrated and NDC-aligned practices that reduce emissions at farm and food system levels (including interventions that reduce milk losses and increase milk collection, decarbonize processing, formalize marketing chains for milk and enhance partnerships with the private sector), as well as interventions which increase rangelands ecosystems resilience and services, and promote circularity. Paradigm shift will be accompanied by measures that strengthen low-carbon climate-resilient dairy policies, regulation and institutional capacities.	Implementing a set of integrated and transformative interventions (animal health, feed and feeding strategies, artificial insemination, herd management, methane inhibitors, manure management, renewable energies etc.), combined with strengthened national capacities (veterinary and extension support services), and enabling conditions (including better access to finance, at concessional terms) that are required to make dairy food system more efficient, climate resilient and on a sustainable pathway to decarbonization. Scaling-up of pasture management practices (rotational grazing, agroforestry, management of grazing intensity etc.) through gender-sensitive public and private extension, ensuring availability of high-quality feeds and fodder, breeds to reduce emission intensities, while sustaining production levels throughout the year, and enhancing soil carbon sequestration. The Programme will advance regional, national and local legislative and regulatory frameworks that encourage investment in low-emission or climate-resilient dairy sector development.

Replicability	Low-carbon climate-resilient practices in the dairy sector are supported by other projects and partially implemented locally through dairy farmers' indigenous/ traditional knowledge, but these practices are currently not adopted and replicated at scale. Post-farm gate adoption of renewable energies (e.g. solar) remains low, despite considerable interest from the private sector in decarbonizing their operations. Dairy farmers and processors have limited incentives to adopt practices and technologies due to institutional, knowledge, social, natural, financial (including government spending and commercial lending) and technological barriers. This is in addition to missing national data on herd, numbers, parameters, production needed to accurately conduct GHG emission assessments, analyze technical improvements, and ascertain measurement, reporting and verification (MRVs). This limitation in reporting prevents leveraging more climate finance, hence replication across the region.	<u>Medium</u>	If well demonstrated (through public and private extension and robust business models), if accompanied by appropriate concessional financing solutions, and if well-promoted locally and regionally, paradigm shift can be replicated by farmers outside the project areas, to other countries in the region, and even internationally. Appropriate reporting on emissions with a well-functioning MRV system for the livestock sector allows to enhance the quality and precision of inventories and mobilize more climate finance for replication across the EAC region, and even other countries.	At local level, the Programme will facilitate knowledge transfer through public (e.g. L-FFS) and private (through cooperatives or aggregators) extension. The GDFF financing and technical assistance facility (with concessional terms) will have a catalytic effect on adoption of climate smart practices. In addition, the establishment of regional Centers of Excellence for technology validation, regional cooperation platforms as well as participation in global networks facilitates the replication of knowledge and practices across the region and internationally. The Programme will generate detailed Tier2 emissions data by enhancing MRV capacities, which are crucial to improving national inventories and informing evidence-based policies, planning, and investments. A regional database on practices, GHG emissions, and adaptation indicators will be established. With the implementation of MRV framework, data gaps are bridged, and climate scenarios and assessments are strengthened, hence allowing more access to climate finance, including private.
Sustainability	The commitment of the governments of Kenya, Rwanda, Tanzania and Uganda for implementing low-carbon and climate-resilient pathways for livestock are evidenced in their NDCs, NAMA and sectoral policies, and livestock master plans (when applicable). Regionally, the East Africa Commission (EAC) also has a framework on low emission livestock.	<u>Medium</u>	Paradigm shift would see sustainable governmental and private sector support to implement transformative interventions for a climate-resilient and low carbon dairy sector. Implement interventions in line with the nationally owned NDCs, as well as regional policies. Given that many dairy farmers are small-scale, farmers cooperatives to better meet the requirements of national financing institutions will be explored, while catering for specific circumstances of the cooperative members. It should also promote key incentives for adoption, leading to increased revenue or reduced production costs.	To ensure sustainability, the Programme will work closely with the relevant government and regional institutions and enhance their capacity and delivery of services (extension, animal health etc.). The Programme will also work closely with farmers organizations (cooperatives for milk collection), the private operators (aggregators, processors, supporting enterprises such as agri-inputs – AI, dairy meals, fodder seed, PV solar solution providers, climate smart and digital tech) to

				demonstrate the switch towards climate-smart practices.		
E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)						
GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁹⁵	
MRA4 Forestry and land use	Core 1: GHG emissions reduced, avoided or removed/sequestered	Surveys at mid-term(years 3) and end-line (year 6) Monitoring with the Global Livestock Environmental Assessment Model, interactive (GLEAM-i) TIER 2 data generated from the MRV system under component 1 (output 1.3) will be used to run GLEAM-i and estimate sequestration in each country.	0	-162,217 tCO ₂ eq. (see Annex 22 for details) <u>Of which:</u> -37,602 tCO ₂ eq. (enteric CH ₄); -4,374 tCO ₂ eq. (CH ₄ manure); -1,296 tCO ₂ eq. (CO ₂); 3,288 tCO ₂ eq. (N ₂ O)	-324,234 tCO ₂ eq. (final) -1,080,780 tCO ₂ eq. (end of lifespan period) (see Annex 22 for details) <u>Of which:</u> -75,204 tCO ₂ eq. (-250,682 tCO ₂ eq., 20 years) (enteric CH ₄); -8,747 tCO ₂ eq. (-29,157 tCO ₂ eq., 20 years) (CH ₄ manure); +6,575 tCO ₂ eq. (+21,918 tCO ₂ eq.,20 years) (N ₂ O);-2,593 tCO ₂ eq. (-8,642 tCO ₂ eq., 20 years) (CO ₂)	Consolidated values from (i) reduced emissions from dairy cattle, (ii) soil carbon sequestration. This indicator will rely on emission and sequestration Tier 2 factors generated under sub-activity 1.3.1.1 (respiratory chambers, flux towers).
MRA1 Energy generation and access	Core 1: GHG emissions reduced, avoided or removed/sequestered	Electricity meters to track amount of grid/ diesel electricity displaced Measure annual biogas production from cattle manure using volumetric flow meters Calculate emission savings based on evidence/ activity data from surveys in years 3 and 6	0	163,306 tCO ₂ eq. (see Annex 22 for details)	326,612 tCO ₂ eq. (end of project) 1,088,705 (end of total lifespan period) (see Annex 22 for details)	Project will successfully displace a significant percentage of electricity consumed in milk collection center, milk processing centers and farms.

⁹⁵ The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

ARA1 Most vulnerable people and communities	Core 2: Direct and indirect beneficiaries reached	Yearly progress reports Mid-term (year 3) and end-line surveys (year 6)	0	Total: 1,016,000 direct beneficiaries (40% women, 30% youth) Reports disaggregated by country, type of beneficiary (direct or indirect) gender and youth. 5.6 million indirect beneficiaries	Total: 2.5 million direct beneficiaries (40% women, 30% youth) Reports disaggregated by country, type of beneficiary (direct or indirect) gender and youth. 15.4 million indirect beneficiaries	Direct beneficiaries include dairy farmers and other value-chain actors (e.g. MCPs, MCCs, SMEs...). Dairy farmers (individuals) calculated based on (i) the number of dairy animals in targeted areas (districts/counties), (ii) number of animals per household and per production system (based on GLEAM database). See Feasibility Study (Chapter V) for more details. Indirect beneficiaries based on the total rural population keeping livestock in the targeted project areas
ARA4 Ecosystems and ecosystem services	Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice	Mid-term and completion surveys in years 3 and 6, using remote sensing (e.g. NDVI) and ground truthing, and data from MIS system.	0	89,181 hectares	178,362 hectares	Rangeland restoration proceeds as planned. This indicator, combined with the sequestration factor under MRA4, will provide an estimate of impacts of the restoration activities.
ARA4 Ecosystems and ecosystem services	Core 2: Direct and indirect beneficiaries reached	Mid-term and completion study in years 3 and 6	0	850,600 direct beneficiaries 1.4 million indirect beneficiaries	About 2,13 million direct beneficiaries About 5.7 million indirect beneficiaries	Beneficiaries are dairy farmers using rangelands and pastures as a main feed source, as well as farmers in other systems practicing the 'cut and carry'. Indirect beneficiaries are based on the total rural population keeping livestock in grazing/ extensive and semi-grazing

						areas (within targeted districts and counties).
ARA4 Ecosystems and ecosystem services	Supplementary 4.2: Number of livestock brought under sustainable management practices	Surveys in years 3 and 6 to monitor the total number of cows under sustainable management practices.	0	Total: 800,000 dairy animals Reports disaggregated by country, type of farming system	Total: about 2,700,000 dairy animals Reports disaggregated by country, type of farming system	Number of dairy cows calculated, with GLEAM.

E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the Programme will contribute	Coverage
Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner	Ministries responsible for livestock, National Dairy Boards, and dairy cooperatives are spearheading the transformation of the dairy sector to a low carbon and climate resilient industry, in line with their mandates and NDCs. However, these championing institutions need more and better understanding of climate change adaptation and mitigation risks, and investment opportunities to address them. Countries' policies and regulations need to be updated (and aligned with regional/ EAC policies/ regulation) to enable access to agro-inputs and technology for more efficiency in dairy production.	<u>highmedium</u>	Ministries responsible for livestock, National Dairy Boards, large-scale processors, and dairy cooperatives develop the required skills and capacities to align their policies and regulations and enhance their capacity to carry out climate vulnerability assessments and run MRV systems and better report their emissions.	A climate-responsive policy and regulatory environment created through the review of the dairy sector's institutional frameworks in a participatory and country-driven manner. Through regional cooperation and knowledge sharing, coordinated strategies will be ensured, individual country efforts will be amplified, and joint learning on CCA and CCM will be fostered. Improving GHG MRV capacities and monitoring climate vulnerabilities.	<u>Multi-countries</u>

	Moreover, activity data on herds, numbers, parameters, and milk production is lacking to conduct accurate GHG emission assessments, analyze technical improvements, improve emissions inventories, and “feed” MRV systems.				
Core Indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination, development or transfer and innovation	Lack of incentives (for dairy farmers, communities, and the broader private sector) and collective learning mechanisms to invest in adaptation and mitigation opportunities. Wide range of emissions intensities (GHG per kg of milk produced), reflecting the wide range in adoption of improved technologies by dairy farmers in intensive, semi-intensive and extensive systems.	<u>highmedium</u>	Dairy farmers are informed and adopt climate adaptation and mitigation practices, based on traditional/ local knowledge, and based on latest evidence from science/ research on low-carbon climate resilient livestock (e.g. ILRI, FAO).	The Programme will invest in proximity veterinary and para veterinary and breeding services, as well as extension services (through L-FFS and other approaches) to reach out to dairy farmers for technology deployment, dissemination, development and transfer and innovation. Participatory learning (including women) and promotion of sustainable grazing practices and land management which will help rehabilitate degraded rangelands, promote fodder access, and address the natural barrier to rangeland degradation. Technologies that increase operations’ efficiencies and decarbonization, organized marketing chains for milk ensuring efficient offtake and development of cold chain infrastructure along the entire supply chain	<u>Multi-countries</u>

				leading to reduced emissions from food loss and waste and better returns. The financing and technical assistance facility (with concessional terms) will have a catalytic effect on the adoption of practices and innovations.	
Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards	There is a need to enhance and update knowledge at local, national and regional levels about practices and technologies for climate change mitigation and adaptation in the dairy sector.	<u>medium</u>	Specialized livestock extension services are strengthened to provide effective support on good practices for greater efficiency, climate resilience and GHG emissions reduction. As mentioned under Core 6, dairy farmers are informed and adopt climate adaptation and mitigation practices, based on traditional/ local knowledge, and based on latest evidence from science.	The Programme will invest in regional cooperation and knowledge sharing, institutional strengthening for improved livestock services, and in farmer-led adaptation and mitigation approaches (FFS).	<u>Multi-countries</u>

E.5. Programme specific indicators (programme outcomes and outputs)

Programme results (outcomes/ outputs)	Programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	

Outcome 1: Systemic and institutional capacities of the livestock sector are strengthened to enable smallholder dairy producers and local value chain actors to reduce CH4 and other GHG emissions	Core Indicator 5: Degree to which GCF projects/ programmes contribute to strengthening institutional and regulatory frameworks for low-emission climate-resilient development pathways in a country-driven manner	Mid-term and final review, qualitative and quantitative assessments in years 3 and 6 Programme completion report.	See E.4 above			
	Core Indicator 6: Degree to which GCF projects/ programmes contribute to market development / transformation at the sectoral, local or national level	Mid-term and final review, qualitative and quantitative assessments in years 3 and 6 Programme completion report.	See E.4 above			
Output 1.1: Climate-responsive policy and regulatory environment established	National policies/ strategic frameworks /regulations revised- Number	Mid-term and final review, qualitative assessments, Completion report	0	8	23	National Governments and National Dairy Boards are willing to adopt a more formal dairy market regulation framework.
	Regional policy dialogues to mainstream CC A&M in dairy policies - Number	Mid-term and final review, qualitative assessments, Completion report	0	2	6	National Governments and Dairy Boards are willing to invest in regional market integration. Partnership with ECA
Output 1.2: Improved service delivery to the dairy sector	Veterinarians and para veterinarians trained – Number by category, age and gender	6-monthly progress reports	0	600	1,800	40% women, 30% youth
	Extension officers / L-FFS facilitators trained – Number by category, age and gender	6-monthly progress reports	0	300	1,000	40% women, 30% youth
Output 1.3: Improved GHG MRV capacities & monitoring climate vulnerabilities	MRV system for livestock in place with Tier2 dairy emission factors generated - Yes/No	6-monthly progress reports	No	Yes	Yes	Willingness of national Governments to engage in capacity building and regional cooperation.
	National feed balances produced per year – Number of FBS	6-monthly progress reports	0	8	8	

Output 1.4: Knowledge management and enhanced regional cooperation	Regional repository on practices, GHG emissions per sources and systems - Number	Mid-term and final review, qualitative assessments, Completion report	0	1	1	Willingness of national Governments to participate
	Functional Regional knowledge sharing platform on climate-smart dairy sector transformation - Number	Mid-term and final review, qualitative assessments, Completion report	0	1	1	Willingness of national Governments to participate
Outcome 2: Smallholder dairy producers and private sector value chain players operate within a low-emission, climate-resilient, and sustainable pathway through increased production and market knowledge and tools	GHG emissions from dairy production relative to BAU - percentage	Global Livestock Environmental Assessment Model, interactive (GLEAM-i) Mid-term and completion study in years 3 and 6	Kenya: 0% Rwanda: 0% Tanzania: 0% Uganda: 0%	Kenya: - 0.4 % Rwanda: - 1.8% Tanzania: - 0.9% Uganda: - 1%	Kenya: - 0.7% Rwanda: - 3.5% Tanzania: - 1.6% Uganda: - 1.9%	TIER 2 data generated from the MRV system under component 1 (see output 1.3) will be used to run the GLEAM-i model in each country.
	Milk production increase relative to BAU - Percentage	Mid-term and completion study in years 3 and 6	Kenya: 0% Rwanda: 0% Tanzania: 0% Uganda: 0%	Kenya: 4.6% Rwanda: 10.9% Tanzania: 5.4 % Uganda: 4.5%	Kenya: 9.3% Rwanda: 21.9% Tanzania: 10.8% Uganda: 9%	Activity data available
Output 2.1: Smallholders' and cooperative technical and business capacity development strengthened.	Dairy cooperatives legally established and functioning - Number	6-monthly progress reports	0	200	650	
	Business alliances / linkages created - Number	6-monthly progress reports	0	50	106	Dairy sector willingness to invest in business alliances.
	Cooperatives, MCCs and MCPs with investments in renewable energy - Number	6-monthly progress reports	0	50	190	
Output 2.2: Breeding, feed and fodder, manure management improved. When relevant, indicators disaggregated by (i) type of recipient, (ii) gender, (iii) age	Farmers using breeding services - Number	6-monthly progress reports	0	661,000	1.6 million	70% of dairy farmers 40% women and 30% youth
	Farmers accessing animal health services (vaccination) - Number	6-monthly progress reports	0	0.94 million	About 2.36 million	All animals will be covered by vaccination campaigns. 40% women and 30% youth
	FFS covering climate-smart forage production, breeding, improved	6-monthly progress reports	0	5,000	9,700	40% women and 30% youth

(youth vs non-youth), (iv) country	feeding and manure management - Number					
	Seed producers trained - Number	6-monthly progress reports	0	3,200	9,500	40% women and 30% youth
	Investments in water points/ water-harvesting systems for cattle – Number	6-monthly progress reports	0	20,000	59,945	Individual and collective types of water points
	Biodigesters installed - Number	6-monthly progress reports	0	2,310	6,930	
Output 2.3: Pasture and grasslands management improved.	Natural resource areas brought under improved low-emission and/or climate-resilient management practices - Ha	Mid-term and completion study in years 3 and 6	0	89,181	178,362	The estimation of the area of rangeland to be rehabilitated was based on the annual amount of biomass (fresh mixture and hay or silage of grass and legumes) required to meet the animal feed intake for improved diet .
	Number of farmers trained in pastures and grasslands management - Number	6-monthly progress reports	0	113,165	339,500	40% women and 30% youth. Farmers in grazing/ semi-grazing systems
Outcome 3: Financial institutions make available the appropriate investment resources to make the necessary transition to low-carbon, climate-resilient, and sustainable dairy production and strengthen value chain linkages and improved productivity.	Number of green dairy loans (DL) and green dairy partnership loans (DP) approved	Mid-term and completion study in years 3 and 6	0 0	350 DL 30 DP	1,000 DL 90 DP	When relevant, indicators disaggregated by (i) type of recipient, (ii) gender, (iii) age (youth vs non-youth), (iv) country
Output 3.1 Effective Green Financing for Dairy Value Chains	Volume of green partnership investments in the dairy value chain – Million US	6 monthly progress reports	0	US\$17.5M	US\$ 17.5 million	The split between partnership investments and smaller dairy loan products is tentative and depends on market demand.

	Volume of smaller dairy loan products – Million US\$	6 monthly progress reports	0	US\$17.5 M	US\$35M	The investments will revolve twice during the 6 years' period. The split between partnership investments and smaller dairy loan products is tentative and depends on market demand
	Performance of the Gdff participating Public Development Banks and of the Gdff Retail Financial Institutions financial services offered to SMEs investing in Green Dairy value chain.	Reports from the PDBs Reports from the Retail Financial Institutions	n/a	Average NPL below 5%	Average NPL below 5%	IFAD will have a comprehensive set of financial performance and development outcome indicators collected from the PDBs and Retail Fls. .
Output 3.2 Capacity Building for Green Dairy Investments	TA assessments for Green Dairy Loan Partnerships Approved Number	6 monthly progress reports	0	30	67	
Programme co-benefit indicators						
Co-benefit 1: Increased food security and food safety and quality for rural/urban consumers	% of households that have increased their food security	Surveys in year 6 (Programme completion report)	0	20%	40%	No major external shocks (i.e political or exogenous) with impact on beneficiaries and programme sites.
Co-benefit 2: Improved women's empowerment in dairy value chains	% of women reporting increased decision-making power in dairy-related investments & revenues (including access to and control over dairy production inputs or assets)	Surveys in year 6 (Programme completion report)	0	20%	60%	Women members interest, ability and agreement to better manage the additional income generated.
E.6. Programme activities and deliverables						
Activities	Description	Sub-activities		Deliverables		
Activity 1.1.1: Review and formulate Dairy Sector policies, strategies and regulations for climate change adaptation and mitigation	Ministries responsible for livestock and National Dairy Boards will support an enabling regulatory environment for climate-sensitive dairy value chain development.	1.1.1.1: Support the review and preparation of national dairy policies, strategies, and regulations for climate-change adaptation and mitigation		23 national dairy policies, strategies and regulations review and amended. 8 regional policies supported.		

	Dairy Boards and local government are increasingly aware of the challenges and opportunities in climate change and the dairy value chain.	1.1.1.2: Support the regional policy dialogue to mainstream climate-change adaptation and mitigation in regional dairy policies, strategies and regulations 1.1.1.3: National dairy value chain governance and awareness creation 1.1.1.4: District/regional/county level dairy value chain governance and planning	In each country, dairy value chain stakeholders have inserted climate change in the dairy value change governance and planning.
Activity 1.1.2: Launch awareness campaigns on nutritional benefits of milk and food safety aspects	Support to milk consumption and nutrition awareness, accompanied by nutrition education and social behavior change, aims at improving nutritionally vulnerable group's consumption of milk, therefore enhancing their individual's health and climate resilience.	1.1.2.1: Public awareness on the contribution of milk to nutrition and resilience, with focus on climate vulnerable populations 1.1.2.2: Build capacity of dairy inspection to ensure milk quality and safety and reduce waste	189 awareness campaigns on milk and nutrition 60 inspectors trained on milk quality and safety
Activity 1.1.3: Improve land tenure security for sustainable dairy farming	The Programme will adopt a fit-for-purpose approach to identify and mitigate tenure risks, which are key barriers to sustainability, and adopt climate resilient and low carbon practices. In Kenya, Tanzania and Uganda, the main risk for intensive/zero-grazing is tenure insecurity, i.e. occupancy rights or land ownership. In Rwanda, the focus will be on optimizing land production and productivity through land consolidation and land use planning.	1.1.3.1: Mapping of local tenure risks and preparation of training materials 1.1.3.2: Developing capacity on land tenure at decentralized level.	Tenure governance capacity assessments carried out in each country Capacity development on tenure carried out at local level (district/county)
Activity 1.2.1: Strengthen veterinary services for better animal health, higher milk yields and reduced losses	Proximity veterinary and para veterinary capacity will be enhanced to be able to reach out to dairy farmers, using the One Health Approach. Reference veterinary laboratories equipped with modern diagnostic technologies in regional centers.	1.2.1.1: Support installation of both private and public veterinarians and veterinary paraprofessionals 1.2.1.2: Strengthening diagnostic capacity of veterinary laboratories and capacity to test quality of animal feed	1,800 veterinarians, para veterinarians and LEWs trained and equipped. Reference laboratories in Kenya (7), Rwanda (4), Tanzania (10) and Uganda (10), equipped for diagnostic analysis of animal diseases and feed testing.
Activity 1.2.2: Strengthen extension services for a low-carbon and climate resilient dairy sector	Improving delivery of public and private extension services at farm and cooperative level, by district or county livestock officers and L-FFS who coordinate the community-based extension system	1.2.2.1 Preparation of training modules for extension in climate mitigation and adaptation measures. 1.2.2.2: Training of master trainers (ToTs) and the L-FFS facilitators in climate-change related topics	Training modules prepared on climate-smart milk production and handling, including fodder production, feeding, reproductive and health management, manure management, and animal welfare.

		1.2.2.3: Promote innovative extension tools.	1,000 extension agents and L-FFS facilitators trained. 9,700 L-FFS supported. 4 extension innovations promoted
Activity 1.3.1: Develop capacity in the country on MRV system	System development for Tier 2 GHG emissions in dairy (e.g. animal numbers, weights, productivity, feed rations, animal health interventions, manure management and grassland management) and develop a data management system	1.3.1.1: Capacity assessment and development in capturing, recording, tracking, and reporting data on dairy GHG emissions. 1.3.1.2: Developing MRV systems and engage in policy dialogue to align with national MRV for the livestock sub-sector	Four (4) operational Tier2 GHG emission system and capacity to manage in each country.
Activity 1.3.2: Improve GHG emission measurements in dairy value chains	Collection on-farm activity data for baseline and “with-project” situation, using digital tools to finetune the GHG emission measurements in each country. Training of national teams to manage their MRV systems and monitoring of Programme Outcomes and Impact.	1.3.2.1: Collect activity data on farm to calculate Tier 2 dairy emissions under various herd, feed, animal health and manure management situations 1.3.2.2: Conduct experiments and run models to generate Tier 2 emission factors (EF) for baseline and improved dairy, under various herd, feed, animal health and manure management situations 1.3.2.3: Training on the Global Livestock Environmental Assessment Model – interactive (GLEAM-i) and IPCC assessment tool to calculate GHG emissions biannually, and to inform the national MRV system and inventories	Four (4) national MRV systems have been finetuned and parameters tested. Four (4) national systems manage their national MRV systems and provide information to the Programme M&E system.
Activity 1.3.3: Climate risk and vulnerability assessments	This will include building the capacity of devolved climate change units and decentralized meteorological offices to develop and disseminate climate information services, including (i) impacts of climate change on livestock/ dairy sector, and (ii) information about the availability of feed, through tools like national feed balance sheets.	1.3.3.1: Climate risk and vulnerability assessments, including impact on the dairy sector, at local level. 1.3.3.2: Review and enhancement of national feed balance mechanisms, and support to early warning systems	4 climate risk assessments undertaken. 8 national feed balances prepared per annum.
Activity 1.3.4: Facilitate low-carbon certification and reduce barriers to access carbon finance	This indicator will give dairy farmers access to carbon finance.	1.3.4.1 Develop a carbon certification system for the dairy sector Yes/No	Qualitative assessment (yes/no)

Activity 1.4.1: Support to Regional Centers of Excellence for technology validation and for maintaining a regional database on practices, GHG emissions and adaptation indicators	The Programme will provide scientific support combined with capacity development to “regional centers of excellence” to export knowledge (regionally and globally) on climate change and GHG emissions for the dairy value-chain.	1.4.1.1: Support to regional Centers of Excellence in technology validation and dissemination 1.4.1.2: Develop and maintain a regional database/ repository on practices, GHG emissions per sources and systems, and climate change adaptation	A regional database on best practices, and their impact on GHG emissions per sources and livestock production systems, and adaptation indicators is operational.
Activity 1.4.2: Establish regional cooperation platforms and participation to regional/ global networks for knowledge exchange on climate-smart dairy sector transformation	A regional platform – involving the above-mentioned entities - will be organized on a rotational basis in each of the participating countries to share experiences and scientific innovations and best practices implemented on farm. The regional platform will function as the knowledge management mechanism for the Programme.	1.4.2.1: Regional platform for knowledge sharing and policy dialogue to enhance low-emission, climate resilient dairy development. 1.4.2.2: Participation in regional and global climate policy mechanisms 1.4.2.3: South-South and Triangular Cooperation	Participation in at least 10 regional and global climate policy events. 23 south-south cooperation events.
Activity 2.1.1: Strengthen dairy groups and cooperatives on organizational management and service delivery	This activity aims to strengthen cooperative, organizational, and technical capacities of dairy cooperatives, and raise their climate-change and gender awareness of their management team. Support will be provided to enhance their network of MCCs and MCPs.	2.1.1.1: Strengthen the governance and management structure of dairy cooperatives. 2.1.1.2: Strengthen capacity to reduce milk losses and ensure milk quality and safety. 2.1.1.3: Expand the outreach and capacity of cooperatives, MCCs and MCPs 2.1.1.4: Create awareness on gender transformative approaches among households of dairy farmers	650 dairy cooperatives are strengthened and ensure that dairy farmers have a stable outlet to less their milk. 650 dairy cooperatives have a better capacity to manage milk in a safer way and upgraded equipment for milk collection. Management of 650 dairy cooperatives is more gender sensitive.
Activity 2.1.2: Build and enhance productive alliances along the dairy value chain.	This activity supports productive alliances between farmer cooperatives and enterprises with farmer-allied engagement models, particularly processors, and scaling-up dairy clusters, that enable farmers to access affordable, high-quality inputs and services.	2.1.2.1: Strengthen cooperative capacity in business and financial management. 2.1.2.2: Build linkages along the dairy value chain. 2.1.2.3: Promote digitalization and traceability of dairy operations.	Management of 650 dairy cooperatives received business training. 300 business links established or strengthened. 15 cooperatives/MCCs (pilot) are testing the digitalization and traceability system to ensure quality of milk.
Activity 2.1.3: Promote renewable energy and waste management technologies	The Programme will focus on resource use efficiency and the substitution of fossil fuels and firewood by green energy sources. Solar is the key mitigation solution and most applicable to relatively larger processors in the short term given high upfront costs.	2.1.3.1: Assess the need and viability of renewable energy and waste management solutions. 2.1.3.2: Support the acquisition of renewable energy and waste management solutions. 2.1.3.3: Organize capacity building in energy efficiency and waste management	190 cooperatives have installed solar equipment. 130 dairy cooperatives have installed wastewater management systems. Management of 650 cooperatives received training in energy efficiency and waste management

Activity 2.2.1: Promote the adoption of climate-smart dairy breeds adapted to climate change	Breeding and genetic improvement will aim at improving productivity and heat resistance of dairy cattle while preserving resilience of livestock keeping systems by crossing indigenous and exotic breeds through artificial insemination (AI). The national systems will be upgraded.	2.2.1.1: Make good quality semen for climate-smart breeds available, adaptable to extreme climatic conditions and to diseases. 2.2.1.2: Invest in breeding and artificial insemination centers. 2.2.1.3: Scale up the insemination of dairy cattle to make the herd more productive and climate resilient. 2.2.1.4: Organize awareness campaigns and support breeders' organizations	Four (4) countries have scaled up the availability of good semen. Four (4) countries have enhanced the capacity of their national centers to store semen and liquid nitrogen. Four (4) countries have scaled up their capacities to inseminate dairy cattle. Awareness campaigns in each of the countries.
Activity 2.2.2: Provide support for better animal care, herd management and disease prevention	Production losses through reduced performance (growth rates, milk yield, suboptimal herd management) or animal loss through morbidity, mortality, and abortion results in cows with relatively high GHG emission intensities. A comprehensive package of interventions will be proposed to improve animal health and herd management.	2.2.2.1: Organize training on animal husbandry, with a specific focus on animal health, animal welfare, herd management and disease prevention. 2.2.2.2: Scale up the use of digital apps to monitor animal health and herd performance. 2.2.2.3 Scale up vaccination campaigns to mitigate the impact of climate change on the spread of diseases.	9,700 FFS implemented, targeting 291,000 dairy farmers, including farmer-to-farmer trainings in animal health, animal welfare, herd management and disease prevention. 5,000 dairy farmers who have been trained in the use of digital apps to monitor animal health and herd performance. 2.7 million dairy cows vaccinated against ECF, PPCB and FMD.
Activity 2.2.3: Promote climate-smart forage production, conservation, and water harvesting	The adoption of climate resilient forage varieties and conservation technologies is essential to reduce enteric methane emissions intensities in the dairy value chain, while increasing milk yields of dairy cattle. A comprehensive package of interventions will be proposed to improve forage production and conservation, and access to water.	2.2.3.1: Make certified forage seed available to private and community multipliers. 2.2.3.2: Support multiplication of climate-resilient seed for fodder crops and pastures 2.2.3.3: Build capacity of dairy farmers on forage production and conservation on intensive and semi-intensive dairy farms. 2.2.3.4: Access to water for animals and forage production in intensive and semi-intensive dairy systems.	9,500 farmers were trained in seed multiplication for forage crops and pastures. 9,700 FFS implemented, targeting 291,000 dairy farmers, including farmer-to-farmer trainings in forage production and conservation. 100,000 farmers received equipment for water harvesting.
Activity 2.2.4: Promote climate-smart feeding practices to reduce enteric methane emissions.	Appropriate diet formulation, based on the available resources, is a key requirement to improve dairy cattle productivity, improve welfare, while reducing methane (CH ₄) emissions. Dairy farmers' capacity will be built.	2.2.4.1: Develop production system specific supplementation plans and feed packages. 2.2.4.2: Build capacity on appropriate feed formulation. 2.2.4.3: Promote appropriate diet formulation that reduces enteric methane emissions	9,700 FFS implemented, targeting 291,000 dairy farmers, including farmer-to-farmer trainings in improved feed packages.

Activity 2.2.5: Increase sustainable nutrient recycling of manure and dairy waste to reduce emissions	Livestock manure is a source of methane (CH ₄) and nitrous oxide (N ₂ O) emissions, ammonia (NH ₃) and nitrate (NO ₃ ⁻). At the same time, manure contains essential nutrients and is often the primary source of crop fertilization on smallholder dairy farms. Dairy farmers' capacity will be built.	2.2.5.1: Manure management in intensive and semi-intensive dairy production systems. 2.2.5.2: Promote adapted animal housing 2.2.5.3: Manure management in extensive systems	9,700 FFS implemented, targeting 291,000 dairy farmers, including farmer-to-farmer trainings in manure management and adapted cattle housing.
Activity 2.2.6: Build capacity for on-farm use of biodigesters and biogas	Manure management using biogas digesters is particularly attractive for intensive and semi-intensive dairy systems. The Programme will promote on-farm adoption of biodigesters for these dairy farms.	2.2.6.1: Support capacity building in biodigester construction and management 2.2.6.2: Support the investments in biodigesters at the household level	6,920 biodigesters installed.
Activity 2.3.1: Strengthening rangeland governing structures for communal grazing areas	To approach net-zero targets of dairy production systems, the Programme considers soil carbon sequestration as an important pathway of GHG emission removals to offset livestock and manure emissions.	2.3.1.1: Participatory rangeland management planning	300,000 hectares of rangeland management plans prepared
Activity 2.3.2: Restoration of degraded pastures, including access to water	Well managed and restored rangelands not only benefit improved productivity but contribute also to improving soil carbon sequestration and the carbon balance of the livestock production systems, increased climate resilience through healthier ecosystems, better access to water, biodiversity conservation, as well as mitigation of environmental risks.	2.3.2.1: Training on rangeland and pasture management 2.3.2.2: Rangeland and pastures restoration 2.3.2.3: Water availability	9,700 FFS implemented, targeting 291,000 dairy farmers, including farmer-to-farmer trainings in rangeland management. 178,362 hectares under improved rangeland management 300 investments in water infrastructure
Activity 3.1.1: GDFF Operationalized	Involves setting up and roll out of investments through the GDFF structure in two windows: Green Dairy Loan Partnerships and Green Dairy Loan Products.	3.1.1.1: GDFF established at the PDBs 3.1.1.2: GDFF wholesale lending provided to the retail financial institutions 3.1.1.3 Retail financial institutions provide loans to the SMEs to invest into green dairy value chains 3.1.1.4 Portfolio management	1,000 smaller (aver. US\$ 50,000) Dairy Loans approved ; capital revolving 3 times over years 90 larger (aver. Loan US\$ 400,000), Green Dairy Partnership Loans approved, capital revolving twice over 12 years Financing of US\$35 million for GDFF green investments
Activity 3.2.1 Awareness Raising and Capacity Development of	Provides appropriate tools and capacity building to the GDFF wholesale and retail financial institutions in the implementation of	3.2.1.1: Green dairy finance lending toolkits developed and implemented	1 toolkit developed and implemented in all countries

Participating Banks in Green Dairy Finance	green finance to agriculture SMEs and dairy value chain details.	3.2.1.2: PFI trainings 3.2.1.3 Coordination by PDBs	4 PDBs and at least 6 retail financial institutions received comprehensive capacity building 4 PDBs provide appropriate coordination
Activity 3.2.2. Green Dairy Investment Support Services to SMEs	Provides appropriate training and capacity building to the GDFF pipeline of borrowing SMEs to invest into	3.2.2.1 Training of MSMEs that are potential pipeline clients of GDFF for the smaller lending product 3.2.2.2 Training of larger SMEs that are potential pipeline clients of GDFF for the Partnership Loans	MSME pipeline of at least 6 retail finance institutions supported by technical assistance at MSME level for the GDFF rollout Larger agribusiness SME pipeline of at least 6 retail finance institutions supported by technical assistance at SME level for the GDFF rollout
Activity 3.2.3. M&E, household surveys and evaluations	This activity will ensure the appropriate M&E system for the GDFF including annual, midterm and final reports.	3.2.3.1 M&E system established and maintained 3.2.3.2 M&E household surveys and evaluations conducted	All M&E indicators for GDFF reported against as per required level of detail and frequency M&E household surveys implemented, and analysis reports submitted

E.7. Monitoring, reporting and evaluation arrangements

280. Each EE will implement and coordinate the overall M&E system building on the existing M&E systems of the related IFAD-financed projects, allocate resources for M&E tasks and deliverables, aggregate M&E data and present them in overarching reports and knowledge documents, verify and ensure quality and validity of the M&E data and information, and organize learning events to facilitate sector wide learning, ensure that funds are effectively managed to deliver expected Outputs and achieve the anticipated Outcomes; ensure the quality of Programme monitoring, Programme closure and final evaluation timelines are complied with. The RSC, with secretarial functions provided by the national EEs on a rotational basis, will be responsible for providing harmonized reporting and methodological guidance on overall M&E implementation, as well as consolidation of timely yet high-quality reports to the GCF. M&E guidance beyond the PIM will be provided by ILRI (e.g. developing templates, methodology, explanatory notes), overseen by the regional steering committee. The data capture and reporting aspects will be handled by the countries, through their PMUs. IFAD will organize and lead annual supervision missions, including mid-term and end term evaluations.

281. The overall M&E for the GDFF will be built around its ToC and Logframe. The system will continuously be built through ongoing adaptive management so that the Programme can regularly assess its performance in supporting planning and reporting, learn and adapt to improve its climate adaptation outcomes. While IFAD will have the overall coordination role to all M&E reporting to the GCF for the GDFF, the Wholesale Banks in each country will have a prominent role in data collection and analysis. Details on the M&E are proposed in Annex 11.

282. Specifically, for the GDFF under Component 3, a senior GDF Coordinator will be recruited at IFAD Kenya Regional Office to support the overall coordination, M&E, reporting and Knowledge Management of the GDFF, for the entire project period. In addition, during the Programme start-up phase, experienced Financial and M&E Specialists will be recruited to assist the DaIMA PMU to develop an appropriate reporting, monitoring and evaluation system for

GDIFF operations, based on a pre-defined reporting schedule, indicators, and formats, to be agreed-upon between IFAD, DaIMA PMU and the Wholesale Banks. The GDIFF reporting and monitoring would be integrated within the reporting, monitoring, and evaluation systems to be developed for the whole regional DaIMA operation. These issues are described in detail in Annex 11 of the Funding Proposal and in the Feasibility Study on Component 3 (Annex 2).

283. **Annual Work Plan and Budgets (AWPBs).** The M&E Officer in each of the national PMUs will prepare the national Programme's consolidated AWPBs, specifying the following: (i) key activities in the period; (ii) specific unit targets and baseline values; (iii) expected outputs the activity is contributing to; (iv) timeline to complete each activity; (v) tabular summary information on the AWPBs of the overall project to easily assess progress; and (vi) responsible reporting parties of each activity and data. The Programme staff and key stakeholders will validate an 18-month AWPB drafted at design during the start-up phase of implementation. As the Programme implements, the preparation of annual AWPBs will be a participatory process that will be based on the activities needed and proposed by the beneficiaries, districts, and implementing partners to achieve the desired changes stated in the Theory of Change and indicator targets. The EEs will be the responsible institution for all planning processes and for inclusion of key stakeholders to guarantee transparency, accountability while ensuring sufficiency of M&E information for informed and timely decision-making. The EEs will receive, review, harmonize and consolidate AWPBs from the lower planning levels in accordance with each country administrative or governing structures while ensuring consensus building among key stakeholders in each planning level. The consolidated AWPB will be reviewed by the **National Programme Steering Committee (NPSC)** in each country for endorsement before submission to the **Regional Steering Committee (RCS)** as the next endorsement level before submission to IFAD. Submission to IFAD will happen at least 60 days before commencement of the next National level Program planning year/cycle. AWPB review meetings will be conducted during each planning cycle in all levels with the involvement of all key stakeholders. This will allow common understanding and consensus building among stakeholders on national annual priorities, expected targets and budgetary adequacy for effective and sustainable implementation of the Program objectives

284. **Reporting System.** Each participating country will facilitate preparation of each quarterly, six-monthly and an annual progress highlighting achieved successes, constraints and, lessons learnt, recommendations and action plans to address the identified constraints in the subsequent plans and implementation periods. Each implementing country's national reports to be submitted to GCF will be consolidated at RSC level, with technical support provided by ILRI and/or other service providers.

285. **Monitoring and Evaluation.** Responsibilities for monitoring and evaluation lie with the national EEs and Component 3 implementation team. Each country has its own Logical Framework that includes GCF core indicators (outcomes and outputs), in line with the GCF requirements. Respective executing lead agencies from each country as the key implementer of the Programme, will play a key role in data collection and analysis and reporting. They will collect and report on baseline data, portfolio information, as well as output and select outcome indicators; participate in Annual Supervision Missions, Midterm and Final Evaluations; as well learning and adaptive management events organized internally, across projects or those supported by IFAD/GCF directly. The RSC, with technical support required, will be responsible for providing guidance on reporting needs requiring regional/Programme level consolidation, as well as progress realized on regional aspects of the Programme, and that regional added value is achieved and sustained beyond the Program life. Specifically, RSC will monitor progress with respect to Output 1.1, Output 1.3 and Output 1.4. At national level, the PMUs will be responsible for monitoring and evaluation, knowledge management of all activities related to the Programme. This should in accordance with the Logical Framework provided that includes GCF core indicators (outcomes and outputs), in line with the GCF requirements. The M&E processes will further be refined during the inception phase of the GDIFF with necessary inputs from the stakeholders. When relevant, indicators will be disaggregated by (i) type of recipient, (ii) gender, (iii) age (youth vs non-youth), (iv) country at least and may further be required at project levels and according to the information needs of each implementing country.

286. **Supervision and implementation support missions** will be organised jointly by IFAD, GCF and the Governments of Kenya, Rwanda, Tanzania and Uganda, to assess and review progress and assist implementing partners and the PMUs improve on Programme implementation based on the mission's support

and recommendation. The supervision missions will assess the overall physical and financial performance, identify implementation challenges and propose possible mitigation measures to help address them. This should be in compliance with all the fiduciary aspects of the financing agreement. The supervision missions will be conducted at least twice a year or more frequently as conditions may warrant, and according to IFAD's GCF own policies and procedures including the provision for implementation support. The outputs of the supervision missions will include: i) Aide Memoire and ii) supervision mission reports with findings and recommendations, focusing on the achievements and constraints for discussion with the lead Ministry. To facilitate the work of the mission team, lead agencies will prepare implementation progress reports ahead of the mission team describing key achievements, lessons learnt, best practices to scale up implementation and constraints encountered during the implementation. The support missions will help identify and address emerging issues, build technical capacities, and to the extent possible, build on experiences and best practices from past and ongoing IFAD - GCF funded projects. The composition of the support mission team will include, but not limited to: The financial Expert, M&E expert, targeting specialist etc. It may also add value and if the budget allows to include a sections of PMU teams on some mission on rotational basis for different projects. This will enhance learning, complement adaptive management and serve as best practices aimed at strengthening knowledge management within the project and government set-up.

287. **Baseline Survey.** The Programme will conduct a baseline survey at the beginning of implementation in the targeted areas. The survey will be designed to capture the following baseline data separately: (i) sampled existing beneficiaries; and (ii) sampled new beneficiaries. The specific GCF core indicators will be included in the baseline survey that will be updated based on the milestones by the Outcome Survey. The study will be contracted to an external service provider (SP) with the involvement of PMUs with ILRI and IFAD providing overall guidance. This involvement is aimed at ensuring harmony in indicator definition and targets refinement. This will be done in a way that ensure data collection appropriates and fir for purpose in the context of Program implementation.

288. **Mid-term Review and Final Evaluation.** Interim and Final Independent Evaluations will be planned and managed by IFAD according to the requirements of the GCF's evaluation criteria and as per the GCF Evaluation Policy. In advance to the end term Evaluation Missions, GCF will conduct a series of more comprehensive household/business surveys including both quantitative and qualitative data collection and analysis. IFAD will lead evaluation process following IFAD's procurement guidelines, supported by IFAD's Research and Impact Assessment (RIA) unit. The evaluations will be supported by an Evaluation Reference Group, made up of IFAD, GCF and the other co-financiers. The MTR and final evaluation will embed an assessment of the Programme's contributions to the paradigm shift and enabling environment.

289. **Knowledge Management, Communication and Learning.** The Programme will elaborate and implement a KM and Communication Strategy that will: (i) provide Project beneficiaries with the necessary material to sustain the technical knowledge acquired with the support of the Project (production of training materials and communication platforms for sensitization) and continuously assess their adoption; (ii) generate shared knowledge acquired from the experience of the past and current Project in various fields and based on the information collected as part of the monitoring of results or from specific thematic studies; (iii) share knowledge generated with the Technical Departments of the Ministry, IFAD, other donors, and implementing partners using various dissemination strategies and channel for learning and visibility. In achieving the Policy Component, the KM and Communication Strategy will strongly aim to develop and create channels of policy influence and dialogue through the development of policy briefs/strategies to be disseminated audience of high-level policy stakeholders in the Dairy Sector. The KM and Communication Strategy should also establish a Community of Practice (CoP) to support the development and coordination of regional implementation creating harmonized y programme requirements in all levels

RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Selected Risk Factor 1 Institutional Capacity for Implementation and Sustainability

Category	Probability	Impact
<u>Technical and operational</u>	<u>High</u>	<u>Medium</u>

Description High

290. Limited skills exist in the dairy value chain in the four countries particularly in areas of social inclusion and gender, dairy technicians, extension services, M&E and functional dairy L-FFS specialists in the implementing ministries, national dairy boards, and organisations to guarantee effective Programme implementation. In Tanzania and Uganda, the local government has limited financial, procurement and human resources to assume their mandate of execution particularly on community service, extension, nutrition, private sector partnerships and infrastructure development). Further, limited M&E skills within PMU and government staff in general is bound to affect the implementation. The four countries are also at different stages of M&E system set up and maturity. Given the scope of DaIMA project, data collection, harmonization and reporting at the regional level may be a challenge.

Mitigation Measure(s)

291. In each of the four countries, a Programme management unit will be established by the IFAD project that will mobilise cofinancing, and staff recruitment will follow a competitive process to ensure quality expertise is in place. The implementation of the Programme will be structured around performance-based contracts which will also be indicated in the Financing Agreement. Service providers will be contracted through competitive government procedures and based on renewable performance-based service contracts to provide advisory services. As part of the support delivered, implementing partners and service providers will ensure that adequate capacity is built among recipients of their services at various levels including local governments to guarantee their exit strategy and overall sustainability. Continuous capacity building will be conducted to both government and PMU staff Exchange visits and joint reviews will also be conducted to enhance learning in addition to routine M&E training to build the capacity of PMUs and government.

Selected Risk Factor 2 Financial Management and Fiduciary Risks

Category	Probability	Impact
<u>Governance</u>	<u>High</u>	<u>Medium</u>

Description

292. The Programme involves multiple line ministries and the Project Management Units (PMU), districts /county governments, all of which hold fiduciary responsibilities. Although the PCMU will consist of qualified personnel with expertise in technical and financial management, there may be a lack of familiarity among staff with the procedures of IFAD and the GCF. The districts /county governments will receive funding from the Ministry of Finance and report to IFAD through the PMU. There is a risk of delays in flow of fund and the consolidation of project financial reports at the PMU. Furthermore, the project involves multiple financiers, including IFAD, the GCF, government contributions, and beneficiary contributions. This creates a risk of commingling funds in budgeting and expenditure allocations among these financiers. Additionally, there is a risk of inadequate audit coverage for the project due to the involvement of various districts/ counties governments located across a wide geographic area. Additionally, there is a possibility that some high-risk expenditure categories may not be adequately addressed during the audits.

Mitigation Measure(s)

293. Formal agreements will be established between the line ministries, the Ministry of Finance, and the participating districts/county governments. These MoUs will clearly outline the requirements for financial reporting with penalties for Non-Compliance. The Financial Management Division (FMD) will conduct capacity-building training for finance staff recruited through a competitive process. This training will focus on: a) Familiarization with IFAD and GCF procedures; b) Understanding financial reporting requirements and timelines; c) Proper categorization of expenditures across different project components; and d) Overall financial management practices related to the project. The risks associated with fund commingling and to facilitate accurate accounting of any advances provided, the following measures will be implemented: a) Funds will be maintained in dedicated project accounts at the Central Bank of each country. This will help ensure that project funds are kept separate from other government funds; b) There will be monthly bank account reconciliations and financial reports generated for these dedicated accounts to ensure transparency and accountability; c) Each district/county receiving project funds will be required to establish sub-project

accounts specifically for segregating the funds received. This will help prevent the mixing of project funds with other county funds; d) Counties will submit monthly financial reports to the PMU to monitor the operations of their sub-accounts and facilitate consolidation of financial data; e) All district /counties will sign MoUs that clearly outline the requirements for maintaining separate bank accounts and the financial reporting obligations associated with the Programme.

294. To mitigate the risks associated with external audits, the following measures will be implemented: a) the auditor will develop a detailed work plan to ensure sufficient coverage of all counties receiving project funds. This plan will specifically address major risk areas to ensure that all critical aspects of the project are audited, b) the IFAD Finance Officer will provide the external auditors with the IFAD audit terms of reference in advance to ensure that all key elements are included in the audit TOR and c) the project finance staff will be informed of the audit requirements as outlined in the IFAD Financial Management and Financial Control (FMFCL) Handbook.

Selected Risk Factor 3 Sector Strategies and Policies

Category	Probability	Impact
<u>Governance</u>	<u>High</u>	<u>Medium</u>

Description High

295. Each of the four countries recently developed a Livestock Sector Transformation Plan, including a Dairy Development Plan. Sector policies including the recently developed national dairy strategy documents in the four countries are formulated in an inclusive manner, involving all organised stakeholders. The policies are aligned with their national strategies, and priorities. The main gap is the low representation of smallholder dairy farmers in these policy fora, due to the lack of professional organisation of this category of actors. Policy dialogue is thus dominated by private sector actors, who are well organised, national dairy boards, larger progressive farmers, NGOs, and public institutions. The other risk is related to the low capacities of public institutions to enforce sector regulations, particularly those related to milk hygiene. This has a significant impact on the value chain as it provides a comparative advantage to the informal raw milk sector.

Mitigation Measure(s)

296. During the design process, the four Governments committed to adhere to IFAD's environmental safeguards and targeting policy, to maximise positive social and environmental impacts, and ensure that DaIMA is in line with IFAD's core principles. The risk mitigation strategies include: a) Participation of the four countries dairy boards in the entire design process, b) The Programme will support the organisation of smallholder farmers at regional, national, and county/district levels, and their participation in policy dialogue fora, to make the policy dialogue more inclusive and c) The Programme will also support the review and update of sector regulations, but also their enforcement. This will entail support to the national Dairy Boards, District milk inspectors, and strengthening of laboratory facilities for milk control.

Selected Risk Factor 4 Project Scope and Approach

Category	Probability	Impact
Technical and operational	Low	Medium

Description

297. The Programme strategic approach responds to the sector priority needs, characterised by a lack of milk supply, dominance of the informal raw milk value chain and inadequate access to services and finance. However, the situation of the dairy value chain is volatile and may evolve during implementation: the supply/demand balance could be reversed, due to changing consumption patterns or emergence of new market segments, and to rapidly increasing production. This would require a slight adaptation of the programme activities, and a re-balancing of activities in favour of either the marketing aspects, or production. The DaIMA design covers a large range of domains due to the overall low performance of the dairy value chain, affected by challenges at all levels (production, services, access to finance, market access, low consumption, policy gaps). The Programme must address all these challenges at the same time to avoid leaving bottlenecks that would impede the overall development of the sector. This leads to a moderate level of complexity of the programme design, which is however common in similar value chain projects.

Mitigation Measure(s)

298. In case the situation of the value chain evolves during implementation, some adjustments may be needed on project activities, including on budget allocation, without modifying the project structure, theory of change, objectives, and overall strategy. This may include for instance increased support for the processing and marketing levels, and reduced emphasis on production and productivity. These adjustments could be made at the MTR stage. Some of the mitigation measures will include: a) Implementation of specific sets of activities will be delegated to implementing partners that have experience and comparative advantage in these domains, b) Design should be flexible and not

overly prescriptive to allow adaptations in course of implementation, based on lessons learned, c) Programme governance should be solid, with mechanisms for effective national and regional coordination of implementing partners and service providers (reporting system, M&E, regular technical meetings, and regional steering committee meetings) and d) Governance organs such as the National and Regional Steering Committees include stakeholders that have very good knowledge and sound analysis of the sector (e.g. ILRI).

Selected Risk Factor 5 Regional Context

Category	Probability	Impact
Governance	Medium	Medium

Description

299. Despite recent protest by the youth, the four countries are characterised by a good socio-political stability, and a rather low occurrence of inter-ethnic tensions or clashes. In the four countries, there is growing concern because young people have become disenchanted with agriculture and often do not have access to (high quality) input and capital. Youth involvement in small-scale / zero-grazing dairy is critical to address the youth bulge. All IFAD projects in the four countries have been designed to reduce rural poverty and consequently inequality and social tensions. There are also risks linked to political commitment, governance and the macroeconomic situation in the 4 countries.

Mitigation Measure(s)

300. As the largest employer in the country, agriculture will remain an entry point for job creation, inclusive growth, and poverty reduction. Small-scale / zero grazing dairy has a lot of potential to reduce poverty, increase food security, improve nutrition, and strengthen resilience. DaIMA will strengthen the livelihoods of the most disadvantaged rural categories including smallholder dairy farmers, poor households without cows, unemployed youth, women, and women-headed households. DaIMA will leverage four national projects that are co-financed by IFAD and other funding mechanisms and will invest in rural areas to increase smallholder productivity. Regarding political commitment, the relevant counterpart government officials have been actively engaged throughout the design of the regional Programme. Three (3) regional preparation workshops were organised, in addition to a series of national events. The IFAD team will continue to work closely with the Government of the four countries during the next phases of the Programme approval and start-up, to ensure Government ownership and alignment with IFAD's and Government's policies is ensured. Regarding governance, IFAD's COSOP in the four countries places the country-level policy engagement (CLPE) at the core of its strategy. Key areas of policy engagement are focusing on increasing investments, improving policy coherence and coordination and improving the business environment for priority value chains. One of the focus areas of DaIMA will be the formulation and implementation, review and update of national policies, strategies and legislations with respect to the dairy sector. It is envisaged to support the Governments to strengthen regulations on milk trade, as well as their enforcement. DaIMA will also contribute to significantly improve the governance of the dairy sector. Finally regarding macroeconomic risks, The Governments of the four countries committed to macroeconomic policies, aimed at not increasing public debt, containing inflation within their target range, and preserving external stability. The authorities have established a track record of sound macroeconomic management, but further reforms to revenue policy and administration, public expenditures, and debt management will be necessary to create adequate space to increase priority social spending and productive investment without jeopardising fiscal sustainability.

Selected Risk Factor 6 Limited adoption/reversal to BAU practices by beneficiaries

Category	Probability	Impact
Other	Medium	Medium

Description

301. Small dairy farmers show a low adoption rates of improved practices or revert to business-as-usual practices

Mitigation Measure(s)

302. The use of participative approaches for capacity development such as farmers field schools, demo farms and learning routes increases the probability of adoption of improved practices. In addition, activities will rely on the support of dairy groups and cooperatives which will be supported by sector specific organisations such as (i) SDLD and KDB in Kenya, (ii) the Rwanda National Dairy Platform ((RNDP)) in Rwanda; (iii) the Dairy Development Authority (DDA), in Uganda; (iv) the Tanzania Dairy Board (TDB) and the Tanzania Cooperative Development Commission (TCDC). Finally, improved practices will not only result in improved resilience and lower emissions but also in better economic return for farmers, which creates economic incentives.

Selected Risk Factor 7 Insufficient access to markets to incentivise the paradigm shift

Category	Probability	Impact
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Other	Low	Medium
Description		
303. Small dairy farmers are interested in improved practices but they do not have sufficient access to market to find the economic return they need.		
Mitigation Measure(s)		
304. The Programme has a strong focus on developing access to market for small holders under Output 2.1: Smallholders' and cooperative capacity development strengthened. It will expand the outreach and capacity of dairy cooperatives, MCCs and MCPs. It will also build productive alliances and linkages along the dairy value chain, including with offtake agreements between cooperatives and private off-takers (Productive Alliance model), as well as public procurement (e.g. school or hospital procurement) for better milk prices guaranteed by contract.		
Selected Risk Factor 8 Risk of low demand for and repayment of Gdff Funds		
Category	Probability	Impact
Other	Medium	Medium
Description		
305. Establishing four relatively small-sized credit lines in a niche sub-sector like dairy poses several challenges. Relatively low funding base can constrain the interest of banks to participate in the scheme. Lack of understanding of the bankable options in green dairy lending can further reduce bank interest. Similarly, lack of knowledge of the advantages of green dairy investments among the dairy businesses can reduce loan demand. Finally, lack of management skills in medium and small businesses can result in low recovery rates particularly in more technically aspects of green dairy investments.		
Mitigation Measure(s)		
306. To address these risks, Gdff includes several design features and instruments. Substantial technical support provided by Gdff aims to substantially increase and improve the understanding of both the banks and borrowing SMEs on the available investment options in greening of the dairy operations. Favourable loan terms, particularly the long grace and repayment periods, will support successful cash flow management in the borrowing companies and support timely loan repayment. At the same time participation in Gdff lending operations will build positive credit histories for small and medium dairy companies, providing a soft start for permanent client relationships with commercial banks to enable larger future investments for participating dairy businesses.		
GCF POLICIES AND STANDARDS		
G.1. Environmental and social risk assessment		
307. The Programme has been reviewed in line with IFAD's Social, Environmental and Climate Assessment Procedures (SECAP) and classified as Moderate for the categorization of Environmental & Social risks as well as Climate Risk. Based on the results of the screening, the activities proposed under the Programme do not pose a high risk of harm to people or the environment and will be located away from environmentally or socially sensitive areas. In addition, the risks associated with the programme are predictable and expected to be temporary or reversible and have a low likelihood of causing serious adverse effects to human health or the environment.		
308. Recognizing that moderate adverse impacts may impinge on the environment and human health during various phases of the Programme, an Environmental, Social and Climate Management Plan (ESCMP) has been prepared as part of the ESMF. The ESMF entails potential adverse environmental and social impacts and risks associated with proposed activities, respective mitigation measures, responsible parties, and cost estimates for implementation of each mitigation measures. To keep track of the implementation of the ESMF, there will be two types of monitoring to be conducted during implementation of the programme i.e. Performance and results monitoring. The Environment and Social Safeguard Specialist will conduct at least one field visit at an interval of six months to verify reports submitted from responsible parties indicated in the ESMF. Additional site visits and/or changes in site visit frequency may be required based on the risk level of the sub-project. As regards Reviews, there will be annual reviews of the ESMF and all associated environmental and social instruments. This activity will be undertaken in a participatory manner and shall involve programme stakeholders at various levels from the sub-project level to the national level in the four countries. Annual reviews are aimed to ensure that implementation of interventions is undertaken in compliance with IFAD SECAP and GCF Environmental and Social Standards.		

309. During proposal development phase, consultation of potential stakeholders in selected sites was done to solicit their views and concerns regarding the proposed interventions. For a continuous engagement during implementation of the program, a stakeholders engagement strategy has been prepared as part of the ESMF. The strategy provides a framework for achieving effective stakeholder involvement and promoting greater awareness and understanding of issues so that the programme is carried out effectively. As part of the stakeholders' engagement and for addressing complaints that may arise during programme implementation, a Grievance Redress Mechanism (GRM) has been established. The GRM will address queries and offer an avenue for clarifications about the program, respond to problems, complaints, and grievances. The GRM also provides an opportunity for appeal if any aggrieved person is not satisfied with a decision or action in response to their issue. To ensure that the Grievance Redress Mechanism is well known and accessible to all program stakeholders, various communication tools will be adopted to disseminate the information about the available opportunity, and how to register complaints and seek redress. The program will also use the opportunities from the existing communication system and program awareness-raising events to disseminate the information on GRM.

310. As regards Information Disclosure, the ESMF will be disclosed 30 days (Category B/I-2 projects) before the GCF or IFAD Board meeting date, whichever is earlier. It will be posted on the IFAD website, on the GCF website together with the funding proposal and at locations convenient to affected peoples, i.e. at all levels at the relevant institutions. The Programme Coordination Teams at the country level will make and distribute copies of the ESMF written in English and working language of the country in compliance with the IFAD's Public Consultation and Disclosure Policy. Thereafter various communication tools will be utilized for the engagement process, such as Project notices published in local newspapers, radio advertisements, direct mailings to communities, presentations with or without focus group sessions, targeted e-mails, one-on-one meetings, social media, seminars, workshops, e-mails, and phone conversations with stakeholders.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

311. The gender assessment and action plan of the Programme provides an overview of the gender inequalities that hinder women, youth, and other vulnerable groups participation in and benefitting from dairy production and climate change mitigation and adaptation interventions. Annex 8 sets out the main parameters of the Gender Equality and Social Inclusion (GESI) assessment and analysis. Part 1 provides a brief GESI analysis of smallholder dairy farmers and MSMEs (dairy cooperatives and processors) i.e., the experiences of women, youth, people with disabilities and other vulnerable in dairy production and the barriers they face Part 2 provides an overview of the GESI action plan setting out actions and targets to achieve the project objectives.

312. The methodology for the gender assessment and action plan was informed by in-country missions in Tanzania, Uganda, Kenya, and Rwanda. The field missions included consultations with key stakeholders and in-depth interviews with key stakeholders in the dairy sector. This was followed by a desk literature review and synthesis, including practical and field examples to inform the needed interventions. How resources (cattle, water, money and family land) are accessed and controlled by different people, and the implications for climate resilience. For example, women and girls are impacted disproportionately from the effects of climate change due to factors such as entrenched discriminatory social and cultural norms compounded by limited mobility and decision-making power exacerbates climate-related impacts on women's poverty.

313. To address these barriers, the Programme will (i) Increase women economic empowerment through dairy production and access and control over productive resources (ii) Reduce time and labour burdens and balance workloads activities (iii) Increase their voice and decision making at individual, household, group, and community levels and (iv) Increase Joint and equitable benefits sharing while promoting positive gender norms and practices. Young women and men, of different groupings, disabilities, ethnicity, and religion will be targeted to access productive assets, services and skills to create jobs and entrepreneurship. The Programme will recruit implementation partner or Service Providers with experience in community centred approaches promoting integration of young men and women, vulnerable groups, persons with disabilities. This may include gender focused, youthful enterprises, NGOs, the academia and researchers. The GESI action plan will be monitored and evaluated through the project M and E system ensuring sex, age and gender disaggregated data is collated to inform project interventions.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

314. **Disbursements/Fund Flow.** Designated accounts (DAs) will be administered following "revolving fund arrangements" as described in Annex 2 chapter 9. The Recipients/Borrowers will enter into MoU with the Implementing Partners (IPs) before any funds transferred to the IPs, except Tanzania where MoUs signed with the IPs only when they are not included in the financing agreement. Each project will open and maintain separate DAs in USD to receive

GCF funds from IFAD. Similarly, separate project accounts in local currency will be opened by PCUs, district offices and IPs to receive funds from the project designated/operating accounts. Countries where single treasury account is used, adequate chart of accounts will be used to segregate GCF sources and uses of funds from other financiers. In case of flow of fund issues to ILRI, direct payments to ILRI may be considered based on the MoUs signed between the Borrower and ILRI. However, all Withdrawal Applications and IFRs need to be submitted by the PCU of each Country.

315. Disbursements will be based on submission of quarterly Interim Financial Reports (IFRs) within 30 days after the period ends and upon availability of funds. IFRs and the Withdrawal Applications will be submitted to IFAD by PCUs through IFAD's Client Portal and will include a cash forecast for the next two quarters. As per GCF rules, there will be no disbursement unless 70% of GCF proceeds is utilized and all other GCF disbursement conditions are met.

316. **Accounting and Financial Reporting.** Rwanda (RDDP-II) and Uganda (ReLIV): Accounting and reporting will be performed through Country's IFMIS system and will follow national regulation, which is in a gradual process of transition to International Public Sector Accounting Standards (IPSAS) accrual and (IPSAS) cash standards respectively. Tanzania (C-SDTP) and Kenya (INReMP): Off-the-shelf accounting software will be used, and adequate chart of accounts will be developed to segregate sources and uses of funds. The Financial Statements will be prepared in accordance with IPSAS-accrual basis and IPSAS-Cash basis of accounting respectively.

317. Quarterly interim financial reports will be submitted to IFAD within 30 days after each quarter end and unaudited financial statements within four months of end of each financial year. In addition, Recipients/Borrowers shall certify semi-annual reports on the use of funds by source of funding, category, components, sub-components and activities in the required GCF budget format will be submitted within 30 days of period end. All implementing partners will be required to prepare the financial reports with standard templates and submit them to the executing agency's PCU. PCUs will prepare the consolidated financial statements and submit them to IFAD.

318. **Internal Controls.** Executing agencies will have the oversight responsibility over IPs. PCUs and all IPs will have adequate internal control arrangements in place to ensure that funds are used for intended purposes. Rwanda (RDDP-II) will be internally audited by an internal Auditor hired under the SPIU and will report to the PSC; Tanzania (C-SDTP) will be internally audited quarterly by the independent Internal Audit (IA) Unit under the aegis of the Internal Auditor General, which is established at MLF. Uganda (ReLIV) will be internally audited quarterly by the independent Internal Audit (IA) Unit under the Ministry of Finance, Planning and Economic Development. Kenya (INReMP) will be internally audited quarterly by the independent Internal Audit (IA) Unit established under the Ministry of Agriculture and Livestock Development.

319. **External Audit.** Financial statements of Rwanda (RDDP-II), Tanzania (C-SDTP), Kenya (INReMP) and Uganda (ReLIV) will be audited annually by the Auditors General in each country as per the International Auditing Standards, and the audit reports will be submitted to IFAD within six months of the Borrower's fiscal year-end. The auditor scope of work will provide a specific opinion on GCF funds.

G.4. Disclosure of funding proposal

☐ **No confidential information:** The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☒ **With confidential information:** The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided confidentially. Accordingly, the accredited entity is providing the Secretariat with the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential info.

ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S documents corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☒ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☒ Others (Indigenous Peoples Planning Framework, IPPF)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☒ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☐ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☐ Annex 16 Map(s) indicating the location of proposed interventions
- ☒ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☐ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☒ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First level AML/CFT (KYC) assessment
- ☒ Annex 21 Operations manual (Operations and maintenance) - OUTLINE
- ☒ Annex 22 Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)⁹⁶
- ☐ Annex X Other references

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⁹⁶ Annex 22 is mandatory for mitigation and cross-cutting projects.

List of main acronyms

AE	Accredited Entity	MLF	Ministry of Livestock and Fisheries
AFOLU	Agriculture, forestry, and other land use	MoALD	Ministry of Agriculture and Livestock Development
AI	Artificial Insemination	MoLG	Ministry of Local Government
AMA	Accreditation Master Agreement	MoW	Ministry of Water
ARCAFIM	Africa Rural Climate Adaptation Finance Mechanism	MRV	Measurement, Reporting and Verification
AU-IBAR	African Union Inter African Bureau for Animal Resources	MWE	Ministry of Water and Environment
AWPB	Annual Work Plan and Budget	M&E	Monitoring and evaluation
BAU	Business-as-Usual	NAGRC&DB	National Animal Genetic Resources Centre and Data Bank
BDS	Business development services	NAP	National Adaptation Plan
BUR	Biennial update reports	NARO	National Research Organization
CC	Climate change	NC	National Communication
COSOP	Country Strategic Opportunity Programme	NDA	National Designated Authority
CSA	Climate Smart Agriculture	NDC	Nationally Determined Contributions
C-SDTP	Climate Smart Smallholder Dairy	NGO	Non-governmental organization
DDA	Dairy Development Authority	PCU	Programme Coordination Unit
EAC	East African Community	PFS	Pastoral field school
EE	Executing Entity	PSC	Programme Steering Committee
EFA	Economic and financial analysis	RAB	Rwanda Agriculture Board
ERR	Economic rate of return	RCVD	Rwanda Council of Veterinary Doctors
ENPV	Economic net present value	RCP	Representative Concentration Pathways
ESADA	East and Southern Africa Dairy Association	RDDP-2	Rwanda Dairy Development Project, ph.
ESMF	Environmental and Social Management	RE	Renewable Energy
FAA	Funded Activity Agreement	REMA	Rwanda Environment Management
FAO	Food and Agriculture Organization of the United Nations	RNDP	Rwanda National Dairy Platform
GALS+	Enhanced Gender Action Learning System	RSC	Regional Steering Committee
GLEAM-i	Global Livestock Environmental Assessment Model – interactive	SDG	Sustainable Development Goal
GCF	Green Climate Fund	SDLD	State Department for Livestock
GDFF	Green Dairy Financing Facility	SME	Small and medium sized enterprise
GHG	Greenhouse gases	SPIU	Single Programme Implementation Unit
GMP	Global Methane Pledge	SSTC	South-South and Triangular Cooperation
IFAD	International Fund for Agriculture	TALIRI	Tanzania Livestock Research Institute
IFI	International financing institution	TDB	Tanzania Dairy Board
ILRI	International Livestock Research Institute	ToC	Theory of Change
INReMP	Integrated Natural Resource Management	UMMB	Urea-molasses multi-nutrient blocks
KDB	Kenya Dairy Board	UNFCCC	United Nations Framework Convention on Climate Change
KeLCoP	Kenya Livestock Commercialization Project	UNMA	Uganda National Meteorological Agency
L-FFS	Livestock farmer field school	UR-CAVM	College of Agriculture, Animal Science and Veterinary Medicine
LSIPT	Livestock Sector Investment Toolkit		
MAAIF	Ministry of Agriculture, Animal Industry and Fisheries		
MCC	Milk collection center		
MCP	Milk collection point		
MINAGRI	Ministry of Agriculture and Animal Resources		

No-objection letter issued by the national designated authority(ies) or focal point(s)



**REPUBLIC OF KENYA
THE NATIONAL TREASURY AND ECONOMIC PLANNING**

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Ref: TNT/CONF/36/021/C/ PART H (9)

Date: September 18, 2024

Ms. Mafalda Duarte
Executive Director
Green Climate Fund
175 Art Centre-daero, Yeonsu-ngu
Incheon City, Republic of Korea

Dear 

Re: Funding proposal for the GCF by International Fund for Agricultural Development (IFAD) regarding “Dairy Interventions for Mitigation and Adaptation (DaiMA)”

We refer to the programme titled *Dairy Interventions for Mitigation and Adaptation (DaiMA)* in the Republic of Kenya as included in the funding proposal submitted by International Fund for Agricultural Development (IFAD) to us on 13 September 2024.

The undersigned is the duly authorized representative of The National Treasury, the National Designated Authority of the Republic of Kenya.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of the Republic of Kenya has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of the Republic of Kenya;

(c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Yours

A handwritten signature in red ink, appearing to read 'Sincerely', followed by a circular stamp or seal that is partially obscured by the signature.

DR. CHRIS K. KIPTOO, CBS
PRINCIPAL SECRETARY/THE NATIONAL TREASURY

Copy to: **Ms. Mariatu Kamara**
Representative and County Director, Kenya
East and Southern Africa Division
NAIROBI



REPUBLIC OF KENYA
THE NATIONAL TREASURY AND ECONOMIC PLANNING

Telegraphic Address: 22921
FAX NO. 310833
Telephone: 2252299

THE NATIONAL TREASURY
P O BOX 30007 – 00100
NAIROBI

When Replying Please Quote

Ref: TNT/CONF/36/021/C/ PART H (9)

Date: September 18, 2024

Ms. Mafalda Duarte
Executive Director
Green Climate Fund
175 Art Centre-daero, Yeonsu-ngu
Incheon City, Republic of Korea

Dear *Manatu*

Re: Funding proposal for the GCF by International Fund for Agricultural Development (IFAD) regarding "Dairy Interventions for Mitigation and Adaptation (DaiMA)"

We refer to the programme titled *Dairy Interventions for Mitigation and Adaptation (DaiMA)* in the Republic of Kenya as included in the funding proposal submitted by International Fund for Agricultural Development (IFAD) to us on 13 September 2024.

The undersigned is the duly authorized representative of The National Treasury, the National Designated Authority of the Republic of Kenya.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of the Republic of Kenya has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of the Republic of Kenya;

(c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Yours

DR. CHRIS K. KIPTOO, CBS
PRINCIPAL SECRETARY/THE NATIONAL TREASURY

Copy to: **Ms. Mariatu Kamara**
Representative and County Director, Kenya
East and Southern Africa Division
NAIROBI

A red ink signature, likely of Dr. Chris K. Kiptoo, is written over a circular stamp. The signature is stylized and appears to be a combination of initials and a surname.



Republic of Rwanda
Ministry of Environment

**RWANDA ENVIRONMENT
MANAGEMENT AUTHORITY
(REMA)**



Kigali, on 09 OCT 2024
N° 2095/DG/2024

**Executive Director
Green Climate Fund**

Dear Madam,

**Re: Funding proposal for the GCF by Rwanda Environment Management Authority regarding
DalMA - Dairy Interventions for Mitigation and Adaptation (ex-PADNET)**

Referring to the programme titled DalMA - Dairy Interventions for Mitigation and Adaptation (ex-PADNET) in Rwanda as included in the funding proposal submitted by International Fund for Agricultural Development to us on 3 October 2024. The undersigned is the duly authorized representative of Rwanda Environment Management Authority the National Designated Authority of Rwanda.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Rwanda has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Rwanda;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed. We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Sincerely,

**Juliet KABERA
Director General**



Cc:

Honourable Minister of Environment
Country Director, International Fund of Agricultural Development (IFAD) Rwanda
KIGALI

THE UNITED REPUBLIC OF TANZANIA
VICE PRESIDENT'S OFFICE

Telegrams: "MAKAMU"
Telephone: +255 026 2352038
Fax: +255 026 2963150
Email: ps@vpo.go.tz
In reply please quote:



Government City- Mtumba,
Vice President's Office Street,
P. O. Box 2502,
DODOMA.

Ref. No. BD.38/90/01F/34

5th September, 2024

To: The Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by International Fund for Agricultural Development (IFAD) regarding DalMA - Dairy Interventions for Mitigation and Adaptation (ex-PADNET)

2. We refer to the programme titled DalMA - Dairy Interventions for Mitigation and Adaptation (ex-PADNET) in the United Republic of Tanzania as included in the funding proposal submitted by IFAD to us on 29 August 2024. The undersigned is the duly authorized representative of Vice President's Office, the National Designated Authority of United Republic of Tanzania.

3. Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

4. By communicating our no-objection, it is implied that:

- (a) The government of United Republic of Tanzania has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of United Republic of Tanzania;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

5. We confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed. We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

6. We acknowledge that this letter will be made publicly available on the GCF website.

Thank you for the continued cooperation.

A handwritten signature in blue ink, appearing to read 'Cyprian J. Luhemeja'.

Eng. Cyprian J. Luhemeja

PERMANENT SECRETARY

Telephone: 256 41 4341305/230487
Fax : 256 41 4233524
Email : finance@finance.go.ug
Website : www.finance.go.ug
Plot No. 2-8 Apollo Kaggwa Road
In any correspondence on
This subject please quote No DARC 79/251/03



Ministry of Finance, Planning &
Economic Development,
P.O Box 8147
Kampala, Uganda

23rd September, 2024

Ms. Mafalda Duarte
Executive Director,
Secretariat of the Green Climate Fund,
175 Art Center-daero,
Yeonsu-gu, Incheon 22004,
REPUBLIC OF KOREA.

**LETTER OF NO OBJECTION TO THE GREEN CLIMATE FUND FOR
A PROJECT TITLED "DAIRY INTERVENTIONS FOR MITIGATION
AND ADAPTATION PROJECT" (DAIMA)**

I refer to the above Project, which has been prepared and submitted to you by the International Fund for Agriculture Development for funding by the Green Climate Fund.

The undersigned is the duly authorized representative of the Ministry of Finance, Planning and Economic Development, the National Designated Authority/focal point of Uganda.

Pursuant to GCF decisions B.08/10 and B.37/22, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no objection, it is implied that:

- a) The Government of the Republic of Uganda has no objection to the Project as included in the funding proposal;
- b) The project as included in the Funding Proposal is in conformity with the National Priorities, Strategies and Plans of Uganda;

Mission

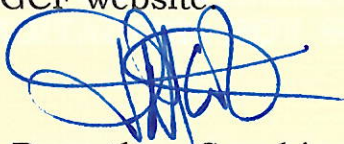
"To formulate sound economic policies, maximize revenue mobilization, ensure efficient allocation and accountability for public resources so as to achieve the most rapid and sustainable economic growth and development"

- c) In accordance with the GCF's Environmental and Social Safeguards, the project as included in the funding proposal is in conformity with relevant laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website



Ramathan Ggoobi

**PERMANENT SECRETARY/ SECRETARY TO THE TREASURY /
NATIONAL DESIGNATED AUTHORITY FOCAL POINT FOR THE
GREEN CLIMATE FUND.**

- Copy to:
- Hon. Minister of Finance, Planning and Economic Development
 - Permanent Secretary, Ministry of Agriculture, Animal Industry and Fisheries
 - Country Representative, International Fund for Agriculture Development

Mission

"To formulate sound economic policies, maximize revenue mobilization, ensure efficient allocation and accountability for public resources so as to achieve the most rapid and sustainable economic growth and development"

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Dairy Interventions for Mitigation and Adaptation (DaIMA)
Existence of subproject(s) to be identified after GCF Board approval	No
Sector (public or private)	Public
Accredited entity	International Fund for Agricultural Development (IFAD)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Kenya, Rwanda, Tanzania and Uganda
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Monday, May 12, 2025
Language(s) of disclosure	English
Explanation on language	English is an official language for all four targeted countries.
Link to disclosure	https://www.ifad.org/documents/d/new-ifad.org/daima-esmf
Other link(s)	N/A
Remarks	Requirements for developing a Category B ESIA are contained in Annex 6 “Environmental and Social Management Framework (ESMF)” of the Funding Proposal.
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	Monday, May 12, 2025
Language(s) of disclosure	English
Explanation on language	English is an official language for all four targeted countries.
Link to disclosure	https://www.ifad.org/documents/d/new-ifad.org/daima-esmf
Other link(s)	N/A
Remarks	An ESMP consistent with the requirements for a Category B project is contained in Annex 6 “Environmental and Social Management Framework (ESMF)” of the Funding Proposal.
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	

Description of report	Indigenous Peoples Planning Framework (IPPF); Stakeholder Engagement Plan (SEP)
Date of disclosure on accredited entity's website	Monday, May 12, 2025
Language(s) of disclosure	English
Explanation on language	English is an official language for all four targeted countries.
Link to disclosure	- IPPF: https://www.ifad.org/documents/d/new-ifad.org/daima-ippf - SEP: https://www.ifad.org/documents/d/new-ifad.org/daima-stakeholder-engagement-plan-annex7
Other link(s)	N/A
Remarks	An Indigenous Peoples Planning Framework (IPPF) and a Stakeholder Engagement Plan (SEP) are included in Annexes 6 and 7 respectively.
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Wednesday, May 28, 2025
Place	Kenya: The National Treasury, Harambee Avenue, Treasury Building, P.O Box 30007-00100, Nairobi. - ESMF: https://www.treasury.go.ke/wp-content/uploads/2025/05/DaIMA-esmf.pdf - IPPF: https://www.treasury.go.ke/wp-content/uploads/2025/05/DaIMA-ippf.pdf Rwanda: RAB/SPIU, Blue Star House, 2° piano, Kacyiru, Gasabo District Kigali, Rwanda. - ESS documents: https://www.rab.gov.rw/index.php?eID=dumpFile&t=f&f=130379&token=21e954ef2b1c371d1c15d35f18e806394a3e20d6 Tanzania Mainland: Ministry of Livestock & Fisheries, Government City - Mtumba, Ulinzi Street, P.O.Box 2870, 40487 Dodoma, Tanzania. Zanzibar: Ministry of Agriculture, Maruhubi, P.O.Box 159, 70471, Mjini Maghreb, Zanzibar. - ESMF: https://www.mifugouvuvu.go.tz/uploads/text-editor/files/DaIMA-esmf_1748345012.pdf - IPPF: https://www.mifugouvuvu.go.tz/uploads/text-editor/files/Annex%206b%20IPPF%20v2_1748345128.pdf

	Uganda: Ministry of Finance, Planning and Economic Development, Plot 2/8 Apollo Kaggwa Road, P.O.Box 8147 Shimon Rd 1, Kampala, Uganda. • ESMF: https://www.finance.go.ug/publications/dairy-interventions-mitigation-and-adaptation-esmf • IPPF: https://www.finance.go.ug/publications/dairy-interventions-mitigation-and-adaptation-ippf
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	N/A
Date of GCF's Board meeting	Monday, June 30, 2025

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP269

Proposal name:	Dairy Interventions for Mitigation and Adaptation (DaiMA)
Accredited entity:	International Fund for Agricultural Development (IFAD)
Countries:	Kenya, Rwanda, Uganda and United Republic of Tanzania
Project/programme size:	Large

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The first GCF dedicated livestock programme, developed in response to the climate risks affecting the sector, particularly drought and increasing temperatures and reduced availability of feed and fodder, while reducing greenhouse gas emissions, especially methane.	Multi-country programme, which might lead to unsynchronized implementation across the four countries. Mitigating measures have been included to decouple implementation in the case of delays.
Sectoral programme designed to build resilience across the dairy value chains across the East Africa region, through implementation by public and private actors, including local financial institutions.	

2. The Board may wish to consider approving this funding proposal in accordance with the terms listed in the term sheet agreed between the Secretariat and the accredited entity (AE) and, if considered appropriate, subject to the conditions set out in annex II to document GCF/B.42/02.

II. Summary of the Secretariat's assessment

2.1 Programme background

3. The dairy sector in East Africa, including Kenya, Rwanda, Uganda and the United Republic of Tanzania, faces significant challenges due to climate change. Dairy farming is a major contributor to greenhouse gas (GHG) emissions in each of these countries and the effects of climate change threaten its long-term sustainability. Inefficiencies in dairy farming lead to high GHG emission intensities per kilogram of milk. Additionally, many dairy farmers struggle with limited market access, challenges in sourcing local feed and fodder, inadequate extension and financial services, and difficulties in risk management. The sector is further impacted by low livestock genetic potential, herd management issues, animal diseases, insufficient extension support, limited awareness of climate-resilient and low-carbon practices and technologies, rangeland degradation and gender inequalities.

4. To address the above-mentioned challenges, this programme, “DaIMA – Dairy Interventions for Mitigation and Adaptation”, will have a comprehensive and holistic approach to improve all aspects of the dairy sector, from improving policy, institutional and service delivery environments to building the capacity of smallholder farmers, including women, youth and marginalized groups, by strengthening climate resilience, reducing emissions across the dairy value chain and improving the livelihoods of dairy farmers in East Africa. It also involves strategies to improve access to finance and innovative technologies that offer both mitigation and adaptation benefits. As a result, dairy farmers and processors will be encouraged to adopt more efficient, climate-resilient practices that enhance incomes and lower production costs.
5. The total budget for the programme is USD 358.26 million. USD 58.20 million is allocated for component 1, USD 234.98 million for component 2 and USD 45.0 million for the Green Dairy Financing Facility (GDFF) (component 3), with the remaining USD 13.36 million allocated for programme management and USD 6.72 million for monitoring and evaluation (M&E).
6. The programme will impact 2.5 million direct and 15.4 million indirect beneficiaries across all four countries. The programme aims to reduce the milk emission intensities by 29 per cent (of which 28 per cent is from Kenya, 50 per cent from Rwanda, 33 per cent from Uganda and 29 per cent from the United Republic of Tanzania) over a 20-year lifespan. The programme is also expected to reduce total GHG emissions on the targeted dairy farms, while significantly increasing milk production by 34 per cent (of which 31 per cent is from Kenya, 73 per cent from Rwanda, 30 per cent from Uganda and 36 per cent from the United Republic of Tanzania).
7. Furthermore, it is projected to reduce total GHG emissions from milk production by 4.01 per cent. Specifically, emissions are expected to decline by 2.02 per cent (61,967 tonnes of carbon dioxide equivalent (t CO₂ eq)) in Kenya, 9.56 per cent (56,507 t CO₂ eq) in Rwanda, 5.22 per cent (82,372 t CO₂ eq) in Uganda and 4.69 per cent (65,718 t CO₂ eq) in the United Republic of Tanzania. Additionally, methane emissions from enteric fermentation are anticipated to drop by 4.6 per cent, while methane from manure management is expected to decrease by 5.93 per cent. The programme will also contribute to estimated emission savings of 54,435 t CO₂ eq per year through improved energy generation and access.
8. Also, the programme contributes to the Food systems (Target 4), Ecosystems (Target 5), Clean Energy (Target 7), Local private sector ventures (Target 10) and National and regional financial institutions (Target 11) objectives of the GCF Updated Strategic Plan (USP-2).
9. The DaIMA Programme strengthens climate resilience in East Africa’s dairy sector by enabling 2.5 million smallholder farmers to adopt adaptive practices. It enhances water access, pasture management, animal health, and disease control, leading to increased milk productivity and improved food and nutrition security, while reducing vulnerability to droughts and climate hazards (Target 4). The programme will also restore and sustainably manage over 178,000 hectares of degraded rangelands (Target 5) and strengthen dairy value chains by upgrading milk collection and processing facilities powered by renewable energy (Target 7). Under Component 3.1, the Green Dairy Financing Facility (GDFF) will provide two loan products to support the adoption of climate-smart technologies and sustainable dairy practices. Through this facility, DaIMA mobilizes concessional loans and technical assistance to promote low-emission investments and institutional capacity building in the dairy sector (Targets 10 and 11).
10. The programme will have three components as set out below.

2.2 Component-by-component analysis

Component 1: Strengthened national capacities and enhanced enabling environment for low emissions climate resilient dairy sectors (total cost: USD 58.2 million; GCF cost: USD 17.28 million)

11. Component 1 will create a supportive environment for transforming East Africa's dairy sector, driving higher productivity, enhancing climate resilience and reducing emission intensity. It will focus on four key outputs: establishing a supportive policy and regulatory framework, including enhanced consumer awareness and nutritional balance (output 1.1); enhancing public service delivery in the dairy sector, particularly through specialized livestock extension and veterinary services (output 1.2); strengthening the capacity to monitor GHG emissions and climate vulnerabilities (output 1.3); and promoting regional cooperation and improving knowledge management (output 1.4).
12. Output 1.1 (Climate-responsive policy and regulatory environment established) aims to establish a climate-responsive policy and regulatory environment to support the sustainable transformation of the dairy sector in East Africa. Activity 1.1.1 focuses on reviewing and formulating dairy sector policies, strategies and regulations to enhance climate change adaptation and mitigation. This includes aligning national policies with nationally determined contributions (NDCs) and national adaptation plans, improving trade and investment regulations and fostering private sector engagement. Sub-activities 1.1.1.1 and 1.1.1.2 will support national-level policy revisions and regional policy dialogue to harmonize dairy sector regulations and enhance technology transfer. Sub-activities 1.1.1.3 and 1.1.1.4 will strengthen dairy value chain governance and local stakeholder participation. Activity 1.1.2 will raise public awareness on the nutritional benefits of milk and food safety, promoting increased consumption of high-quality dairy products through educational campaigns and school milk programmes. Sub-activity 1.1.2.1 will focus on awareness initiatives for climate-vulnerable populations, while sub-activity 1.1.2.2 will enhance dairy inspection systems to ensure quality standards and reduce milk waste. Activity 1.1.3 will work towards improving land tenure security for sustainable dairy farming, identifying tenure risks and supporting land-use planning to enhance dairy productivity. Sub-activities 1.1.3.1 and 1.1.3.2 will provide training, public awareness campaigns and capacity-building efforts to secure land rights, particularly for women and youth. These activities collectively aim to create a more climate-resilient, inclusive and sustainable dairy sector in East Africa.
13. Output 1.2 (Enhancing public service delivery in the dairy sector, particularly through specialized livestock extension and veterinary services) aims to enhance service delivery in the dairy sector through improved veterinary and extension services. Activity 1.2.1 focuses on strengthening veterinary services to improve animal health, boost milk yields and reduce losses caused by climate-induced diseases. This will involve expanding veterinary outreach, investing in diagnostic laboratory capacities and equipping veterinarians, paraveterinarians and community animal health workers with better tools and training. Sub-activity 1.2.1.1 will support the deployment of veterinarians and paraveterinary professionals, particularly young and female practitioners, while sub-activity 1.2.1.2 will enhance the diagnostic capacity of veterinary laboratories and improve feed quality testing. Activity 1.2.2 focuses on strengthening extension services to support a low-carbon and climate-resilient dairy sector. This will involve innovative advisory approaches such as digital tools, Farmer Field Schools and participatory extension models to enhance farmers' knowledge of climate-smart dairy practices. Sub-activity 1.2.2.1 will develop training modules on climate mitigation and adaptation, including topics like fodder production, biogas management and animal welfare. Sub-activity 1.2.2.2 will train master trainers and facilitators to scale up climate-smart practices, with a focus on women and youth participation. Sub-activity 1.2.2.3 will promote innovative extension tools, including digital platforms, text messaging service based advisories, social media and mobile apps like iCow in the United Republic of Tanzania and Kenya Agricultural Observatory Platform. These initiatives will increase farmers' access to knowledge, improve productivity and enhance climate resilience in the dairy sector.
14. Output 1.3 (Strengthening the capacity to monitor GHG emissions and climate vulnerabilities) will enhance GHG emission measurement, reporting and verification (MRV) capacities and improve climate vulnerability monitoring in the dairy sector. Activity 1.3.1

focuses on developing national MRV systems to track dairy emissions accurately and support climate commitments. Activity 1.3.2 will improve GHG emission measurement by collecting on-farm data, refining emission factors and using advanced modelling tools. Activity 1.3.3 will conduct climate risk and vulnerability assessments, strengthening early warning systems and national feed balance mechanisms. Activity 1.3.4 will facilitate low-carbon certification and enable dairy stakeholders to access carbon finance by developing verification standards and certification systems. These efforts will enhance climate reporting, attract climate finance and strengthen dairy sector resilience.

15. Output 1.4 (Promoting regional cooperation and improving knowledge management) will aim to enhance knowledge management and regional cooperation to promote the replication of climate-resilient and low-carbon dairy practices across East Africa. Activity 1.4.1 will support regional centres of excellence and national research institutions to build capacity in GHG measurement, livestock productivity and climate adaptation. This includes developing a regional database to document best practices and emissions data. Activity 1.4.2 will establish regional cooperation platforms for knowledge exchange, involving public and private stakeholders, research institutions and international organizations. These platforms will facilitate policy discussions, scientific innovations and best practice sharing through workshops, conferences and study tours. Additionally, participation in global climate policy events and South-South and Triangular Cooperation will strengthen regional and international collaboration on dairy sector sustainability and climate finance solutions.

Component 2: Enabling low-emission and climate-resilient dairy value chain development (total cost: USD 234 million; GCF cost: USD 100.7 million)

16. Component 2 focuses on enhancing incomes and productivity for smallholder dairy farmers, cooperatives and small and medium-sized enterprises (SMEs) by promoting sustainable, climate-resilient and low-emission growth pathways. The three associated outputs outlined in the following paragraphs will also contribute to gender equality, improved food security and nutrition for rural and urban consumers and greater empowerment and inclusion of women and marginalized groups.

17. Output 2.1 (Smallholders' and cooperative capacity development strengthened) aims to improve market access, strengthen dairy cooperatives and promote sustainable practices for smallholder dairy farmers. Activity 2.1.1 will enhance the governance, capacity and service delivery of dairy cooperatives, ensuring fair prices, reduced milk losses and improved quality standards. Activity 2.1.2 will build productive alliances along the dairy value chain, supporting business management, financial literacy and digitalization for better efficiency, traceability and access to credit. Activity 2.1.3 will promote renewable energy and waste management solutions by introducing solar-powered milk cooling, energy-efficient equipment and sustainable waste recycling. These initiatives will boost farmer incomes, enhance climate resilience and reduce emissions in the dairy sector.

18. Output 2.2 (Breeding, animal health, feed and fodder, manure management improved) will reduce farm-level GHG emissions and enhance productivity through climate-smart breeding, improved animal health, better feeding practices and sustainable manure management. Activity 2.2.1 will promote climate-resilient dairy breeds by scaling up artificial insemination and breeding programmes to enhance milk yields while reducing emissions. Activity 2.2.2 focuses on disease prevention, improved herd management and digital tools for animal health monitoring. Activity 2.2.3 will support forage production, water harvesting and certified seed multiplication to improve feed availability. Activity 2.2.4 promotes climate-smart feeding strategies to lower enteric methane emissions, while activity 2.2.5 will enhance manure management and nutrient recycling for soil health and emission reduction. Lastly, activity 2.2.6 will scale up biogas digesters to convert manure into clean energy, reducing deforestation and providing sustainable fuel alternatives. These initiatives will boost dairy sector resilience, increase milk production and contribute to climate change mitigation.

19. Output 2.3 (Pasture and grasslands management improved) aims to improve pasture and grassland management to enhance climate resilience, biodiversity and carbon sequestration while reducing GHG emissions from dairy production. Activity 2.3.1 will strengthen rangeland governance by supporting community-led pasture management planning, securing land tenure rights and resolving resource conflicts. Activity 2.3.2 will focus on restoring degraded rangelands through rotational grazing, integration of leguminous trees for carbon sequestration, and soil and water conservation practices. Additionally, the programme will increase water availability for dairy cattle by investing in water harvesting systems, such as micro-dams and rock catchments, with a view to ensuring sustainable access to water in extensive dairy systems. These interventions will support sustainable livestock production, reduce land degradation and enhance dairy sector resilience.

Component 3: Green Dairy Financing Facility (total cost: USD 45 million; GCF cost: USD 17.5 million loan and USD 10 million grant)

20. Component 3 will establish GDFF in the four target countries to enhance access to finance for dairy value chain actors, including farmers, cooperatives, SMEs and technology providers. By addressing financial and technological barriers, GDFF will support the adoption of climate-smart technologies and sustainable practices, facilitating the transition to low-carbon and climate-resilient dairy value chains. Businesses identified under component 2 will be encouraged to apply for GDFF financing to further strengthen the sector's sustainability. This component is structured in two parts – output 3.1 (Effective green financing for dairy value chains) and output 3.2 (Capacity-building for green dairy investments).

21. Under output 3.1, GDFF will offer two loan products to support the adoption of climate-smart technologies and sustainable dairy practices. The Green Dairy Investment Partnership Loan is a long-term financing option for larger, well-established dairy businesses with annual turnovers above USD 800,000, offering loans between USD 250,000 and USD 550,000 with tenors of five to nine years. These loans will help businesses to expand smallholder networks, acquire productive cows, solarize equipment and improve milk transport. The Green Dairy Loan Product targets smaller SMEs and dairy cooperatives, providing short-term working capital loans of up to USD 250,000 (average USD 50,000) with a three-year tenure. Both loan types require businesses to demonstrate clear climate-smart investment ambitions and strong smallholder partnerships. Additionally, technical assistance will be provided to help businesses to prepare investment plans and optimize their operations post-financing.

22. Under output 3.2, GDFF will use a USD 10 million technical assistance facility to enhance the capacity of banks and dairy sector businesses in implementing climate-smart projects. The technical assistance facility will address financial and non-financial barriers, ensuring effective green financing. Key activities include (i) capacity development for banks – wholesale and retail banks will receive training and support to integrate green dairy finance products, improve risk management and enhance reporting on sustainability impacts; (ii) support for SMEs – dairy businesses will receive tailored assistance in business planning, investment readiness, financial management and climate adaptation strategies to improve loan access and investment success; and (iii) regional coordination and knowledge-sharing – best practices, lessons learned and technical expertise will be shared among stakeholders to strengthen green finance in the dairy sector. This initiative aims to scale up climate-smart investments, improve operational efficiency and support sustainable growth in the dairy value chain.

Monitoring and evaluation components (total cost: USD 6.7 million; GCF cost: USD 1.4 million)

23. Each executing entity (EE) will manage and coordinate the overall M&E system, leveraging existing project frameworks financed by the International Fund for Agricultural Development (IFAD). Responsibilities include allocating resources, aggregating and verifying data, producing reports and organizing learning events to enhance sector-wide knowledge. The Regional Steering Committee, supported by rotating national EEs, will ensure harmonized

reporting and methodological guidance. International Livestock Research Institute (ILRI) will provide additional M&E support, such as templates and methodologies, while data capture and reporting will be handled by country-level project management units (PMUs). IFAD will oversee the process through annual supervision missions, including mid-term and final evaluations.

24. For component 3, a senior coordinator will be recruited at IFAD Kenya Regional Office to oversee GDFF coordination, M&E, reporting and knowledge management. During the start-up phase, financial and M&E specialists will assist the DaIMA PMU in establishing a reporting and evaluation system for GDFF, aligned with policy of IFAD, the DaIMA PMU and wholesale banks. GDFF monitoring and reporting will be integrated into the overall DaIMA regional M&E framework.

Project management (total cost: USD 13.35 million; GCF cost: USD 3.049 million)

25. IFAD will serve as the AE for the programme, overseeing its overall management. This includes programme appraisal and administrative, fiduciary and technical supervision to ensure the achievement of the expected outputs and outcomes, as well as programme closure and final evaluation. IFAD will fulfil these responsibilities in line with the accreditation master agreement with GCF, coordinating efforts through its headquarters in Rome, the East and Southern Africa Regional Office in Nairobi and country offices in Kenya, Rwanda, Uganda and the United Republic of Tanzania. IFAD will also set up a dedicated programme task force consisting of IFAD experts from within its headquarters and regional and country offices.

26. Additionally, a Regional Steering Committee will oversee regional alignment and value addition, while governments in each country will implement the programme through designated EEs. National programme steering committees will provide oversight, and programme implementation units will handle coordination. International and national entities, including ministries of agriculture and specialized organizations like ILRI, will offer technical support to national teams.

III. Assessment of performance against investment criteria

3.1 Impact potential

Scale: High

27. The programme will impact 2.5 million direct and 15.4 million indirect beneficiaries across all four countries. As detailed in paragraph 6 above, the programme aims to reduce the milk emission intensities by 29 per cent over a 20-year lifespan and is expected to reduce total GHG emissions on the targeted dairy farms while increasing milk production by 34 per cent.

28. The programme is projected to reduce total GHG emissions from milk production by 4.01 per cent (see para. 7 above). Additionally, methane emissions from enteric fermentation are anticipated to drop by 4.6 per cent, while methane from manure management is expected to decrease by 5.93 per cent. The programme will also contribute to estimated emission savings of 54,435 t CO₂ eq per year through improved energy generation and access.

29. The programme will directly contribute to the results of the updated Strategic Plan for the GCF 2024–2027 in the areas of food systems, ecosystems, clean energy, local private sector early-stage ventures and micro, small and medium-sized enterprises innovation by supporting resilient market chains and GDFF.

3.2 Paradigm shift potential

Scale: High

30. The programme takes a holistic approach to reshaping the dairy sector in East Africa, aiming to lower GHG emissions while enhancing the well-being of dairy producers. By

strengthening policy frameworks, institutional capacity and service delivery, alongside expanding financial access and climate-smart solutions for smallholder farmers, it seeks to drive a comprehensive transformation across the dairy value chain. Its strategy integrates private sector engagement and financial mechanisms to promote sustainable, low-carbon growth that extends well beyond the programme's initial investment. The scalability of its interventions, coupled with knowledge-sharing, improved regulatory frameworks and the promotion of climate-resilient practices, ensures a long-term impact on the dairy industries.

31. The programme enhances access to finance and alternative technologies, creating opportunities for local private sector actors, including dairy cooperatives and milk processors, to overcome financial and technological limitations that have historically hindered their growth and sustainability. By addressing these key constraints, the programme fosters innovation, efficiency and resilience within the dairy value chain. This approach enhances productivity and profitability and also serves as a catalyst for systemic change, promoting the adoption of sustainable, climate-smart practices that contribute to a low-emission, climate resilient livestock sector. The programme supports the adoption of more resilient animal breeds, improved livestock husbandry practices and windrow composting techniques to reduce methane emissions from enteric fermentation in livestock. It also promotes more efficient manure management systems, which help to lower GHG emissions while improving soil health and nutrient recycling.

32. Additionally, it lays the groundwork for long-term sectoral transformation, ensuring that the benefits extend beyond individual businesses to influence policy frameworks, market structures and broader economic development. Through these targeted interventions, the programme demonstrates a scalable and far-reaching impact, positioning the dairy sector as a model for sustainable agricultural development in the region.

3.3 Sustainable development potential

Scale: High

33. DaIMA contributes to several Sustainable Development Goals (SDGs) by enhancing financial access, increasing productivity, improving food security and nutrition, fostering innovation in climate-smart technologies, promoting sustainable dairy production and supporting biodiversity conservation and rangeland restoration.

34. The programme contributes to SDG 3 (Good health and well-being) by promoting safe, high-quality dairy consumption and raising nutrition awareness. Strengthening veterinary services supports SDG 12 (Responsible consumption and production) by improving animal health, increasing productivity and reducing emissions. Additionally, by ensuring better food safety and access to nutritious dairy products, it advances SDG 2 (Zero hunger), fostering food security and improved nutrition in both rural and urban communities.

35. It also supports environmental sustainability by enhancing soil conservation, biodiversity and ecosystem health. It aligns with SDG 15 (Life on land) through rangeland restoration and biodiversity conservation, combating climate change and maintaining ecosystem balance. By promoting SDG 12 (Responsible consumption and production), it integrates circular economy principles, improves manure management for biogas production, and enhances soil health through nutrient recycling. It also drives economic growth by creating jobs, reducing poverty (SDG 1) and enhancing productivity through climate-smart technologies. It strengthens dairy farmers' incomes and financial access, fostering a resilient, low-carbon dairy sector while promoting innovation and infrastructure development (SDG 9).

3.4 Needs of the recipient

Scale: Medium to high

36. The sustainability of dairy farming in the region is increasingly at risk owing to rising temperatures, erratic weather patterns and resource constraints, all of which negatively affect milk production, feed availability and animal health. Smallholder farmers, who constitute the majority of dairy producers, are particularly vulnerable as they operate on a small scale with limited resources and infrastructure. Their productivity remains low as a result of insufficient access to quality feed, veterinary services and climate-smart technologies. Additionally, limited training and technical knowledge prevent farmers from adopting more efficient and sustainable dairy farming methods. Financial barriers further hinder investments in improved breeding practices, better manure management and energy-efficient dairy processing technologies. The programme seeks to address these issues by providing technical assistance, financial support and capacity-building initiatives to enable smallholder dairy farmers to transition to climate-resilient and low-emission dairy production. By equipping farmers with the necessary skills, resources and innovative solutions, the programme aims to enhance productivity, reduce GHG emissions and ensure the long-term sustainability of the dairy sector in East Africa.

3.5 Country ownership

Scale: High

37. DaiMA serves as a comprehensive framework to address the impacts of climate change on the dairy sector and the sector's GHG emissions. It aligns with the NDCs, nationally appropriate mitigation actions, national adaptation plans for agriculture and national agriculture policies, contributing to national development plans for climate-smart dairy value chains. By promoting climate-smart agricultural practices and technologies, the programme aims to drive resilient, low-carbon growth in the dairy sector, with a particular focus on methane reduction.

38. The NDCs and other national strategies define each country's approach to reducing GHG emissions and adapting to climate change. The proposed programme aligns with these national strategies by incorporating various measures aimed at enhancing climate resilience in the agriculture and livestock sectors. These measures include the promotion of climate-resilient crops that can withstand extreme weather conditions, the development of climate-resilient livestock breeds that are better adapted to heat stress and disease, and the expansion of crop and livestock insurance schemes to protect farmers against climate-related risks.

39. Overall, the programme will provide technical and policy support to the ministries responsible for the dairy sector in reviewing and formulating national policies, strategies and regulations that are critical for the sustainable transformation of the dairy industry. These efforts will ensure that national policies are fully aligned with NDC commitments, driving a shift towards low-carbon, climate-resilient dairy production that benefits both the environment and rural communities.

3.6 Efficiency and effectiveness

Scale: High

40. The programme's economic and financial analyses assess profitability from both a societal and a participant perspective, using the Livestock Sector Investment and Policy Toolkit to model dairy production across four countries over a 20-year period. Five dairy herd models, reflecting different production systems, were developed per country based on national herd structures. Financial models were also created for milk collection points, chilling centres, processing units (all solar powered) and service providers. The programme's GHG mitigation impact was evaluated using the Global Livestock Environmental Assessment Model.

41. The overall financial analysis of the programme indicates the profitability of the programme's investments for the farmers and cooperatives. It uses a 17.5 per cent discount rate

(reflecting average lending rates in the target countries) based on incremental net cash flows comparing scenarios with and without the programme.

42. At the regional level, the DaIMA programme is economically viable, with an expected net present value of USD 181.20 million and an economic rate of return (ERR) of 25.0 per cent under the baseline scenario. Country-level analyses for the United Republic of Tanzania, Uganda, Rwanda and Kenya also confirm viability. When the economic value of GHG emissions is considered, the expected net present value rises to USD 217.87 million (ERR 27.9 per cent) under a low-carbon price and to USD 254.64 million (ERR 30.3 per cent) under a high carbon price scenario.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

43. The DaIMA programme to be implemented across four countries in East Africa is assessed as category B with a moderate environmental and social risk rating. The programme has been screened against IFAD Social, Environmental and Climate Assessment Procedures and the GCF Environmental and Social Policy. The AE shared an environmental and social management framework and stakeholder engagement plan as part of the funding proposal. Anticipated environmental and social risks are likely to be related to pollution associated with the intensification of dairy production, lack of appropriate waste management and child labour associated with engaging youths in the dairy value chain. The programme also proposes to improve pasture productivity and quality by enhancing the composition of forage varieties, fodder conservation and better pasture management, coupled with low-carbon sources of energy in milk processing.

44. During the review process, information was limited on the resilient seed variety fodder crops to be adopted to eliminate invasive species, specific national legislation and policies for agriculture and livestock for the beneficiary countries and information on envisaged infrastructure works that are likely to include the construction of veterinary laboratories, centres for artificial insemination, water harvesting systems such as micro-dams and cattle sheds. Following the GCF appraisal mission to Rome in March 2025 and various review iterations, the AE provided specifications on the type of infrastructure works to be adopted on the programme and the promotion of locally proven fodder varieties to eliminate the potential of introducing invasive species.

45. **GCF Indigenous Peoples Policy and environmental and social safeguard 7 (Indigenous Peoples).** An Indigenous Peoples planning framework is provided as an annex to the environmental and social management framework. The AE has confirmed that the Indigenous Peoples planning framework will not be confined to one or two pastoralist groups and that it will be applied thoroughly. In line with its roles and functions, the Indigenous Peoples Advisory Group is available to provide advice to the AE and EEs. In line with the GCF Indigenous Peoples Policy, the GCF Indigenous Peoples focal point will be available for assistance at any stage, including before a claim has been made.

46. **Sexual exploitation, abuse and harassment.** The revised GCF Environmental and Social Policy adopted by decision B.BM-2021/18 requires safeguarding from sexual exploitation, abuse and harassment (SEAH) in GCF-financed activities. The AE provided SEAH safeguarding in its submission to this funding proposal. During the operational phases, especially construction related, it will be import for the project to ensure that there is a comprehensive SEAH policy and overall continuum of care with respect to gender-based violence (GBV). The AE has therefore highlighted potential SEAH/GBV risks for all activities, including activity risk assessment in alignment with the GCF SEAH safeguards requirement.

SEAH mitigation measures have been included, including post-mitigation, to ensure that any residual SEAH/GBV impacts are addressed and monitored accordingly. Since the project is targeting four countries in the region with dynamic SEAH/GBV and gender inequalities contexts, it is highly recommended that the AE also take a closer look (during implementation and monitoring) at the differentiated approaches where applicable.

4.2 Gender policy

47. In compliance with the GCF Gender Policy, the AE prepared and submitted a gender assessment together with its gender equality and social inclusion (GESI) action plan. Informed by the IFAD gender analysis approach, the assessment covers key areas required for gender mainstreaming and specific interventions to this project. The gender assessments discuss the targeting criteria and provide sex and gender/social disaggregation targets for its interventions. The GESI action plan will be carried out through an implementation partner and a range of service providers included in the action plan. These may include youth enterprises, non-governmental organizations, academia and researchers. The line ministries will also be key during implementation of the GESI action plan as these already have gender focal points. In addition, a GESI specialist will be appointed for each country's operation. Additional expertise will be procured under GDFF and technical assistance sub-activities to ensure implementation of the gender action plan. The specific technical assistance will include (i) capacity strengthening and technical support provided to partner staff and agents to help them to deliver the gender action plan; (ii) the recruitment of experienced service providers to support outreach, gendered approaches and capacity-building at the community level; and (iii) ongoing GESI technical support during implementation.

48. DaIMA aims for 40 per cent of project beneficiaries to be women, 30 per cent youth and 5 per cent comprising persons with disabilities and Indigenous Peoples. Women, youth and persons with disabilities will be targeted with technological solutions to reduce methane and GHG emissions, and with financial support services to sustain efficient production and marketing processes. The objective of the DaIMA GESI action plan is to ensure that women, youth, people with disabilities and marginalized groups have equal opportunities to participate in, and benefit from, the programme in these communities. Social inclusion considerations for these groups will be achieved by increasing their participation in project activities, economic empowerment opportunities and access to information and skills, as well as market linkages and value addition.

4.3 Risks

4.3.1. Overall project/programme assessment (high risk)

4.3.2. Accredited entity/executing entity capability to execute the current project/programme (low risk)

49. IFAD is an international AE with nearly 50 years of development work in global agriculture and food security. To date, IFAD has provided approximately USD 24 billion in grants and loans for agricultural and rural development projects in 109 developing countries. The project EEs are the finance ministries and development banks in each of the host countries.

4.3.3. Project/programme-specific execution risks (high risk)

50. **Sovereign credit risk.** The GCF loan will be extended to the governments of four host countries in the programme. Proceeds will then be channelled to public development banks and to retail banks or microfinance banks, which will on-lend to small dairy farmers and

cooperatives. GCF credit exposure will be to the Governments of Kenya, Rwanda, Uganda and the United Republic of Tanzania with low sovereign credit rating between B- to B+.

51. **Programme implementation risk.** Within the four target countries, there is limited institutional capacity and skill set in the dairy industry including dairy technicians, social / gender specialists and national dairy boards, etc. which could lead to low adoption or project delays. This is being mitigated by competitive recruitment of qualified project management units in each country with performance-based contracts.

52. **Programme's concessional pricing.** To ensure that full benefit of concessional loan pricing under Component 3 is passed on to end borrowers, the facility agreements will include provisions capping the margin of financial intermediaries. Lending rates to end borrowers will be reduced by the lowered funding cost from this programme compared to the retail banks' own funding cost.

4.3.4. **Compliance risk (medium risk)**

53. The inherent money laundering/terrorist financing (ML/TF) risk associated with certain programme activities, such as the launch of a Green Dairy Financing Facility to provide green loans, is relatively elevated.

54. The AE will oversee the programme through its regional office in Kenya, as well as its country offices in the four target countries. The four target country governments will act as EEs for Components 1 & 2 through their respective ministries and state-owned development financial institutions. The AE's financial management risk assessments of the recipient governments did not flag elevated risks related to ML/TF. The ministries of finance in the target countries will serve as EEs for Component 3 and will disburse local currency loans to local Public Development Banks in each target country to execute green dairy sector financing operations.

55. Anti-money laundering/countering the financing of terrorism (AML/CFT) controls to be implemented by the AE, which is responsible for overall fiduciary management, include training conducted during project preparation and implementation. In addition, the AE assesses counterparty integrity risk taking a risk-based approach, incorporating integrity due diligence measures including daily screening of project counterparties including EEs, payment beneficiaries, and procurement counterparties, to identify ML/TF and other prohibited practices risks. Furthermore, capacity-building training for finance staff will be conducted as an additional safeguard against financial management and fiduciary risks. In addition, the four PDBs are subject to AML/CFT regulations in their respective countries, and the retail banks that the PDBs will on-lend to will be selected based on criteria that includes whether they operate under prudential supervision with solid institutional and financial conditions per central bank standards.

56. Based on the elevated ML/TF risk posed by some of the proposed programme activities and the various controls and mitigation measures to be applied during project preparation and implementation, the overall compliance risk is determined to be medium.

4.3.5. **GCF portfolio concentration risk**

57. In the case of approval, the impact of this proposal within monitoring threshold on the GCF portfolio concentration in terms of results area, AE concentration and single proposal.

4.3.6. **Recommendation**

58. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment	
Overall project/programme	High
Accredited entity/executing entity capability	Low
Project-/programme-specific execution	High
Compliance	Medium
GCF portfolio concentration	Within monitoring threshold

4.4 Fiduciary

59. IFAD serves as the AE and is responsible for overall programme management, including appraisal, fiduciary and technical oversight, supervision and closure, in line with its accreditation master agreement with GCF. IFAD will execute these functions through its headquarters in Rome, regional office in Nairobi and country offices in Kenya, Rwanda, Uganda and the United Republic of Tanzania. Implementation will be carried out by national EEs in each country, including key ministries, public development banks and agricultural agencies. Programme coordination units will be established or integrated within existing structures and staffed with competitively recruited government personnel, with IFAD involved in the recruitment process through a no-objection mechanism. The programme coordination units will oversee day-to-day coordination, implementation and reporting.

60. Component 3 centres on operationalizing GDFF, a regional initiative to expand inclusive green finance for climate-smart dairy investments. Each country will implement a dedicated line of credit managed by public development banks, which act as wholesale banks and report regularly to the programme coordination units and IFAD. These banks will on-lend to retail banks (commercial or prudentially supervised microfinance institutions), which will extend loans to end beneficiaries using their existing frameworks and digital platforms.

61. Disbursement will follow a revolving fund arrangement, with each country operating separate USD Designated Accounts and local currency accounts for downstream transfers. Where treasury single account systems are used, GCF funds will be tracked via a dedicated chart of accounts. Disbursements will be based on quarterly interim financial reports and withdrawal applications, conditional on 70 per cent utilization of prior tranches. Rwanda and Uganda will use integrated financial management information systems aligned with International Public Sector Accounting Standards, while Kenya and the United Republic of Tanzania will apply commercial accounting software with GCF-specific coding. Programme coordination units will consolidate financial reports from implementing partners and submit them to IFAD. Internal controls will be enforced through national audit systems, with Rwanda using a project-specific auditor and the other countries relying on established internal audit units. Annual external audits will be conducted by each country's Auditor General, with specific audit opinions on GCF funds due within six months of fiscal year-end.

4.5 Results monitoring and reporting

62. The DaIMA funding proposal presents a well-structured results architecture anchored in a robust theory of change. The programme is designed to deliver transformative impacts in the East African dairy sector through a combination of mitigation and adaptation measures across Kenya, Uganda, Rwanda and the United Republic of Tanzania. The causal logic is clearly articulated and is supported by a coherent results framework that links activities and outputs to expected outcomes and long-term impact, in line with the GCF Integrated Results Management Framework.

63. The programme contributes to three GCF results areas: mitigation through forestry and land use, adaptation through the areas of most vulnerable people and communities, and ecosystems and ecosystem services. It commits to reporting on GCF core indicator 1 (GHG emissions reduced, avoided or removed/sequestered), with an expected reduction in emissions of approximately 2.17 million t CO₂ eq; core indicator 2 (direct and indirect beneficiaries reached), estimated at 2.5 million and 15.4 million respectively; and core indicator 4 (hectares of natural resource areas brought under improved low-emission and/or climate-resilient management practice), with a target area of 178,362 hectares. These indicators are complemented by programme-specific and supplementary indicators.

64. The monitoring, evaluation and learning (MEL) system is comprehensively described in annex 11 to the funding proposal and is well aligned with GCF guidance and IFAD evaluation standards. Given the multi-country nature of the programme, the MEL plan appropriately outlines mechanisms for country-level disaggregation of data and indicator targets. The MEL system includes baseline assessments, mid-term and final evaluations, and annual performance reviews. Outcome surveys will be harmonized across countries, and the programme plans to establish a regional Community of Practice to facilitate ongoing coordination and methodological consistency. Particular attention is given to the capacity-building of implementing partners and EEs to ensure the long-term sustainability of the MEL system and its integration with national frameworks. IFAD and ILRI will provide technical support in areas such as geographic information systems, MRV systems and carbon accounting.

65. The programme has allocated approximately USD 6.72 million for MEL activities. This is considered sufficient to support the ambitious and technically demanding results monitoring, particularly given the scale and scope of the interventions across four countries. The proposed MEL arrangements meet the requirements of the GCF Evaluation Policy and Integrated Results Management Framework, and the Secretariat is satisfied that the programme has the systems and resources in place to support rigorous tracking of results, evaluation of impact and learning for scale-up.

4.6 Legal assessment

66. The Accreditation Master Agreement was signed with the Accredited Entity on 24 September 2018 (the “AMA”), and it became effective on 9 November 2018. The Accredited Entity’s five-year accreditation term lapsed on 8 November 2023 but was extended by Decision B.37/18, paragraph (q) until the earlier of three years from the date of lapse of the five-year accreditation term or the date on which a revised accreditation framework is adopted by the Board. The Board approved the re-accreditation of the Accredited Entity pursuant to Decision B.37/18, and negotiation of the Amended and Restated Accreditation Master Agreement is ongoing.

67. The Accredited Entity has not provided a legal opinion/certificate confirming that it has obtained all internal approvals, and it has the capacity and authority to implement the programme.

68. The proposed programme will be implemented in four host countries. The GCF has signed a bilateral agreement on privileges and immunities with one of these host countries, namely, Rwanda. In the case of the remaining three host countries, namely Kenya, Tanzania and Uganda, GCF is not provided with privileges and immunities. This means that, amongst other things, GCF is not protected against litigation or expropriation in these countries, which risks need to be further assessed. Moreover, the ability of GCF to undertake redress activities and/or investigations in such countries may be hindered due to the absence of privileges and immunities for relevant GCF personnel.

69. Therefore, it is recommended that the Board considers whether disbursements of GCF proceeds should only be made after GCF has obtained satisfactory protection against litigation and expropriation in those countries, or has been provided with appropriate privileges and immunities for GCF and its personnel.

70. With regard to the activities being financed by the GCF loans, the Accredited Entity has proposed that it will enter into a financing agreement with each host country, acting through its ministry of finance (each a “**MoF**”). Each MoF will then on-lend the GCF proceeds to the wholesale bank in that host country under the terms of an on-lending agreement between MoF and the wholesale bank. Although it has been agreed that each MoF and each wholesale bank are executing entities for the programme, under the Accredited Entity’s proposed contractual structure, the Accredited Entity will not have a direct contractual relationship with each wholesale bank which is responsible for the channelling of GCF proceeds and programme implementation. For the Accredited Entity to pass down and enforce the relevant requirements of the AMA and funded activity agreement (“**FAA**”) on each wholesale bank, the Accredited Entity has agreed in the term sheet that it will ensure that each MoF passes down, monitors, supervises and enforces such requirements against each wholesale bank in a manner that allows the Accredited Entity to comply with the AMA and FAA requirements as if it had a direct contractual relationship with that wholesale bank.

71. Since the Accredited Entity’s direct counterparty is a sovereign state (as one of the executing entities), the Secretariat has agreed to disapply the provisions of the AMA allowing GCF to step in to the Accredited Entity’s contractual arrangements with the Executing Entity, i.e., each MoF as, due to the sovereign nature of the counterparty and the contracts, it is not feasible for the Accredited Entity to allow GCF or any other party to take over its contractual relationship with the executing entities under the subsidiary agreements. Accordingly, and similar to the arrangements for FP143, FP162, FP183, FP236, SAP012 and SAP017, as a condition for execution of the FAA, the Accredited Entity is required to inform the National Designated Authority (NDA) of each host country of this disapplication and its implications. An express provision has been included in the term sheet to provide for such disapplication.

72. To address the matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Submission by the Accredited Entity to the Fund of a certificate or legal opinion, in form and substance satisfactory to the GCF Secretariat, within 120 days after Board approval, confirming that the Accredited Entity has obtained all final internal approvals needed by it and has the capacity and authority to implement the proposed programme;
- (b) Signature of the FAA in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval, or the date the Accredited Entity has provided a certificate or legal opinion confirming that it has obtained all final internal approvals, whichever is later; and
- (c) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's review of FP269

Proposal name:	Dairy Interventions for Mitigation and Adaptation (DaiMA)
Accredited entity:	International Fund for Agricultural Development (IFAD)
Countries:	Kenya, Rwanda, Uganda and United Republic of Tanzania
Project/programme size:	Large

I. Assessment of the independent Technical Advisory Panel

1. This funding proposal for a large, public-sector cross-cutting programme in environmental and social safeguards category B is submitted by the International Fund for Agricultural Development (IFAD) as accredited entity (AE).¹ The Dairy Interventions for Mitigation and Adaptation (DaiMA) programme will operate over six years in four contiguous East African countries: Kenya, Rwanda, Uganda and the United Republic of Tanzania (see figure 1). The total cost of the programme is USD 358.26 million, of which USD 150 million is proposed as GCF grant finance and USD 208.26 million as co-finance from IFAD and two of the governments (see section 1.7 below).

2. The proposed executing entities (EEs) for the implementation of components 1 and 2 are as follows: in Kenya, the Government of Kenya through its National Treasury, the Ministry of Agriculture and Livestock Development and the Agricultural Finance Cooperation; in Rwanda, the Government of Rwanda through its Ministry of Finance and Economic Planning, Rwanda Agriculture Board within the Ministry of Agriculture and Animal Resources and the Development Bank of Rwanda; in Uganda, Government of Uganda through its Ministry of Finance, Planning and Economic Development, the Ministry of Agriculture, Animal Industry and Fisheries and Uganda Development Bank; and in the United Republic of Tanzania, the Government of the United Republic of Tanzania through its Ministry of Finance, the Ministry of Livestock and Fisheries and Tanzania Agricultural Development Bank. No overall executing entity is identified for component 3 in the funding proposal, section A.20, but this is understood to be IFAD itself.

3. The aim of the proposed programme is:

“to reduce methane and other GHG emissions from the dairy sector while increasing the resilience of livestock-dependent communities. The programme will support a shift from low production efficiency to improved, gender-sensitive, integrated and NDC [nationally determined contribution]-aligned practices that reduce emissions at farm and food system levels (including interventions that reduce milk losses and increase milk collection, decarbonize processing with renewable energies, formalize dairy value chains and strengthen partnerships with the private sector), as well as interventions that promote circularity, and increase rangelands ecosystems resilience and services (including carbon sequestration in grasslands)” (funding proposal, para. 2).

4. The programme's theory of change is also summarized in paragraph 2.

5. The programme's interventions, geographically targeted as shown in figure 1 below, will address climate change adaptation and mitigation issues across the various dairy production systems and along the dairy value chain, while addressing a number of barriers. It will follow a

¹ This assessment is based on the full and final funding proposal package received by the iTAP on 24 March 2025.

holistic approach and will simultaneously scale up: (i) **farm and herd-level practices** to increase productivity, reduce emission intensity, and decrease absolute emissions compared to a business-as-usual scenario; (ii) **farm-level solutions** for manure management and nutrient recycling, to improve circularity, resilience and reduce emissions; (iii) **landscape-level interventions** to improve the resilience and services of rangelands ecosystems, particularly soil carbon sequestration, and water and biomass supply for animal feed; and (iv) **value-chain level interventions**, to reduce milk losses (and corresponding greenhouse gas (GHG) emissions) and increase the use of renewable energy. A useful detailed table, linking the climate drivers with interventions and their benefits, is available in annex 2 to the feasibility study (table 23).

Figure 1: Dairy Interventions for Mitigation and Adaptation programme intervention areas in the four targeted countries (source: funding proposal figure 14)



6. The programme structure, under three components, is shown in table 1:

Table 1: Structure of the Dairy Interventions for Mitigation and Adaptation programme

Component 1: Strengthened national capacities and enhanced enabling environment for low-emission climate-resilient dairy sectors	
Outcome 1: Systemic and institutional capacities of the livestock sector are strengthened to enable smallholder dairy producers and local value chain actors to reduce methane and other greenhouse gas emissions	Output 1.1: Climate-responsive policy and regulatory environment established
	Output 1.2: Improved public-service delivery to the dairy sector
	Output 1.3: Improved greenhouse gas measurement, reporting and verification capacities & monitoring of climate vulnerabilities
	Output 1.4: Knowledge management and enhanced regional cooperation
Component 2: Enabling low-emission and climate-resilient dairy value chain development	
Outcome 2: Smallholder dairy producers and private-sector value chain players operate within a low-emission, climate-resilient, and	Output 2.1: Smallholders' and cooperative technical and business capacity development strengthened
	Output 2.2: Breeding, animal health, feed and fodder, and manure management improved

sustainable pathway through increased production and market knowledge and tools	Output 2.3: Pasture and grassland management improved
Component 3: Green Dairy Financing Facility	
Outcome 3: Financial institutions make available the appropriate investment resources to make the necessary transition to low-carbon, climate-resilient and sustainable dairy production, and strengthen value-chain linkages and improved productivity	Output 3.1 Effective green financing for dairy value chains
	Output 3.2 Capacity-building for green dairy investments

1.1 Impact potential

Scale: Medium to high

7. The programme will target two of the GCF results areas for adaptation, aiming to enhance the resilience of: (i) most vulnerable people and communities (adaptation result area (ARA) 1), and (ii) ecosystems and ecosystem services (ARA4). Despite the core goal of enhancing dairy productivity and production, the programme does not target the GCF results area of increased resilience in relation to health, food and water security, but highlights co-benefits in this regard in section D.3 of the funding proposal.

8. In addition, the programme will target two of the GCF results areas for mitigation, aiming to reduce GHG emissions from (i) energy generation and access (mitigation result area (MRA) 1); and (ii) forests and land use (MRA4, which includes livestock farming), both by reducing emissions from dairy farming, particularly release of methane; and by increasing the capacity of carbon sinks through pasture and rangeland restoration.

9. **Climate-change risk and impacts.** The funding proposal highlights the climate context of the four countries, and undertakes a systematic analysis of climate hazards. This covers observed changes historically in four key parameters across the region – a mostly declining volume of precipitation, increased annual average maximum temperature, declining relative humidity in some areas, and increased rainfall variability throughout the region, as well as increased frequency of dry days, very wet days, and very warm days. Future projections on rainfall vary across the region, with arid and semi-arid regions expected to receive less precipitation, while the highlands and lake regions may receive more rainfall. Mean maximum and minimum temperature are projected to increase across the region, as is mean relative humidity in Kenya and along Lake Victoria. All areas are projected to experience rainfall variability, with some experiencing more dry days and widespread increase in very wet days and very warm days. The independent Technical Advisory Panel (iTAP) notes that the AE chose not to include analysis of duration hot days, given the complexity and variety of definitions of a heatwave², thus missing an opportunity to explore the link between prolonged heat stress and declines in animal health, milk production, fertility and calving complications.

10. The funding proposal demonstrates how these observed and projected changes in climate are impacting the four countries, causing increased exposure to droughts and floods, and increased vulnerability to rangeland degradation (compounding direct anthropogenic impacts). A clear analysis is provided of the vulnerability of the dairy sector and the agroecosystems on which it depends to these climate change and other challenges, with limited access to quality inputs, limited water availability, frequent disease outbreaks, inadequate

² Written questions from the iTAP with IFAD responses, 11 April 2025

infrastructure, energy supply and access to markets, limited access to credit and finance, and land tenure insecurity.

11. The funding proposal correctly identifies heat stress as a critical climate hazard affecting dairy productivity and health. However, it defines heat stress primarily through temperature thresholds ($>35^{\circ}\text{C}$), and does not use the temperature-humidity index (THI), which is a scientifically recognized indicator for livestock stress, based on the combined effects of temperature and humidity. The absence of explicit THI consideration constitutes a methodological gap, leading to potential for: (i) underestimation of dairy-sector vulnerability; (ii) inaccurate assessment of productivity and mortality risks; and (iii) suboptimal targeting of adaptation measures.

12. Impacts of climate change at the dairy-farm level are shown to include:

- (a) A decline in milk yields, especially of high milk-yielding breeds;
- (b) Heat stress in dairy cattle;
- (c) Increased development of pathogens, diseases, parasites and vectors in dairy cattle;
- (d) Decreased feed production and productivity, due to reduced or excess soil moisture and fertility;
- (e) A decrease in the quantity and quality of pasture; and
- (f) Deterioration of fodder in unprotected warehouses due to increased humidity (see funding proposal, paras 1 and 35).

13. At the rangeland level, livestock owners face a decrease in rangeland and pasture productivity and forage quality. At the dairy value chain level, high temperatures compromise the safety of milk and dairy products, increase the risk of milk losses, and elevate costs per unit of produce.

14. **Adaptation impact potential.** The intended adaptation impact of the programme is not stated clearly in section D.1, which speaks broadly about efficiency, productivity and resilience of dairy value chains, this leads also to closely interwoven mitigation benefits. References to reducing vulnerability to particular climate hazards are scattered throughout the funding proposal, but are stated explicitly only in the feasibility study (see table 23) and even there, not comprehensively. The iTAP would have expected to see a clear statement in the funding proposal itself on how the programme interventions reduce vulnerability or exposure, or enhance adaptive capacity (see annex VI to decision B.33/12), in relation to the six key impacts of climate change identified in paragraphs 1 and 35. The iTAP recommends that this gap be addressed in the programme implementation manual, to avoid a situation in which adaptation impacts are tracked with less rigour than mitigation impacts.

15. Table 23 of the feasibility study (annex 2 to the funding proposal) provides the clearest analysis of how the specific exposure and vulnerability discussed above will be addressed through particular interventions. For example, the table shows how rangeland restoration interventions will build resilience against drought – increasing vegetation cover and enhancing water infiltration, as well as setting aside fodder reserves.

16. Measures to decrease cattle's exposure and vulnerability to increased heat, humidity or combined heat and humidity, are not fully covered in the funding proposal, and are mentioned only once in table 23. In response to a question from the iTAP about this, the AE indicated its intention to include a range of measures:

“to attenuate heat-stress effect on animal's reproduction and production parameters (e.g. improvement of livestock housing and animal welfare; trees for shade to reduce the exposure of livestock to heat stress / solar radiation in grazing fields; water harvesting and recycling for cattle watering; crossbreeding with heat-tolerant, disease-resilient and

productive cross-breeds;...improved herd management – including observation on secondary signs of delayed oestrus due to heat stress).”³

17. The iTAP recommends that this aspect be brought out strongly in the project implementation manual to be developed, with practical recommendations for management protocols on very hot days and during heatwaves.

18. Overall the programme aspires to directly benefit about 2.5 million individual dairy farmers and dairy value chain actors in 94 districts/counties that are vulnerable to climate change – approximately 467,000 people in Kenya, 614,000 in Rwanda, 665,000 in Uganda and 758,000 in the United Republic of Tanzania. Of the beneficiaries, at least 40 per cent will be women, 30 per cent youth and 5 per cent from marginalized groups (including people with disabilities). An estimated 15.4 million indirect beneficiaries will benefit from strengthened veterinary services reducing the spread of diseases exacerbated by climate change, enhanced extension and breeding services, and improved access to climate information. A total of about 2.7 million dairy cattle will be reached, and approximately 178,362 ha of rangelands and natural pastures will be brought under improved management. The funding proposal shows the following combination of criteria used to select intervention areas: (i) climate-risk hotspots; (ii) absolute methane emissions from the dairy sector; (iii) soil carbon sequestration potential in pastures and rangelands; and (iv) milk production by district/county.

19. **Mitigation impact potential:** The DaiMA programme aims to achieve total mitigation results of 2,169,485 tonnes of carbon dioxide equivalent (tCO₂eq) over its 20-year lifespan. This results from a combination of emissions reductions (from the introduction of renewable-energy solutions and better dairy-cattle management) and increased carbon sequestration (from the rehabilitation of pastures and rangelands).

20. Estimates for these mitigation results were calculated using the best practice Global Livestock Environmental Assessment Model-interactive (GLEAM-i) developed by the Food and Agriculture Organization of the United Nations (FAO). GLEAM-i uses an embedded herd dynamic model, and estimates of feed requirements to predict direct emissions resulting from enteric methane and manure, based on digestibility and nitrogen content. Indirect emissions from the production of feed resources, and direct and embedded energy use are also included. Using this methodology, milk production is projected to increase by 34 per cent (through productivity increases and loss reductions), while reducing the milk emission intensity (i.e. the volume of methane emissions generated per kg of milk) by a significant 29 per cent,⁴ and reducing the total feed intake by 5.23 per cent, compared with the without-project scenario.

21. The contributions of the three major clusters of interventions to the projected mitigation results are summarized in table 2.

Table 2: Major programme interventions and how they contribute to mitigation results (created by the independent Technical Advisory Panel)

Intervention area		Type of mitigation result	Total result (over 20 years in t)	Summary of how mitigation results are to be achieved
1	Renewable energy	Emission reductions	1,088,705	Substitution of grid and diesel electricity (which would be expanded in the without-project scenario) with biogas-generated electricity and solar photovoltaic (PV) systems, through grants to dairy

³ Written questions from the iTAP with IFAD responses, 11 April 2025.

⁴ This average comes from the following intended reductions per country: 28 per cent in Kenya, 50 per cent in Rwanda, 33 per cent in Uganda and 29 per cent in the United Republic of Tanzania.

				cooperatives for solar PV-powered milk coolers, milk transfer tanks, solar water heaters, water-recycling systems, heat exchangers, milk pasteurizers, butter churners and cheese makers, as well as loans to dairies for other technology such as insulated/refrigerated trucks
2	Dairy-herd management	Emission reductions	266,560	Reduced emissions in the with-project scenario, compared to the without-project scenario, after successful implementation of interventions – enabling access to better quality pasture and feed sources that reduce enteric fermentation; reduced mortality and higher productivity through access to better veterinary care and vaccination; improved fertility and productivity rates through support on artificial insemination with high-yielding cross-breeds; and effective management of manure and use for biogas
3	Pasture and rangeland restoration	Carbon sequestration	814,220	Significant increase in soil organic carbon, and related above-ground and below-ground biomass, compared with the without-project scenario – as a result of restoration and management interventions such as rotational grazing, earthworks for infiltration, reseeding of palatable species, and removal of encroacher bush
Total			2,169,485	

22. **Limitations to achieving mitigation potential:** A challenge with achieving emission reductions related to dairy-herd management is that the reductions depend not only on successful implementation of all the improved practices (on breeding, animal health, fodder and feed, and manure management), but critically also on an assumption that herd sizes can be stabilized – an assumption which may not hold true in reality. The with-project scenario assumes a very low rate of herd size growth, and compare this with a slightly higher rate in the without-project scenario. The iTAP acknowledges that the programme interventions aim to increase milk yields without increasing head count, as a result of several efficiency and productivity-raising interventions as well as improving herd composition. But in reality, the programme and its proponents will have little control over the decisions made by the millions of farmers and producers involved over whether and to what extent to increase their cattle head count and business size in response to increasing demand,⁵ as well as the incentives provided by the programme itself (access to cold-chain technology, finance and markets).

23. In response to a question from the iTAP on whether herd stability is a realistic assumption, the AE pointed to evidence from Kenya showing that the dairy herd had remained around 5 million animals since 2002, while production of milk had increased by 60 per cent in the same period. According to IFAD, this is explained by various factors, including the use of mobile payment system (M-pesa) and offtake of unproductive animals. The AE also pointed out that “the feed base and the collection capacity are not expendable, which means that farmers cannot grow a significantly larger numbers of animals, even if they wanted to”. They also pointed to the livestock-farmer field schools training modules on herd management, focusing on

⁵ Discussed in ‘Selected Risk Factor 4’ in the funding proposal, with mention also in paragraph 33 of projected dairy-herd growth in the United Republic of Tanzania to satisfy future demand for dairy products, which is expected to grow from 1.4 billion litres of raw milk in 2020, to 2 billion litres in 2030, and 8 billion litres in 2050.

the reproduction calendar, performance recording using technology, an enhanced female-to-male ratio through sexed-semen and reduction/off-take of unproductive animals, as well as production efficiencies (through breeding, feeding and animal health), and interventions to strengthen value chains and enhance market access “allowing shifting the value from their livestock to the milk”.⁶ The iTAP recommends tracking of herd sizes (as a combined result of births, deaths and acquisitions of new livestock) during project implementation.

24. A further factor which could undermine project results, in both mitigation and adaptation, is the uncertainty around how climate-change impacts will evolve during the programme implementation. For example, exposure of cattle to extreme climate conditions resulting from the combined effect of high temperature and high humidity may turn out to be even higher than projected. It is well documented that high THI conditions lead to higher mortality, lower fertility and reduced calving rates, which could counteract programme interventions to improve productivity, thus limiting both adaptation and mitigation results.

25. **Monitoring landscape impacts:** The iTAP notes the solid monitoring system (set out in annex 22 to the funding proposal) for rangeland restoration interventions, with clear indications of responsibilities for site-level ground-truthing and data collection on: (i) GPS coordinates of rangelands under intervention, and areas of rangelands where improved practices were implemented; (ii) surveys on species composition and improved biomass yield; (iii) the type of improved grazing management applied at the community level (rotational grazing, agroforestry, improved pasture management, management of grazing intensity, etc.); and (iv) the quantity and type of legume seeds used, date of planting or reseeded. This will be combined with remote sensing analysis through high-resolution satellite images (using, for example, the normalized difference vegetation index) to monitor the rangelands under restoration. On the mitigation side, a system is also in place for measurement, reporting and verification (MRV) of soil carbon sequestration in areas under restoration, including modelling for estimating the impact of management practices. Using baseline information on soil carbon stocks, soil texture, carbon inputs from vegetation and manure, as well as climate (all collected through the baseline survey), process-based models (e.g. RothC) will be used to estimate actual results. The team will also leverage data from the national MRV systems enhanced under component 1 and make estimations using tier 2 sequestration factors generated under sub-activity 1.3.1.1 (e.g. with flux towers such as eddy covariance towers – see feasibility study for more details).

26. Overall, the impact potential is rated as medium to high.

1.2 Paradigm shift potential

Scale: Medium to high

27. Paradigm shift relates to the degree to which the proposed programme can successfully catalyse impact beyond the GCF one-off investment of USD 150 million over six years, together with complementary IFAD projects. The programme, which will work with approximately 6.5 per cent of all dairy farmers in Kenya, 30 per cent in Rwanda, 11.3 per cent in Uganda and 9.1 per cent in the United Republic of Tanzania, aims to facilitate the transition from a dairy sector characterized by rising emissions, low production efficiency, market informality and climate vulnerability, with limited access to services and finance, to a more efficient, formal and low-carbon, climate-resilient sector. By improving policy, institutional and service-delivery environments, building climate adaptation capacity, and enabling access to finance and climate-smart solutions for smallholder dairy farmers, the programme seeks to transform all aspects of the dairy value chain.

⁶ Written questions from the iTAP with IFAD responses, 11 April 2025.

28. **Enabling environment for scaling up.** Component 1 of the programme focuses on creating a climate-responsive policy and regulatory environment in the four countries, through work to review and formulate dairy-sector policies, strategies and regulations. Awareness campaigns will be run on the nutritional benefits of milk and food safety aspects. The programme also seeks to improve land-tenure security and management for sustainable dairy farming. Public-sector capacity for livestock support will be increased, enhancing veterinary services for better animal health, higher milk productivity and reduced losses, and extension services will be strengthened. At a regional scale, GCF financing supports the transformation of the dairy sector through interventions such as regional policy dialogues, cooperation platforms and centres of excellence; cross-country cooperation, and knowledge exchange; MRV systems using cutting-edge methods; climate risk and vulnerability assessments, including feed balance assessments; and provision of finance through the Green Dairy Financing Facility (GDFF).
29. **Focus on national MRV:** The programme has the potential to make an important contribution to building MRV capacities in the four countries' governments and research institutions, for both climate risk and vulnerability assessments, improving greenhouse gas emission measurements in dairy value chains, as part of national MRV systems. The programme also aims to facilitate the development of a carbon certification system for the dairy sector, and reduce barriers to accessing carbon finance. Although the funding proposal does not mention the issue of benefit-sharing of potential proceeds from the sale of carbon credits in the future, in response to a question on this from the iTAP, IFAD stated: "To ensure fair and inclusive access to carbon finance under pilot Activity 3.4, the Programme will support preparedness by establishing transparent mechanisms for beneficiary identification, benefit-sharing, and participation in governance structures across the dairy value chain".⁷
30. **Retail bank uptake:** The intention of the programme is to de-risk private investment and accelerate the adoption of low-emission, climate-resilient technologies and solutions by smallholders and small and medium-sized enterprises (SMEs), catalysing private-sector engagement and scaling-up climate-finance flows. The programme is based on a thorough dairy market study (funding proposal, annex 2, appendix 1), and effectively targets many of the key barriers to technology dissemination and strengthening dairy value chains. At the same time, there is a possibility that demand for and uptake of loans through the GDFF will be lower than anticipated, limiting the paradigm shift achieved in financial markets. In the experience of the iTAP, a risk with providing lines of credit via commercial banks is that they end up not being drawn upon, because the banks fail to actively engage new customers in unfamiliar market sectors. Cheap liquidity alone has often proven insufficient to entice local commercial banks to broaden their lending portfolios to include higher-risk clients or enterprises with new business models. Risk participation agreements, which share a portion of credit losses between a concessional finance provider and the banks, may represent a more effective incentive mechanism but remain unexplored in the current programme design.
31. The funding proposal highlights the technical assistance programme that will work with the wholesale and retail banks, as well as the favourable loan terms, including long grace and repayment periods, but it is unclear what the programme can/will do to incentivize these retail banks to go out and actually build business with this particular client base. This is highlighted in Selected Risk Factor 8 in the funding proposal, which acknowledges that "[e]stablishing four relatively small-sized credit lines in a niche sub-sector like dairy poses several challenges. Relatively low funding base can constrain the interest of banks to participate in the scheme. Lack of understanding of the bankable options in green dairy lending can further reduce bank interest". In response to a request to the AE to explain the approach to mitigating this risk in a bit more depth, IFAD stated that "The financing will be made in tranches and only against proven success concerning the level and quality of the GDFF portfolio implementation. The

⁷ Written questions from the iTAP with IFAD responses, 11 April 2025.

meetings held with candidate retail banks indicate strong demand by SMEs in the dairy sector. The low cost of capital to the retail banks, together with the TA grant, further incentivizes the utilization of resources”.⁸ Nonetheless, experience shows that proactive measures beyond concessional pricing, such as risk-sharing facilities or partial credit guarantees, are often necessary to overcome entrenched risk aversion in commercial banks and to ensure that climate finance flows reach the intended underserved market segments.

32. Taking all these factors into account, paradigm shift potential is rated as medium to high.

1.3 Sustainable development potential

Scale: High

33. The programme is expected to have a number of social, economic and environmental co-benefits, as outlined below.

34. **Social and health co-benefits.** The programme will promote improved food safety and nutrition, and thus human health – using market formalization, awareness campaigns and product traceability to improve nutrition and health awareness, with a focus on the increased consumption of safe and high-quality dairy products (Sustainable Development Goal (SDG) 3: Good health and well-being). The programme contributes to ending hunger, achieving food security and improving children’s nutritional status (SDG 2: Zero hunger). In addition, by strengthening veterinary services, it aims to improve animal health and welfare, contributing to higher milk productivity and reduced losses, as well as decreased GHG emissions and increased farmers’ incomes, which will, in turn, incentivize farmers to adopt better measures on animal health and welfare (SDG 12: Responsible consumption and production).

35. **Gender and social inclusion co-benefits.** The programme includes strengthening of dairy groups and cooperatives, enhancing social cohesion. It has clear targets for the involvement of women, youth and marginalized groups (see paragraph 15 of this assessment). Project activities take into account the distinct traditional roles that women and men (and youth) play in rearing cattle, milking cows and producing dairy products in households and small businesses. These roles, as well as constraints on women’s access to land, financing and other resources, are analysed in annex 8 (Gender assessment and action plan). The programme aims to reduce gender inequalities (SDG 5: Gender equality) by actively promoting women’s participation in livestock-farmer field schools, and opening space for women and youth in traditionally male-dominated roles – including milk aggregation, animal health services, manure management, fodder production and cooperative leadership. The Gender Action Learning System Plus will be used – a participatory tool that engages households and communities in rethinking power dynamics, promoting joint decision-making, benefit-sharing, and reshaping socially acceptable roles through gender champions and peer influence.

36. **Environmental co-benefits.** The programme’s interventions to restore and better manage rangelands and pastures will not only have core adaptation and mitigation results, but will also produce environmental co-benefits – such as an increase in biodiversity, with native species reintroduced or allowed to regenerate (SDG 15: Life on land). Although planted pastures with exotic grasses may be used more extensively in humid and semi-humid grasslands/rangelands in Uganda and Rwanda, indigenous grasses and trees will be allowed to regenerate in arid and semi-arid grasslands/rangelands in Kenya and the United Republic of Tanzania (see funding proposal, annex 2, feasibility study, textbox 14). The programme follows circular economy principles, improving manure management for biogas production and nutrient cycling, which in turn improves soil health (SDG 12: Responsible consumption and production). It is noted that sub-activity 2.1.3.3 will include strategies to avoid waste (including

⁸ Written questions from the iTAP with IFAD responses, 11 April 2025.

e-waste) through the repair of the installed renewable energy-based equipment and the establishment of an equipment maintenance account in a local financial institution by each supported rural organization.

37. **Economic co-benefits.** The programme is expected to generate significant positive economic impacts, including job creation, poverty reduction, increasing productivity, strengthening income-generating capacities, and hopefully attracting additional private finance into the dairy sector. Smallholder dairy producers and local private-sector value chain actors will have access to essential financial services to improve production processes. This approach aims to increase dairy farmers' incomes, motivating them to adopt further efficient practices and climate-smart technologies (SDG 9: Industry, innovation and infrastructure). Capacity-building and adoption of integrated solutions will improve sustainable and resilient production, thereby improving the incomes and productivity of smallholder dairy farmers (SDG 1: No poverty).

38. **Semi-nomadic pastoralists:** Although the programme focuses primarily on intensive and semi-intensive livestock systems for the value-chain interventions, extensive grazing systems (rangelands on which multipurpose cattle are 100 per cent dependent for foraging) are also part of the activities on rangeland restoration and management, as well as manure management. In response to a question from the iTAP about how the programme will ensure the involvement of extensive pastoralists who are semi-nomadic, including some who self-identify as Indigenous Peoples, the AE confirmed that such groupings will be engaged through local community governance structures, and that free, prior and informed consent processes will be conducted where appropriate. Pastoralists who so wish will be integrated into supply chains through dairy training and the establishment of new milk collection centres.⁹ Although traditional knowledge is mentioned in the funding proposal only in the logframe (section E.3), the AE confirmed its centrality in the approaches to be followed by the livestock-farmer field schools, alongside science, especially in relation to rangeland management and restoration.

39. **Seed sovereignty:** In response to a question from the iTAP, the AE confirmed that activity 2.2.3, which makes certified forage/fodder seed available for multiplication, will be aligned with the principle of seed sovereignty:

"IFAD supports seed sovereignty, recognizing farmers' rights to save, use, exchange, and sell their own seeds, while promoting access to diverse, open-source, non-GMO seed systems that strengthen resilience and local control. Under Activity 2.2.3, DaIMA will align with these principles by promoting the use, multiplication, and dissemination of certified or quality-declared forage and fodder seeds, focusing specifically on open-pollinated, locally adapted varieties — not hybrids or genetically modified organisms (GMO). In doing so, the programme will engage local seed businesses, farmer cooperatives, and community seed systems, in accordance with national regulatory frameworks and farmer preferences."

40. Sustainable development potential is rated as high.

1.4 Needs of the recipient

Scale: High

41. Three of the four DaIMA countries are classified as least developed countries¹⁰, and all of them are recognized as being highly vulnerable to climate change. Observed and projected climate-change scenarios for the region indicate significant changes in precipitation and temperature patterns, as well as heat stress, increased frequency and severity of droughts and other extreme weather events. The effects will negatively impact the dairy sector, resulting in

⁹ Written questions from the iTAP with IFAD responses, 11 April 2025.

¹⁰ Rwanda, Uganda and the United Republic of Tanzania.

substantial losses in dairy livestock, and loss of productivity of pastures and rangelands, with profound implications for the dairy industry and dairy farmers, leading to lower milk yields and production, reduced cattle health, deterioration in pasture and feed quality, and increased risks in milk losses along the value chains. These impacts compound the sector's existing vulnerabilities, such as farmers' lack of access to knowledge and new skills, insufficient milk collection facilities, and limited access to quality inputs and finance.

42. On the mitigation side, methane emissions from the dairy sector dominate the GHG emission inventories of Kenya, Rwanda, Uganda and the United Republic of Tanzania. The expected growth of the dairy sector means that both absolute emissions and emission intensities will continue to rise if no action is taken to reduce emission intensity per litre of milk produced, and to enhance productivity such that yields can increase without growing herd sizes. The urgency of reducing the sector's GHG emissions and increasing its resilience to climate change is highlighted in national policies and plans, more particularly in the countries' NDCs. As pointed out in the funding proposal (para. 1), the Intergovernmental Panel on Climate Change recognizes "efficient livestock systems" as a feasible adaptation option that has synergy with mitigation.¹¹

43. Each of these countries suffers from a significant development deficit, which negatively impacts their ability to steer the dairy sector on a path to low-emission and climate-resilient development. In all four countries, public fiscal space is limited (see section B.5), and governments rely heavily on official development assistance and concessional financing to fund their dairy sector programmes. As a result, vulnerable populations, such as smallholders, women and youth in the dairy sector are highly exposed to climate risks due to technological, organizational and policy constraints.

44. The needs of the recipient are rated as high.

1.5 Country ownership

Scale: High

45. The Governments of Kenya, Rwanda, Uganda and the United Republic of Tanzania are fully supportive of the proposed DaIMA programme aiming to improve resilience to climate change in dairy production and value chains and to reduce methane and other GHG emissions from the dairy sector. Integrating the dairy sector into national climate actions will also have multiple benefits for the environment, nutrition, livelihoods and social inclusion. Partners supporting the programme design and development process include IFAD, FAO, the Global Dairy Platform, the International Livestock Research Institute (ILRI), and Bain & Company.

46. According to annex 10 to the funding proposal, the programme was initially conceptualized as mitigation only and was entitled "Pathways to Dairy Net Zero (PADNET)". This project idea and the concept of net zero itself, however, were perceived as too complex, unclear and difficult to understand by many government and development partners. In addition, country-level stakeholders asserted that a key priority in East Africa is the high vulnerability of smallholder farmers and pastoralists and their adaptation to climate change, leading to the shift to a cross-cutting programme called "Dairy Interventions for Mitigation and Adaptation (DaIMA)". The word "daima" means "always" or "forever" in Swahili, a language spoken in all four countries, highlighting the importance of sustainability and a regional identity to the programme.

¹¹ This is shown in the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, Synthesis Report, Figure SPM.7 (fourth row).

47. **Alignment with policy.** The DaIMA programme aligns with the national strategies of the participating countries by supporting and complementing existing plans, policies and strategies related to agriculture, livestock and climate change. In Kenya, the proposed programme aligns with the Agriculture Sector Development Strategy (2019–2029), National Climate Change Action Plan, Kenya Climate Smart Agriculture Implementation Framework (2018–2027), and the Kenya National Strategy on Climate Change and Low Carbon Development. In Uganda, the programme aligns with the Livestock Development Strategy and Investment Plan, National Adaptation Plan, and National Climate Change Strategy. In Rwanda, it aligns with Rwanda's NDC, National Climate Change and Environment Policy, Rwanda Livestock Master Plan, National Adaptation Plan, and National Livestock Policy. The funding proposal is silent on alignment with the relevant general policies of the United Republic of Tanzania, though these are shown in table 11 of the feasibility study (annex 2 to the funding proposal). Relevant regional policy frameworks are also highlighted in section 2.11 of the feasibility study.

48. **Engagement with stakeholders.** The stakeholder engagement plan (annex 7 to the funding proposal) provides a record of an extensive consultation process over the course of 2023 and 2024. The national designated authorities in the four target countries were directly involved in all stages of programme ideation, conceptualization and preparation. In each country, National Programme Preparation Teams (comprising a wide range of stakeholders from ministries of agriculture/livestock and dairy boards) were established in 2023, at the regional workshop in Kigali (September 2023). Through regional (Arusha, February 2023; Kigali, September 2023) and national workshops (July and September 2023), stakeholder engagement meetings were organized to involve a wide array of national and regional public institutions, private-sector operators, civil society, research institutions and development partners in designing the programme. Four design missions were held in 2023, involving visits to farms and dairy centres, and meetings with government agencies (ministries of agriculture/livestock, ministries of the environment, meteorological services, national agricultural research institutes, dairy development boards/authorities), with private dairy processors, with milk collection centres, dairy cooperatives and farmers, with pastoral communities, with financial services providers (savings and credit cooperative societies, commercial banks, fintech companies), and with input providers (feed and fodder, machinery).

49. **Capacity of accredited and executing entities:** As AE for the programme, IFAD will be responsible for the overall management of the programme, building on its experience with 11 other GCF projects/programmes under implementation. This role includes programme appraisal, fiduciary and technical oversight, fund management, quality monitoring, reporting to the GCF, and overseeing final evaluation, engaging IFAD headquarters in Rome and its regional office in Nairobi, as well as its country offices in Kenya, Rwanda, Uganda and the United Republic of Tanzania. A Regional Steering Committee – constituted of Permanent Secretaries (most senior civil servants) from the EEs, 12 government entities listed in paragraph 2 above – will ensure regional alignment and added value, with national governance structures in each country overseeing implementation. International and national entities, including ministries of agriculture and specialized entities (e.g. ILRI), will provide targeted services to the national implementation teams, with preference given to those that have both proven capacity and regional presence.

50. Component 1's work on policies, awareness and vulnerability assessment will be implemented by the relevant EEs in each country, through a wide range of additional government agencies and service providers. Contracted implementing partners, over which the EEs will have oversight responsibility, will include local governments, national government institutions, financial institutions, universities and knowledge institutions, local and national non-governmental organizations, ILRI, FAO and private-sector entities. The same applies to work in component 2 on cooperative strengthening, promoting renewable energy and waste management, investment in cattle-breeding centres, support to animal health and forage production. The programme will build on the fiduciary capacity of the key EE in each of the four

countries (National Treasury of Kenya, Ministry of Agriculture and Animal Resources - Rwanda Agriculture Board in Rwanda, Ministry of Finance, Planning and Economic Development, in Uganda, and Ministry of Livestock and Fisheries in the United Republic of Tanzania), and the existing expertise of the wide range of implementing partners in the four countries, further developing this capacity over the six years of programme implementation.

51. Country ownership is rated as high.

1.6 Efficiency and effectiveness

Scale: Medium

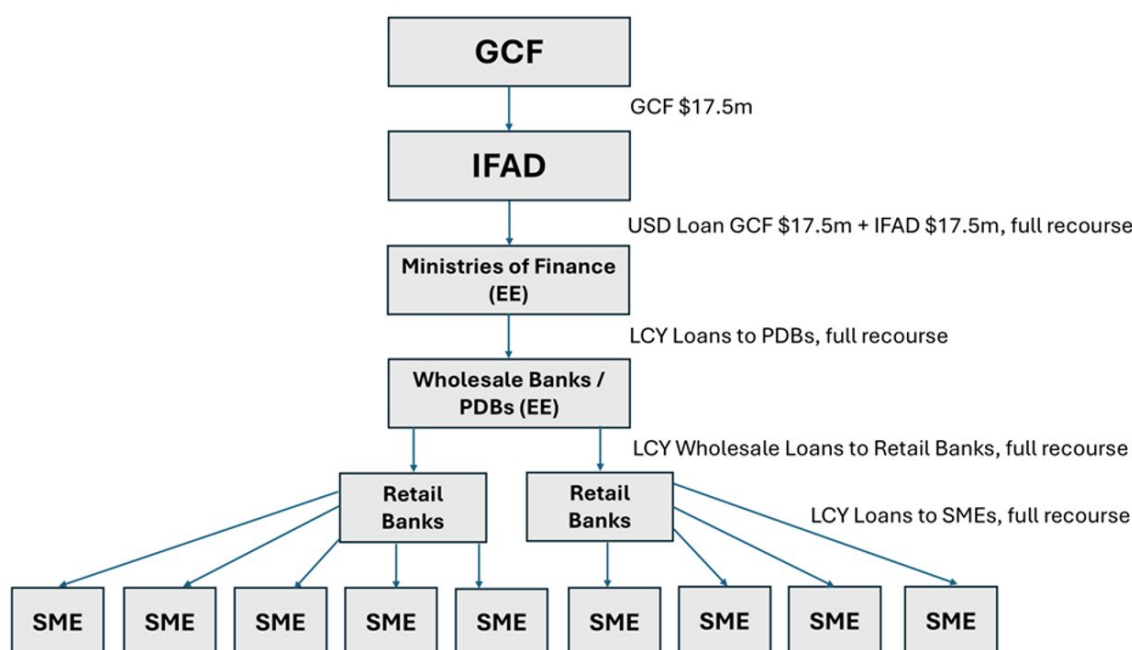
52. With a total budget for the proposed DaIMA programme of USD 358.26 million, IFAD is seeking USD 150 million from GCF – 42 per cent of the overall budget. IFAD is investing a total of USD 193.28 million – made up of USD 17.5 million in loans from its general programme of loans and grants, and USD 175.78 million in senior loans through four existing IFAD-to-government loan programmes: in Rwanda – the Rwanda Dairy Development Project, phase 2, in the United Republic of Tanzania – Climate Smart Smallholder Dairy Transformation Project, in Kenya – the Integrated Natural Resource and Environmental Management Project, and in Uganda – the Resilient Livestock Value Chain project.

53. The proposed financial structure for the DaIMA programme includes slightly concessional loans to dairy producers (12 per cent of GCF investment, 63 per cent of the total programme) and grants to governments, service providers and producer beneficiaries (88 per cent of the proposed GCF investment, 37 per cent of total programme). Maximum concessionalism is sought from GCF for component 1 activities to strengthen the enabling environment and national capacities for low-emission climate-resilient dairy sectors, and also for component 2 investments in strengthening smallholders' and cooperative technical and business capacity, in improving breeding, feed and fodder, and manure management in the value chain, and in improving management of pasture and grasslands.

54. **Loan facility:** In addition to the four complementary IFAD-government loans serving as co-finance to the GCF investment, a programme-wide investment facility is being set up through component 3 – the Green Dairy Financing Facility. The GDFF will de-risk and enable affordable loans to producers for renewable energy-driven productivity enhancements, with a total of USD 45 million allocated to the facility. Of this amount, USD 35 million will be for on-lending loan capital (GCF loan of USD 17.5 million, IFAD loan of USD 17.5 million), with USD 10 million for a technical assistance facility that will improve investment demand and readiness among pipeline enterprises, and provide support post-deal to ensure successful implementation.

55. The Ministries of Finance in the participating countries, acting as EEs, will receive this combined loan package. These ministries will then disburse local currency loans to wholesale banks or public development banks (PDBs) within their respective countries. Following this, the wholesale banks provide local currency wholesale loans to retail banks operating at the national or local level. Finally, the retail banks on-lend the funds directly to SMEs, specifically targeting dairy value chain actors. An annual survey of market interest rates will help to define an appropriate range of acceptable interest rates for end-borrower lending, within which the retail banks will operate. The entire amount revolves across the GDFF financing chain until the loans to IFAD and GCF have been repaid. As per the repayment schedule, the retail banks are fully responsible for repaying the total loan amounts back to the wholesale banks, regardless of the performance of the underlying portfolio, and the wholesale banks are required to repay the full loan amounts to the Ministries of Finance for repayment back to IFAD and GCF.

Figure 2: Green Dairy Financing Facility financing chain (from funding proposal, para. 146)



Abbreviations: EE = executing entity, IFAD = International Fund for Agricultural Development, LCY = local currency, m = million, PDB = public development bank, SMEs = small and medium-sized enterprises

56. The programme thus includes a complex on-lending structure across four steps: 1) from GCF to the host government; 2) from the host government to a PDB; 3) from the PDB to participating retail banks; and 4) from retail banks to borrowers. The iTAP acknowledges the value of the host governments taking the hard currency risk and thus enabling on-lending in local currency. Nonetheless, this long chain of institutions involved, each taking a cut, potentially undermines the concessional nature of GCF funds provided, and seems likely to make coordination of the facility (e.g. ensuring use of proceeds clauses are properly implemented) very challenging.

57. **Economic analysis:** The ex-ante financial and economic viability of the programme is demonstrated by an economic and financial analysis conducted by the AE (annex 3 to the funding proposal). Economic performance indicators for the four countries individually confirm the economic viability of the investment for society as a whole. At regional level, the aggregated economic net present value (ENPV) of the countries' incremental net benefit streams, discounted at 13.25 per cent (the average social discount rate), is USD 181.20 million, with an economic rate of return (ERR) of 25 per cent for the baseline scenario. Two additional economic performance scenarios are included, by integrating the economic value of the GHG emissions balance with a low vs a high carbon market price assumption. Under the low scenario, the ERR is 27.9 per cent and the ENPV is US\$ 217.87 million. Under the high scenario, the ERR is 30.3 per cent and the ENPV is US\$ 254.64 million. The sensitivity analysis confirms that DaIMA represents a robust economically worthwhile investment.

58. **Financial analysis:** The financial analysis examines the profitability of programme investments from the perspective of direct participants. The Livestock Sector Investment Policy Toolkit was used to develop dairy production models in all four countries, using five different herd types: (i) small-scale intensive system/zero grazing with cross-breed and exotic animals; (ii) small-scale intensive system/zero grazing with exotic animals; (iii) semi-intensive with cross-breeds; (iv) semi-intensive with local breeds; and (v) extensive systems with local breeds. Combinations of these herd models with likely investments through dairy farmers, milk collection points and centres, and dairy-processing units were used as the basis for the analysis.

59. The financial analysis used the financial discount rate for each of the four countries (i.e. the rate at which commercial banks can borrow from the central bank): 11.25 per cent in Kenya,

6.5 per cent in Rwanda, 9.75 per cent in Uganda and 5.5 per cent in the United Republic of Tanzania. The analysis indicates that the financial internal rate of return (FIRR) is above the opportunity cost of capital in all four countries and thus that the proposed investments can be profitable. The analysis suggests: for Kenya, an FIRR of 31 per cent and net present value (NPV) of USD 172.82 million; for Rwanda, an FIRR of 19 per cent and NPV of USD 41.62 million; for Uganda, an FIRR of 17 per cent and NPV of USD 47.48 million; and for the United Republic of Tanzania, an FIRR of 21 per cent and NPV of USD 87.11 million. A regional average rate of return is also calculated for the proposed programme as a whole, using an average discount rate of 8.25 per cent, and yielding a regional FIRR of 23 per cent and a NPV of USD 349.2 million.

60. **Mitigation cost-effectiveness:** The estimated cost per tonne of carbon dioxide equivalent for the programme, defined as total investment cost/expected lifetime emission reductions, is USD 165 per tonne when considering the entire programme (with co-finance, including the existing IFAD loan programmes), and USD 85 per tonne when considering only the 50 per cent of the programme cost nominally associated with GCF mitigation results (see funding proposal, section A.4).

61. Given the high unit cost of the mitigation results and the complexity of the on-lending structure that reduces financial efficiency, efficiency and effectiveness are rated as medium.

II. Overall remarks from the independent Technical Advisory Panel

62. This proposed DaIMA programme is expected to make a significant contribution to helping the smallholder dairy farming and dairy production sectors in these four East African countries reduce their GHG emissions and enhance carbon removals, while simultaneously building resilience of vulnerable smallholder farmers and SMEs.

63. To enhance the robustness of adaptation actions related to heat stress, the iTAP recommends that IFAD, during the inception phase and early implementation, systematically integrate the Temperature-Humidity Index into operational climate-risk assessments and adaptation monitoring frameworks. This will refine the targeting of livestock heat stress interventions (e.g. housing, shade, water management, breed selection) across intervention areas.

64. The iTAP further recommends that IFAD include in the programme implementation manual:

- (a) A clear statement of the intended adaptation impact of the project – in reducing vulnerability or exposure, or enhancing adaptive capacity, in relation to the key climate change impacts identified: (i) a decline in milk yields; (ii) heat stress in dairy cattle; (iii) increased development of pathogens, diseases, parasite and vectors in dairy cattle; (iv) decreased feed production and productivity, due to reduced soil moisture and fertility; (v) a decrease in the quantity and quality of pasture; and (vi) deterioration of fodder in unprotected warehouses due to increased humidity.
- (b) A detailed section on measures to be implemented for heat stress reduction in cattle, taking into account the combined effects of temperature and humidity, including both the broader project strategies like introducing shade trees in pastures and cross-breeding with heat-tolerant breeds, and also a management protocol for very hot days – involving, for example, avoiding animal handling, ensuring water availability, bringing cattle under a shade roof with solar-powered fans, and spraying cattle with water.
- (c) An addition to the project's monitoring and evaluation plan to track herd size (including mortality rate, birth rate and acquisition) over the six years of programme implementation, in order to test the assumption on which the programme's mitigation



results depend (i.e. that herd sizes will remain stable), and to put in place adaptive management measures where growth trends are emerging.

65. The iTAP recommends that the Board approve this funding proposal.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP269)

Proposal name:	Dairy Interventions for Mitigation and Adaptation (DaiMA)
Accredited entity:	International Fund for Agricultural Development (IFAD)
Countries:	Kenya, Rwanda, Uganda and United Republic of Tanzania
Project/programme size:	Large

Impact potential
IFAD takes note of the positive assessment.
Paradigm shift potential
IFAD takes note of the positive assessment.
Sustainable development potential
IFAD takes note of the positive assessment.
Needs of the recipient
IFAD takes note of the positive assessment.
Country ownership
IFAD takes note of the positive assessment.
Efficiency and effectiveness
IFAD takes note of the positive assessment. IFAD takes note of the iTAP comment in reference to the high unit cost of the mitigation results and the required three layers of on-lending structure which potentially increase the risk of reduced financial efficiency, and that the efficiency and effectiveness are rated as medium. IFAD will ensure that the Financing Agreements at all layers are designed to facilitate funds flow and eligible use of resources, offer substantial TA for the implementation, and closely monitor the same.
Overall remarks from the independent Technical Advisory Panel:
IFAD thanks the iTAP for its positive review of the proposal.

DAIRY INTERVENTIONS FOR MITIGATION AND ADAPTATION (DAIMA)

FUNDING PROPOSAL

ANNEX 8: GENDER ASSESSMENT AND PROGRAMME-LEVEL ACTION PLAN

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ACRONYMS

AGYW	Adolescent Girls and Young Women
AI	Artificial Insemination
ASDP-L	Agricultural Sector Development Programme–Livestock
BBT-YAI	Build a Better Tomorrow – Youth Agribusiness Initiative
BDS	Business Development Services
CAHW	Community Animal Health Workers
CC	Climate Change
C-SDTP	Climate Smart Smallholder Dairy Transformation Project
CVIL	Central Veterinary Investigation Laboratory of Zanzibar
DaIMA	Dairy Interventions for Mitigation and Adaptation
DHS	Demographic and Health Survey and Malaria Indicator Survey
DVC	Dairy Value Chain
EAC	East African Community
EADD	East African Dairy Development Project
ELISA	Enzyme Linked Immunosorbent Assay
ESADA	East and Southern Africa Dairy Association
FDA	Food and Drugs Authority
FGM	Female Genital Mutilation
FGM/C	Female Genital Mutilation/Cutting
GALS	Gender Action Learning System
GBV	Gender-Based Violence
GDFF	Green Dairy Financing Facility
GESI	Gender and Social Inclusion
GEWE	Gender Equality Women’s Empowerment
GHG	Green House Gas(ses)
GoT	Government of Tanzania
HDI	Health Development Initiative
HHs	Households

HIV	Human immunodeficiency virus
HPI	Heifer Project International
ICPALD	Center for Pastoral Areas and Livestock Development
ICT	Information and communication technology
IFAD	International Fund for Agricultural Development
IGAD	Inter-Governmental Authority on Development
KALRO	Kenya Agriculture and Livestock Research Organization (KALRO)
KAOP	Kenya Agricultural Observatory Platform
KDB	Kenya Dairy Board
KDHS	Kenya Demographic and Health Survey
KeLCoP	Kenya Livestock Commercialization Project
LEWs	Livestock Extension Workers
L-FFS	Livestock Farmer Field School (L-FFS)
MCC	Milk Collection Centre
MCP	Milk Collection Plant
MDAs	Ministries, Departments, and Agencies
MoA	Ministry of Agriculture
MRV	Measurement, Reporting and Verification
MSMES	Micro, small and medium enterprises
NGO	Non-Governmental Organization
NVL	National Veterinary Laboratory
OHRECA One Health Research and Education Centre in Africa	
OUT	Open University of Tanzania
PCR	Polymerase Chain Reaction
PFF	Pastoralist Field Schools
PFI	Participating Financial Institutions
PICSA	Participatory Integrated Climate Services for Agriculture
PSEAH	Preventing Sexual Exploitation, Abuse and Harassment
PSTA5	Strategic Plan for Agriculture Transformation 5
PWDs	People Living with Disabilities

RNDP	Rwanda National Development Plan
RSB	Rwanda Standards Board
SACCO	Savings and Credit Cooperative
SDLD	State Department of Livestock Development
SEAH	Sexual Exploitation, Abuse and Harassment
SGBV	Sexual Gender-Based Violence
SOA	The Sexual Offenses Act
SOPs	Standard Operating Procedures
SSTC	South-South and Triangular Cooperation
SUA	Sokoini University of Agriculture
TAD	Transboundary Animal Diseases
TDB	Tanzania Dairy Board
TMR	Total Mixed Rations
ToT	Training of Trainers
TVET	Technical and Vocational Education and Training
TVLA	Tanzanian Veterinary Laboratories Agency
UDSM	University of Dar-es-Salaam
UMMB	urea-molasses multi-nutrient blocks
VCs	Value Chains
VMGs	Vulnerable and Marginalized groups
YLP	Youths Livelihood Programme

INTRODUCTION

This annex sets out the main parameters of the Gender Equality and Social Inclusion (GESI) assessment and action plan for implementation of DaIMA's interventions in Tanzania, Kenya, Rwanda and Uganda. Part 1 provides a contextual GESI analysis of smallholder dairy farmers and MSMEs (dairy cooperatives and processors) in each respective country. Identifying the key challenges for gender equality and women empowerment, youth empowerment, nutrition, marginalized and vulnerable groups and persons living with disability (PWDs) in dairy production and associated services enterprises and proposes required interventions. Part 2 provides the gender action plan setting out proposed actions to achieve the Gender Equality and Social Inclusion (GESI) targets. These will be further reviewed and revised during project start-up.

The DaIMA theory of change is "IF technological solutions to reduce methane and GHG emissions and financial services to efficiently sustain production and marketing processes, which increase revenue and/or reduce production costs, are supported for dairy farmers and milk processors to adopt in East Africa, THEN farm-level methane emissions and supply chain GHG emissions will be reduced, because a cross-cutting transition to net-zero dairy production will be economically attractive to farmers, pastoralists and dairy sector processors". The key objectives of DaIMA are: (i) To strengthen systemic and institutional capacities of the livestock sector to enable smallholder dairy producers and local value chain actors to reduce CH₄ and other GHG emissions (ii) To increase production, market knowledge and tools for smallholder dairy producers and private sector value chain operators for a low-emission, climate-resilient, and sustainable pathway (iii) To enable smallholder dairy producers and local private sector value chain actors access to required financial services to make the necessary transition to low-carbon, climate-resilient, and sustainable livestock production and value chain development.

PART 1: GESI ANALYSIS

Agriculture plays a significant role in the economy, providing employment and income to millions of people. As a result, the impact of climate change on agriculture may have long-reaching consequences that extend well beyond individual farmers and their families. According to the United Nations, this output is mostly generated by family farms, where women supply around 60 to 80 percent of the labor input. East Africa has demonstrated the positive effects of dairy production on income, food security and dietary diversity. Most of the EAC countries have ratified international frameworks and progressive national legal and policy frameworks promoting gender equality and women empowerment. However, women in agriculture face gender-specific barriers, including limited access to land, financing, and other resources¹. Policy implementation remains weak in country and regional disparities persist. Each country, therefore, presents a unique context with varying statistics and progress towards closing the gender gaps. Nevertheless, present data reveals some progress towards achieving the stated policy commitments for the four DaIMA project focus countries and the need to address barriers in the dairy sector as outlined below.

Rwanda

In Rwanda, maternal and child health has improved significantly over the last two decades and exceeded Millennium Development Goals (MDGs) ambitions. Maternal mortality reduced by 80% between 2000 and 2014 while infant and child-mortality decreased by over 70% in the same period.² Free universal basic education was initiated and scaled up

¹ FAO 2019. ESA Working Paper No. 11-02 March 2011 Agricultural Development Economics Division. The Food and Agriculture Organization of the United Nations www.fao.org/economic/esa

² The National Institute of Statistics of Rwanda (NISR), 2015. Demographic and Health Survey (2014/15). Kigali, National Institute of Statistics of Rwanda.

to 12 years. This resulted in a net enrolment of nearly 100% in primary school for both boys and girls. Gender parity was achieved with more girls than boys in primary school.³

Women outnumber men (92% compared to 77%, as per 2012 population census) in the rural population working in agriculture.⁴ About 25% of the (rural) households in Rwanda are headed by women and working age youth (ages 15-34) constitute 77% of the rural population. Women account for only 24% of the agro-dealers in Rwanda. Women and men farmers in dual households are characterized by unequal power relations, and women have limited decision-making powers. Within the dairy value chain, women have distinct and significant roles, although those roles vary by production system. In extensive systems, men and boys predominately bring the animals to graze and find water, while women care for the calves and process the milk. In zero grazing systems, which are required for participation in the GIRINKA⁵ program, women are responsible for feeding the animals, cleaning the stalls and milking utensils, while men are more involved in the milking of cows as well as the transport and sale of milk. These social and gender dynamics related to both women's roles in the production as well as consumption of livestock, if not fully considered, limit the effectiveness of such programs, or worse, increase gender inequalities within the household. Such cases may occur, for instance, when men take over decision-making and control of income from a livestock product, such as dairy, that has become more lucrative following the intervention⁶. Additionally, women in agriculture are more vulnerable to climate change and land degradation because they generally do not have other livelihood options to support their families.⁷

Uganda

The Government of Uganda has identified gender equality and women's empowerment as critical to achieving accelerated socio-economic transformation. The country has ratified key gender equality instruments and commitments and has put in place a legal framework to promote Gender Equality and Women's Empowerment (GEWE). Although the Government of Uganda has established the Gender Equality Commission, which produces an annual report on the state of gender equality in the country, and signs Gender Equality Compacts with Ministries, Departments and Agencies to track the country's progress in improving the livelihoods and well-being of the most vulnerable, gaps remain in the economic, political and social inclusion of women. Men generally own cattle while women have ownership over the milk with women tending to engage more in dairy processing. Women are hindered in dairy development by a lack of ownership, access to resources and assets, which also results in lack of collateral for financial instruments. Generally, at household or farm level women's decision-making power can be assumed low as also seen in farmers groups, in which they seldom hold leadership positions. The proportion of women in the cabinet rose from 9.4 percent in 1996 to 32.8% in 2003 but decreased to 19.4 percent in 2006. However, most women in Uganda are engaged in livestock yet have limited access to agricultural assets, including large-livestock, land, labor and credit. In most communities, land and other productive assets belong to men, and women's access is mediated through men.

Tanzania

Women constitute 52 percent of the population in Tanzania. 75 percent of these women live in rural areas. Of these, 90.4 percent work in the agriculture sector, 0.1 percent in the mining sector and 1.2 percent in small and medium enterprises (SMES). 44.1 percent of women in the urban centres work in SMES and mainly sell clothes, food and non-

³ NISR, 2017. *Statistical yearbook 2016*. Kigali, National Institute of Statistics of Rwanda.

⁴ Agriculture Policy of Rwanda 2018

⁵ Rwanda National Government programme where a poor family receives a heifer by way of donation. When a cow gives birth, the first female calf is given to a neighbour or community member. Once that cow gives birth, its first female calf is given to the next selected beneficiary and so on

⁶ Umuzigambeho, J.F. 2017. Gender assessment of dairy value chains: evidence from Rwanda, Rome, FAO. <http://www.fao.org/3/a-i6845e.pdf>

⁷ ibid

alcoholic and alcoholic beverages. Among the challenges experienced by women doing business include lack of capital, materials, markets, expertise / skills, knowledge and physical spaces to conduct business.⁸ The constitution recommends 30 percent among all appointed and elected officers.⁹ Women and youth tend to have limited access to and control over the resources and information needed for dairy production, value addition, and marketing. Men generally own cattle while women have ownership over the milk. Women are hindered in dairy development by a lack of ownership and access to resources and assets, which also results in lack of collateral for credit. Generally, at household or farm level women's decision-making power is low as also seen in farmers groups, in which they seldom hold leadership and decision-making positions.

Mostly, gender relations vary with diverse cultural contexts, and this diversity should inform analysis and interventions. In traditional dairy production systems, gender roles vary with geographies. For example, in the Southern Highlands of Tanzania, milking is a man's activity, but not a taboo for women, whereas in Tanga both women and men can milk. In both these areas, men own the cattle and women the milk, but the man can sell the milking cow and therefore the woman loses the milk. Youth too often have limited rights over household assets, control and ownership of commodities. As value chains commercialize and markets become sophisticated, moving further away from the farm gate, women tend to lose control of the commodities in question. For as long as the production of milk is low and the markets rudimentary, milk shall remain a woman's commodity. Once the productivity increases to profitable magnitudes and the value chain becomes commercialized the ownership of milk is likely to change to men. To address these, lessons learnt from the Heifer Project International (HPI) Tanzania, which DaIMA foresees as implementing partner, uses participatory sensitization meetings with women and men at household and farmer group level to develop agreed gender strategies and action plans for mainstreaming and transformation. Couple training in the Gender Action Learning System (GALS) where women have multiple livelihood options, such as dairy production, crop residue preservation for fodder production and fodder harvesting "cut and carry" from communal lands, raise poultry, collect and trade with non-timber forest products such as medicinal herbs and gums (for food, medicine and other uses). With support for the various livelihood options, the women can still survive the worst-case scenario of losing milk business to men. The DaIMA and the IFAD Climate Smart Smallholder Dairy Transformation Project (C-SDTP) interventions have much complementarity including areas of geographical coverage. DaIMA will, support additional geographical areas such as the peri-urban dairy communities in the coastal region as well as some pastoral communities not included in the C-SDTP.

Kenya

women have a right to own land according to law, however socio-cultural practices favour male inheritance. Only about 7 percent women own land alone. Without secure rights to land, women's participation in and benefit from land uses such as dairy is inhibited. Yet women provide up to 75% of agricultural labour in addition to heavy workloads related to unpaid childcare and other domestic chores. They produce food crops for domestic consumption and provide labour for commercial farming where they are paid a pittance. Women also have less access to information, extension services and climate smart technologies. The situation is compounded by the delineation of agriculture as a function of the county government necessitating coordination and cooperation between the national and county governments.

For the various activities within the livestock sub-sector, women contribute between 60-80 percent of the labour-force, especially in ASAL areas. On average, women work longer hours (12.9 hours) than men (8.2 hours), yet they earn less because most of these hours are taken by unremunerated work despite contribution to mainstream economic activities. Enrolment of both boys and girls in primary and secondary schools has been increasing from 2009

⁸ The policy for gender equality and women empowerment (Tanzania sera ya maendeleo ya wanawake na jinsia) 2000.

⁹ The Constitution of the United Republic of Tanzania 1977 and Zanzibar 1984.

to 2018 and is almost at par with 49.1 percent boys and 47.3 percent girls. Females also account for 41.3 percent of public and private university enrolments, 43 percent of technical and TVET institutions and 69.2 percent adult education learners. Performance of girls has shown remarkable improvements especially in Modern Agriculture, Manufacturing, Wholesale finance and insurance. Generally, boys perform better than girls, but girls also perform well.

Maternal mortality is improving, the Kenya Demographic and Health Survey (KDHS) 2014 the ratio stood at 362 deaths per 100,000 live births compared to a high ratio of 488 deaths per 100,000 in the 2008-09 KDHS. The prevalence of HIV is higher among women compared to men. In 2017, the National adult HIV prevalence rate had reduced and was estimated at 4.9% with prevalence higher among women (5.2%) than men (4.5%). Retrogressive and harmful cultural practices continue to limit women's empowerment. Prevalence of FGM in Kenya was 32% in 2003; 27% in 2008/9; and 21% in 2014 for women 15 – 24yrs. Prevalence of FGM varies among ethnicities was highest among the Somali – 94%; Samburu – 86%; Kisii 84; and Maasai – 78%.

Climate change is one of the greatest environmental and development challenges facing the world today. Women bear most burden related to gender roles such as in access to water for domestic use and livestock. Dairy provides an opportunity for women to earn income from the sale of dairy products, have decent jobs, and improve their nutrition and provide for their household's general wellbeing.

Youth empowerment. Youth in East Africa accounts for over 40 percent of those between the ages of 15-35. Youth unemployment remains high and is higher than among young women than men. In rural areas youths are informally employed (and underemployed) in subsistence agriculture and family-based livelihood activities. This demographic dividend has tremendous potential to transform the supply and demand of food and will impact the agri-food industry. As the largest employer in the country, the agriculture sector remains a key entry point for job creation, inclusive growth and poverty reduction.

However, most young people face barriers to participate in dairy due to huge capital and operational investments required. Involving the youth in dairy farming is crucial as it provides a platform to tap into the potential of a young, educated population that can be trained and skilled to meet input and services delivery gaps necessary to develop sustainable agri-value chains and food systems. Youth are resourceful, innovative and crucial for the future of the dairy sector. Youth's openness towards new practices can be key in development of, for example, new and environmentally responsible dairy practices and to fully utilize the potential of new technologies such as ICT for the dairy sector. For example, it is evident that the 23% growth of the Kenyan dairy sector triggered by increased demand of milk has resulted to many employment opportunities along in support activities for agriculture. The youth mainly involved in the informal jobs, which are majorly self-employment opportunities, include mobile milk traders, milk bars and shops and kiosks. Youth participation in dairy is hampered by limited access to productive resources, including land, capital, limited entrepreneurial skills, poor rural infrastructure, mindset barriers and drudgery of farm labour due to limited access to modern technologies. Addressing these constraints will raise the industry's productivity, which in turn would create more jobs for the young women and men.

The 2003 Rwandan Constitution enshrines gender equality and social inclusion and sets out the obligations of the State. In 2019, MINAGRI launched its Gender and Youth Mainstreaming Strategy aimed at providing women and youth with increased knowledge and access to services, as well as enabling them to participate equally in the value chain. The youth population in Uganda comprises of both the educated/skilled and the unskilled living majority in rural areas. Two-thirds of the labour force is below 35 years. Although 65.5 per cent of the population works in agriculture, its contribution to national GDP is only 29 per cent. For the youth, having access to productive assets, being linked to a market, have an opportunity for access to credit and finance, and working with modern and digital technologies are important. Specially, creating jobs and decent employment for youth in dairy and beef value chain, can reduce rural

poverty, improve food security and nutrition in the country. In Tanzania. Youth (15-35 years) and children (0-15 years) account for 75% of the population, with a median age of 18 years old. Two-thirds of Tanzania's labour force is younger than 35 years. Around half of the 16 million youth are, however, engaged in farming (off-farm agribusiness: 24%). Young men more often than young women find non-agriculture employment, approximately 60% versus 40%.

The Government of Tanzania (GoT), specifically the Ministry of Agriculture (MoA), is implementing the Build a Better Tomorrow – Youth Agribusiness Initiative 2022-2030 (BBT-YAI), with the following strategic objectives: i) inspire youth through implementing behaviour/attitude-changing communication strategy; ii) empower youth through training mentoring and coaching; iii) engage youth in profitable and sustainable management of agribusinesses; iv) enable youth-led enterprises by improving business environment; and v) coordinate effectively youth agribusiness support initiatives for synergy and efficiency. Through partnerships between the private sector, NGOs and government institutions, the MoA anticipates achieving 12,000 new agribusiness enterprises in villages across the country, for which 200,000 youth are trained via internship and 15,000 youth are mentored through incubation programmes. For livestock specifically, youth involvement is less well documented. There are eight Youth Incubation Centres for beef and goat fattening that will take 240 and 1,000 youth in 2022/23 and 2023/24 respectively. Collective action for the youth has been explored and has successfully worked in Zanzibar, as a result of the Agricultural Sector Development Programme–Livestock (ASDP-L) project. Communal Cowsheds for dairy are a type of Livestock Farmer Field School (L-FFS) that is dependent on social cohesion from which youth learn and farm dairy together¹⁰. The Communal Cowshed Concept could serve as a dairy production incubator for youth, having the young farmers passing through and starting their own dairy business from there.

Kenya's population is largely young, with 35.7 million Kenyans (75.1%) below the age of 35 years and offers a dynamic workforce that is highly adaptable, with a high uptake of technological innovations. Yet, youth record a high unemployment rate of 35%. The key challenges for youth participation in agriculture include negative perception and attitude to agriculture, inadequate skills, knowledge, and information such as in dairy farming, limited agricultural innovation, limited access to finance and resources such as land. The GoK has put in place various legal, institutional and policy frameworks to empower the youth at national level and county levels. However, most counties are yet to enact youth policies.

Marginalized and vulnerable groups. In Rwanda, the 2003 Constitution does not recognize the economic, social and cultural rights of the Batwa and it only refers indirectly to them under the term “historically marginalized communities”, a term that during the constitution-making process was generally understood to refer to the Batwa. The Constitution of Uganda provides a list of indigenous communities under its Third Schedule and bears no compatibility with the definitions of indigenous peoples under international and regional human rights law – in particular the emphasis on the right to self-identification – and can therefore not be interpreted as a similar legal protection of the rights of indigenous peoples within the meaning of international and regional human rights law. However, the National land policy does make considerations for pastoralists such as the Karamoja community living in the North. Tanzania consists of over 100 distinct ethnic groups (in majority indigenous of Bantu and Nilotic origin) and with no group dominating. However, Tanzania policy is conservative in specifically identifying indigenous peoples. Ethnic controversy and conflict are low in Tanzania compared to neighbouring countries. Tension between peoples of sedentary smallholder mix farming and pastoralism is, however, increasing with climate change. Sector transformations, private financial investments, and policies favour sedentary over pastoral livelihoods.

Protection of pastoral peoples' rights can be difficult, since they often migrate (in marginal areas) across national borders and are not seen as indigenous to a nation, region or district. In the Southern Highlands and Zanzibar, indigenous people are sedentary smallholder farmers combining agriculture, horticulture and livestock farming. From

¹⁰ IFAD 2022. https://www.ifad.org/documents/38714170/47116743/Stocktaking_ESA_web.pdf/cc8f6720-6dc1-5a92-f61b-4bccc4d1d15?t=1671547955602

Tabora and Singida, indigenous pastoralists migrate to Mbeya and Iringa in search for grazing. In Tanga pastoralists and smallholders, both indigenous to Tanga, coexist with their free-grazing livestock, sharing communal pastures. Cross-border migration from Kenya and Rwanda respectively takes place. GoT allocated land owned by the National Ranching Company to new farmers (including pastoralists deciding to become farmers) through tripartite loans. Consultation with selected members of the Maasai community in Matebete Village, Mbarali district in Mbeya region, revealed that 12,750 hectares of land that they currently use for their settlements and livestock grazing used to be the government ranch in the 1970s. The beneficiaries of this land, however, still move their cattle to other areas in search for fodder and water, especially during dry season. C-SDTP is not expected to target indigenous people's territories and will not alter their culture nor marginalize them, neither should DaIMA.

An estimated 79,000 people identify as indigenous in Kenya, and are found among the pastoralist communities of Turkana, Rendille, Borana, Maasai, Samburu, Ilchamus, Somali, Gabra, Pokot and Endorois communities. Some of these groups may be found living in the project areas of Nandi, Elgeyo Marakwet and West Pokot. They experience land and resource tenure insecurity, poor service delivery, weak political representation, as well as general discrimination and exclusion. It is also important to recognize that they are valuable partners and holders of local traditional knowledge that can improve nutrition and enhance biodiversity. Marginalised groups also include Persons Living with HIV and AIDS (PLWHAs) and Persons with Disabilities (PWDs), who are estimated to be about 4.6% of the population.

People living with Disability. The EAC Secretariat has developed a regional EAC Policy on Persons with Disabilities (PWD) 2009 and each country has a related Policy and institutional framework to manage affairs of the PWDs. People with disabilities are among the most vulnerable groups in society. They are often undereducated, untrained, unemployed or underemployed. They are also likely to be extremely poor – especially women, youth and those living in rural areas. In Tanzania, movements supporting / protecting the rights of people with disability are quite well established. In the four countries, different disabled persons' organisations and the umbrella organisation regularly take part in discussions with the government on issues affecting the lives of people with disabilities. Therefore, there is need to systematically equity, and ultimately equality of people with disability and living in rural areas where they rely on agriculture and related activities with the rest of the rural population. This can be achieved by providing them with opportunities to engage in agri-business and supporting them with access to training, finance and productive resources to participate in the dairy sector.

Nutrition. There has been modest progress towards achieving global nutrition targets in the East Africa region. However, not a single country in the subregion is on course to meet the targets for anemia in women of reproductive age (aged 15 to 49 years), low birth weight, diabetes among men, diabetes among women, obesity among men, and obesity among women. The latest data shows that anaemia affects an estimated 31.9% of women of reproductive age. Some 13.4% of infants have a low weight at birth in the Eastern Africa subregion. East Africa still experiences a malnutrition burden among children aged under 5 years. The average prevalence of overweight is 4.0% - the second lowest compared to other subregions in Africa with sufficient data. The prevalence of stunting is 32.6%, which is significantly higher than the global average of 22.0%. Conversely, the East Africa subregion's prevalence of wasting is 5.2%, which is lower than the global average of 6.7%. In addition, the region's adult population also faces a malnutrition burden: an average of 7.0% of adult (aged 18 and over) men live with diabetes, compared to 6.5% of women. Meanwhile, 12.1% of women and 3.5% of men live with obesity¹¹.

In Rwanda, anaemia among children went down from 52% in 2005 to 38% in 2010 but this prevalence is still high. Among children under one year of age, seven out of ten are suffering from anaemia. Among pregnant women, (20%) were found to be anaemic. Seven percent of Rwandan women are too thin and 16 percent are overweight or obese.

¹¹ <https://globalnutritionreport.org/resources/nutrition-profiles/africa/eastern-africa> assessed 13 December 2023

Overweight and obesity is higher in urban areas with 30% prevalence for women in the City of Kigali. Vitamin A and Iodine deficiency has reduced more than anaemia, which mainly affects pregnant women and children six to 24 months. During pregnancy and from 6-24 months are periods of rapid growth and more iron is needed. The demand cannot be met by diet alone. In 2010, anaemia prevalence in Rwanda was 25% during pregnancy and 17% among women of reproductive age. More than 70% of children 6-12 months were found to be anaemic. Malnutrition of young children adds permanent risks to their health, growth and cognitive development as well as a loss to their economic productivity estimated at up to 10% over a lifetime. Cumulating these negative effects can result in economic losses of up to 3% of GDP.

In Uganda, gender relations play an important role in food security and nutrition management in the community. Women produce most of the food and with limited resources, but they have no control over this food. Food and nutrition data available for Uganda are over two decades old and need to be updated. Even so, in 2003, the infant mortality rate was at 88.4 per 1,000 live births while the under-five mortality rate was 152 per 1000 live births. The Maternal Mortality Ratio (MMR) was estimated to be 495 per 100,000 live births.¹² Pockets of malnutrition, famine and hunger exist and childhood under-nutrition still persists. 40 percent of deaths among children were due to malnutrition. Over thirty-eight per cent of the children below 5 years were stunted, 4.0 per cent were wasted and 22.5 per cent were under weight.¹³ Micro-nutrient deficiencies are common, especially vitamin A deficiencies which had a prevalence rate of 5.4%, iron deficiency anaemia was slightly more than 50%, while 10% of the women population were undernourished. The total goitre rate ranged between 60-70%¹⁴. Under-nutrition is widespread, affecting mainly children less than 5 years old, school-going-age children, adolescents, women in the reproductive age, internally displaced persons and prisoners.¹⁵

Malnutrition in Tanzania remains high. Over 34% of children under the age of five are stunted and nearly 45% of women of reproductive age suffer from anaemia. The most affected regions, with a prevalence of stunting exceeding 40%, include the semi-arid areas of Ruvuma (41%), Iringa (47%), Rukwa (48%), Kigoma (42%), Njombe (54%), and Songwe (43%). In Pemba North, 39%, and, in Pemba South, 31%, of children are stunted. In Zanzibar, stunting rates range from 20 to 24% in Stone Town and Unguja North, respectively. 14% of children under five are underweight (too thin for age). Inadequate caring and feeding practices, low nutrition education on dietary needs of different age groups (young children, adolescent girls, women of reproductive age, pregnant and lactating mothers, etc.) remain an issue. Studies also show that 59% of the Tanzanian population cannot afford a nutritionally adequate diet. To improve access to milk, a nutritious (high protein animal source food), dairy farmers need to be trained on the importance of allocating some of the milk (mainly evening milk) for household consumption and the importance of making other animal source foods such as chicken and eggs available for consumption at home, rather than selling. For the sake of nutrition, protection of women's livelihood options such as poultry, and collective action in the milk cooperative will empower these women on knowledge on nutrition. DaIMA, partners e.g. GoT, HPI etc can support school milk feeding programs by applying lessons from past successful interventions such as in Kenya whereby providing school milk increased nutrition, attendance and enrolment.

In Kenya, lack of access to adequate and diversified diet results in various forms of nutrition problems. In 2011, over 2 million children (35%) were reported to be stunted. Children under five years are particularly affected by deficiencies in vitamin A (84% of children), iron (73.4%), and zinc (51%). Women, especially pregnant women, are among the most vulnerable with a high risk of iron deficiency (60% among pregnant woman) and vitamin A. An estimated 16% of adult

¹² UDHS, 2000/2001

¹³ *ibid*

¹⁴ Uganda government 2003. The Uganda Food and Nutrition Policy

¹⁵ *ibid*

males suffer from iron deficiency (anaemia). Kenya is also increasingly faced with diet-related non-communicable diseases, especially in urban areas. These are mainly caused by excessive energy intake associated with purchased meals and processed foods and decreasing levels of physical activity in urban settings. Additionally, millions of children and adults suffer from the ill health effects of food-borne diseases.

Gender Based Violence (GBV) and Sexual Exploitation Abuse and Harassment (SEAH). In Rwanda, cultural and patriarchal stereotypes and attitudes perpetuate GBV. Negative gendered attitudes, roles and expectations such as a culture of silence, traditional settlement of GBV cases within families is common. This compromises justice for survivors. Men and boys are often left out of the dialogue and seen as perpetrators rather than allies in GBV prevention. For male survivors, service seeking is limited mainly by the fear of ridicule, loss of influence and space in the community of men. Some forms of GBV are culturally accepted complicates and compromises response measures. Coupled with gendered power imbalances, the culture of silence and the high levels of tolerance and non-reporting, the practice of GBV in Rwanda has created an internal self-sustaining cycle which can only be broken through combined efforts of transformative and strategic engagement with communities and their leaders and law enforcement¹⁶.

In Uganda, the 2016 Demographic and Health Survey revealed that seven in 10 women who had suffered sexual violence neither sought help nor told anyone. Factors that contribute to under-reporting of GBV include social stigma and shame, fear of reprisal, concerns about confidentiality and being believed, and, in some cultural settings, a high tolerance for violence. Most Ugandans think the police take GBV cases seriously, but majorities also see domestic violence as a private rather than a criminal matter and think that women who report GBV to the authorities will face negative reactions from the community. The negative social norms often accept violence such as wife battering, child marriage, FGM, and in some cases rape as normal societal occurrences.¹⁷ Sexual gender-based violence predisposes survivors to HIV, unwanted teenage pregnancies, and mental illnesses like depression. This leads to dropouts among adolescent girls and young women (AGYW), increased risk of HIV transmission, and socio-economic challenges related to child pregnancies. Uganda has a high prevalence of SGBV (22%) among women 15-49 years. While SGBV among AGYW does not necessarily lead to pregnancy, an analysis of District Health Information System (DHIS2) data indicates that an increase in SGBV among AGYW occurred with an increase in teenage pregnancies.¹⁸

Tanzania has a supportive policy and legal framework for GBV. However, many forms of GBV, including intimate partner violence and rape, are tolerated by both men and women—although the justifications for acceptance differ. Women and girls are also frequently blamed for causing or provoking GBV. In part due to blame and shame, women and girls rarely report GBV to authorities or seek other kinds of treatment or support. For example, President Kikwete publicly stated that GBV should be included as one of the Millennium Development Goals. Furthermore, Tanzania's National Strategy for Growth and Poverty Reduction lists violence against women as one of its indicators of poverty—which is rare among poverty reduction strategy papers in other countries.¹⁹

The Kenya Demographic and Health Survey, shows that 45 percent of women aged 15-49 have experienced either physical or sexual violence, while nearly half (47 percent) of those ever married experienced either emotional, physical or sexual violence. Women are also largely at risk of FGM/C, the prevalence rate in Kenya being 27 percent. Furthermore, children also experience sexual violence which are usually perpetrated by persons the victims know.

¹⁶ Republic of Rwanda 2011. National Policy against Gender-Based Violence.

http://197.243.22.137/migeprof/fileadmin/_migrated/content_uploads/GBV_Policy-2_1_.pdf

¹⁷ UNFFPA 2024. Ending Gender Based Violence and Harmful Practices. <https://uganda.unfpa.org/en/topics/ending-gender-based-violence-and-harmful-practices>

¹⁸ Sexual Gender-Based Violence among Adolescent Girls and Young Women 10-24 Years during COVID-19 Pandemic, Bukedi Region, Eastern Uganda. Uganda National Institute of Public Health. Quarterly Epidemiological Bulletin: October–December, 2022 Volume 7 / Issue 4 / Article No. 4.

¹⁹ USAID 2008. Gender-based Violence in Tanzania: An Assessment of Policies, Services, and Promising Interventions. http://www.healthpolicyplus.com/archive/ns/pubs/hpi/Documents/1454_1_TZ_GBV_Summary_FINAL_6_8_11_acc.pdf

However, social stigma and family shame result in the knowledge of the abuse being kept within the family. At times, families are paid “damage money” to buy their silence. This culture of secrecy serves to aid the impunity of the perpetrators.²⁰ In regard to GBV and Sexual exploitation, abuse and harassment, DaIMA will seek to support prevention of GBV and SEAH, by creating awareness, supporting reporting and referral of cases through establishment of Grievance Redress Mechanisms in the IFAD baseline projects as detailed in the Gender Action Plan.

Programme beneficiaries

DaIMA will benefit about 590,000 households (HHs) of which 145,000 HHs will be involved in dairy value chains. The project will target 40 %women, 20 % youth (50 % female) and 5 % vulnerable groups including minorities and PWDs. The majority are dairy farmers in intensive and semi-intensive systems who currently sell or aim to sell milk in formal and informal markets. These smallholder dairy farmers have generally between 1 to 5 dairy cows (6 to 10 kg of milk/cow/day) in zero-grazing/intensive or semi-zero grazing/semi-intensive systems. They are the essential segment of producers who need to be scaled to support the formalization of the dairy sector and foster sustainable intensification based on sound investments and viable business development (see Feasibility Study and Market Studies). Dairy value chain support will also benefit medium and large commercial farmers (6 or more dairy cows, 5-9 kg of milk/cow/day in semi-intensive systems, 12 or more kg/cow/day in intensive systems). Under Output 2.3, the Programme will also benefit (i) pastoralists and agro-pastoralists, and (ii) medium- and large-scale extensive farmers through the rehabilitation/ improved management of 268,000 hectares of rangelands. Entry points to reach dairy farmers will also benefit from the Programme’s support, including dairy cooperatives and milk collection centres (MCCs), L-FFS and Pastoralist field schools (PFS), farmers and community-based organisations. These smallholder farmers will be mainly targeted through Component 1 and Component 2. Beneficiaries of Component 3 will be the larger formal stakeholders in the value chain (dairy cooperatives, private enterprises, processors, input suppliers) who have the capacity (with the support of technical assistance) to prepare business plans for green investments, submit funding applications, and manage implementation. These formal stakeholders will provide backward and forward linkages, extension services and agro supplies to the smallholders.

²⁰ 2012. Sexual and Gender Based Violence in Kenya: A Call For Action Overview of Gender Based violence
<https://ncpd.go.ke/wp-content/uploads/2021/02/Policy-Brief-26-Sexual-and-Gender-Based-Violence-in-Kenya.pdf>

The target beneficiaries for DalMA will include:

- smallholder dairy farmers/ entrepreneurs, pastoralists, input and service providers including dairy processors and their communities, including women, youth, children and vulnerable groups such as people with disabilities and indigenous people involved in dairy keeping.

Others who will benefit in order to facilitate services to smallholder dairy farmers will include;

- ***financial service providers:*** Participating Financial Institutions (PFIs) which will be offering financial services to the small holder dairy entrepreneurs and MSME markets e.g., dairy cooperatives and SACCOs and other micro-finance institutions (MFIs).
- ***other service providers:*** such as advisory and extension staff, input suppliers, agrovets, value chain actors;
- ***program management:*** DalMA Facility Coordination Unit at IFAD and implementing partners or focal points at country subsidiaries.

Targeting strategy and approaches

The targeting strategy will comprise of: (i) geographic targeting, to identify participating Districts and regions and sites in Uganda, Kenya, Tanzania and Rwanda, where dairy has been developed and there is potential to develop the identified VC interventions, and where mature clustered farmers exist for marketing potential. The programme intervention areas are identified based on the integration of: (i) the climate risk hotspots; (ii) absolute methane emissions from the dairy sector; (iii) soil carbon sequestration potential in pastures and rangelands; (iv) milk production by district/county; and (v) complementarity with other international financing institutions (IFI)-funded interventions, more particularly from IFAD (see figure 1 below); (ii) direct targeting of the vulnerable households, including those women headed, youth, vulnerable and marginalized and PWDS to reach the 40 per cent and 30 per cent quota for youth (iii) self-targeting, with activities geared towards the needs of small and medium producers that are engaged in dairy and iv) Enabling and capacity-building measures to develop the capacity and self-confidence of those with less of a voice and less power so that they can articulate their needs and participate in planning, decision-making and project activities.

PART 2: GENDER AND SOCIAL INCLUSION ACTION PLAN

DalMA will target 40 per cent women, 30 per cent youth, and 5 per cent comprising of persons living with disabilities (PWDs), indigenous people. Women, youth and PWDs will be targeted with technological solutions to reduce methane and GHG emissions, and the financial support services to sustain efficient production and marketing processes. Indigenous people will be protected by ensuring that they will not be displaced from their ancestral land; children will benefit from nutrition interventions, which will include milk provision through the school feeding program and other interventions such as mother nutrition education.

The objective of this DalMA GESI action plan is to ensure that women, youth, people with disabilities and marginalized groups have equal opportunities to participate in, and benefit from, the program as men in these communities. Social inclusion considerations for these groups will be achieved by increasing their participation in project activities, economic empowerment opportunities, access to information and skills as well as linkage to market and value addition. In this regard, the action plan primarily proposes to integrate GESI actions within the project activities and sub-activities and only proposes separate GESI actions such as those to address GBV and SEAH where a direct fit is not guaranteed with core DalMA activities.

Purpose

The purpose of the DaIMA Programme is to increase smallholder dairy production and associated input and service providers' revenue, and/or reduce their production costs through reduced methane and other GHG emissions by adopting the GHG emission reduction technologies and financial services to efficiently sustain production and marketing processes.

Objectives of the Programme

- i. To strengthen systemic and institutional capacities of the livestock sector to enable smallholder dairy producers and local value chain actors to reduce CH₄ and other GHG emissions
- ii. To increase production, market knowledge and tools for smallholder dairy producers and private sector value chain operators for a low-emission, climate-resilient, and sustainable pathway
- iii. To enable smallholder dairy producers and local private sector value chain actors access to required financial services to make the necessary transition to low-carbon, climate-resilient, and sustainable livestock production and value chain development.

Theory of change, assumptions and risks

The programme will integrate GESI considerations into DaIMA Components and activities, complemented by targeted interventions for women and youth to overcome the specific barriers identified. This approach will enable programme opportunities to be expanded to women, young women and men, vulnerable groups and persons with disabilities, while also addressing the main gaps and barriers they face to inclusion.

The GESI theory of change is founded on the need to identify and address systemic barriers and social and negative gender norms that perpetuate inequalities and exacerbate barriers for women, youth and vulnerable groups that preventing from equally participating in and benefitting from dairy production. Therefore, the integration of GESI activities within the Programme Components is critical. Component 1 is about strengthened national capacities and enabling environment for low-emission climate-resilient dairy sector, whereas Component 2 addresses low emission and climate resilient primary dairy production and Component 3 on Green Dairy Financing Facility (GDFF).

DaIMA aims to address the following challenges: (i) strengthening climate resilience, (ii) reducing emissions across the dairy value chain; (iii) the reduction of GHG emissions per kg of milk; and (iv) improving the livelihoods of dairy farmers in East Africa. This requires a holistic approach to improving all aspects of the dairy sector, from improving policy, institutional and service delivery environments to building the capacity of smallholder farmers including women, youth and the marginalized. It also includes measures to facilitate access to finance and innovative technologies with mitigation and adaptation benefits. As a result, female and youth dairy farmers and processors will be motivated to adopt more efficient, climate-resilient practices that will increase incomes, reduce production costs, and help combat climate change. Central to this transformative approach is the symbiosis of private sector support and financial access within the dairy sector. This integration aims to foster market growth while promoting climate resilience and reducing GHG emissions.

To address these gendered barriers, DaIMA will contribute to gender equality and women empowerment women by: (i) Increasing women economic empowerment through access to improved dairy breeds, better access to productive resources and services, marketing and value addition and access to finance by targeting women dairy smallholders in dairy production; (iii) Balancing and reducing women workloads related to dairy keeping through improved technologies, climate-resilient crop and sustainable productions and greater efficiency; (iv) increased women participation, voice and decision-making in dairy farmer organizations and cooperatives (v) Increasing positive gender and social norms such as joint benefits sharing through the GALS methodology and (vi) Contributing to policy

engagement by undertaking gender analyses or developing policy briefs in the planned related policy reviews. These interventions will target women involved in dairy production, young females and males and vulnerable groups including PWDs among beneficiary communities. Addressing barriers to gender equality and women empowerment, youth, nutrition and social inclusion will ensure that project benefits will be equally distributed. This will be integrated throughout the project activities strengthening capacity of those responsible for programme delivery and strengthening of the monitoring and evaluation system to ensure sex and gender disaggregated data. The section below highlights specific GESI interventions that will be integrated under each of the project activities for alignment and ease of implementation by teams.

COMPONENT 1: STRENGTHENED NATIONAL CAPACITIES AND ENHANCED ENABLING ENVIRONMENT FOR LOW EMISSIONS CLIMATE RESILIENT DAIRY SECTORS

1. Output 1.1: Climate-responsive policy and regulatory environment established.

Activity 1.1.1: Review and formulate Dairy Sector policies, strategies and regulations for climate-change adaptation and mitigation. In the four countries, the Programme will support the Ministries, Departments and aligned institutions in charge of the dairy sector to review and formulate national policies, strategies, and regulations that are gender and socially inclusive and essential for a sustainable transformation of the dairy sector, in line with the NDCs and NAPs. This includes aspects related to mainstreaming CC adaptation and mitigation, and adopting export and import regulations that favour sustainable development and private sector engagement and investment in the dairy value chain (taxation of imports of dairy produce, specialized equipment, inputs, such as germplasm, fodder seeds, feed additives, solar cooling, biodigesters, etc.). Resources will be provided to (i) mobilize national and international expertise for policy review and formulation; (ii) facilitate stakeholder consultations to ensure that the policy review and dialogues are inclusive; (iii) validation processes, and (iv) the preparation of policy briefs and communication materials for popularization.

Sub-activity 1.1.1.1 Support the review and preparation of national dairy policies, strategies, and regulations to mainstream climate-change adaptation and mitigation. Priorities in the four countries will be the following. In **Kenya**, (i) the Dairy Industry Strategy and Investment Plan to guide the sustainable transition of the dairy sector; (ii) the draft National Strategy and legal framework for Animal Feed and Fodder; (iii) a National Strategy for Animal Breeding and breeding regulations (iv) the draft Dairy Industry Act (1958); (v) the Kenya Agricultural Sector Extension Policy (KASEP), (vi) the development of compost and bio-slurry standards; and (vii) milk and milk product standards on food safety and quality. In **Rwanda**, the upcoming Strategic Plan for Agriculture Transformation 5 (PSTA5) provides a framework for a low-carbon and climate-resilient agriculture sector. Within this framework, the Programme will support the review of: (i) national climate and sectorial policies that are regulating the dairy sector; a new agriculture and livestock policy for Rwanda is currently under development which includes climate resilient livestock production; (ii) the regulatory framework and standard operating procedures (SOP's) for dairy value chain operators in direct collaboration with Rwanda Food and Drugs Authority (FDA) and Rwanda Standards Board (RSB), for milk quality, standards and safety, feed formulation, energy efficiency, and CC; (iii) the 2016 Ministerial Order to address new dairy value chain realities, including the MCP-model and the need to regulate the raw milk market, whose actors are mainly women and young men. Support will be given to the formulation of a national regulatory framework for the private veterinary practice including for Community Animal Health Workers (CAHWs) and Livestock Farmer Field Schools (L-FFS) facilitators with considerations of increasing women and youth participation; (iv) support will also be given to RNDP for the formulation of the dairy value chain (DVC) regulatory law empowering the platform to conduct overall DVC regulation as in EAC member countries. In **Tanzania**, the Programme will support finalizing the draft national Livestock Policy, as well as the review of: (i) the Animal Health Act (2003) to update aspects related to private practitioners; (ii) the Dairy Act to update regulations on raw milk trade considering that raw milk is mainly traded by women and young men; (iii) the

formulation of disease contingency and control plans; (iii) the draft Tanzania Dairy Development Strategy and roadmap to become carbon neutral, sustainable and inclusive. In **Uganda**, the Programme will support: (i) finalizing the National Livestock Development Policy, integrating CC priorities (breeding, feeding), enabling policies for private sector investments, research strategies on climate-resilient feed germplasm to offset predicted yield declines, as well as rangeland ecology; (ii) the review of the Dairy Industry Act, the Animal Disease and Health Act, and the Animal Breeding Act; (iii) support drafting regulations for animal identification, dairy cattle breeding, livestock insurance) to integrate CC mitigation; (iv) review land policies and regulation to mitigate land tenure insecurity for dairy farmers.

The gender activity will review of existing IFAD investment project baselines to analyse the barriers for women, youth and vulnerable groups and recommend actions where DaIMA will be operating. This will be undertaken using the Pro WEAI approach adapting the IFAD Empowerment Indicator. And will include an analysis of (i) Existing laws, sectoral policies (in review), regulations, and institutional practices (ii) Cultural norms and beliefs related to women and men in targeted communities in relation to dairy production (iii) Gender roles, responsibilities and Time Used in dairy activities (iv) Access to and Control over Assets and Resources and (v) Patterns of power and decision-making to identify gaps and opportunities for advancing gender equality and women empowerment. This analysis will recommend and inform targeted interventions to address identified gender barriers and challenges specific to the dairy sector if not already complimented by ongoing IFAD investments.

Sub-activity 1.1.1.2: Support the regional policy dialogue to mainstream climate-change adaptation and mitigation in regional dairy policies, strategies, and regulations. In coordination with regional and national dairy platforms and stakeholders, DaIMA will support a region-level EAC assessment and proposals for further regional alignment on dairy sector regulations, including export and import regulations, to accelerate regional convergence in dairy standards and market integration, and to enhance climate responsiveness. National stakeholders will be supported to participate in regional events. Under this activity the programme will facilitate women, youth and vulnerable groups participation in the events. This will include supporting women with childcare, special provisions for PWDs access and participation.

Sub-activity 1.1.1.3 National dairy value chain governance and awareness creation. The aim of this activity is to enhance dairy value chain governance and to mainstream the importance of climate-change in this process. At national level, the national dairy boards (KDB, DDA, TDB, RNDP) and other relevant professional bodies of private stakeholders, will be supported to create gender awareness in the dairy policy dialogue, to create awareness and education/capacity building on climate, feed and dairy policies and regulations and how they affect women, youth and the vulnerable.

Sub-activity 1.1.1.4 District/regional/county level dairy value chain governance and planning. Local-level multi-stakeholder platforms involving delegates from dairy value chain stakeholders will participate in the annual planning and review of Programme activities. DaIMA will ensure participants women, youth dairy cooperative members, milk collectors, milk processors, service providers of artificial insemination (AI), veterinary services, extension staff and local government.

Activity 1.1.2 Launch awareness campaigns on nutritional benefits of milk and food safety aspects. This will be a targeted support to milk consumption and nutrition awareness, accompanied by nutrition education and social behaviour change among smallholder producer and consumer households, informal dairy value chain actors and milk handlers in the school milk program. The aim is to improve nutritionally vulnerable groups' consumption of milk, therefore enhancing their individual health and nutrition. The Programme will support milk consumption and nutrition awareness through increased availability, accessibility, affordability, and consumption of high-quality and safe dairy products, accompanied by nutrition education and social behaviour change and communication (SBCC) to reach out to informal dairy value chain actors, milk handlers at home and institutions and nutritionally vulnerable groups. This

activity will target 40 per cent women ensuring male engagement to increase the participation of men in improving household nutrition. Women will be trained as ToTs and facilitated to roll out community outreach campaign sessions and food exhibitions and demonstrations.

Sub-activity 1.1.2.1 Public awareness on the contribution of milk to nutrition and resilience, with focus on climate vulnerable population. The awareness raising will use various community platforms such as extension, L-FFS, MCC/cooperatives will include women and other vulnerable and marginalized groups meeting quotas. Topics will include improved milk treatment, food safety, handling and nutritional benefits of dairy produce, pasteurized and processed milk. In **Rwanda**, it will include participation in dairy fairs/expos under the auspices of RNDP to showcase climate-friendly and energy efficient local processing and consumption of dairy products. In **Tanzania and Uganda**, the Programme will scale up school milk programs for improved nutrition of vulnerable children especially those under five years.

Sub-activity 1.1.2.2 Build capacity of dairy inspection to ensure milk quality and safety and reduce waste. Dairy inspection is essential for sector-level milk quality and safety, thus encouraging formal market growth and reduction of milk rejection and waste (less GHG emission intensities). The Programme will provide technical training and equipment to the actors in the informal dairy sector, especially women and male youth, strengthen capacities of inspections to ensure development and enforcement of milk quality and safety standards. Women and youth will be considered among the inspectors to increase their capacity development and execution of inspection, creating jobs.

Activity 1.1.3 Improve land tenure security for sustainable dairy farming. The Programme will adopt a fit-for-purpose approach to identify and mitigate tenure risks. Activities to mitigate this risk will vary depending on the type of land, type of governance and sex and age of the person accessing and controlling the land. For pastoral and agropastoral systems, the focus will be on the identification of use of rangelands by gender, their protection and supporting gender and social inclusive participatory planning and management. In **Rwanda**, all land is registered and individually owned. The focus will be on optimizing land productivity through land consolidation and land use planning. Activities will include capacity building and support to participatory planning processes to create local-level capacities.

Sub-activity 1.1.3.1: Mapping of local tenure risks and preparation of training materials. The Programme will mobilize specialized technical assistance to: (a) implement a rapid tenure governance capacity assessment, including a tenure risk assessment for the Programme target areas analysing gender barriers for women; (b) develop training material to increase local authorities' capacity to secure gender and social inclusive tenure rights for dairy production and the public's capacity to identify and secure legitimate tenure rights. Based on the national tenure context, each dairy production system's tenure risks in terms of gendered access, use and security of land will be identified and assessed, and appropriate gendered mitigating strategies will be proposed.

Sub-activity 1.1.3.2: Developing capacity on land tenure at decentralized level. The focus will be on capacity building of local authorities and raising public awareness on options to incrementally increase tenure security for women and men of all ages and the vulnerable such as allocation of government land to youth groups. The Programme will support public awareness campaigns on tools available to secure tenure rights for women, youth and the vulnerable.

Sub-activity 1.1.3.3: Participatory planning of tenure for dairy farms on communal and customary land at local level. The Programme will scale-up best practices in gender and social inclusive communal rangelands participatory planning and management. This activity will support the participation of women as community members and leaders in the planning of tenure for dairy farms on communal and customary land for example ensuring the community committees include 40 percent women.

2. Output 1.2 Improved service delivery to the dairy sector

Activity 1.2.1 Strengthen veterinary services for better animal health, higher milk yields and reduced losses. A key barrier to reduce GHG emissions and sustainably increase the productivity of dairy animals is animal health and welfare. Climate change exacerbates the emergence and re-emergence of diseases and infestation (East Coast Fever – ECF, helminths infestation, etc.). Healthier animals are more resilient to climate hazards and more productive, with less milk to be discarded, leading to a reduction of “unproductive” GHG emissions. The higher revenues provide incentives for farmers to adopt animal health and welfare measures, especially if the benefits are equitably distributed among the men, women and youth in the household. To expand the outreach and capacity of veterinary services, the Programme will invest in proximity animal health support, as well as in national diagnostic capacity of veterinary laboratories while supporting equitable capacity development targeting women service providers.

Sub-activity 1.2.1.1 Support the installation of both private and public veterinarians and veterinary paraprofessionals. The Programme will implement the animal health support and disease control system, using the “One Health” approach. The approach will have three levels: (i) support private veterinarians and para-veterinarians, in particular young female and male practitioners; (ii) recruit and support livestock extension workers, while ensuring gender (sex and age) equity in participation at local level; (iii) provide additional training to lead female, male and youthful farmers from the L-FFS for coaching their peers and assist private female and male para-veterinarians and livestock extension workers. In **Kenya**, veterinary education, and training programs to women and men from both private and public veterinarians and veterinary paraprofessionals in the Programme area will be enhanced. In **Rwanda**, support will be provided for the installation of young female and male veterinary doctors and veterinary technicians as private practitioners to enhance accessibility of services while creating jobs. In **Tanzania**, female and male Livestock Extension Workers (LEWs) will be equipped and trained, and private male and female paravets will be incentivised to ensure the last mile animal health and extension services. In **Uganda**, the Programme will build capacity of female and male veterinarians, para-veterinarian technicians, and community-based animal health workers (CAHWs) on One Health principles, disease diagnosis, emerging and re-emerging diseases because of CC.

Sub-activity 1.2.1.2. Strengthening diagnostic capacity of veterinary laboratories and capacity to test quality of animal feed. In **Kenya**, the Programme will improve disease control (surveillance and monitoring, contingency planning, and treatment) of targeted diseases including modern diagnostic equipment in regional centres (located in Nairobi i.e. Kabete and Embakasi, Nakuru, Kericho, Eldoret, Karatina, Garissa and Mariakani), testing kits, telemedicine. In **Rwanda**, four (4) satellite laboratories will receive additional equipment for serology (ELISA) and molecular diagnosis (PCR) to enable rapid detection of infectious diseases, as well as equipment for handling biological waste and the establishment of environmentally friendly practices. The Programme will also strengthen the national veterinary laboratory (NVL) to handle advanced diagnostic capacity for viral diseases affecting livestock. In **Tanzania**, the Programme will: (a) strengthen the Tanzanian Veterinary Laboratories Agency (TVLA) and the Central Veterinary Investigation Laboratory in Zanzibar; (b) support the establishment of district mobile laboratories that will serve LEWs local (para)vets; (c) train female and male laboratory technicians for each mobile laboratory. In **Uganda**, veterinary laboratories will be strengthened (equipment, reagents, consumables for sample collection and mobile laboratory facility) for detection of the diseases related to CC. It will also establish a national data management system on emerging and re-emerging diseases because of CC (TAD and zoonosis). In each country, the Programme will: (i) enhance the capacity to undertake feed testing in the national research institutions (KALRO in Kenya, RAB and UR-CAVM in Rwanda, NARO in Uganda, TALIRI in Tanzania); this includes equipment, training of young female and male researchers, and inputs for testing feed samples; (ii) undertake on-farm feed sampling and laboratory quality testing (e.g., dry matter, carbohydrates, protein, fat, minerals, vitamins).

Activity 1.2.2 Strengthen extension services for low-carbon and climate resilient dairy sector. To reach out to men, women and youth dairy farmers, raise awareness and build their capacity in the field of CC mitigation and adaptation, the Programme will adopt innovative extension models and tools (including digital), and enhance the capacity of extension services. This activity will involve improving delivery of public and private extension services at farm and cooperative/ aggregation levels, by female and male local (district/county) livestock officers who coordinate the community-based extension system. To provide proximity advisory services on climate-smart practices, the Programme will provide support through Livestock Farmer Field Schools (L-FFS), affirmatively ensuring the involvement of female lead and participant farmers, youth, in mixed crop-livestock systems and Pastoralist Field Schools (PFS) in grazing and pastoralist systems.

Sub-activity 1.2.2.1. Preparation of training modules for extension in climate mitigation and adaptation measures.

If required, specific topics that are part of the proposed pathways under Outputs 2.2 and 2.3 and related to CC will be added to the modules of the L-FFS (or other extension approaches). The topics in the training modules related to climate-smart milk production and handling could include: fodder production, harvesting and conservation; use of conserved silage and urea-molasses multi-nutrient blocks (UMMB); supplementation (use of non-conventional feed resources e.g. sweet potato vines or tannin-rich plants reducing and inhibiting CH₄ production) reproductive management; health management including One-Health aspects and management of anti-microbials, biogas and manure management, use of drought-resistant fodder, application of crop residues, seed production (certification, traceability, quality control), improved diet formulation to reduce emission intensities (use of feed formulation software, such as the innovative SNV-developed Rumen8® tool), and animal welfare, including housing in zero-grazing practices. The guidelines will be prepared on improved fodder cultivar and climate-smart technologies, innovative management practices for pasture and fodder production, and agroforestry practices (e.g. leguminous fodder trees as standalone or line trees). The L-FFS modules will be developed with the support of national or international technical assistance. The L-FFS modules will include gender modules with strategies and guidelines to support women, youth and vulnerable groups participation DaIMA will support the selection of female and young trainers.

Sub-activity 1.2.2.2 Training of master trainers (ToTs) and facilitators of extension systems in climate-change related topics. The Programme will support the strengthening of existing L-FFS and the establishment of new L-FFS (aiming at 40 percent women and 30 percent youth). To support the sustainability and scaling up of the L-FFS, existing L-FFS/ PFS male and female master trainers and facilitators will be trained on the specific topics listed above. The training modules will include Gender and related topics.

Sub-activity 1.2.2.3 Promote innovative extension tools. These innovative extension tools could include: (i) participatory extension approaches, such as the Participatory Integrated Climate Services for Agriculture (PICSA), which has been piloted in **Kenya, Rwanda and Tanzania**, (ii) e-extension tools by using bulk SMS and internet to reach more farmers, provide highly captivating and interactive packages, partner with other stakeholders including targeting youth in the development of content and capacity and to improve ability to access and share knowledge and skills on farming technologies, (iii) TV, radio and social media networks. In addition, other innovative extension tools could include: in Kenya, the use of the agro-weather platform namely Kenya Agricultural Observatory Platform (KAOP), a digital platform developed through a joint initiative between the KALRO and the KMD, designed to provide weather information to support decision-making in the agricultural sector in Kenya to develop capacity of extension officers and producers on climate change adaptability. In **Tanzania**, the iCow app and or other innovative extension tool will be developed and promoted. In **Rwanda** digital agriculture solutions such as the *smart-nkunganire* system provides services to farmers.

3. Output 1.3: Improved GHG MRV capacities & monitoring of climate vulnerabilities

To reliably track progress on the ambitious climate goals of the national NDCs, state-of-the-art systems of Measurement, Reporting and Verification (MRV) of GHG emissions are needed within the participating countries, along with national capacity to manage such systems, more particularly in the livestock sector. This Output will enable countries to report with accuracy on emissions in the dairy sub-sector in their national GHG inventories at Tier 2 level, and address the institutional, planning and knowledge barriers identified in the ToC. It will also enable countries to capture the impact of investments in dairy on GHG emissions in their NDC and therefore contribute to their climate commitments. Improved MRV systems will ensure that participating countries can conduct their own GHG calculations with cutting-edge methods, develop their own mitigation pathways and policies, and attract more public and private climate finance, including through carbon credits. This Output will also address the issues related to the data for climate vulnerability assessment and national feed balances, all contributing to improved climate resilience. Countries will receive implementation support from ILRI (Mazingira Centre and research stations) to execute Activities 1.3.1 and 1.3.2.

Activity 1.3.1: Develop capacity in the country on MRV system. The Programme will identify critical capacity needs at national and local levels to improve the operation of MRV systems by young and older women and men. This will help developing consistent MRV systems for the dairy sub-sector that allows systematic updates, so that the impact of interventions of the Programme, and beyond, can be monitored over time, and progress can be assessed against an established baseline.

Sub-activity 1.3.1.1: Capacity assessment and development in capturing, recording, tracking, and reporting data on dairy GHG emissions. This sub-activity will carry out in depth capacity assessments of existing MRV tools and frameworks in the four countries, and the place of dairy in them. Based on these assessments, capacity development will be provided to design activity data questionnaires for Tier 2 GHG emissions in dairy (e.g. animal numbers, weights, productivity, feed rations, animal health interventions, manure management and grassland management) and develop a data management system to ensure reliable flow of information between the various national and local agencies involved in the MRV system. ILRI's Mazingira Centre will provide trainings to young women and men from the countries on data collection protocols adjusted to their needs to relevant stakeholders (technicians, university staff, staff of national research institutes and ministries, and other relevant actors). ILRI will provide dedicated digital data collection tools (e.g. ODK), data quality control, and calculation of GHG emissions for inventory reporting, following established protocols.

Sub-activity 1.3.1.2: Developing MRV systems for the livestock sub-sector. This intervention will include baseline, measurement of individual mitigation activities, impact assessment on absolute emissions and emission intensities and institutional arrangements. To ensure sustainability and impact at scale, the sub-activity will update of national livestock census questionnaires to include activity data requirements for GHG inventories outlined by IPCC on other cattle and livestock. This will allow countries to track livestock productivity and to report on enteric CH₄ and manure CH₄ and N₂O emissions, and soil carbon storage in grasslands/ rangelands, following international inventory guidelines, using IPCC Tier 2 methodology.

Activity 1.3.2: Improve GHG emission measurements in dairy value chains. On-farm data like animal live weights and milk yields are necessary to calculate Tier2 GHG emissions. Emission factors also need to be improved to reflect local conditions (e.g., animal metabolism, feed composition). This activity will comprise the following sub-activities:

Sub-activity 1.3.2.1: Collect activity data on farm to calculate Tier 2 dairy emissions under various herd, feed, animal health and manure management situations. The Programme will finance the collection on-farm activity data for baseline and “with-project” situation, using digital tools (e.g. livestock Tier 2-specific ODK tool). This includes data on animal numbers, weights, productivity, feed rations and manure management system. The measurements will be disaggregated by sex and age of the owner if the animal. This sub-activity will be performed annually with the support of ILRI, in direct partnership with national research institutions and academia of the four countries, to ensure sustainability and replication through the transfer of capacity (see output 1.4). This activity will inform the Programme’s M&E system and provide data for baseline and “with-project” situation.

Activity 1.3.4: Climate risk and vulnerability assessments. Together with improvements to extension services under output 1.2, the Programme will seek to improve the accuracy and use of information (climate, feed availability) to last mile-users in the dairy sector especially women as livestock caretakers. This will include building the capacity of devolved CC units and decentralized meteorological offices to develop and disseminate climate and usage of information services by different groups, including (i) impacts of CC on the dairy sector; and (ii) information on availability of feed, through tools like national feed balance sheets.

Sub-activity 1.3.4.1: Climate risk and vulnerability assessments, including impacts on the dairy sector, at local level. It will consist of carrying out dairy sector climate risk and vulnerability assessments at local level, based on historical and projected climate data. This will involve the development of agro-climatic zone maps and Dairy Sector suitability maps based on the current climate (1991-2020 period). The assessments will include gendered aspects related to vulnerabilities faced by women and other groups such as pastoralists etc.

4. Output 1.4 Knowledge management and regional cooperation enhanced.

The Programme will facilitate transfer of knowledge through regional and South-South and Triangular Cooperation (SSTC), with the objective to replicate the transformative practices for climate-resilient and low-carbon dairy adopted by the Programme. The EAC / AU-IBAR will contribute to the implementation, by facilitating regional platforms, sharing experiences, exporting best practices, and spearheading policy dialogue. ILRI Mazingira Centre, as a State-of-the-Art environmental and regional research and education Centre, will also contribute. This Output will address the institutional, planning and knowledge barriers identified in the ToC. Shared knowledge will enhance future low-carbon planning in the dairy sector, minimize climate risks and enable participating countries to take advantage of available climate smart opportunities within the region.

Activity 1.4.1. Support to Regional Centres of Excellence for technology validation and for maintaining a regional database on practices, GHG emissions and adaptation indicators. To ensure sustainability and transfer of capacity to national institutions, the Programme will provide scientific support combined with capacity development targeting female and young officers in the “regional centres of excellence” to export knowledge regionally and globally on CC and GHG emissions for the dairy value chain. National Centres of Excellence (or “knowledge hubs”) will include national agricultural research institutions (e.g., KALRO in Kenya, NARO in Uganda, TALIRI in Tanzania, RAB in Rwanda), local universities (e.g., Makerere in Uganda; University of Nairobi, Eldoret or Maseno; University of Dar es Salaam or Sokoine University of Agriculture (SUA) in Tanzania). It will build on the legacy of the East Africa Agricultural Productivity Programme (EAAPP, World Bank-funded), which has built Regional Centre of Excellence (RCoE) in the dairy sector in Kenya.

Sub-activity 1.4.1.1: Support to regional Centres of Excellence in technology validation and dissemination. The Programme will assist the above-mentioned institutions with (i) technical trainings and protocols for GHG measurements (emission factors, soil carbon storage) and estimation methods to build their own capacity for research, MRV and technology validation linked to climate change mitigation, (ii) research capacity for conducting innovative research such as optimizing the biogas process for enhanced energy and nutrient recovery and building efficient biogas technologies, (iii) vocational training and development of materials, standards, university-level educational curricula and guidelines integrating youth and women, and (iv) knowledge, information and visibility products, related to the four focus areas feeding, animal health, manure management and biogas, and grassland management and soil carbon sequestration.

Sub-activity 1.4.1.2: Develop and maintain a regional database/ repository on practices, GHG emissions per sources and systems, and climate change adaptation. This sub-activity will consist in developing, maintaining, and promoting a regional database on best practices, and their impact on GHG emissions per source and livestock production system, and adaptation indicators. ILRI has a large repository of training materials on its open-access “CGSpace” platform, which will be leveraged, and made available to all actors in the Programme, with the support of the EAC.

Activity 1.4.2: Establish regional cooperation platforms and participation to regional/ global networks for knowledge exchange and policy dialogue on climate-smart dairy sector transformation. The regional collaboration platforms will include various public and private stakeholders, including the EAC, AU-IBAR, IFAD, FAO, ILRI, the East and Southern Africa Dairy Association (ESADA), local government partners, centres of excellence, international organisations and NGOs, and involve private sector platforms and networks (e.g. One Health Research and Education Centre in Africa – OHRECA; Global Research Alliance Manure Management Network; Global Dairy Platform; Nutrient Circularity Network; National Biogas Stakeholders Network in Kenya). The participants will include women and youth. The Inter-Governmental Authority on Development (IGAD), through its Center for Pastoral Areas and Livestock Development (ICPALD), will be invited to share its experience in facilitating appropriate and applicable knowledge, research, and technology development, including their domestication, adoption, and transfer through regional platforms.

Sub-activity 1.4.2.1: Regional platform for knowledge sharing and policy dialogue to enhance low-emission, climate resilient dairy development. A regional platform – involving the above-mentioned entities - will be organised on a rotational basis in each of the participating countries to share experiences and scientific innovations and best practices implemented on farm. The regional platform will function as the knowledge management mechanism for the Programme. It will capture and disseminate good practices and lessons learned from both public sector, private sector, networks and the research and academic community. This sub-activity will provide evidence – generated from the Programme M&E system - on the GHG efficiency gains (from breeding, artificial insemination/ herd management, feed, animal health), circularity (manure management, recycling) and carbon sequestration. Success stories for replication (e.g. Kenya for Tier 2 MRV), scientific innovations (animal nutrition / TIMPs, manure management for continuous improvement) will be documented. It will finance annual regional workshops, conferences, seminars, tours (exchange visits), meetings and policy briefs ensuring participation quotas for women and youth are met.

Sub-activity 1.4.2.2: Participation in regional and global climate policy mechanisms. In practice, the sub-activity will consist of a joint participation of the 4 countries in global and regional events (global science and policy mechanisms on dairy and climate change) such as the Greenhouse Gas and Animal Agriculture Conference – GGAA 2025, the African Climate Summit, the African Dairy Conference, and the UNFCCC COP). The objective will be to present the progress and results from the Programme for scaling-up and replication, increase its visibility and engage with other countries

in sharing experience and learning. Participants in these conferences will include women, youth leaders and include other intersectional / marginalizing attributes such as disability.

Sub-activity 1.4.2.3: South-South and Triangular Cooperation (SSTC). This sub-activity will support the emergence of a global network of experience and expertise on reducing GHG emissions, especially methane, in dairy production and value chains, as well as financing solutions to support these reductions. This sub-activity will finance the participation in knowledge-exchange programs, including through IFAD's "Learning Routes" with e.g. Argentina, Brazil (Embrapa), Costa Rica and Uruguay on Tier2 analysis and climate impact assessments in the dairy sector. Other initiatives for SSTC will be identified and assessed by the Regional Steering Committee in collaboration with the EAC. It will finance study tours and gender and socially inclusive country delegations to workshops to share innovations and scalable practices generated by the Programme.

COMPONENT 2: LOW EMISSION AND CLIMATE RESILIENT PRIMARY DAIRY PRODUCTION

5. Output 2.1: Smallholders' and cooperative capacity development strengthened.

Activity 2.1.1 Strengthen dairy cooperatives on organizational management and service delivery. This activity aims to strengthen cooperative, organizational, and technical capacities of dairy cooperatives, and raise their climate-change and gender and social inclusion awareness of their management team. Also enhance the enrolment of women and youth dairy value chain actors in the cooperative membership and leadership. Support will also be provided to establish or enhance their network of MCCs and MCPs, and their business linkages.

Sub-activity 2.1.1.1 Strengthen the governance and management structures of dairy cooperatives. This sub-activity will include: (i) training on cooperative governance and management practices for all management, supervisory and general committees of dairy cooperatives, (ii) the business coaching of cooperatives by business development service providers, (iii) where desirable, assistance for the registration as a formal dairy cooperative; and (iv) the participation in national and regional events that are essential for dairy value chain governance, promoting inclusive policy, and business dialogues.

Sub-activity 2.1.1.2 Strengthen capacity to reduce milk losses and ensure milk quality and safety. At the aggregation stage, up to approximately 8 percent of the milk is lost via spillage and spoilage. Aiming at reducing unproductive GHG emissions due to milk loss or quality loss, this sub-activity is to assist dairy cooperatives, MCCs and MCPs in upgrading their technical capacity in milk quality control covering raw milk testing at milk collection, hygiene, and initial processing beyond the sale of raw milk, in line with national and regional food safety standards. The support will assist women and youth to acquire (i) basic equipment for milk collection (motorcycles, milk cans and can holders, testing equipment), (ii) insulated milk transport vehicles (demonstration purpose); and (iii) training on milk quality, food safety and hygiene including informal milk vendors, transporters and technical staff of cooperatives, MCCs and MCPs ensuring women and youth participate.

Sub-activity 2.1.1.3 Expand the outreach and capacity of dairy cooperatives, MCCs and MCPs. The targeted MCCs and MCPs will be supported to develop gender and youth responsive action plans to ensure opportunities for youth employment such as inclusive and equal employment opportunities ensuring women can work such as 'zero tolerance' to GBV and SEAH and other inclusive policies and practices. This may include trainings or technical assistance for such institutional capacity building.

Sub-activity 2.1.1.4 Create gender awareness and roll out GALS among households of dairy farmers. Gender transformative approaches will be integrated in the Programme. Capacity building and trainings for stakeholders on

gender equality and women empowerment will be rolled out across the four countries. This will be done for national, regional and community level stakeholders. The enhanced Gender Action Learning System (GALS+) methodology will be used to address power relations and inequalities especially targeting households and at group or community levels. As men and women have diverse ways of coping and adapting to the adverse impacts of CC, GALS+ includes (i) an integrated climate component to strengthen the mitigation measures and adaptive capacities of households and communities to CC; (ii) nutrition and food safety aspects. DaIMA will support: (i) gender awareness and targeting workshops for staff of implementation units and contractors; (ii) training lead farmers and cooperative committees in the GALS methodology; (iii) rolling out of GALS in L-FFS households including support for community outreach sessions; and (iv) national gender sensitization and training workshops for the management of dairy cooperatives. This activity will include trainings on the prevention of GBV and SEA. Specifically, DaIMA will ensure awareness on GBV and prevention of SEA training for each new employee, intern/fellow, volunteer, contractor, board member or partner as part of their orientation. This may include development of Project guidelines or policy with clear reporting channels and redress mechanisms.

Activity 2.1.2 Build and enhance Productive Alliances along the dairy value chain. The Programme will provide support to productive alliances between farmer cooperatives and enterprises with farmer-allied engagement models, and scaling-up of gender and social inclusive dairy clusters. This will enable female and young farmers to access affordable, high-quality inputs and services.

Sub-activity 2.1.2.1 Strengthen cooperatives' capacity in business and financial management. The aim is to ensure that dairy cooperatives and their MCCs are run as viable business enterprises. Therefore, the Programme will support cooperatives in building out business plans and partnerships for expanding and enhancing the services they offer member farmers (e.g., facilitated feed and AI service purchases checked-off against milk sales) – serving as “dairy hubs”. These dairy hubs will build capacity of young farmer leads as mentors to increase youth engagement. For implementation, providers of business development services (BDS) will be mobilized. In addition, the members of dairy cooperatives and MCCs will receive: (i) financial literacy training, financial management training and equipment (software and IT equipment); (ii) gender-sensitive and inclusive business management and business plan development training (using participatory tools such as RuralInvest, developed by FAO). This activity will support job placements for trained young people in participating MCCs and MCPs as part of increasing their employability and creating jobs. Further, youth trainees will receive business mentorship and coaching on business and financial management in well performing dairy cooperatives and enterprises over an agreed period after completing their job placements.

Sub-activity 2.1.2.2. Build linkages along the dairy value chain. The suppliers of BDS will also provide capacity building for negotiation of offtake agreements between cooperatives and private off takers (Productive Alliance model), as well as public procurement (e.g. school or hospital procurement) for better milk prices guaranteed by contract. The aim is to support establishing Productive Alliances with cooperatives and in designing, enhancing, and scaling key components of their farmer engagement models, such as quality-based or volume-based price incentive terms offered by processors, extension provision, and direct or facilitated provision of inputs and services. Through this support, a pre-investment pipeline will be developed, and related financing needs identified. This support will facilitate youth and women led alliances.

Sub-activity 2.1.2.3. Promote digitalization and traceability of dairy operations. The Programme will promote digitalization and traceability to ensure quality and entry into formal supply arrangements that come with services, and improved accounting (money and stock) for access to credit. Young technicians will be trained in these digital solutions to scale up uptake and usage especially ensuring female farmers benefit. Use of transformative end-to-end (E2) digital solutions is currently limited. Digitalized systems for monitoring on-farm activities, milk production

disaggregated by farm (gender of farmer), aggregation, transportation, and marketing will increase the efficiency and transparency in the dairy value chain and cold chain. It will support off-takers and cooperatives to improve timely payments to farmers, traceability of the quality of milk and linkage to advisory services where farmers might need advisory interventions and access to finance. In the four countries, the Programme will enhance the capacity of milk aggregators and cooperatives to improve their record-keeping and management through a digital cooperative and MCC management information system (MIS) that will: (a) record the production of farmers, (b) track milk flow/volumes and quality, (c) manage records of farmer payments allowing the development of digital profiles and credit scores. Digital solutions developed by local agritechs and agri-fintechs (e.g. Emata in Uganda) or FAO (Collect Mobile App) and youth led fintechs will be scaled-up to enable business operations management and credit-scoring for better access to finance. A pilot will start with a limited number of cooperatives in each country and gradually upgraded. The Programme will finance the required IT equipment and services.

Activity 2.1.3 Promote renewable energy (RE) and waste management technologies. The decarbonization of the dairy value chain will be promoted. The Programme will focus on resource use efficiency and the substitution of fossil fuels and firewood by green energy sources such as solar especially targeting female headed households or less wood-intensive energy sources for milk cooling and processing. Insulated vehicles and solar energy supply are typically more suitable solutions for larger MCCs (>3m³ Litres capacity). For processing, cold chain and quality preservation solutions (e.g., cooling tanks, insulated / refrigerated trucks) are fundamental to business operations, yet not always used by small-scale processors. Solar is the key mitigation solution and most applicable to relatively larger processors in the short-term given high upfront costs (*See Feasibility Study*). Also, the management of waste and wastewater will be promoted. There is limited adoption of waste management solutions along the value chain today. Processors only re-use waste (e.g., by-product whey as feed) selectively and recycling is not a priority. Beneficiaries of the support would be MCCs and SMEs.

Sub-activity 2.1.3.1 Assess the need and viability of renewable energy and waste management solutions. The aim will be to assess the clean energy needs of dairy cooperatives and propose RE solutions (photovoltaic – PV, biogas systems, etc.), energy-efficient equipment, and solid and water waste management solutions. The assessment will look at scale and viability of such investments not just needs and potential tech solutions.

Sub-activity 2.1.3.2 Support the acquisition of renewable energy and waste management solutions. This would include the upgrading or equipping of dairy cooperatives with suitable RE, PV-powered technology for milk coolers for milk aggregation, milk transfer tanks, solar water heaters, water recycling systems, heat exchangers, milk pasteurizers, butter churners and cheese makers. Solar energy will complement the existing electricity from national grid or generated by standby generators to cool down milk to 4 C°. This activity will facilitate support for women to acquire energy and labour-saving equipment for use in dairy at home or dairy value chain activities.

Sub-activity 2.1.3.3 Organize capacity building in energy efficiency and waste management. This training will focus on the management of the RE solutions in dairy cooperatives, SMEs and MCCs. Such will include trainings for women as lead trainers and participants in waste management activities.

6. Output 2.2: Breeding, feed and fodder, manure management improved.

Activity 2.2.1 Promote the adoption of climate-smart dairy breeds adapted to climate change. Breeding and genetic improvement will aim at improving productivity and heat resistance of dairy cattle while preserving resilience of livestock keeping systems by crossing indigenous and exotic breeds through artificial insemination (AI). Exotic breeds, such as Holstein-Friesian or Jersey, are more vulnerable to high temperatures, and consequently to climate change,

but present a higher productive potential. Indigenous breeds, despite their lower productive potential and small size, are a reservoir for resilience traits such as hardiness, mobility, rendering them more adaptable to extreme climatic conditions and to certain diseases. Different generations of crossbreeds (exotic x native breeds) would contribute to maintaining the resilience traits of indigenous breeds, while also adding productive traits to the gene pool, increasing the offspring's productivity in milk production, therefore contributing to reducing GHG emission intensity. Available studies shows that improved dairy animals have a significantly lower GHG emission intensity per kg of milk due to their higher production when compared to indigenous breeds. This activity will target women and youth to benefit from distribution and 'pass on' of improved breeds to increase their economic empowerment.

Sub-activity 2.2.1.1 Make good quality semen for climate-smart breeds available, adaptable to extreme climatic conditions and to diseases. The Programme will enhance availability and accessibility of germplasm of exotic that will be crossed with local. This will be achieved through (i) the identification of potential climate-smart cattle breeds; (ii) the import of sexed semen and/or embryos, if needed; (iii) the procurement of high genetic merit bulls to produce sexed semen or no-sexed semen for national and regional use; (iii) the production and conservation of climate-smart breeds' germplasm, using laboratory techniques like in-vitro fertilization, or simply artificial insemination (AI); and (iv) capacity development of laboratory personnel including youth to produce sexed semen to increase production of dairy heifers.

Sub-activity 2.2.1.2 Invest in breeding and AI centres. To solve bottlenecks in AI and embryo transfer, the Programme will support scaling up capacities for production and distribution of liquid nitrogen, semen, and embryos. This includes the availability of liquid nitrogen, storage and quality control equipment, dry ice pilots for conservation of semen, and other specialized equipment. In addition, the following investment will be supported. In **Kenya**, the Programme will involve the Kenya Livestock Breeders Association, which is specialised in extension and training, breed inspection and registration, performance recording and reporting and feedback. In **Rwanda**, support will be given for the establishment of semen and liquid nitrogen storage and distribution centers, as well as to the existing bull centre in Songa Station to acquire micro-manipulators to enhance semen production operations. Moreover, RAB will be strengthened in record keeping and continuous updates of the Studbook to identify resilient and high productive crossbreeds. In **Tanzania**, support would be given to establish an AI centre in Southern Highland. Also, the feasibility of an LN unit in Zanzibar will be analysed. In **Uganda**, this investment will strengthen the NAGRC and Data Bank to improve the production and access to high-quality semen in small and medium size farms. DaIMA will facilitate women and youth led farms to access high-quality semen at subsidized costs.

Sub-activity 2.2.1.3 Scale up the insemination of dairy cattle to make the herd more productive and climate resilient. The Programme will (i) train AI technicians, who can be private inseminators, para-veterinarians, lead farmers of L-FFS and District livestock officers (at least 30 percent youth and 40 percent women); (ii) scale up the insemination of dairy cattle in the Programme intervention area; (iii) include aspects of the interaction between dairy cattle husbandry and climate change in the L-FFS curriculum. In **Kenya**, the Programme will support the National Breeding Strategy and Action Plan, in collaboration with the Kenya Livestock Breeders Association. Public and private partnerships (PPP) will be promoted to have AI services accessible to the male and female farmers. In **Rwanda**, the breeding programme of RAB will be scaled up including the training of more private AI technicians. In **Tanzania**, additional technicians will be trained and equipped with start-up kits. In addition, bull centers will be established in regions where AI is not available. In **Uganda**, more AI technicians will be trained by the NAGRC and will be equipped with AI kits focusing on the regions where AI services are still poorly developed.

Sub-activity 2.2.1.4 Organize awareness campaigns and support breeders' organizations. This activity will facilitate female and youth members of dairy cooperatives and L-FFS groups to create awareness on the use of sexed-semen,

thus impacting the herd's sex ratio, and further reduce absolute GHG emissions and the emission intensity per kg of milk produced mainly by replacing many low-productive local cows with a few adapted and high performant crossbreeds, thus reducing the herd size. In the four countries, support will be given to the breeders' associations and societies (that include women and youth in their membership) for animal registration and maintenance of breeding standards.

Activity 2.2.2 Provide support for better animal care, herd management and disease prevention. In East Africa, infectious diseases and parasites rank among the primary causes of livestock productivity and production loss. Production losses through reduced performance (growth rates, milk yield) or animal loss through morbidity, mortality, and abortion results in cows with relatively high GHG emission intensities. These losses affect people's resilience to CC, livelihoods and nutrition security and can affect human health through zoonotic disease transmission. In addition, the growing importance of zero-grazing systems has implications for animal welfare, thus farmers will be encouraged to consider animal well-being aspects. Day to day feeding and management is mainly a female and male youth gender role in the four project countries. Although they may not own the dairy cattle, they should constitute a major target group in this intervention.

Sub-activity 2.2.2.1 Organise trainings on animal husbandry, with a specific focus on animal health, animal welfare, herd management and disease prevention. Updated modules of the L-FFS training on animal health and welfare, and herd and reproduction management will be implemented. Female and young dairy farmers, managers and handlers (feeding and milking) will receive training to better care for the animals and to adopt biosecurity measures to prevent common diseases and pests that impact the production or suitability of the milk for human consumption. They will also receive support to better manage their herds, reproduction calendar, and animal nutrition to reduce animal mortality, heat stress and increase productivity, thus further reducing absolute CH₄ emissions and the emission intensity per unit of milk produced through targeted pro women and youth extension activities.

Sub-activity 2.2.2.2 Scale up the use of digital apps to monitor animal health and herd performance. Digital record keeping on-farm (e.g., using iCow, DigiCow Africa, My Fugo) will be scaled up to monitor the productive and health performances of the herd (e.g., records of calf mortality and causes, number of lame animals), to plan artificial insemination, and guide herd management. The use of the app will be promoted through the L-FFS in interested groups using youth as digital leads. The support includes: (i) training of L-FFS lead farmers including women on the use of the apps; (ii) guiding L-FFS farmers in the use of the technology; (iii) equip the (youthful) farmers in the pilot with digital tools e.g. by providing start-up gadgets and mobile phones. To implement the sub-activity, the Programme will work with agritech companies including some youth led ones and start-ups to develop digital solutions that will enhance transparency in farm data and operations, monitoring of CH₄ emissions reduction and traceability of milk products increase create job opportunities.

Sub-activity 2.2.2.3 Scale-up vaccination campaigns to mitigate the impact of climate change on the spread of diseases. The Programme will scale up and extend the vaccination campaigns to target in priority dairy farmers in the Programme area, who are members of various cooperatives and L-FFS. Youth will be supported to lead creative campaigns such as using road shows and other community media. The vaccination campaign will focus on: (i) the spread of East Coast Fever (ECF), Contagious Bovine Pleuropneumonia (CBPP) and Heartwater disease to new production areas through an extended vaccination campaign, in **Kenya**; (ii) ECF, RVF and FMD in **Rwanda**; (iii) ECF, FMD, RVF and CBPP in **Tanzania**; (iv) trans boundary animal diseases (TADs) like FMD, CBPP, and LSD in **Uganda**.

Activity 2.2.3 Promote climate-smart forage production and conservation. The adoption of climate resilient forage varieties and conservation technologies is essential to reduce enteric methane emissions intensities in the dairy value

chain, while increasing milk yields of dairy cattle. The benefits are multiple and encompass CC adaptation through adoption of a resilient forage variety (*Lablab purpureus*, *Brachiaria* sp., *Cenchrus purpureus* (Napier grass), *Leucaena*, *Calliandra*, *Panicum maximum* (Giant Panicum)), improvements in soil fertility and increased capacity of farmers, managers and handlers (feeding and milking) to meet forage production needs during dry periods and to respond to climate-induced pests and diseases.

Sub-activity 2.2.3.1 Make certified forage seed available to private and community multipliers. This sub-activity will (i) support the multiplication of breeder and foundation seeds for climate-smart, emission-reducing fodder by the national mandated public services; (ii) facilitate a seed certification scheme and sensitize the private sector to produce certified forage and fodder seeds.

Sub-activity 2.2.3.2 Support multiplication of climate-resilient seed for fodder crops and pastures. This sub-activity will include contracting community and private seed multipliers (including the youth and women) for multiplication, conservation of forage and pasture seeds. In each country, standard approaches will be used to multiply seeds and make them available to L-FFS, PFS, cooperatives, and other groups.

Sub-activity 2.2.3.3 Build capacity of dairy farmers on forage production and conservation on intensive and semi-intensive dairy farms. The L-FFS will be the knowledge-sharing vehicle to promote improved forage production and conservation. The L-FFS will focus on (i) training on good forage management practices (e.g., grass and legume intercropping, agroforestry, weed management, (ii) training and demonstration on small-scale irrigation to produce forage crops; (iii) training and demonstrations of mechanization of forage production with fodder choppers/cutters and conservation equipment; (iv) training and demonstrations on forage conservation practices, to respond to seasonal feed availability and quality (e.g., protein content, digestibility) variations. Youth and women groups will be supported to venture into commercial fodder production, which may include production using hydroponic technology²¹.

Sub-activity 2.2.3.4 Access to water for animals and forage production in intensive and semi-intensive dairy systems. The Programme will provide small-scale equipment for rainwater harvesting and recycling (e.g. individual dam sheets, 3m³ rooftop catchment water tanks and water distribution kits). This activity will target to provide green solutions for women aiming to reduce workloads related to accessing water.

Activity 2.2.4 Promote climate-smart feeding practices to reduce enteric methane emissions. Appropriate diet formulation is key to improve dairy cattle productivity, improve animal welfare, while reducing methane (CH₄) emissions. Balanced diets (more energy-dense or more digestible) decrease retention time in the rumen and reduce ruminal fermentation and methane yield (kg CH₄/kg dry matter intake). Better nutrient utilization (e.g., avoiding deficiencies or surpluses) reduces the environmental impact of livestock.

Sub-activity 2.2.4.2 Build capacity on appropriate feed formulation. In each country, the Programme will support the capacity development of technical advisors, cooperative management, small feed processing plants and L-FFS lead farmers on sustainable animal feed formulations such as total mixed rations (TMR) to enhance productive efficiency and to mitigate GHG emissions.

Sub-activity 2.2.4.3 Promote appropriate diet formulation that reduces enteric methane emissions. In the four countries, the L-FFS and the dairy cooperatives will be the vehicle to reach out to dairy farmers and promote

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appropriate feeding of dairy cattle. The Programme will: (i) provide start-up kits for youth and women in L-FFS to monitor diet formulation to reduce methane yield (e.g., Rumen8 tool by SNV); (ii) run L-FFS of feed formulation, including on the use of agroforestry leguminous fodder trees with methane inhibiting properties such as *Calliandra sp.* for improved feed supplementation and carbon sequestration. The Programme will also pilot and leverage on the existing innovative enteric methane inhibitors used as mixtures in feed to be adopted in zero-grazing and commercial/intensive systems.

Activity 2.2.5 Increase sustainable nutrient recycling of manure and dairy waste to reduce emissions. Livestock manure is a source of methane (CH₄) and nitrous oxide (N₂O) emissions, as well as of pollutants such as ammonia (NH₃) and nitrate (NO₃⁻). At the same time, manure contains essential nutrients and is often the primary source of crop fertilization on smallholder dairy farms, as mineral fertilizers are too expensive or unavailable. In addition, its costs have further increased due to recent events (COVID-19 pandemic, Ukraine war). Many smallholder farms are limited in nutrients, which limits crop yields. Circular management of agricultural nutrients via livestock manure is the best scalable option to overcome these limitations, improve soil health, and improve on-farm productivity. This has a positive effect on livestock productivity, as better manure management also improves the nutrient availability for feed and forages, which increases their growth, biomass yield and nutritional quality and ultimately improves milk production. In addition, manure application reduces soil nutrient mining and soil degradation and prevents additional CO₂ emissions from loss of soil organic carbon.

Sub-activity 2.2.5.1 Manure management in intensive and semi-intensive dairy production systems. The Programme will create awareness and build capacity on low-cost manure management for soil health and soil microbiome improvement. The sub-activity will consist in providing trainings through L-FFS and demonstration farms, using existing manuals developed by NARO in Uganda, ILRI, etc. on scalable and already tested manure management practices (pit storage, manure ponds, manure covering with banana leaves, compaction, etc..) and manure use applications offering livelihoods opportunities. This activity will target to include women and youth led farms as demonstration centres to attract more women and youth to participate.

Sub-activity 2.2.5.2 Promote adapted animal housing. The Programme will promote adapted animal housing with the possibility of separating solid and liquid manure. Good example of animal housing promoted by Heifer International can be scaled up targeting women and youth dairy farmers.

Sub-activity 2.2.5.3 Manure management in extensive systems (pastoral and agro-pastoral dairying regions). The Programme will: (i) run L-FFS/PFS for promoting manure collection from communal kraals for use, sale/ income generation at community level; and (ii) establish community manure sheds to promote manure collection, storage and covering for better manure management and reduce emissions. This activity will target to support women and youth led groups.

Activity 2.2.6 Build capacity for on-farm use of biodigesters and biogas. Biogas digesters utilize microorganisms to break down organic matter such as livestock manure or other organic waste, during which biogas and a nutrient-rich by-product (called digestate or bio-slurry) are produced. This technology offers a clean source of energy for people who do not have access to clean cooking fuels. In addition, it creates a biofertilizer (for use as organic fertiliser on nearby fields), reduces methane emissions (up to 70-90 percent compared to liquid manure storage) and ammonia and nitrous oxide emissions. Biodigesters can reduce deforestation and empower women by reducing the time they spent collecting firewood. As part of the low-carbon pathways, the Programme will encourage the uptake of on-farm biodigesters for intensive and semi-intensive dairy farms. Women are the main consumers of biogas and should be

involved in trainings on how the setup works as well as identifying where outside and inside the house the equipment should be placed not to curtail the functionality of the kitchen.

Sub-activity 2.2.6.1 Support capacity building in biodigester management. The Programme will support the training and accreditation of biodigester technicians and consumers, with a particular focus on women and youth, in the target districts. In addition to installation-related aspects of biogas valorisation and quality management, including safe working standards.

Sub-activity 2.2.6.2 Support investments in biodigesters at the household level. The Programme will support investments in biodigesters for dairy farms ensuring women and jointly owned farms are supported in the intervention area.

7. Output 2.3: Pasture and grasslands management improved

Activity 2.3.1 Strengthen rangeland governance structures for communal grazing areas. This activity will finance interventions to improve rangeland management (including access to water), mainly in **Kenya, Tanzania and Uganda**. Engagement and awareness raising among local stakeholders involved in landscape management is essential for locally led adaptation and people-centred planning.

Sub-activity 2.3.1.1 Participatory rangeland management planning. In selected districts/counties, the Programme will (i) map and conduct a capacity assessment of existing community governance structures (including status of women participation and opportunities to increase representation of women), formal (e.g. village land committees, environmental committees, water use committees, local government, district) and informal entities (grazing committees, traditional committees, clan leaders); (ii) support community governance structures to produce pasture and rangelands management plans, including land tenure to secure tenure rights, water access, as well as possible conflict resolution over land and water use, and (iii) support the development, revitalization and strengthening of laws/charters and by-laws governing the use of communal resources (including water). The groups to be supported will be require meeting the women and youth quotas in representation.

Activity 2.3.2. Restoration of degraded pasture lands, including access to water. This activity will finance interventions to restore rangelands.

Sub-activity 2.3.2.1 Training on rangeland and pasture management. The Programme will facilitate L-FFS/PFS aimed at guiding and engaging farmers towards improving the management of rangelands and pastures. L-FFS will focus on (i) rotational grazing, (ii) management of grazing intensity (destocking and restocking i.e. sale of animals during the dry season and restocking during seasons of high food availability to reduce pressure on pastures), (iii) integration of sylvo-pastoral practices (multipurpose leguminous trees) for better carbon sequestration and reduction of livestock exposure to heat stress conditions, by providing shade for animals (more particularly around water harvesting points), and (iv) biochar production, elaborated from invasive bush species. This activity will support female and youth farmers with animal shades and biochar production.

Sub-activity 2.3.2.2 Rangelands restoration. This sub-activity, focusing on communal grazing areas, will focus on: (i) reseedling and integration of legumes with grasses in natural pastures, (ii) integration of multipurpose leguminous trees and shrubs (improved agro-forestry systems) and control of invasive tree species, (iii) preservation of traditional non-forest tree products (medicinal and other) known to women mainly and (iv) implementation of soil and water

conservation practices to increase water availability and biomass productivity. Women and youth groups will be supported with tree seedlings

Sub-activity 2.3.2.3 Water availability. Depending on the country, to enhance access to water for dairy cattle in extensive systems, the Programme will invest in water harvesting systems such as micro-dams, small-scale valley-dams, construction/ rehabilitation of charco dams and rock-catchments schemes. Feasibility studies will be carried out before the start of the work to assess sustainability. The Programme will also enhance the capacity of the community-based water supply organizations (COWSOs) to better manage and operate all water harvesting and storage infrastructures. The design of water infrastructure will follow the guidelines of the local government authorities and integrate gender sensitive facilities such as toilet and sanitation blocks and washing bays.

COMPONENT 3: GREEN DAIRY FINANCING FACILITY (GDFF)

Component 3 aims to enhance access to finance and support the implementation of climate-smart solutions and practices, thereby facilitating the transition to sustainable, low-carbon, climate-resilient dairy value chains. Component 3 will have two key outputs: (3.1) the establishment of an investment window, and (3.2) the establishment of a Technical Assistance (TA) Facility. Eligible businesses for GDPs will need to exhibit clear ambitions to invest, develop or distribute climate-smart technologies, productivity-enhancing technologies or SHF extension services and have a track record of SHF allyship. Investment decisions will be made on a case-by-case basis depending on business need. The GDFF's credit decisions will be based on a thorough assessment of potential investees' ability to repay and to invest in products tailored for women and youth. Dairy Loan Products will target smaller businesses, and investees will be eligible to receive light-touch pre-deal TA to translate investment needs into investment requests, and they can be connected to other DaIMA components to access any post-deal TA support needed.

Through the (3.1) Investment window established, the main objective of the GDFF is to facilitate the larger-scale transition to sustainable, low-carbon, climate-resilient dairy value chains in Kenya, Uganda, Rwanda and Tanzania. To do so, the GDFF will improve access to finance for underserved dairy value chain SMEs that operate as farmer-allied enterprises and are therefore well-integrated—or have the ambition to be well-integrated—across the value chain. These enterprises will be required to show considerations for women and youth farmers in their business plans and business operations.

Output 3.1 on the Technical Assistance Facility, low-emission cold chains and inefficient, low-loss milk processing units will be established and upgraded. This will overcome the limited processing infrastructure and inadequate utilization capacity barriers. The eligible facilities will receive technical assistance (TA) to integrate Gender action plans into their business plans. Key expected outputs expected from the units will be to (i) create jobs for male and female youth aged 18-35 years as part of their workforce (ii) Be gender-responsive and create an inclusive policy environment whereby women and men have equal opportunities. Only units with gender (sex and age) sensitive and socially inclusive policies and practices shall participate in the project. These units will target to employ 40% women, 30% youth and vulnerable groups.

Activity 3.1.1 Fund set up. This activity aims to establish the GDFF fund as a functioning entity with the required policies, structures and staffing, and to crowd in private commercial capital in pursuit of the programme's climate impact ambitions. The GDFF staffing and structure will include provisions for gender and youth expertise.

Sub-Activity 3.1.1.2 Investor engagement and negotiation (led by Fund Manager, in coordination with IFAD). The Fund Manager will be responsible for identifying, engaging and carrying out due diligence on prospective investors. The Fund Manager will develop criteria for screening investors demonstrate forward and backward linkages with women and youth enterprises.

Activity 3.1.2 Identify eligible businesses for investment. While this will focus on developing a screening framework to assess prospective investees for potential impact and financial return. This activity will ensure identification of women and youth led businesses by making gender and youth provisions in the screening framework.

Sub-Activity 3.1.2.1 Develop screening framework. The Fund Manager will establish a screening framework to ensure consistency in investment decisions. This screening framework will be underpinned by a taxonomy of investment categories which are justified and eligible for investment including those that facilitate women and youth empowerment by creating jobs and employment opportunities.

Sub-Activity 3.1.2.2 Pipeline development and screening. The pipeline for the GDFF will be sourced through three main channels; (i) GDFF Investment Advisors, (ii) DaLMA's PCUs, and (iii) TA providers under Component 2. Investment Advisors will screen prospective investees against pre-defined screening criteria, which will include women and youth considerations and quotas to ensure uptake.

Activity 3.1.3 Climate-smart lending products and services are rolled out by GDFF to eligible businesses. These will include labour and energy saving products aimed at reducing workloads for women. The two products: Green Dairy Partnerships and Dairy Loans will ensure the partnerships promote business working with women and youth farmers. The Dairy loan tickets will be developed to ensure the ticket sizes are affordable and accessible to these groups. Provisions for community outreach will be made through different media to ensure uptake. The Fund Manager's gender and youth expert will be responsible for monitoring this.

Output 3.2 Technical Assistance Facility established. The Green Dairy Financing Facility will invest in climate-resilient dairy producers, off-takers, processors, and livestock businesses. Tailored financial products with technical assistance will increase access to finance and affordable credit for on-farm investments especially for youth, women and vulnerable groups. Women and youth will receive tailored BDS and financial literacy modules that incorporate gender transformative approaches such as confidence building, communication skills and agency building. The GDFF will use a technical assistance (TA) facility to maximise both impact and returns by enhancing the capacity of investees (including on gender and youth sensitive approaches) to implement effective strategies, improve operational efficiencies and achieve growth.

Activity 3.2.1 Prospective investees will receive targeted pre-deal TA under Output 3.2. Noting that Component 2 also provides funding to assist dairy cooperatives in acquiring renewable energy and waste management solutions (see Sub-Activity 2.1.3.2), Investment Advisors will liaise with the TA providers of Component 2 to ensure targeting of women and youth linked enterprises.

Sub-Activity 3.2.1.1 Pre-Deal Development for Dairy Loan Products. These businesses will receive light touch TA to translate their investment needs, into a well-defined investment request prior to the deal. Tailored support will be developed to support women and youth friendly loan products.

Activity 3.2.2 Post-deal TA: Providing ongoing support post-deal to ensure successful implementation, monitor progress and ensure outcomes are met. Activity 3.2.2 focuses on providing post-deal TA to Green Dairy Partnership investees, ensuring that outcomes are achieved. The type and level of support will be tailored based on the specific needs identified for each business. This TA will coach, mentor and profile good case examples of women and youth investments for learning exchanges and scale-up. Considering the Dairy Loans will be short-term, Dairy Loan investees especially women, youth and vulnerable groups may be considered for follow-ups in place of post-deal TA.

Implementation arrangements

The GESI action plan shall be implemented through implementation partner or Service Providers activities as defined by the project and IFAD supported investment projects.

DalMA will seek implementing partners with experience working on gender equality and women empowerment, youth sensitive programmes, nutrition and social inclusion, with emphasis on community centred approaches promoting integration of young men and women, vulnerable groups, persons with disabilities. This may include youthful enterprises, NGOs, the academia and researchers.

Component 3: Programme management. Actions to strengthen the GESI dimensions of programme management:

- Support for GESI Action Plan implementation will be provided through the Technical Assistance, which includes capacity building at different levels including for implementing partners and end beneficiaries.
- A **GESI specialist** for each Country operation shall ensure the roll out of the GESI action plan for each country. Additional expertise will be procured under the GDFF and Technical assistance sub activities to ensure access and uptake for women and youth. The specific TA will include; (i) capacity strengthening and technical support to partner staff and agents; (iii) the recruitment of experienced service providers to support outreach, gendered approaches and capacity building at the community level; (iv) liaison with the Monitoring and Evaluation team; and (v) ongoing GESI technical support during implementation.
- Include a specific commitment to gender-responsive planning in the terms of reference of the implementation teams from partner organizations. This shall be backed by capacity strengthening on identifying and addressing gender issues. Annual performance reviews, aligned to the Logframe targets will track performance on delivering GEWE and GESI goals.
- Select implementing partners (input and service providers) with a proven commitment to community participatory approaches and gender equality, and staff with the relevant GESI skills. The implementing partners shall be held accountable for delivering on GESI targets.
- Ensure that annual work plans and budgets are prepared, delivered and tracked from a gender perspective, and that the funds allocated and/ or locked to implement gender-related activities are appropriate.
- Ensure the DalMA project and implementing partners promote facilities, processes and policies are gender-responsive, supporting gender parity in recruitment, promotion, and retention of staff; equal pay between women and men for the same grade of employment; life-work balance; policies and communications on the prevention of sexual exploitation, discrimination, harassment and the abuse of authority.²²
- Ensure the systematic integration of GESI considerations – including attention to representation and expertise – in the programme’s governance, operational procedures, technical expert advisory bodies and decision-making structures.

Monitoring, evaluation and knowledge management will:

- Support to planned any additional gender and social norms assessments for the dairy interventions where the baselines of IFAD country projects may not be adequate. Gender and social norms assessment studies on dairy may precede the alignment of the DalMA GESI strategy and IFAD investment gender strategies. In order to increase the inclusion of women, youth and people with disabilities and facilitate their economic empowerment in the dairy interventions, assessment questions will be integrated into baselines, mid-terms and end term reviews of the IFAD country projects to understand gender and social status and monitor improvements over the lifecycle of DalMA. The findings will help inform the detailed GESI strategies for each of the IFAD project to ensure that the

²² Detailed in the DALMA grievance procedures.

implementation of DaIMA will lead to more effective interventions to increase inclusion in climate smart dairy enterprises.

- Ensure all data in the logframe are disaggregated by sex, age, location and disability, where appropriate (including targets and data on youth²³) using standard formats at all levels of implementation.
- Ensure the integration of the IFAD empowerment index to measure gender transformation which applies the principles of the Women's Empowerment in Agriculture Index (Pro-WEAI) at baseline, mid-term and end-term studies and adjust project interventions as required to address any negative impacts.
- Train staff on inclusive collection, analysis and interpretation of disaggregated data at various levels: from the communities, service providers to DaIMA focal points at country subsidiaries considering female enumerators where possible at household.
- Track performance indicators, including by sex, age, location of the dairy entrepreneurs, input and service providers including MSME. Establish and monitor mechanisms for tracking progress on targets and outputs between the different levels of programme management and service providers.
- Support IFAD baseline investment annual outcome surveys to monitor and analyse benefits for women and men of different ages. Develop recommendations for project actions if required to narrow the gender gaps.
- Undertake studies on good practices leading to gender-responsive dairy enterprise outcomes and impacts and organize knowledge management and sharing events to disseminate findings.
- Ensure GESI issues and climate smart dairying actions are addressed during supervision missions and mid-term review. Make provision for additional TA if it is recognized that the programme is failing to deliver on GESI targets during the project implementation.

²³ The youth target of 20% of beneficiaries needs to be disaggregated by sex; that is, 40% of the youth target is young women.

GENDER AND SOCIAL INCLUSION ACTION PLAN MATRIX

Impact Statement: Outcome Statement:	IF technological solutions for reducing methane and other greenhouse gas (GHG) emissions and increasing climate resilience are adopted by dairy farmers and milk processors in East Africa, and IF financial services that enhance revenue or lower production costs are utilized, THEN a decrease in farm-level methane emissions and a reduction in supply chain GHG emissions will be observed. This is BECAUSE a transition to net-zero dairy production will be perceived as economically beneficial to farmers, pastoralists, and dairy sector processors. DAIMA will secure climate change-resilient dairy enterprise dependent livelihoods with the benefits of food and nutrition security for women, men, youth and vulnerable groups. Business and employment opportunities and incomes will improve with more equitable benefits to all key players DAIMA intends to achieve the following gender-responsive outcomes: 1. Strengthened systemic and institutional capacities of the livestock sector, which will enable smallholder dairy producers and local value chain actors to reduce CH₄ and other GHG emissions 2. Increased production, market knowledge and tools for smallholder dairy producers and private sector value chain operators for a low-emission, climate-resilient, and sustainable pathway 3. Enabled access to required financial services to make the necessary transition to low-carbon, climate-resilient, and sustainable livestock production and value chain development, for smallholder dairy producers and local private sector value chain actors. The overall budget required to implement the GESI responsive DAIMA program is estimated to be USD 2,399,608				
Outputs, activities and sub-activities	Indicators and Targets	Time line	Responsibilities	Budget in USD for the 4 countries combined	Gender sensitive based on women participation
COMPONENT 1: STRENGTHENED NATIONAL CAPACITIES AND ENHANCED ENABLING ENVIRONMENT FOR LOW EMISSIONS CLIMATE RESILIENT DAIRY SECTORS			MLF, MoALD, MAAIF, RAB	520,818	Yes

Output 1.1: Climate-responsive policy and regulatory environment established. Activity 1.1.1: Review and formulate Dairy Sector policies, strategies and regulations for climate-change adaptation and mitigation. In the four countries, the Programme will support the Ministries, Departments and aligned institutions in charge of the dairy sector to review and formulate national policies, strategies, and regulations that are gender and socially inclusive and essential for a sustainable transformation of the dairy sector, in line with the NDCs and NAPs	Number of policies and strategies prioritized for gender analysis review (Target at least one per country)	By Year 3			
Sub-activity 1.1.1.1 Support the preparation of gender and social analysis for the national dairy policies, strategies, and regulations to mainstream climate-change adaptation and mitigation. In the four countries, Rwanda, Uganda, Tanzania and Kenya. For all four countries, there will be (i) (Dairy Act) regulation of the raw milk market, (ii) Support will be given to the formulation of a national regulatory framework for the private veterinary practice including for Community Animal Health Workers (CAHWs) and Livestock Farmer Field Schools (L-FFS) facilitators	Number gender analysis and recommendations supported per country (at least one per country) Existing/new laws, regulations, policies or strategies proposed to policy makers for approval, ratification or amendment	By Year 3	MLF, MoALD, MAAIF, RAB	525,636	Yes
Sub-activity 1.1.1.3 National dairy value chain governance and awareness creation. The aim of this activity is to enhance dairy value chain governance and to mainstream the importance of climate-change in this process. At national level, the national dairy boards (KDB, DDA, TDB, RNDP) and other relevant professional bodies of private stakeholders, will be supported to participate in the dairy policy dialogue, to create awareness and education/capacity building on climate, feed and dairy policies and regulations. Support women participation (4 countries)	Number of female participants in the national policy dialogues held (40% women in each country)	Annually	MLF, RAB, KDB, MAAIF	515,000	Yes
Sub-activity 1.1.1.4 District/regional/county level dairy value chain governance and planning. Local-level multi-stakeholder platforms involving delegates from dairy value chain stakeholders will participate in the annual planning and review of Programme activities. Participants will include dairy cooperative female and male members, milk collectors, milk processors, service providers of artificial insemination (AI), veterinary services, extension staff and local government. Support women participation in District/local platforms.	Number of female participants in the district policy dialogues held (<40% women) Functioning multi-stakeholder platforms	Annually	MLF, RAB, KDB, MAAIF	320,000	Yes

	supported including women farmers				
Activity 1.1.2 Launch awareness campaigns on nutritional benefits of milk and food safety aspects. This will be a targeted support to milk consumption and nutrition awareness, accompanied by nutrition education and social behaviour change among smallholder producer and consumer households, informal dairy value chain actors and milk handlers in the school milk program. The aim is to improve nutritionally vulnerable groups' consumption of milk, therefore enhancing their individual health and climate resilience. The Programme will support milk consumption and nutrition awareness through increased availability, accessibility, affordability, and consumption of high-quality and safe dairy products, accompanied by nutrition education and social behaviour change and communication (SBCC) to reach out to informal dairy value chain actors, milk handlers at home and institutions and nutritionally vulnerable groups. Support food/nutrition demos in events.	No of women in milk campaign events per county/ district and nationally per year (<40%)	Year 2-5	MLF, RAB, KDB, MAAIF	412,818	Yes
Sub-activity 1.1.2.1 Public awareness on the contribution of milk to nutrition and resilience, with focus on climate vulnerable population. The awareness raising in all four countries will use various community platforms such as extension, L-FFS, MCC/cooperatives that should include women and marginalized groups. Topics will include improved milk treatment, food safety and nutritional benefits of dairy produce, pasteurized and processed milk. In Rwanda, it will include women, youth participation in dairy fairs/expos under the auspices of RNDP to showcase climate-friendly and energy efficient local processing and consumption of dairy products. In Tanzania, Uganda and Kenya the Programme will scale up school milk programs for improved nutrition of vulnerable children.	No. of women, men participating in forums (<40% women)		MLF, RAB, KDB, MAAIF	612,818	Yes
Sub-activity 1.1.2.2 Build capacity of dairy inspection to ensure milk quality and safety and reduce waste in all four countries. Dairy inspection is essential for sector-level milk quality and safety, thus encouraging formal market growth and reduction of milk rejection and waste (less GHG emission intensities). The Programme will provide technical training and equipment for women and male youth, strengthen capacities of inspections to ensure development and enforcement of milk quality and safety standards. Gender inclusion will be considered among the inspectors identified for capacity development and execution of inspection. Women and youth will be supported to attend/travel to trainings.	Number and gender (sex and age) of trained inspectors (<40% women)		MLF, RAB, KDB, MAAIF	512,818	Yes

Activity 1.1.3 Improve land tenure security for sustainable dairy farming. The Programme will adopt a fit-for-purpose approach to identify and mitigate tenure risks. This activity will support in country study to document land tenure risks, challenges and opportunities for women and produce policy briefs. For pastoral and agropastoral systems, the focus will be on the identification of use of rangelands by gender, their protection and supporting gender and social inclusive participatory planning and management. In Rwanda, all land is registered and individually owned. The focus will be on optimizing land productivity through land consolidation and land use planning. The project will include capacity building and support to participatory planning processes to create awareness of land rights for women and youth local-level capacities.	% no. of studies supported on women and youth land tenure risks		MLF, RAB, KDB, MAAIF	712,818	Yes
Sub-activity 1.1.3.1: Mapping of local tenure risks and preparation of training materials. The Programme will mobilize specialized technical assistance to: (a) develop training material to increase local authorities' capacity to secure gender and social inclusive tenure rights for dairy production and the public's capacity to identify and secure legitimate tenure rights. Based on the national tenure context, each dairy production system's tenure risks in terms of gendered access, use and security of land will be identified and assessed, and appropriate mitigating strategies will be proposed.	Number of identified land tenure capacity development materials with gender modules developed (at least one per country) Beneficiaries (M/F) gaining increased secure access to land		MLF, RAB, KDB, MAAIF	320,818	Yes
Sub-activity 1.1.3.2: Developing capacity on land tenure at decentralized level. In all four countries the focus will be on capacity building of local authorities and raising public awareness on options to incrementally increase tenure security for women, men and youth and the vulnerable (e.g. disability, ethnicity). The Programme can support public awareness campaigns on tools available to secure tenure rights.	No. of local authorities trained (M/F) to secure land tenure for women, youth per country (<40% women)		MLF, RAB, KDB, MAAIF	612,818	Yes
Sub-activity 1.1.3.3: Participatory planning of tenure for dairy farms on communal and customary land at local level. In the four countries, the Programme will document and scale-up best practices including women and youth in communal rangelands participatory planning and management.	No. female and male participants per country (<40% women) Increased participation of beneficiaries in the forums (M/F)		MLF, RAB, KDB, MAAIF	212,818	Yes

Output 1.2 Improved service delivery to the dairy sector Activity 1.2.1 Strengthen veterinary services for better animal health, higher milk yields and reduced losses. In all four countries interventions in disease control (focus on prevention). Healthier animals are more resilient to climate hazards and more productive, with less milk to be discarded, leading to a reduction of “unproductive” GHG emissions. The higher revenues provide incentives for farmers to adopt animal health and welfare measures, especially if the benefits are equitably distributed among the men, women and youth in the household. To expand the outreach and capacity of veterinary services to women, the Programme will support community outreach to female farmers and train women service providers at community levels.	No. of women trained per country (<40% women) Households reporting adoption of new/improved inputs, technologies or practices		MLF, RAB, KDB, MAAIF	591,770	Yes
Sub-activity 1.2.1.1 Support the installation of both private and public veterinarians and veterinary paraprofessionals by providing travel grants. The Programme will implement the animal health support and disease control system, using the “One Health” approach in all four countries. The approach will have three levels: (i) support private veterinarians and para-veterinarians, in particular young female and male practitioners; (ii) recruit and support livestock extension workers, while ensuring gender (sex and age) equity in participation at local level; (iii) provide additional training to lead female, male and youthful farmers from the L-FFS for coaching their peers and assist private female and male para-veterinarians and livestock extension workers. In Kenya, veterinary education, and training programs to women and men from both private and public veterinarians and veterinary paraprofessionals in the Programme area will be enhanced. In Rwanda, support will be provided for the installation of young female and male veterinary doctors and veterinary technicians as private practitioners to enhance accessibility of services while creating jobs. In Tanzania, female and male Livestock Extension Workers (LEWs) will be equipped and trained, and private male and female paravets will be incentivised to ensure the last mile animal health and extension services. In Uganda, the Programme will build capacity of female and male veterinarians, para-veterinarian technicians, and community-based animal health workers (CAHWs) on One Health principles, disease diagnosis, emerging and re-emerging diseases because of CC.	Number of female trained laboratory personnel on use of the facilities in the lab and in the field Number of female technicians trained (<40%) Number of CAHWs trained Target: 40% women, 30% youth		MLF, RAB, KDB, MAAIF	859,770	Yes
Activity 1.2.2 Strengthen extension services for low-carbon and climate resilient dairy sector. In all four countries, reach out to men, women and youth dairy farmers, raise awareness and build their capacity in the field of CC mitigation and	Number of women receiving extension		MLF, RAB, KDB, MAAIF	459,770	Yes

adaptation, the Programme will adopt innovative extension models and tools (including digital), and enhance the capacity of extension services. This activity will involve improving delivery of public and private extension services at farm and cooperative/ aggregation levels, by female and male local (district/county) livestock officers who coordinate the community-based extension system. To provide proximity advisory services on climate-smart practices, the Programme will provide support through Livestock Farmer Field Schools (L-FFS), including male and female lead and youth, in mixed crop-livestock systems and Pastoralist Field Schools (PFS) in grazing and pastoralist systems.	services through LFFS (<40%) Households reporting an increase in production				
Sub-activity 1.2.2.1. Preparation of training modules for extension in climate mitigation and adaptation measures. If required, specific topics that are part of the proposed pathways under Outputs 2.2 and 2.3 and related to CC will be added to the modules of the mixed gender L-FFS (or other extension approaches). The L-FFS modules will be developed with the support of national or international technical assistance. Also gender transformative training modules on the implementation of guidelines will be developed. Trainers will be constituted by young and middle-aged women and men.	No. of gender transformative modules developed for L-FFS (Target one per country) No. of female trainers trained (<40%)		MLF, RAB, KDB, MAAIF	659,770	Yes
Sub-activity 1.2.2.2 Training of master trainers (ToTs) and facilitators of extension systems in climate-change related topics. The Programme will support the strengthening of existing L-FFS and the establishment of new L-FFS (aiming at 40 percent women and 30 percent youth). To support the sustainability and scaling up of the L-FFS, existing L-FFS/ PFS male and female master trainers and facilitators will be trained on the specific topics listed above.	Number of women TOTs trained (<40%) Households reporting adoption of environmentally sustainable and climate- resilient technologies and practices (disaggregated)		MLF, RAB, MoALD, MAAIF	579,770	Yes
Output 1.3: Improved GHG MRV capacities & monitoring of climate vulnerabilities Sub-activity 1.3.1.1: Capacity assessment and development in capturing, recording, tracking, and reporting data on dairy GHG emissions. This sub-activity will carry out in depth capacity assessments of existing MRV tools and frameworks	Number of young women trained by Mazingira (<40%)		MLF, RAB, MoALD, MAAIF	643,250	Yes

in the four countries, and the place of dairy in them. Based on these assessments, capacity development will be provided to design activity data questionnaires for Tier 2 GHG emissions in dairy (e.g. animal numbers, weights, productivity, feed rations, animal health interventions, manure management and grassland management) and develop a data management system to ensure reliable flow of information between the various national and local agencies involved in the MRV system. ILRI's Mazingira Centre will provide trainings to young women and men from the four countries on data collection protocols adjusted to their needs to relevant stakeholders (technicians, university staff, staff of national research institutes and ministries, and other relevant actors). ILRI will provide dedicated digital data collection tools (e.g. ODK), data quality control, and calculation of GHG emissions for inventory reporting, following established protocols.	Number of women (<40%) as iii. AI tech Trainees ii. Lead farmers iii. New AI technology				
Activity 1.3.4: Climate risk and vulnerability assessments. Together with improvements to extension services conducted by women and men under output 1.2, the Programme will seek to improve the accuracy and use of information (climate, feed availability) to last mile-users in the dairy sector. This will include building the capacity of devolved CC units and decentralized meteorological offices to develop and disseminate climate information services, including (i) impacts of CC on the dairy sector; and (ii) on availability of feed, through tools like national feed balance sheets.	No. of women trained in devolved CC units (<40%)		MLF, RAB, MoALD, MAAIF	34,250	Yes
Activity 1.4.2: Establish regional cooperation platforms and participation to regional/ global networks for knowledge exchange and policy dialogue on climate-smart dairy sector transformation. Sub-activity 1.4.2.2: Participation in regional and global climate policy mechanisms. In practice, the sub-activity will consist of a joint participation of the 4 countries in global and regional events (global science and policy mechanisms on dairy and climate change) such as the Greenhouse Gas and Animal Agriculture Conference – GGAA 2025, the African Climate Summit, the African Dairy Conference, and the UNFCCC COP). The objective will be to present the progress and results from the Programme for scaling-up and replication, increase its visibility and engage with other countries in sharing experience and learning. Participants include women, youth, those with disability and marginalized	Number of women and youth farmers supported to attend Oregional campaign activities (<40% women) Households reporting they can influence decision-making of local authorities (disaggregated)		MLF, RAB, MoALD, MAAIF	824,250	Yes

Sub-activity 1.4.2.3: South-South and Triangular Cooperation (SSTC). This sub-activity will support the emergence of a global network of experience and expertise on reducing GHG emissions, especially methane, in dairy production and value chains, as well as financing solutions to support these reductions. This sub-activity will finance the participation in knowledge-exchange programs, including through IFAD's "Learning Routes" with e.g. Argentina, Brazil (Embrapa), Costa Rica and Uruguay on Tier2 analysis and climate impact assessments in the dairy sector. Other initiatives for SSTC will be identified and assessed by the Regional Steering Committee in collaboration with the EAC. It will finance study tours and participation of country delegations (including women and youth) to workshops to share innovations and scalable practices generated by the Programme.	No. of women supported to take part in the South-to-South exchange learnings (<40%)		MLF, RAB, MoALD, MAAIF	1,142,500	Yes
COMPONENT 2: LOW EMISSION AND CLIMATE RESILIENT PRIMARY DAIRY PRODUCTION Output 2.1: Smallholders' and cooperative capacity development strengthened. Activity 2.1.1 Strengthen dairy cooperatives on organizational management and service delivery. This activity aims to strengthen cooperative, organizational, and technical capacities of dairy cooperatives, and raise their climate-change and gender and social inclusion awareness of their management team. Support the enrolment of women and youth dairy value chain actors in the cooperative membership and leadership. Support will also be provided to establish or enhance women only networks in MCCs and MCPs, and their business linkages.	No. of women in executive committees leadership in cooperatives (<40%)		MLF, RAB, KDB, MAAIF	520,500	Yes
Sub-activity 2.1.1.2 Strengthen capacity to reduce milk losses and ensure milk quality and safety. At the aggregation stage, up to approximately 8 percent of the milk is lost via spillage and spoilage. Aiming at reducing unproductive GHG emissions due to milk loss or quality loss, this sub-activity is to assist dairy cooperatives, MCCs and MCPs in upgrading their technical capacity in milk quality control covering raw milk testing at milk collection, hygiene, and initial processing beyond the sale of raw milk, in line with national and regional food safety standards. The support will assist to acquire (i) basic equipment for milk collection (motorcycles, milk cans and can holders, testing equipment), (ii) insulated milk transport vehicles (demonstration purpose); and (iii) training on milk quality, food safety and hygiene for informal milk vendors (often women), transporters (often men) and technical staff of cooperatives, MCCs and MCPs.	No of women supported with milk handing equipment (<40%) No. of women trained on milk quality and safety (<40%) Percentage of people/households reporting an increase in production /milk sales (M/F)		MLF, RAB, KDB, MAAIF	1,420,500	Yes

Sub-activity 2.1.1.3 Expand the outreach and capacity of dairy cooperatives, MCCs and MCPs. The targeted MCCs and MCPs will be supported to develop gender and youth responsive action plans e.g. ensure opportunities for youth employment as well as inclusive and equal employment opportunities ensuring women can work such as 'zero tolerance' to GBV and SEAH and other inclusive policies and practices. This may include trainings or technical assistance for such institutional capacity building. This activity will include trainings on the prevention of GBV and SEA	No of MCCs and MCPs supported on gender and youth approaches Beneficiaries reporting reduction of GBV (M/F)		MLF, RAB, KDB, MAAIF	920,500	Yes
Sub-activity 2.1.1.4 Create gender awareness and roll out GALS among households of dairy farmers. The GALS trainings will first train TOTs in each country and then roll out regional, community trainings. In each community GALS facilitators will be trained. The GALS facilitators will train households and identify gender champions who will undertake community outreach and sessions. Gender transformative approaches will be integrated in the Programme activities as much as possible. However specific GALS sessions and trainings for stakeholders on gender equality and women empowerment will be rolled out across the four countries. This will be done for national, regional and community level stakeholders. This will include exchange visits and learning events. Support trainings in GALS to encourage men sharing household and farm workloads and positive gender norms.	No of women and men TOTs trained on GALS (<40% women) Beneficiaries reporting reduced or more balanced workloads for women		MLF, RAB, KDB, MAAIF	2,920,500	Yes
Activity 2.1.2 Build and enhance Productive Alliances along the dairy value chain. The Programme will provide support to productive alliances between farmer cooperatives and enterprises with farmer-allied engagement models, and scaling-up of gender and socially inclusive dairy clusters. This activity will support women and youth led farmer groups to participate in the Alliances	No of women and youth led groups participating (<40% women)		MLF, RAB, KDB, MAAIF	425,500	Yes
Activity 2.1.2.3 Sub-activity 2.1.2.1 Strengthen cooperatives' capacity in business and financial management. The aim is to ensure that dairy cooperatives and their MCCs) are run as viable business enterprises. This activity will support job placements for trained young people in participating MCCs and MCPs as part of increasing their employability and creating jobs. Further, youth trainees will receive business mentorship and coaching on business and financial management through paid for placements in well performing dairy cooperatives and enterprises over an agreed period after completing their job placements.	No. of jobs/placements for youth (M/F) percentage of rural producer organizations engaged in formal partnerships/agreements or contracts with public or private entities supporting youth		MLF, RAB, KDB, MAAIF	1,920,500	Yes
Output 2.2: Breeding, feed and fodder, manure management improved.			MLF, RAB, KDB, MAAIF	923,502	Yes

Sub-activity 2.2.1.3 Scale up the insemination of dairy cattle to make the herd more productive and climate resilient. The Programme will (i) train AI technicians, who can be private inseminators, para-veterinarians, lead farmers of L-FFS and District livestock officers (at least 30 percent youth and 40 percent women); (ii) scale up the insemination of dairy cattle in the Programme intervention area; (iii) include aspects of the interaction between dairy cattle husbandry and climate change in the L-FFS curriculum. In Kenya, the Programme will support the National Breeding Strategy and Action Plan, in collaboration with the Kenya Livestock Breeders Association. Public and private partnerships (PPP) will be promoted to have AI services accessible to the male and female farmers. In Rwanda, the breeding programme of RAB will be scaled up including the training of more private AI technicians. In Tanzania, additional technicians will be trained and equipped with start-up kits. In addition, bull centers will be established in regions where AI is not available. In Uganda, more AI technicians will be trained by the NAGRC and will be equipped with AI kits focusing on the regions where AI services are still poorly developed. Campaigns and support breeders' organizations. The campaigns will target male, female and youth members of dairy cooperatives and L-FFS groups to create awareness on the use of sexed-semen, thus impacting the herd's sex ratio, and further reduce absolute GHG emissions and the emission intensity per kg of milk produced mainly by replacing many low-productive local cows with a few adapted and high performant crossbreeds, thus reducing the herd size. In the four countries, support will be given to the breeders' associations and societies for animal registration and maintenance of breeding standards.	No. of female technicians trained (<40% women) No. of women equipped with start-up kits (<40%) percentage of supported rural producers who are members of rural producer organizations reporting new or improved services provided by their organization				
Activity 2.2.2 Provide support for better animal care, herd management and disease prevention. Day to day feeding and management is mainly a female and male youth gender role in the four project countries. Women will be trained.	Number of women trained (< 40% women and 30% youth)		MLF, RAB, MoALD, MAAIF	327,123	Yes
Sub-activity 2.2.2.1 Organize trainings on animal husbandry, with a specific focus on animal health, animal welfare, herd management and disease prevention targeting women and youth. Updated modules of the L-FFS training on animal health and welfare, and herd and reproduction management will be implemented. Dairy farmers, managers and handlers (feeding and milking) will receive training to better care for the animals and to adopt biosecurity measures to prevent common diseases and pests that impact the production or suitability of the milk for human consumption. They will also receive support to better manage their herds, reproduction calendar, and animal nutrition to reduce animal mortality, heat	Number of women and youth trained (< 40% women and 30% youth)		MLF, RAB, KDB, MAAIF	525,500	Yes

stress and increase productivity, thus further reducing absolute CH4 emissions and the emission intensity per unit of milk produced.					
Sub-activity 2.2.2.2 Scale up the use of digital apps to monitor animal health and herd performance. Digital record keeping on-farm (e.g., using iCow, DigiCow Africa, My Fugo) will be scaled up to monitor the productive and health performances of the herd (e.g., records of calf mortality and causes, number of lame animals), to plan artificial insemination, and guide herd management. The use of the app will be promoted through the L-FFS in interested groups. The support includes: (i) training of L-FFS lead mixed gender farmers on the use of the apps; (ii) guiding L-FFS farmers in the use of the technology; (iii) equip the (youth) farmers in the pilot with digital tools. To implement the sub-activity, the Programme will work with agritech companies and start-up to develop digital solutions that will enhance transparency in farm data and operations, monitoring of CH4 emissions reduction and traceability of milk products.	Number of women and youth supported with digital tools (< 40% women and 30% youth)		MLF, RAB, KDB, MAAIF	1,720,500	Yes
Activity 2.2.3 Promote climate-smart forage production and conservation. The adoption of climate resilient forage varieties and conservation technologies is essential to reduce enteric methane emissions intensities in the dairy value chain, while increasing milk yields of dairy cattle. The benefits are multiple and encompass CC adaptation through adoption of a resilient forage variety (Lablab purpureus, Brachiaria sp., Cenchrus purpureus (Napier grass), Leucaena, Calliandra, Panicum maximum (Giant Panicum)), improvements in soil fertility and increased capacity of farmers, managers and handlers (feeding and milking) to meet forage production needs during dry periods and to respond to climate-induced pests and diseases. Sub-activity 2.2.3.3 Build capacity of dairy farmers on forage production and intensive dairy farms. The L-FFS will be the gender inclusive knowledge-sharing vehicle to promote improved forage production and conservation. The L-FFS will focus on (i) training on good forage management practices (e.g., grass and legume intercropping, agroforestry, weed management, (ii) training and demonstration on small-scale irrigation to produce forage crops; (iii) training and demonstrations of mechanization of forage production with fodder choppers/cutters and conservation equipment; (iv) training and demonstrations on forage conservation practices, to respond to seasonal feed availability and quality (e.g., protein content, digestibility) variations. Mixed gender Youth groups will be supported to venture into commercial fodder production, which may include production using hydroponic technology.	Number of women benefitting from climate and energy saving technologies tools (< 40%) percentage of people/households reporting a significant reduction in time spent collecting cooking fuel		MLF, RAB, KDB, MAAIF	920,500	Yes

Activity 2.2.6 Build capacity for on-farm use of biodigesters and biogas. Biogas digesters utilize microorganisms to break down organic matter such as livestock manure or other organic waste, during which biogas and a nutrient-rich by-product (called digestate or bio slurry) are produced. This technology offers a clean source of energy for people who do not have access to clean cooking fuels. In addition, it creates a biofertilizer (for use as organic fertiliser on nearby fields), reduces methane emissions (up to 70-90 percent compared to liquid manure storage) and ammonia and nitrous oxide emissions. Biodigesters can reduce deforestation and empower women by reducing the time they spent collecting firewood. As part of the low-carbon pathways, the Programme will encourage the uptake of on-farm biodigesters for intensive and semi-intensive dairy farms. Including women are the main consumers of biogas. Sub-activity 2.2.6.1 Support capacity building in biodigester management. The Programme will support the training and accreditation of biodigester technicians and consumers, with a particular focus on women and youth, in the target districts. In addition to installation-related aspects of biogas valorisation and quality management, including safe working standards.	Number of women supported with biodigesters, biogas tools (< 40%) Number of youths trained as biodigesters, biogas technicians (< 30%)		MLF, RAB, KDB, MAAIF	2,500,000	Yes
Sub-activity 2.3.2.2 Rangelands restoration. This sub-activity, focusing on communal grazing areas, will focus on: (i) reseedling and integration of legumes with grasses in natural pastures, (ii) integration of multipurpose leguminous trees and shrubs (improved agro-forestry systems) and control of invasive tree species, (iii) preservation of traditional non-forest tree products (medicinal and other) known to women mainly and (iv) implementation of soil and water conservation practices to increase water availability and biomass productivity.	Number of women supported with biodigesters, biogas tools (< 40%)		MLF, RAB, KDB, MAAIF	2,000,000	Yes
COMPONENT 3: GREEN DAIRY FINANCING FACILITY (GDFF)					
Activity 3.1.1 Fund set up. This activity aims to establish the GDFF fund as a functioning entity with the required policies, structures and staffing, and to crowd in private commercial capital in pursuit of the programme's climate impact ambitions. The GDFF staffing and structure will include provisions for gender and youth expert.	Gender and youth expert identified and in post	Annual (Yr2-6)	Regional	50,000	Yes
Sub-Activity 3.1.1.2 Investor engagement and negotiation (led by Fund Manager, in coordination with IFAD). The Fund Manager will be responsible for identifying, engaging and carrying out due diligence on prospective investors. The Fund Manager will develop criteria for screening investors demonstrate forward and backward linkages with women and youth enterprises. Expertise may be needed to support youth and women screening	Gender and youth expert identified and in post	Annual (Yr2-6)	Regional	50,000	Yes

Activity 3.1.2 Identify eligible businesses for investment. While this will focus on developing a screening framework to assess prospective investees for potential impact and financial return. This activity will ensure identification of women and youth led businesses by making gender and youth provisions in the screening framework.	Number of women and youth businesses identified (< 40% women, youth < 30%)	Annual (Yr2-6)	Regional	50,000	Yes
Sub-Activity 3.1.2.1 Develop screening framework. The Fund Manager will establish a screening framework to ensure consistency in investment decisions. This screening framework will be underpinned by a taxonomy of investment categories which are justified and eligible for investment including those that facilitate women and youth empowerment by creating jobs and employment opportunities.	No. of taxonomies suitable for women and youth	Annual (Yr2-6)	Regional	312,000	Yes
Sub-Activity 3.1.2.2 Pipeline development and screening. The pipeline for the GDFF will be sourced through three main channels; (i) GDFF Investment Advisors, (ii) DaIMA's PCUs, and (iii) TA providers under Component 2. Investment Advisors will screen prospective investees against pre-defined screening criteria, which will include women and youth considerations and quotas to ensure uptake. Undertake mobilization drives for women and youth investees	No. of women and youth qualifying for investments (< 40% women, youth < 30%)	Annual (Yr2-6)	Regional	525000	Yes
Activity 3.1.3 Climate-smart lending products and services are rolled out by GDFF to eligible businesses. These will include labour and energy saving products aimed at reducing workloads for women. The two products: Green Dairy Partnerships and Dairy Loans will ensure the partnerships promote business working with women and youth farmers. The Dairy loan tickets will be developed to ensure the ticket sizes are affordable and accessible to these groups. Undertake community outreach will be made through different media to ensure uptake by these groups	No. of women and youth accessing climate smart products and services (< 40% women, youth < 30%) percentage of people/households reporting using financial services	Annual (Yr2-6)	Regional	3,150,000	Yes
Output 3.2 Technical Assistance Facility established. The Green Dairy Financing Facility will invest in climate-resilient dairy producers, off-takers, processors, and livestock businesses. Tailored financial products with technical assistance will increase access to finance and affordable credit for on-farm investments especially for youth, women and vulnerable groups. Women and youth will receive tailored BDS and financial literacy modules that incorporate gender transformative approaches such as confidence building, communication skills and agency building. The GDFF will use a technical assistance (TA) facility to maximise both impact and returns by enhancing the capacity of investees (including on gender and youth sensitive approaches) to implement effective strategies, improve operational efficiencies and achieve growth.	No. of women and youth receiving targeted BDS. Financial literacy (< 40% women, youth < 30%) percentage of financial actors offering women and youth centric products	Annual (Yr2-6)	Regional	4,150,000	Yes

Activity 3.2.1 Prospective investees will receive targeted pre-deal TA under Output 3.2. Noting that Component 2 also provides funding to assist dairy cooperatives in acquiring renewable energy and waste management solutions (see Sub-Activity 2.1.3.2), Investment Advisors will liaise with the TA providers of Component 2 to ensure targeting of women and youth linked enterprises.	No. of women and youth linked enterprises (< 40% women, youth < 30%) Percentage of people/households reporting improved physical access to processing and milk storage facilities	Annual (Yr2-6)	Regional	550,000	Yes
Sub-Activity 3.2.1.1 Pre-Deal Development for Dairy Loan Products. These businesses will receive light touch TA to translate their investment needs, into a well-defined investment request prior to the deal. Tailored support will be developed to support women and youth friendly loan products.	No. of women and youth linked enterprises benefiting from loan products developed (< 40% women, youth < 30%)	Annual (Yr2-6)	Regional	550,000	Yes
Activity 3.2.2 Post-deal TA: Providing ongoing support post-deal to ensure successful implementation, monitor progress and ensure outcomes are met.	No. of women and youth linked enterprises (< 40% women, youth < 30%)	Annual (Yr2-6)	Regional	5,150,000	Yes
3.2.2 focuses on providing post-deal TA to Green Dairy Partnership investees, ensuring that outcomes are achieved. The type and level of support will be tailored based on the specific needs identified for each business. This TA will coach, mentor and profile good case examples of women and youth investments for learning exchanges and scale-up. Considering the Dairy Loans will be short-term, Dairy Loan investees especially women, youth and vulnerable groups may be considered for follow-ups in place of post-deal TA.	No. of women and youth products developed (< 40% women, youth < 30%)	Annual (Yr2-6)	Regional	15,150,000	Yes
Programme management and coordination DaIMA Gender and Social Inclusion expert (full time)		Yr 2-7	Regional	500,000	Yes
M and E (support Baseline, mid-term and end term) to include Empowerment indicator (for gender) and new and decent jobs indicator (youth)		Yr1. Yr2 and Yr6	Per country	4,123,320	
Grand Total				62,429,725	

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